

Modulhandbuch Computer Science Master 2025 (Master of Science (M.Sc.))

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KIT-FAKULTÄT FÜR INFORMATIK



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5.136. Privacy Enhancing Technologies - T-INFO-110989	385
5.137. Probability and Computing - T-INFO-114196	386
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5.139. Python for Computational Risk and Asset Management - T-WIWI-110213	388
5.140. Reinforcement Learning - T-INFO-111255	389
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5.142. Research Focus Class: Blockchain & Cryptocurrencies - T-INFO-113400	391
5.143. Research Focus Class: Blockchain & Cryptocurrencies - Seminar - T-INFO-113401	392
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5.146. Research Practical Course: Explainable Artificial Intelligence - T-INFO-113114	395
5.147. Research Project Deep Learning for Robotics - T-INFO-114203	396
5.148. Research Project: Generative AI for Autonomous Agents - T-INFO-114189	397
5.149. Resilient Networking - T-INFO-111209	398
5.150. Robotic Intelligence: Navigation and Mobile Manipulation - T-MACH-114952	399
5.151. Robotic Intelligence: Perception and Scene Understanding - T-MACH-114813	400
5.152. Robotics I - Introduction to Robotics - T-INFO-114190	401
5.153. Robotics II - Humanoid Robotics - T-INFO-114152	402
5.154. Robotics III - Sensors and Perception in Robotics - T-INFO-114155	403
5.155. Robotics Institute Germany - Lecture Series - T-INFO-114895	404
5.156. Sampling Methods for Machine Learning - T-INFO-114133	405
5.157. Sampling Methods for Machine Learning - Pass - T-INFO-114134	406
5.158. Scientific Methods to Design and Analyze Secure Decentralized Systems - T-INFO-111568	407
5.159. Selbstverbuchung-HOC-SPZ-FORUM-PEBA-benotet - T-INFO-111475	408
5.160. Selbstverbuchung-HOC-SPZ-FORUM-PEBA-benotet - T-INFO-115027	409
5.161. Selbstverbuchung-HOC-SPZ-FORUM-PEBA-benotet - T-INFO-111476	410
5.162. Selbstverbuchung-HOC-SPZ-FORUM-PEBA-benotet - T-INFO-111474	411
5.163. Selbstverbuchung-HOC-SPZ-FORUM-PEBA-benotet - T-INFO-115028	412
5.164. Selbstverbuchung-HOC-SPZ-FORUM-PEBA-unbenotet - T-INFO-111478	413
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5.166. Selbstverbuchung-HOC-SPZ-FORUM-PEBA-unbenotet - T-INFO-111477	415
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5.168. Selbstverbuchung-HOC-SPZ-FORUM-PEBA-unbenotet - T-INFO-115030	417
5.169. Selbstverbuchung-HOC-SPZ-FORUM-PEBA-unbenotet - T-INFO-115029	418
5.170. Seminar Dependable Computing - T-INFO-105577	419
5.171. Seminar in Privacy - T-INFO-114268	420
5.172. Seminar on Behavior Foundation Models - T-INFO-115119	421
5.173. Seminar: Advanced Topics in Natural Language Processing - T-INFO-114433	422
5.174. Seminar: Advanced Topics on SAT Solving - T-INFO-114231	423
5.175. Seminar: Algorithm Engineering - T-INFO-112312	424
5.176. Seminar: Applications and Extensions of Timed Systems - T-INFO-113132	425
5.177. Seminar: Artificial Intelligence for Energy Systems - T-INFO-113110	426
5.178. Seminar: Bionic Algorithms and Robot Technologies - T-MACH-113842	427
5.179. Seminar: Continuous Software Engineering - T-INFO-110794	428
5.180. Seminar: Critical Topics in AI - T-INFO-113915	429
5.181. Seminar: Current Trends in Theoretical Computer Science - T-INFO-114091	430
5.182. Seminar: Deep Learning for Robotics - T-INFO-114204	431
5.183. Seminar: Embedded Systems I - T-INFO-114255	432
5.184. Seminar: Embedded Systems II - T-INFO-114256	433
5.185. Seminar: Exoskeletons & Motion Capture - T-INFO-113892	434
5.186. Seminar: Explainable Artificial Intelligence - T-INFO-113115	435
5.187. Seminar: Exploring Robotics - Insights from Science Fiction, Research and Society - T-INFO-113398	436
5.188. Seminar: Failing Forward: Robotics, Critical Reading and the Making of Good Research - T-INFO-114604	437
5.189. Seminar: Fine-Grained Complexity Theory & Algorithms - T-INFO-113392	438
5.190. Seminar: Hot Topics in Artificial Intelligence & Security 1 - T-INFO-113761	439
5.191. Seminar: Hot Topics in Artificial Intelligence & Security 2 - T-INFO-113762	440
5.192. Seminar: Hot Topics in Bioinformatics - T-INFO-101287	441
5.193. Seminar: Hot Topics in Decentralized Systems - T-INFO-109922	442
5.194. Seminar: Hot Topics in Explainable Artificial Intelligence (XAI) - T-INFO-112917	443
5.195. Seminar: Information Systems - T-INFO-115055	444
5.196. Seminar: Interpretability and Causality in Machine Learning - T-INFO-114237	445
5.197. Seminar: Law and Legal Studies - T-INFO-114094	446
5.198. Seminar: Machine Learning in Climate and Environmental Sciences - T-INFO-113519	447
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5.200. Seminar: Post-Quantum Cryptography - T-INFO-111200	449

5.201. Seminar: Practical Graph Algorithms - T-INFO-114297	450
5.202. Seminar: Privacy and Security - T-INFO-114236	451
5.203. Seminar: Proofs from THE BOOK - T-INFO-106604	452
5.204. Seminar: Quantum Information Theory - T-INFO-110904	453
5.205. Seminar: Recent Highlights in Algorithms - T-INFO-114201	454
5.206. Seminar: Secure Multiparty Computation - T-INFO-111501	455
5.207. Seminar: Software Architecture, Security and Privacy - T-INFO-114260	456
5.208. Seminar: Speech-to-Speech Translation - T-INFO-114208	457
5.209. Seminar: Type Systems and Static Program Analysis - T-INFO-114955	458
5.210. Service Design Thinking - T-WIWI-102849	459
5.211. SIL Entrepreneurship Projekt - T-WIWI-110166	460
5.212. Simulation and Optimization in Robotics and Biomechanics - T-INFO-113123	461
5.213. Simulation and Optimization in Robotics and Biomechanics - Pass - T-INFO-114838	462
5.214. Software Architecture and Quality - T-INFO-114261	463
5.215. Software Engineering II - T-INFO-114259	464
5.216. Software Product Line Engineering - T-INFO-114234	465
5.217. Software Security Engineering - T-INFO-112862	466
5.218. Software-Evolution - T-INFO-101256	467
5.219. Softwaretest and Quality Management (SQM) - T-INFO-114263	468
5.220. Systems and Software Engineering - T-ETIT-100675	469
5.221. Systems Engineering for Automotive Electronics - T-ETIT-100677	470
5.222. Telematics - T-INFO-114269	471
5.223. Testing Digital Systems I - T-INFO-101388	472
5.224. Testing Digital Systems II - T-INFO-105936	473
5.225. Text-Indexing - T-INFO-114226	474
5.226. Text-Indexing Project/Experiment - T-INFO-114227	475
5.227. Theory of Satisfiability Algorithms - T-INFO-114982	476
5.228. Timed Systems - T-INFO-112754	477
5.229. Type Systems and Program Analysis - T-INFO-115016	478
5.230. Ubiquitous Computing - T-INFO-114188	479
5.231. Wearable Robotic Technologies - T-INFO-114145	480

1 Introduction

The module handbook is the official guide for students regarding their academic progress and program of study. It describes the modules and the corresponding exams. It also explains their interdependencies, for example the need to pass one module before taking another. For each module its content and the qualification goals are provided. Furthermore, the module handbook explains the means of measuring academic success, such as the type of exam or course work (in German: "Studienleistung") the students must pass. The scope of each module is indicated by credit points (CP), which are credited after the successful completion of the module. CP = ECTS (European Credit Transfer System) = Leistungspunkte in German (LP).

The course catalog (in German: "Vorlesungsverzeichnis", <https://campus.studium.kit.edu/events/catalog.php>) complements the module handbook and provides up-to-date information on various event data (e.g. time and place of the course) for each semester.

Every semester the module handbook is updated. The new module handbook is available about a month before the semester begins. Prior to that students can refer to the current module handbook but should expect changes, especially with elective modules.

The most important information regarding modules is:

- Are there prerequisites or other requirements for this module?
- How many credits are you awarded for the module?
- What term is the module held in? E.g. summer / winter term
- Which language is the module held in?
- Who is responsible for the module? E.g. which professor

The purpose of this introduction is to give an overview of the study program in computer science at KIT and provide additional regulations, not specified in the Study and Examination Regulations (in German Prüfungsordnung). It provides students with a more well-rounded idea of the requirements of the modules and their field of study. It also gives information on elective modules, study minors and interdisciplinary qualifications (= soft skills), helping the students to make a more personal plan of study, taking into consideration factors like the turns of the modules.

1.1 Program and Qualification Profile

The Master of Science in Computer Science continues the education and development of the scientific competences that the students have acquired during their Bachelor program of study. The Master Program of Study provides students with the knowledge and skills necessary for scientific work and research. The program stands out due to the variety and broad range of the courses. This Master Program of Study is structured to provide well-grounded and a broad based education at the same time. Students are required to take two graduate specializations. The program offers a wide variety of specializations in computer science areas including Theoretical Foundations, Algorithm Engineering, Cryptography and Security, Parallel Computing, Software Engineering and Compiler Construction, Design of Embedded Systems and Computer Architectures, Telematics, Anthropomatics and Cognitive Systems and Robotics and Automation.

The core focus of the program is the two graduate specializations. For each specialization, students have to select courses of at least 15 ECTS. For a comprehensive education the program offers a broad variety of compulsory courses. Students must also select a minor in a related interdisciplinary field (minor studies). Key competences such as social and teamwork skills are also mandatory (key competences).

Graduates of the Master of Science in Computer Science are equipped with essential skills:

- *Computer Science Competences (Core Competences)*
Graduates are able to independently apply and enhance their scientific knowledge and methods in computer science. They can assess the relevance and consequences of different computer science methods in solving complex scientific and social problems. Graduates have the necessary skills needed to successfully solve applied as well as scientifically complex problems in the field of computer science and related interdisciplinary fields.
- *Communication Skills*
Graduates can present and explain computer science ideas clearly and convincingly, both orally and in writing. They are able to communicate effectively to and with technical and non-technical audiences.
- *Team-work and Project Work*
Graduates are able to work in multidisciplinary teams. They have project planning and organizing skills.
- *Commitment to Society (Commitment to the Civil Society)*
Graduates recognize the impact of computer science in a societal context. They have the understanding of professional and ethical responsibilities and are able to act accordingly.
- *Personal and Professional Development*
Graduates are able to adapt to the newest technologies and use their knowledge for further development.

1.2 Additional certificates within the Master's program

The KIT Department of Informatics offers eight different certificates within the Master's program. The Master Computer Science can be completed with or without a certificate. The regulations for completing a certificate are meant as guidelines for structuring the study plans, no additional effort should be necessary. The study and examination regulations for the Master's degree program apply unchanged, when students aim to achieve a certificate. Certificates are awarded at the end of the studies in addition to the Master's certificate in computer science, e.g. "Master of Computer Science with an IT Security profile". An

overview of the certificates and the associated guidelines and regulations can be found at: <https://www.informatik.kit.edu/english/9378.php>

1.3 Modularization of computer science studies

The Computer Science program has a structure based on modules. A module may consist of several courses or only one course. Modules themselves are classified into ten areas of specialization. The module hand book only contains those areas of specialization, that can be studied completely in english. For further areas of specialization, you may refer to the german module hand book of "Master of Science in Informatik". The structure of the Master Program of Study is:

- 2 Areas of specialization
- Elective Studies in Computer Science
- Minor Studies
- Interdisciplinary Qualifications

Further constraints students must fulfill are:

- to choose four advanced mandatory modules (in German Stammmodule, see 2.1) in their areas of specialization or their elective studies
- to choose 12 – 18 CP of practical courses and seminars (see 2.2) in their areas of specialization or their elective studies.

1.3.1 Credit points

To keep track of the students' performance, study and examination achievements are evaluated with credit points (CP = ECTS European Credit Transfer System, in German Leistungspunkte = LP). One credit point corresponds to approximately 30 hours of workload for an average student.

1.3.2 Modules and partial achievements

Modules contain one or more partial achievements and their description. Partial achievements are abstract descriptions of the examinations or of the type of course work required for passing the module. Prerequisites and recommendations are also specified for each partial achievement.

Modules and partial achievements are assigned credit points. On the one hand the credits indicate the amount of work necessary in order to pass the examinations and fulfill all requirements in a module. On the other hand, the credits usually also indicate the weight of a partial achievement in a specific module. Exceptions from this rule are specified accordingly. Details on the calculation of the module grades or of the Bachelor final grade are published on the department website: <https://www.informatik.kit.edu/faq-wiki/doku.php?id=notenberechnung>.

The section field of study (see section 3) in the handbook contains the structure of the master program of study and the specific modules to be chosen in each area. From there you can navigate to each module and their partial achievements. The courses are also linked to the partial achievements, making it easier to understand modules.

Modules are dynamic constructs, in which updates and changes can occur on a regular basis. Some modules are no longer offered, some partial achievements and associated courses or prerequisites change (e.g. if the type of the exam changes). The module handbook of the current semester is therefore always binding for students. If the requirements of a module change, students usually may still complete the module in the version they started with (e.g. if they already took an exam). This means, that the exams in the modules can be taken, even if the course is not being offered anymore, provided that students have already begun the module. Changes and associated rules for changing between module versions are usually announced in advance. In case of problems with the online registration for examinations, the Informatics Study Program Service (ISS) (e-mail: beratung-informatik@informatik.kit.edu) can assist. ISS should also be contacted, if a module has been started, but can no longer be completed.

The completed module will appear in the individual study plan (in German Studienablaufplan).

1.3.3 Academic levels

The Master Program of Study in Computer Science is structured for a period of two years with two semesters each. All modules have level four, i.e. master level.

1.4 Exams

Exams can be written, oral or they can be in form of an examination of another type (in German: Prüfungsleistung anderer Art). These exams are graded. Partial achievements can also be in form of ungraded course work (in German: Studienleistung). For further information, please refer to the Study and Examination Regulations (in German: Studien- und Prüfungsordnung) §4. Each partial achievement is linked to the corresponding courses (exercise, lecture, seminar, practical course etc.) and the exam event (in German Prüfungsveranstaltung, see next section).

1.4.1 Registration for / withdraw from exams and repetition of exams

Registration and withdraw (or sometimes de-registration, in German: Abmeldung) from exams or course works take place in the Campus-Management Portal (CMS <https://campus.studium.kit.edu>). Deadlines on registration and withdraw are provided on the website of the course and in CMS. Most courses use the ILIAS E-Learning-Platform at KIT for posting and exchanging information (<https://www.zml.kit.edu/lms-ilias.php>).

In order to register for an exam (or course work) students have to access the exam event (in German Prüfungsveranstaltung). Students must ensure that they first select the module and the partial achievement in their individual study plan in CMS. An exam event specifies not only the time and place when and where an exam takes place, but also provides students with further information regarding the exam or course work as well as deadlines.

Students are encouraged to verify that they are actually registered for the exam or course work and that the status is registered (in German: angemeldet). If there are concerns regarding the registration, students should contact the ISS (e-mail: beratung-informatik@informatik.kit.edu). Participating in exams without registration is not permitted! Further information regarding registration and withdrawal can be found in <https://www.informatik.kit.edu/faq-wiki/doku.php?id=start>

Each exam (oral, written or of another type) can usually be repeated once. In the case of a written exam, after failing twice, an oral re-examination takes place promptly (usually in the same examination period). This exam can only be "passed" (4.0) or "not passed" (5.0). If an exam is not passed after the re-examination, students lose their right to study computer science. Participation in further exams is not permitted. There is the possibility to request a second re-examination, which has to be approved by the examination board. If the request is approved, further exams can be taken. However, students will not receive the credits for these exams, until the failed exam has been passed in the second re-examination. The second re-examination consists in the participation in a written exam and if this exam is not passed either, in the participation in a further oral re-examination. If the second re-examination is failed, the credits passed in between are discarded.

Course work can be repeated until they are being passed, if no further regulations are provided in the module handbook.

1.5 Student advising

The Department of Informatics at KIT offers help with questions regarding studies, applications or student organization through the Informatics Study Program Services (ISS) (e-mail: beratung-informatik@informatik.kit.edu). ISS is an official service and provides official information.

The student representative body for informatics (Fachschaft Mathe-Info, FSMI) also gives helpful and in some cases more tailored advice. They offer assistance with queries and give advice regarding your studies.

2 Structure of the Master Program of Study

To complete the masters' degree in computer science students need to complete 120 CP. The credits are mostly achieved through taking different modules but also through the master thesis (30 CP). Students are allowed to exceed the maximum of 120 CP by one single module. Credits should be spread out evenly over all semesters.

The table in Figure 1 contains an overview of the Master Program of Study.

Sem.	Area of Specialization I min. 15 CP	Area of Specialization II min. 15 CP	Elective Studies max. 49 CP	Minor Studies 9 – 18 CP	Key Competences 2 – 6 CP
1	<u>Advanced Mandatory</u> Module I Advanced Mandatory Module II Advanced Mandatory Module IV <u>Advanced Mandatory</u> Module III			Electrical Engineering/ Mathematics/ Mechanical Engineering/ Economics	General Studies / Language Courses / Soft Skills
2	Seminars (min. 3 CP) + Practical Courses (min. 6 CP) = in total min. 12 CP / max. 18 CP				
3	min. 10 CP <u>Lecture modules</u> min. 10 CP <u>Lecture modules</u> <u>Elective modules</u>				
4	Master Thesis 30 CP				

Figure 1: Structure of the Master Program of Study

During the studies at least four advanced mandatory modules (with 6 CP each) must be completed. They can be chosen in one of the specialization areas or in the elective studies area.

There are ten areas of specializations in English. This module handbook only features ten areas of specialization that can be studied completely in English. Further two specialization: Cryptography and System Architecture can be chosen by students with sufficient German language skills.

Students must choose two specialization areas. In each area of specialization students must choose modules of at least 15 CP. 10 CP must be of modules containing lectures. The advanced mandatory modules cannot be chosen to achieve 10 CP on lectures. In the specialization areas Telematics and System Architecture only 8 CP of module containing lectures are required.

A total of overall 73 CP in both specialization areas can not be exceeded.

Once exams have been taken in a specialization area, this specialization area is set. A change of the specialization area is possible with the permission of the examination board. For that students have to fill in a form (for details see <https://www.informatik.kit.edu/faq-wiki/doku.php?id=start>) and submit it to ISS via E-mail (berating-informatik@informatik.kit.edu). In the section 3 the list of modules for each specialization is listed.

This module handbook contains only modules in English. **Students may choose a maximum of 30 CP of the 120 CP in German, if they have the necessary language skills.** The german module handbook should be consulted, in order to know which german modules can be chosen in which areas (specialization, elective studies etc.) and their specific regulations regarding the exams.

2.1 Advanced mandatory modules (in German Stammmodule)

Advanced mandatory modules cover important basic themes in computer science. They assure, that students have a broad education and prepare them for the area of specialization. They are offered once each year. This cannot be guaranteed for other elective modules. Students may also choose advanced mandatory modules in German as part of the four advanced mandatory modules.

Students must take four advanced mandatory modules in their master studies. Advanced mandatory modules, that have been completed during the bachelor studies, cannot be repeated during master studies. The table in Figure 2 provides the list of advanced mandatory modules.

The advanced mandatory modules can be taken either in one of the specialization areas or in the elective studies area.

	Advanced Mandatory	CP	Term	Language
M-INFO-107201	Algorithms II	6	Winter term	English
M-INFO-100818	Computer Architecture	6	Summer term	German
M-INFO-100856	Computer Graphics	6	Winter term	German
M-INFO-100799	Formal Systems	6	Winter term	German
M-INFO-107166	Human Computer Interaction	6	Summer term	English
M-INFO-106998	IT Security	6	Winter term	English
M-INFO-107169	Machine Learning - Foundations and Algorithms	6	Summer term	Englisch
M-INFO-107676	Multimodal Artificial Intelligence	6	Summer term	English
M-INFO-107162	Robotics I- Introduction to Robotics	6	Winter term	English
M-INFO-107235	Software Engineering II	6	Summer term	English
M-INFO-107243	Telematics	6	Winter term	English

Figure 2: List of advanced mandatory modules

2.2 Seminars and practical courses in computer science

Students have to take at least 3 CP in seminars. They also have to take at least 6 CP in practical courses. A minimum of 12 CP of seminars and practical courses must be achieved in total. A maximum of 18 CP of seminars and practical courses is permitted.

Seminars and practical courses in minor studies are not subject to these constraints. These constraints regard only seminars and practical courses to be chosen in the areas of specialization and elective studies.

2.3 Elective Studies in computer science

Elective modules are not necessarily offered regularly. The current ones can be found in section 3. All modules of all specialization areas can be chosen in the elective studies area. A maximum of 49 CP is permitted (120 CP minus the minimum of credit points to be achieved in the specialization areas, the minor studies, interdisciplinary qualifications and the master thesis).

2.4 Minor Studies

To give the students a broader education, minor studies provide knowledge of an adjacent field of study. Students must choose at least 9 CP and a maximum of 18 CP. Additional credits will be discarded. Minor Studies have a significant importance for the future career, to have learned of other fields outside of the computer science core. The minor modules are listed in section 3. Some minor studies have only one module, others consist of several. Students can only choose one minor subject.

Students may also choose an individual compiled list of modules for their minor. The constraints can be read in the FAQ: <https://www.informatik.kit.edu/faq-wiki/doku.php?id=ergaenzungsfach>.

2.5 Interdisciplinary Qualifications

Another part of the studies are the interdisciplinary qualifications (2 – 6 CP). This area includes key competences and soft skills on social topics, interdisciplinary academic topics as well as foreign languages.

All courses from the House of Competence (HoC), FORUM Science and Society (except computer science or minor studies courses) and from the Language Center (SpZ), but also special courses of the informatics department, can be validated in this area. The courses are not listed in the module handbook, but can be found on the webpages of these institutions. In order to be accepted, English language certificates must demonstrate a level above B2.

These qualifications are not graded. Although some exams are graded and the partial achievement may also be listed in the individual study plan with a grade, the interdisciplinary qualifications can only be passed/failed, so these grades does not contribute to the overall grade of the Master program.

Participation certificates (in German: Teilnahmebescheinigungen) cannot not be validated as interdisciplinary qualifications. An exam or coursework must be done.

2.6 Additional modules (in German: Zusatzleistungen)

During the master studies, students may take additional courses (max. 30 CP). These achievements are not included in the overall grade. If the corresponding partial achievements cannot be selected in the individual study plan, students should contact ISS.

3 Aufbau des Studiengangs

Wahlinformationen

Please note that two specialization subjects and one minor subject must be taken in the Master's program. You can select these in your study plan when clicking on the elective button "Select areas" next to the study program identifier Informatics / Computer Science Master 2025.

Pflichtbestandteile	
Master's Thesis	30 LP
Areas of Specialization (Wahl: 2 Bestandteile)	
Area of Specialization: Algorithm Engineering	15-73 LP
Area of Specialization: Cryptography and Security	15-73 LP
Area of Specialization: Data Science	15-73 LP
Area of Specialization: Design of Embedded Systems and Computer Architectures	15-73 LP
Area of Specialization: Human-centred Machine Intelligence	15-73 LP
Area of Specialization: Robotics and Automation	15-73 LP
Area of Specialization: Software Engineering and Compiler Construction	15-73 LP
Area of Specialization: Telematics	15-73 LP
Area of Specialization: Theoretical Foundations	15-73 LP
Pflichtbestandteile	
Elective Studies in Informatics	6-49 LP
Minor Studies (Wahl: 1 Bestandteil)	
Minor Studies: Biologically Inspired Robotics	9-18 LP
Minor Studies: Electrical Engineering	9-18 LP
Minor Studies: Mathematics	9-18 LP
Minor Studies: Economics	9-18 LP
Minor Studies: Law	9-18 LP
Pflichtbestandteile	
Interdisciplinary Qualifications	2-6 LP

3.1 Master's Thesis

Leistungspunkte
30

Pflichtbestandteile				
M-INFO-106828	Master's Thesis	EN	WS+SS	30 LP

3.2 Area of Specialization: Algorithm Engineering**Leistungspunkte**
15-73

Specialization Coordinator: Prof. P. Sanders

Wahlinformationen

Students must choose at least 10 CP of lectures (no practical courses, no seminars, no advanced mandatory courses).
In total students must choose at least 15 CP in each specialization.

Elective Modules: Algorithm Engineering (Wahl: zwischen 15 und 73 LP)				
M-INFO-107200	Advanced Data Structures	EN	SS	5 LP
M-INFO-100795	Algorithm Engineering	EN	SS	5 LP
M-INFO-106960	Algorithmic Graph Theory	EN	Unregelm.	5 LP
M-INFO-106961	Algorithms for Visualization of Graphs	EN	Unregelm.	5 LP
M-INFO-107201	Algorithms II	EN	WS	6 LP
M-INFO-104447	Automated Planning and Scheduling	EN	Unregelm.	5 LP
M-INFO-107599	Beating the Worst-Case in Practice: Unexpectedly Efficient Algorithms	EN	Unregelm.	6 LP
M-INFO-107228	Computational Geometry	EN	Unregelm.	6 LP
M-INFO-106644	Fine-Grained Complexity Theory & Algorithms	EN	Unregelm.	6 LP
M-INFO-107211	Graph Partitioning and Graph Clustering in Theory and Practice	EN	SS	5 LP
M-INFO-101573	Hands-on Bioinformatics Practical	EN	Unregelm.	3 LP
M-INFO-100749	Introduction to Bioinformatics for Computer Scientists	EN	WS	3 LP
M-INFO-107199	Parallel Algorithms	EN	WS	5 LP
M-INFO-107167	Parameterized Algorithms	EN	Unregelm.	6 LP
M-INFO-107203	Practical Course: Efficient Parallel C++	EN	Unregelm.	6 LP
M-INFO-106784	Practical Course: Fine-grained Algorithm Design and Engineering	EN	Unregelm.	6 LP
M-INFO-107600	Practical Course: Route Planning	EN	Unregelm.	6 LP
M-INFO-107168	Probability and Computing	EN	WS	5 LP
M-INFO-107209	Seminar: Advanced Topics on SAT Solving	EN	WS	3 LP
M-INFO-106086	Seminar: Algorithm Engineering	EN	Unregelm.	4 LP
M-INFO-107027	Seminar: Current Trends in Theoretical Computer Science	EN	Unregelm.	4 LP
M-INFO-106645	Seminar: Fine-Grained Complexity Theory & Algorithms	EN	Unregelm.	4 LP
M-INFO-100750	Seminar: Hot Topics in Bioinformatics	EN	SS	3 LP
M-INFO-107264	Seminar: Practical Graph Algorithms	EN	Unregelm.	4 LP
M-INFO-107172	Seminar: Recent Highlights in Algorithms	EN	Unregelm.	4 LP
M-INFO-107202	Text Indexing	EN	WS	5 LP
M-INFO-107639	Theory of Satisfiability Algorithms neu	EN	Unregelm.	4 LP

3.3 Area of Specialization: Cryptography and Security**Leistungspunkte**
15-73

Specialization Coordinator: Prof. J. Müller-Quade

Wahlinformationen

Students must choose at least 10 CP of lectures (no practical courses, no seminars, no advanced mandatory courses).
In total students must choose at least 15 CP in each specialization.

Elective Modules: Cryptography and Security (Wahl: zwischen 15 und 73 LP)				
M-INFO-106303	Access Control Systems: Models and Technology	EN	SS	5 LP
M-INFO-106810	Artificial Intelligence & IT-Security	EN	WS	6 LP
M-INFO-107435	Cryptographic Protocols II: Universal Composability	EN	WS	3 LP
M-INFO-105334	Decentralized Systems: Fundamentals, Modeling, and Applications	EN	SS	6 LP
M-INFO-106998	IT Security	EN	WS	6 LP
M-INFO-107218	Network Security: Architectures and Protocols	EN	SS	4 LP
M-INFO-107642	Post-quantum Cryptography from Lattices neu	EN	Unregelm.	3 LP
M-INFO-106996	Practical Course: Application Security	EN	WS+SS	4 LP
M-INFO-106867	Practical Course: Artificial Intelligence & Security Lab (AISEC-Lab)	EN	SS	4 LP
M-INFO-107412	Practical Course: Building PETs	EN	WS	4 LP
M-INFO-106627	Practical Course: Real-world Vulnerability Discovery and Exploits	EN	WS+SS	4 LP
M-INFO-105453	Practical Course: Security, Usability and Society	EN	Unregelm.	4 LP
M-INFO-107241	Practical Introduction to Hardware Security	EN	WS+SS	6 LP
M-INFO-105452	Privacy Enhancing Technologies	EN	SS	6 LP
M-INFO-106654	Research Focus Class: Blockchain & Cryptocurrencies	EN	Unregelm.	6 LP
M-INFO-106866	Research Practical Course: Artificial Intelligence & Security	EN	WS+SS	6 LP
M-INFO-105591	Resilient Networking	EN	WS	6 LP
M-INFO-105780	Scientific Methods to Design and Analyze Secure Decentralized Systems	EN	WS	5 LP
M-INFO-107242	Seminar in Privacy	EN	SS	4 LP
M-INFO-106868	Seminar: Hot Topics in Artificial Intelligence & Security 1	EN	WS	4 LP
M-INFO-106869	Seminar: Hot Topics in Artificial Intelligence & Security 2	EN	SS	4 LP
M-INFO-104891	Seminar: Hot Topics in Decentralized Systems	EN	Unregelm.	3 LP
M-INFO-106392	Seminar: Hot Topics in Explainable Artificial Intelligence (XAI)	EN	SS	4 LP
M-INFO-105585	Seminar: Post-Quantum Cryptography	EN	Unregelm.	3 LP
M-INFO-107216	Seminar: Privacy and Security	EN	WS	4 LP
M-INFO-105408	Seminar: Quantum Information Theory	EN	Unregelm.	3 LP
M-INFO-105761	Seminar: Secure Multiparty Computation	EN	Unregelm.	3 LP
M-INFO-106344	Software Security Engineering	EN	SS	3 LP

3.4 Area of Specialization: Data Science**Leistungspunkte**
15-73

Specialization Coordinators: Prof. K. Böhm, Prof. G. Neumann

Wahlinformationen

Students must choose at least 10 CP of lectures (no practical courses, no seminars, no advanced mandatory courses).
In total students must choose at least 15 CP in each specialization.

Elective Modules: Data Science (Wahl: zwischen 15 und 73 LP)				
M-INFO-106812	Advanced Bayesian Data Analysis	EN	WS	5 LP
M-INFO-106655	Data Science and Artificial Intelligence for Energy Systems	EN	SS	6 LP
M-INFO-106302	Explainable Artificial Intelligence	EN	SS	3 LP
M-INFO-106959	Machine Learning for Natural Sciences	EN	SS	6 LP
M-INFO-106470	Machine Learning in Climate and Environmental Sciences	EN	SS	6 LP

3.5 Area of Specialization: Design of Embedded Systems and Computer Architectures**Leistungspunkte**
15-73

Specialization Coordinators: Prof. J. Henkel, Prof. W. Karl

Wahlinformationen

Students must choose at least 10 CP of lectures (no practical courses, no seminars, no advanced mandatory courses).
In total students must choose at least 15 CP in each specialization.

Elective Modules: Design of Embedded Systems and Computer Architectures (Wahl: zwischen 15 und 73 LP)				
M-INFO-107230	Design and Architectures of Embedded Systems (ESII)	EN	WS	3 LP
M-INFO-105775	Embedded Machine Learning Lab	EN	WS+SS	4 LP
M-INFO-100807	Low Power Design	EN	SS	3 LP
M-INFO-107229	Optimization and Synthesis of Embedded Systems (ESI)	EN	WS	3 LP
M-INFO-103706	Practical Course: Internet of Things (IoT)	EN	WS+SS	4 LP
M-INFO-107265	Practical Course: Chip Design I	EN	WS+SS	3 LP
M-INFO-107266	Practical Course: Chip Design II	EN	WS+SS	3 LP
M-INFO-102570	Practical Course: Digital Design & Test Automation Flow	EN	WS+SS	3 LP
M-INFO-102661	Practical Course: FPGA Programming	EN	WS+SS	3 LP
M-INFO-104031	Practical Course: Low Power Design and Embedded Systems	EN	WS+SS	4 LP
M-INFO-107241	Practical Introduction to Hardware Security	EN	WS+SS	6 LP
M-INFO-100850	Reliable Computing I	EN	Unregelm.	3 LP
M-INFO-102662	Seminar Dependable Computing	EN	WS+SS	3 LP
M-INFO-107231	Seminar: Embedded Systems I	EN	WS+SS	3 LP
M-INFO-107232	Seminar: Embedded Systems II	EN	WS+SS	3 LP
M-INFO-100851	Testing Digital Systems I	EN	Unregelm.	3 LP
M-INFO-102962	Testing Digital Systems II	EN	SS	3 LP
M-INFO-107448	Hardware-efficient Artificial Intelligence	EN	WS	3 LP

3.6 Area of Specialization: Human-centred Machine Intelligence**Leistungspunkte**
15-73

Specialization Coordinator: Prof. R. Stiefelhagen

Wahlinformationen

Students must choose at least 10 CP of lectures (no practical courses, no seminars, no advanced mandatory courses).
In total students must choose at least 15 CP in each specialization.

Elective Modules: Human-centred Machine Intelligence (Wahl: zwischen 15 und 73 LP)				
M-INFO-106810	Artificial Intelligence & IT-Security	EN	WS	6 LP
M-INFO-106655	Data Science and Artificial Intelligence for Energy Systems	EN	SS	6 LP
M-INFO-107197	Deep Learning and Neural Networks	EN	SS	6 LP
M-INFO-105775	Embedded Machine Learning Lab	EN	WS+SS	4 LP
M-INFO-106302	Explainable Artificial Intelligence	EN	SS	3 LP
M-INFO-106237	Geometric Deep Learning	EN	WS	6 LP
M-INFO-106650	HRI and Social Robotics	EN	SS	6 LP
M-INFO-107166	Human Computer Interaction	EN	SS	6 LP
M-INFO-106649	Humanoid Robots - Locomotion and Whole-Body Control	EN	SS	6 LP
M-INFO-107152	Humanoid Robots - Seminar	EN	WS	3 LP
M-INFO-107176	Lab Project: Speech Translation	EN	WS	6 LP
M-INFO-107663	Localization of Mobile Agents neu	EN	SS	6 LP
M-INFO-107169	Machine Learning - Foundations and Algorithms	EN	SS	6 LP
M-INFO-106959	Machine Learning for Natural Sciences	EN	SS	6 LP
M-INFO-106470	Machine Learning in Climate and Environmental Sciences	EN	SS	6 LP
M-INFO-102555	Motion in Human and Machine - Seminar	EN	SS	3 LP
M-INFO-107676	Multimodal Artificial Intelligence neu	EN	SS	6 LP
M-INFO-107178	Natural Language Processing	EN	WS	6 LP
M-INFO-105329	Optimization Methods for Machine Learning and Engineering	EN	WS	5 LP
M-INFO-107170	Participatory Technology Design	EN	WS	6 LP
M-INFO-106867	Practical Course: Artificial Intelligence & Security Lab (AISEC-Lab)	EN	SS	4 LP
M-INFO-106646	Practical Course: Human-Centred Robotics	EN	WS+SS	6 LP
M-INFO-107609	Practical Course: Mathematical and Computational Methods in Robotics and AI	EN	WS+SS	6 LP
M-INFO-106648	Practical Course: Movement and Technology	EN	SS	6 LP
M-INFO-107177	Practical Course: Natural Language Dialog Systems	EN	Unregelm.	6 LP
M-INFO-105623	Reinforcement Learning	EN	WS	6 LP
M-INFO-107728	Research Practical Course on Behavior Foundation Models neu	EN	SS	6 LP
M-INFO-106866	Research Practical Course: Artificial Intelligence & Security	EN	WS+SS	6 LP
M-INFO-106495	Research Practical Course: Explainable Artificial Intelligence neu	EN	WS	6 LP
M-INFO-107174	Research Project Deep Learning for Robotics	EN	WS	6 LP
M-INFO-107163	Research Project: Generative AI for Autonomous Agents	EN	SS	6 LP
M-INFO-107155	Robotics - Practical Course	EN	SS	6 LP
M-INFO-107123	Robotics II - Humanoid Robotics	EN	SS	3 LP
M-INFO-107090	Sampling Methods for Machine Learning	EN	SS	6 LP
M-INFO-107729	Seminar on Behavior Foundation Models neu	EN	SS	3 LP
M-INFO-107353	Seminar: Advanced Topics in Natural Language Processing	EN	WS+SS	3 LP
M-INFO-106490	Seminar: Artificial Intelligence for Energy Systems	EN	WS	4 LP
M-INFO-106958	Seminar: Critical Topics in AI	EN	SS	3 LP
M-INFO-107175	Seminar: Deep Learning for Robotics	EN	WS	3 LP
M-INFO-106497	Seminar: Explainable Artificial Intelligence neu	EN	WS	3 LP
M-INFO-106651	Seminar: Exploring Robotics - Insights from Science Fiction, Research and Society	EN	SS	3 LP
M-INFO-107436	Seminar: Failing Forward: Robotics, Critical Reading and the Making of Good Research	EN	WS	3 LP
M-INFO-106868	Seminar: Hot Topics in Artificial Intelligence & Security 1	EN	WS	4 LP
M-INFO-106869	Seminar: Hot Topics in Artificial Intelligence & Security 2	EN	SS	4 LP
M-INFO-106392	Seminar: Hot Topics in Explainable Artificial Intelligence (XAI)	EN	SS	4 LP
M-INFO-107217	Seminar: Interpretability and Causality in Machine Learning	EN	SS	3 LP
M-INFO-106719	Seminar: Machine Learning in Climate and Environmental Sciences	EN	Unregelm.	3 LP

M-INFO-107179	Seminar: Speech-to-Speech Translation	EN	SS	3 LP
M-INFO-107113	Wearable Robotic Technologies	EN	SS	4 LP

3.7 Area of Specialization: Robotics and Automation

Leistungspunkte
15-73

Specialization Coordinator: Prof. T. Asfour

Wahlinformationen

Students must choose at least 10 CP of lectures (no practical courses, no seminars, no advanced mandatory courses). In total students must choose at least 15 CP in each specialization.

Elective Modules: Robotics and Automation (Wahl: zwischen 15 und 73 LP)				
M-INFO-106608	Autonomous Learning for Intelligent Robot Perception	EN	WS	4 LP
M-INFO-106190	Computational Imaging	EN	WS	5 LP
M-INFO-106650	HRI and Social Robotics	EN	SS	6 LP
M-INFO-106649	Humanoid Robots - Locomotion and Whole-Body Control	EN	SS	6 LP
M-INFO-107152	Humanoid Robots - Seminar	EN	WS	3 LP
M-INFO-107663	Localization of Mobile Agents neu	EN	SS	6 LP
M-INFO-106959	Machine Learning for Natural Sciences	EN	SS	6 LP
M-INFO-106646	Practical Course: Human-Centred Robotics	EN	WS+SS	6 LP
M-INFO-107609	Practical Course: Mathematical and Computational Methods in Robotics and AI	EN	WS+SS	6 LP
M-INFO-106648	Practical Course: Movement and Technology	EN	SS	6 LP
M-INFO-107174	Research Project Deep Learning for Robotics	EN	WS	6 LP
M-INFO-107155	Robotics - Practical Course	EN	SS	6 LP
M-INFO-107162	Robotics I - Introduction to Robotics	EN	WS	6 LP
M-INFO-107123	Robotics II - Humanoid Robotics	EN	SS	3 LP
M-INFO-107130	Robotics III - Sensors and Perception in Robotics	EN	SS	3 LP
M-INFO-107598	Robotics Institute Germany - Lecture Series	EN	WS	3 LP
M-INFO-107090	Sampling Methods for Machine Learning	EN	SS	6 LP
M-INFO-107175	Seminar: Deep Learning for Robotics	EN	WS	3 LP
M-INFO-106927	Seminar: Exoskeletons & Motion Capture	EN	WS	3 LP
M-INFO-106651	Seminar: Exploring Robotics - Insights from Science Fiction, Research and Society	EN	SS	3 LP
M-INFO-107436	Seminar: Failing Forward: Robotics, Critical Reading and the Making of Good Research	EN	WS	3 LP
M-INFO-107113	Wearable Robotic Technologies	EN	SS	4 LP

3.8 Area of Specialization: Software Engineering and Compiler Construction

Leistungspunkte
15-73

Specialization Coordinators: Prof. A. Koziol, Prof. R. Reussner

Wahlinformationen

Students must choose at least 10 CP of lectures (no practical courses, no seminars, no advanced mandatory courses). In total students must choose at least 15 CP in each specialization.

Elective Modules: Software Engineering and Compiler Construction (Wahl: zwischen 15 und 73 LP)				
M-INFO-106966	Compiler Design	EN	Unregelm.	9 LP
M-INFO-107685	Empirical Software Engineering neu	EN	Unregelm.	4 LP
M-INFO-106626	Engineering Self-Adaptive Systems	EN	WS	3 LP
M-INFO-106102	Logical Foundations of Cyber-Physical Systems	EN	WS	6 LP
M-INFO-106931	Model-Driven Software Development	EN	WS	3 LP
M-INFO-107677	Practical Course: Continuous Software Engineering neu	EN	WS+SS	6 LP
M-INFO-107203	Practical Course: Efficient Parallel C++	EN	Unregelm.	6 LP
M-INFO-107678	Practical Course: Engineering Approaches to Software Development neu	EN	WS+SS	6 LP
M-INFO-106932	Practical Course: Model-Driven Software Development	EN	SS	6 LP
M-INFO-106512	Seminar: Applications and Extensions of Timed Systems	EN	Unregelm.	4 LP
M-INFO-105309	Seminar: Continuous Software Engineering	EN	WS	4 LP
M-INFO-107236	Seminar: Software Architecture, Security and Privacy	EN	Unregelm.	4 LP
M-INFO-107625	Seminar: Type Systems and Static Program Analysis	EN	WS	4 LP
M-INFO-107237	Software Architecture and Quality	EN	WS	3 LP
M-INFO-107235	Software Engineering II	EN	SS	6 LP
M-INFO-107212	Software Product Line Engineering	EN	SS	3 LP
M-INFO-106344	Software Security Engineering	EN	SS	3 LP
M-INFO-100719	Software-Evolution neu	EN	WS	3 LP
M-INFO-107239	Softwaretest and Quality Management (SQM)	EN	WS	5 LP
M-INFO-106293	Timed Systems	EN	SS	6 LP
M-INFO-107664	Type Systems and Program Analysis neu	EN	Unregelm.	6 LP

3.9 Area of Specialization: Telematics**Leistungspunkte**
15-73

Specialization Coordinators: Prof. S. Abeck, Prof. H. Hartenstein, Prof. M. Zitterbart

Wahlinformationen

Students must choose at least 8 CP of lectures (no practical courses, no seminars, no advanced mandatory courses). In total students must choose at least 15 CP in each specialization.

Elective Modules: Telematics (Wahl: zwischen 15 und 73 LP)				
M-INFO-106303	Access Control Systems: Models and Technology	EN	SS	5 LP
M-INFO-106655	Data Science and Artificial Intelligence for Energy Systems	EN	SS	6 LP
M-INFO-105334	Decentralized Systems: Fundamentals, Modeling, and Applications	EN	SS	6 LP
M-INFO-107215	Distributed Computing	EN	WS	4 LP
M-INFO-107244	Practical Course on Network Security Research	EN	Unregelm.	3 LP
M-INFO-107166	Human Computer Interaction	EN	SS	6 LP
M-INFO-100800	Internet of Everything	EN	WS	4 LP
M-INFO-107245	Mobile Communication	EN	WS	4 LP
M-INFO-107218	Network Security: Architectures and Protocols	EN	SS	4 LP
M-INFO-100784	Next Generation Internet	EN	SS	4 LP
M-INFO-107220	Practical Course on Telematics Research	EN	Unregelm.	3 LP
M-INFO-105870	Practical Course: Advanced Topics in High Performance Computing, Data Management and Analytics	EN	WS+SS	6 LP
M-INFO-105453	Practical Course: Security, Usability and Society	EN	Unregelm.	4 LP
M-INFO-105955	Practical Course: Smart Energy System	EN	WS+SS	6 LP
M-INFO-107221	Practical Course: Software Defined Networking	EN	SS	6 LP
M-INFO-105452	Privacy Enhancing Technologies	EN	SS	6 LP
M-INFO-106654	Research Focus Class: Blockchain & Cryptocurrencies	EN	Unregelm.	6 LP
M-INFO-105591	Resilient Networking	EN	WS	6 LP
M-INFO-105780	Scientific Methods to Design and Analyze Secure Decentralized Systems	EN	WS	5 LP
M-INFO-106490	Seminar: Artificial Intelligence for Energy Systems	EN	WS	4 LP
M-INFO-104891	Seminar: Hot Topics in Decentralized Systems	EN	Unregelm.	3 LP
M-INFO-107243	Telematics	EN	WS	6 LP
M-INFO-107161	Ubiquitous Computing	EN	Unregelm.	5 LP

3.10 Area of Specialization: Theoretical Foundations**Leistungspunkte**
15-73

Specialization Coordinators: Prof. P. Sanders, Prof. S. Erdweg

Wahlinformationen

Students must choose at least 10 CP of lectures (no practical courses, no seminars, no advanced mandatory courses).
In total students must choose at least 15 CP in each specialization.

Elective Modules: Theoretical Foundations (Wahl: zwischen 15 und 73 LP)				
M-INFO-107200	Advanced Data Structures	EN	SS	5 LP
M-INFO-106960	Algorithmic Graph Theory	EN	Unregelm.	5 LP
M-INFO-106961	Algorithms for Visualization of Graphs	EN	Unregelm.	5 LP
M-INFO-107201	Algorithms II	EN	WS	6 LP
M-INFO-104447	Automated Planning and Scheduling	EN	Unregelm.	5 LP
M-INFO-107599	Beating the Worst-Case in Practice: Unexpectedly Efficient Algorithms	EN	Unregelm.	6 LP
M-INFO-107228	Computational Geometry	EN	Unregelm.	6 LP
M-INFO-106256	Constructive Logic	EN	SS	5 LP
M-INFO-106644	Fine-Grained Complexity Theory & Algorithms	EN	Unregelm.	6 LP
M-INFO-107211	Graph Partitioning and Graph Clustering in Theory and Practice	EN	SS	5 LP
M-INFO-106102	Logical Foundations of Cyber-Physical Systems	EN	WS	6 LP
M-INFO-107167	Parameterized Algorithms	EN	Unregelm.	6 LP
M-INFO-106784	Practical Course: Fine-grained Algorithm Design and Engineering	EN	Unregelm.	6 LP
M-INFO-107494	Practical Course: Interactive Theorem Proving	EN	Unregelm.	3 LP
M-INFO-107238	Practical SAT Solving	EN	Unregelm.	5 LP
M-INFO-107168	Probability and Computing	EN	WS	5 LP
M-INFO-107209	Seminar: Advanced Topics on SAT Solving	EN	WS	3 LP
M-INFO-106086	Seminar: Algorithm Engineering	EN	Unregelm.	4 LP
M-INFO-106512	Seminar: Applications and Extensions of Timed Systems	EN	Unregelm.	4 LP
M-INFO-107027	Seminar: Current Trends in Theoretical Computer Science	EN	Unregelm.	4 LP
M-INFO-106645	Seminar: Fine-Grained Complexity Theory & Algorithms	EN	Unregelm.	4 LP
M-INFO-103306	Seminar: Proofs from THE BOOK	EN	Unregelm.	3 LP
M-INFO-107172	Seminar: Recent Highlights in Algorithms	EN	Unregelm.	4 LP
M-INFO-107625	Seminar: Type Systems and Static Program Analysis	EN	WS	4 LP
M-INFO-107639	Theory of Satisfiability Algorithms neu	EN	Unregelm.	4 LP
M-INFO-106293	Timed Systems	EN	SS	6 LP
M-INFO-107664	Type Systems and Program Analysis neu	EN	Unregelm.	6 LP

3.11 Elective Studies in Informatics

Leistungspunkte
6-49

Elective Studies (Wahl: zwischen 6 und 49 LP)				
M-INFO-106303	Access Control Systems: Models and Technology	EN	SS	5 LP
M-INFO-106812	Advanced Bayesian Data Analysis	EN	WS	5 LP
M-INFO-107200	Advanced Data Structures	EN	SS	5 LP
M-INFO-100795	Algorithm Engineering	EN	SS	5 LP
M-INFO-106960	Algorithmic Graph Theory	EN	Unregelm.	5 LP
M-INFO-106961	Algorithms for Visualization of Graphs	EN	Unregelm.	5 LP
M-INFO-107201	Algorithms II	EN	WS	6 LP
M-INFO-106810	Artificial Intelligence & IT-Security	EN	WS	6 LP
M-INFO-104447	Automated Planning and Scheduling	EN	Unregelm.	5 LP
M-INFO-106608	Autonomous Learning for Intelligent Robot Perception	EN	WS	4 LP
M-INFO-107599	Beating the Worst-Case in Practice: Unexpectedly Efficient Algorithms	EN	Unregelm.	6 LP
M-INFO-106966	Compiler Design	EN	Unregelm.	9 LP
M-INFO-107228	Computational Geometry	EN	Unregelm.	6 LP
M-INFO-106190	Computational Imaging	EN	WS	5 LP
M-INFO-106256	Constructive Logic	EN	SS	5 LP
M-INFO-107435	Cryptographic Protocols II: Universal Composability	EN	WS	3 LP
M-INFO-106655	Data Science and Artificial Intelligence for Energy Systems	EN	SS	6 LP
M-INFO-105334	Decentralized Systems: Fundamentals, Modeling, and Applications	EN	SS	6 LP
M-INFO-107197	Deep Learning and Neural Networks	EN	SS	6 LP
M-INFO-107230	Design and Architectures of Embedded Systems (ESII)	EN	WS	3 LP
M-INFO-107215	Distributed Computing	EN	WS	4 LP
M-INFO-107234	Edge-AI in Software and Sensor Applications	EN	WS+SS	3 LP
M-INFO-105775	Embedded Machine Learning Lab	EN	WS+SS	4 LP
M-INFO-107685	Empirical Software Engineering neu	EN	Unregelm.	4 LP
M-INFO-106626	Engineering Self-Adaptive Systems	EN	WS	3 LP
M-INFO-106302	Explainable Artificial Intelligence	EN	SS	3 LP
M-INFO-106644	Fine-Grained Complexity Theory & Algorithms	EN	Unregelm.	6 LP
M-INFO-106237	Geometric Deep Learning	EN	WS	6 LP
M-INFO-107211	Graph Partitioning and Graph Clustering in Theory and Practice	EN	SS	5 LP
M-INFO-107448	Hardware-efficient Artificial Intelligence	EN	WS	3 LP
M-INFO-106650	HRI and Social Robotics	EN	SS	6 LP
M-INFO-107166	Human Computer Interaction	EN	SS	6 LP
M-INFO-106649	Humanoid Robots - Locomotion and Whole-Body Control	EN	SS	6 LP
M-INFO-107152	Humanoid Robots - Seminar	EN	WS	3 LP
M-INFO-100800	Internet of Everything	EN	WS	4 LP
M-INFO-100749	Introduction to Bioinformatics for Computer Scientists	EN	WS	3 LP
M-INFO-106998	IT Security	EN	WS	6 LP
M-INFO-107176	Lab Project: Speech Translation	EN	WS	6 LP
M-INFO-107663	Localization of Mobile Agents neu	EN	SS	6 LP
M-INFO-106102	Logical Foundations of Cyber-Physical Systems	EN	WS	6 LP
M-INFO-100807	Low Power Design	EN	SS	3 LP
M-INFO-107169	Machine Learning - Foundations and Algorithms	EN	SS	6 LP
M-INFO-106959	Machine Learning for Natural Sciences	EN	SS	6 LP
M-INFO-106470	Machine Learning in Climate and Environmental Sciences	EN	SS	6 LP
M-INFO-107245	Mobile Communication	EN	WS	4 LP
M-INFO-106931	Model-Driven Software Development	EN	WS	3 LP
M-INFO-107676	Multimodal Artificial Intelligence neu	EN	SS	6 LP
M-INFO-107178	Natural Language Processing	EN	WS	6 LP
M-INFO-107233	Natural Language Processing and Software Engineering	EN	WS	3 LP
M-INFO-107218	Network Security: Architectures and Protocols	EN	SS	4 LP

M-INFO-100784	Next Generation Internet	EN	SS	4 LP
M-INFO-107229	Optimization and Synthesis of Embedded Systems (ESI)	EN	WS	3 LP
M-INFO-105329	Optimization Methods for Machine Learning and Engineering	EN	WS	5 LP
M-INFO-107199	Parallel Algorithms	EN	WS	5 LP
M-INFO-107167	Parameterized Algorithms	EN	Unregelm.	6 LP
M-INFO-107170	Participatory Technology Design	EN	WS	6 LP
M-INFO-107642	Post-quantum Cryptography from Lattices neu	EN	Unregelm.	3 LP
M-INFO-107244	Practical Course on Network Security Research	EN	Unregelm.	3 LP
M-INFO-107220	Practical Course on Telematics Research	EN	Unregelm.	3 LP
M-INFO-103706	Practical Course: Internet of Things (IoT)	EN	WS+SS	4 LP
M-INFO-105870	Practical Course: Advanced Topics in High Performance Computing, Data Management and Analytics	EN	WS+SS	6 LP
M-INFO-106800	Practical Course: AI for Climate and Weather Predictions	EN	Unregelm.	6 LP
M-INFO-106996	Practical Course: Application Security	EN	WS+SS	4 LP
M-INFO-106867	Practical Course: Artificial Intelligence & Security Lab (AISEC-Lab)	EN	SS	4 LP
M-INFO-107412	Practical Course: Building PETs	EN	WS	4 LP
M-INFO-107265	Practical Course: Chip Design I	EN	WS+SS	3 LP
M-INFO-107266	Practical Course: Chip Design II	EN	WS+SS	3 LP
M-INFO-107677	Practical Course: Continuous Software Engineering neu	EN	WS+SS	6 LP
M-INFO-102570	Practical Course: Digital Design & Test Automation Flow	EN	WS+SS	3 LP
M-INFO-107203	Practical Course: Efficient Parallel C++	EN	Unregelm.	6 LP
M-INFO-107678	Practical Course: Engineering Approaches to Software Development neu	EN	WS+SS	6 LP
M-INFO-106784	Practical Course: Fine-grained Algorithm Design and Engineering	EN	Unregelm.	6 LP
M-INFO-102661	Practical Course: FPGA Programming	EN	WS+SS	3 LP
M-INFO-100724	Practical Course: General-Purpose Computation on Graphics Processing Units	EN	WS+SS	3 LP
M-INFO-106646	Practical Course: Human-Centred Robotics	EN	WS+SS	6 LP
M-INFO-107494	Practical Course: Interactive Theorem Proving	EN	Unregelm.	3 LP
M-INFO-104031	Practical Course: Low Power Design and Embedded Systems	EN	WS+SS	4 LP
M-INFO-107609	Practical Course: Mathematical and Computational Methods in Robotics and AI	EN	WS+SS	6 LP
M-INFO-106932	Practical Course: Model-Driven Software Development	EN	SS	6 LP
M-INFO-106648	Practical Course: Movement and Technology	EN	SS	6 LP
M-INFO-107177	Practical Course: Natural Language Dialog Systems	EN	Unregelm.	6 LP
M-INFO-106627	Practical Course: Real-world Vulnerability Discovery and Exploits	EN	WS+SS	4 LP
M-INFO-107600	Practical Course: Route Planning	EN	Unregelm.	6 LP
M-INFO-105453	Practical Course: Security, Usability and Society	EN	Unregelm.	4 LP
M-INFO-105955	Practical Course: Smart Energy System	EN	WS+SS	6 LP
M-INFO-107221	Practical Course: Software Defined Networking	EN	SS	6 LP
M-INFO-101567	Practical Course: Visual Computing	EN	WS+SS	6 LP
M-INFO-107241	Practical Introduction to Hardware Security	EN	WS+SS	6 LP
M-INFO-107238	Practical SAT Solving	EN	Unregelm.	5 LP
M-INFO-105452	Privacy Enhancing Technologies	EN	SS	6 LP
M-INFO-107168	Probability and Computing	EN	WS	5 LP
M-INFO-105623	Reinforcement Learning	EN	WS	6 LP
M-INFO-100850	Reliable Computing I	EN	Unregelm.	3 LP
M-INFO-106654	Research Focus Class: Blockchain & Cryptocurrencies	EN	Unregelm.	6 LP
M-INFO-107728	Research Practical Course on Behavior Foundation Models neu	EN	SS	6 LP
M-INFO-106866	Research Practical Course: Artificial Intelligence & Security	EN	WS+SS	6 LP
M-INFO-106495	Research Practical Course: Explainable Artificial Intelligence neu	EN	WS	6 LP
M-INFO-107174	Research Project Deep Learning for Robotics	EN	WS	6 LP

M-INFO-107163	Research Project: Generative AI for Autonomous Agents	EN	SS	6 LP
M-INFO-105591	Resilient Networking	EN	WS	6 LP
M-INFO-107155	Robotics - Practical Course	EN	SS	6 LP
M-INFO-107162	Robotics I - Introduction to Robotics	EN	WS	6 LP
M-INFO-107123	Robotics II - Humanoid Robotics	EN	SS	3 LP
M-INFO-107130	Robotics III - Sensors and Perception in Robotics	EN	SS	3 LP
M-INFO-107598	Robotics Institute Germany - Lecture Series	EN	WS	3 LP
M-INFO-107090	Sampling Methods for Machine Learning	EN	SS	6 LP
M-INFO-105780	Scientific Methods to Design and Analyze Secure Decentralized Systems	EN	WS	5 LP
M-INFO-102662	Seminar Dependable Computing	EN	WS+SS	3 LP
M-INFO-107242	Seminar in Privacy	EN	SS	4 LP
M-INFO-107729	Seminar on Behavior Foundation Models neu	EN	SS	3 LP
M-INFO-107353	Seminar: Advanced Topics in Natural Language Processing	EN	WS+SS	3 LP
M-INFO-107209	Seminar: Advanced Topics on SAT Solving	EN	WS	3 LP
M-INFO-106086	Seminar: Algorithm Engineering	EN	Unregelm.	4 LP
M-INFO-106512	Seminar: Applications and Extensions of Timed Systems	EN	Unregelm.	4 LP
M-INFO-106490	Seminar: Artificial Intelligence for Energy Systems	EN	WS	4 LP
M-INFO-105309	Seminar: Continuous Software Engineering	EN	WS	4 LP
M-INFO-106958	Seminar: Critical Topics in AI	EN	SS	3 LP
M-INFO-107027	Seminar: Current Trends in Theoretical Computer Science	EN	Unregelm.	4 LP
M-INFO-107175	Seminar: Deep Learning for Robotics	EN	WS	3 LP
M-INFO-107231	Seminar: Embedded Systems I	EN	WS+SS	3 LP
M-INFO-107232	Seminar: Embedded Systems II	EN	WS+SS	3 LP
M-INFO-106927	Seminar: Exoskeletons & Motion Capture	EN	WS	3 LP
M-INFO-106497	Seminar: Explainable Artificial Intelligence neu	EN	WS	3 LP
M-INFO-106651	Seminar: Exploring Robotics - Insights from Science Fiction, Research and Society	EN	SS	3 LP
M-INFO-107436	Seminar: Failing Forward: Robotics, Critical Reading and the Making of Good Research	EN	WS	3 LP
M-INFO-106645	Seminar: Fine-Grained Complexity Theory & Algorithms	EN	Unregelm.	4 LP
M-INFO-106868	Seminar: Hot Topics in Artificial Intelligence & Security 1	EN	WS	4 LP
M-INFO-106869	Seminar: Hot Topics in Artificial Intelligence & Security 2	EN	SS	4 LP
M-INFO-100750	Seminar: Hot Topics in Bioinformatics	EN	SS	3 LP
M-INFO-104891	Seminar: Hot Topics in Decentralized Systems	EN	Unregelm.	3 LP
M-INFO-106392	Seminar: Hot Topics in Explainable Artificial Intelligence (XAI)	EN	SS	4 LP
M-INFO-107686	Seminar: Information Systems neu	EN	WS+SS	3 LP
M-INFO-107217	Seminar: Interpretability and Causality in Machine Learning	EN	SS	3 LP
M-INFO-106719	Seminar: Machine Learning in Climate and Environmental Sciences	EN	Unregelm.	3 LP
M-INFO-107205	Seminar: Operating Systems	EN	Unregelm.	3 LP
M-INFO-105585	Seminar: Post-Quantum Cryptography	EN	Unregelm.	3 LP
M-INFO-107264	Seminar: Practical Graph Algorithms	EN	Unregelm.	4 LP
M-INFO-107216	Seminar: Privacy and Security	EN	WS	4 LP
M-INFO-103306	Seminar: Proofs from THE BOOK	EN	Unregelm.	3 LP
M-INFO-105408	Seminar: Quantum Information Theory	EN	Unregelm.	3 LP
M-INFO-107172	Seminar: Recent Highlights in Algorithms	EN	Unregelm.	4 LP
M-INFO-105761	Seminar: Secure Multiparty Computation	EN	Unregelm.	3 LP
M-INFO-107236	Seminar: Software Architecture, Security and Privacy	EN	Unregelm.	4 LP
M-INFO-107179	Seminar: Speech-to-Speech Translation	EN	SS	3 LP
M-INFO-107625	Seminar: Type Systems and Static Program Analysis	EN	WS	4 LP
M-INFO-106504	Simulation and Optimization in Robotics and Biomechanics	EN	WS	6 LP
M-INFO-107237	Software Architecture and Quality	EN	WS	3 LP

M-INFO-107235	Software Engineering II	EN	SS	6 LP
M-INFO-107212	Software Product Line Engineering	EN	SS	3 LP
M-INFO-106344	Software Security Engineering	EN	SS	3 LP
M-INFO-100719	Software-Evolution	EN	WS	3 LP
M-INFO-107239	Softwaretest and Quality Management (SQM)	EN	WS	5 LP
M-INFO-107243	Telematics	EN	WS	6 LP
M-INFO-100851	Testing Digital Systems I	EN	Unregelm.	3 LP
M-INFO-102962	Testing Digital Systems II	EN	SS	3 LP
M-INFO-107202	Text Indexing	EN	WS	5 LP
M-INFO-107639	Theory of Satisfiability Algorithms neu	EN	Unregelm.	4 LP
M-INFO-106293	Timed Systems	EN	SS	6 LP
M-INFO-107664	Type Systems and Program Analysis neu	EN	Unregelm.	6 LP
M-INFO-107161	Ubiquitous Computing	EN	Unregelm.	5 LP
M-INFO-107113	Wearable Robotic Technologies	EN	SS	4 LP

3.12 Minor Studies: Biologically Inspired Robotics

Leistungspunkte
9-18

Elective Modules (Wahl: zwischen 9 und 18 LP)				
M-MACH-106903	Biologically Inspired Robots neu	EN	SS	3 LP
M-MACH-106905	CAD Engineering Project for Intelligent Systems neu	EN	WS+SS	6 LP
M-MACH-106904	Practical Course: Software Development and Application of Mobile, Bio-Inspired Robots neu	EN	WS	6 LP
M-MACH-107621	Robotic Intelligence: Navigation and Mobile Manipulation neu	EN	SS	6 LP
M-MACH-107558	Robotic Intelligence: Perception and Scene Understanding neu	EN	WS	6 LP
M-MACH-106902	Seminar: Bionic Algorithms and Robot Technologies neu	EN	WS+SS	3 LP

3.13 Minor Studies: Electrical Engineering

Leistungspunkte
9-18

Elective Modules (Wahl: zwischen 9 und 18 LP)				
M-ETIT-101834	Business Innovation in Optics and Photonics	EN	WS	4 LP
M-ETIT-105616	Channel Coding: Algebraic Methods for Communications and Storage	EN	SS	3 LP
M-ETIT-105617	Channel Coding: Graph-Based Codes	EN	WS	6 LP
M-ETIT-100449	Hardware Modeling and Simulation	EN	WS	4 LP
M-ETIT-106963	Hardware Synthesis and Optimization	EN	SS	6 LP
M-ETIT-105971	Mobile Communications	EN	WS	4 LP
M-ETIT-105604	Nano- and Quantum Electronics	EN	SS	6 LP
M-ETIT-100456	Optical Engineering	EN	WS	4 LP
M-ETIT-105073	Student Innovation Lab	EN	WS+SS	15 LP
M-ETIT-100537	Systems and Software Engineering	EN	WS	6 LP
M-ETIT-100462	Systems Engineering for Automotive Electronics	EN	SS	4 LP

3.14 Minor Studies: Mathematics

Leistungspunkte
9-18

Elective Modules (Wahl: zwischen 9 und 18 LP)				
M-MATH-101336	Graphentheorie	EN	s. Anm.	9 LP
M-MATH-102950	Kombinatorik	EN	s. Anm.	9 LP
M-MATH-106957	Moderne Methoden der Kombinatorik	EN	Unregelm.	6 LP

3.15 Minor Studies: Economics**Leistungspunkte**
9-18

Elective Modules (Wahl: zwischen 9 und 18 LP)				
M-WIWI-105659	Advanced Machine Learning and Data Science	EN	WS+SS	9 LP
M-WIWI-105032	Data Science for Finance	EN	WS	9 LP
M-WIWI-106258	Digital Marketing	EN	WS+SS	9 LP
M-WIWI-101503	Service Design Thinking	EN	WS	9 LP

3.16 Minor Studies: Law**Leistungspunkte**
9-18

Elective Modules (Wahl: zwischen 9 und 18 LP)				
M-INFO-107030	EU Data Protection Law	EN	WS	3 LP
M-INFO-107029	Public International Law with an Economic Law Focus	EN	SS	3 LP
M-INFO-107028	Seminar: Law and Legal Studies	EN	WS+SS	3 LP

3.17 Interdisciplinary Qualifications**Leistungspunkte**
2-6

All courses from the House of Competence (HoC), FORUM Science and Society (except computer science or minor studies courses) and from the Language Center (SpZ), but also special courses of the informatics department, can be validated in this area. The courses are not listed in the module handbook, but can be found on the webpages of these institutions. In order to be accepted, English language certificates must demonstrate a level above B2.

Wahlinformationen

All courses from the House of Competence (HoC), FORUM Science and Society (except computer science or minor studies courses) and from the Language Center (SpZ), but also special courses of the informatics department, can be validated in this area. The courses are not listed in the module handbook, but can be found on the webpages of these institutions. In order to be accepted, English language certificates must demonstrate a level above B2.

Interdisciplinary Qualifications (Wahl: zwischen 2 und 6 LP)				
M-INFO-107254	Interdisciplinary Qualifications	EN	WS+SS	6 LP

4 Module

M

4.1 Modul: Access Control Systems: Models and Technology [M-INFO-106303]

Verantwortung: Prof. Dr. Hannes Hartenstein
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
5 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-112775	Access Control Systems: Models and Technology	5 LP	Hartenstein

Erfolgskontrolle(n)

See Partial Achievements (Teilleistung).

Voraussetzungen

See Partial Achievements (Teilleistung).

Qualifikationsziele

- The student understands the challenges of access control in the era of hyperconnectivity.
- The student understands that an information security model defines access rights that express for a given system which subjects are allowed to perform which actions on which objects. The student understands that a system is said to be secure with respect to a given information security model, if it enforces the corresponding access rights.
- The student is able to derive suitable access control models from scenario requirements and is able to specify concrete access control systems. The student is able to decide which concrete architectures and protocols are technically suited for realizing a given access control model.
- The student knows access control protocols using cryptographic methods and is able to compare protocol realizations based on different cryptographic building blocks.
- The student is aware of the limits of access control models and systems with respect to their analyzability and performance and security characteristics. The student is able to identify the resulting tradeoffs.
- The student knows the state of the art with respect to current research endeavors, e.g., access control in the context of decentralized and distributed systems, Trusted Execution Environments, AI, robotics, or hash-chain based systems.

Inhalt

Access control systems are everywhere and the backbone of secure services as they incorporate who is and who is not authorized: think of operating systems, information systems, banking, vehicles, robotics, cryptocurrencies, or decentralized applications as examples. The course starts with current challenges of access control in the era of hyperconnectivity, i.e., in cyber-physical or decentralized systems. Based on the derived needs for next generation access control, we first study how to specify access control and analyze strengths and weaknesses of various approaches. We then focus on up-to-date proposals, like IoT and AI access control. We look at current cryptographic access control aspects, blockchains and cryptocurrencies, and trusted execution environments. We also discuss the ethical dimension of access management. Students prepare for lecture and exercise sessions by studying previously announced literature and by preparation of exercises that are jointly discussed in the sessions.

Arbeitsaufwand

Lecture workload:

1. Attendance time
 Lecture: 2 SWS: 2,0h x 15 = 30h
 Exercises: 1 SWS: 1,0h x 15 = 15h
 2. Self-study (e.g., independent review of course material, work on homework assignments)
 Weekly preparation and follow-up of the lecture: 15 x 1h x 3 = 45h
 Weekly preparation and follow-up of the exercise: 15 x 2h = 30h
 3. Preparation for the exam: 30h
- $\Sigma = 150h = 5 \text{ ECTS}$

Empfehlungen

Basics according to the lectures "Information Security" and "IT Security Management for Networked Systems" are recommended.

M**4.2 Modul: Advanced Bayesian Data Analysis [M-INFO-106812]**

Verantwortung: Prof. Dr. Nadja Klein
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Data Science](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
5 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113673	Advanced Bayesian Data Analysis	5 LP	Klein

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

- Develop a deep understanding of Bayesian statistical principles and computational techniques.
- Master the application of Bayesian regression models to real-world data.
- Gain proficiency in Markov Chain Monte Carlo (MCMC) methods, including Metropolis-Hastings and Gibbs sampling.
- Acquire skills in implementing Bayesian models using relevant software tools such as Stan.

Inhalt

This course deepens students' understanding of Bayesian methods and introduces the latest advancements in Bayesian computation. It is designed for Master students in Computer Science, Mathematics, Econometrics, Techno-Mathematics, Business Informatics, or similar programs seeking to enhance their expertise.

Examples of topics covered are the review of key Bayesian concepts including Bayes' Theorem, conjugate prior distributions, and posterior inference. For instance, students may explore the Beta-Binomial conjugacy, where a Beta prior pairs with a Binomial likelihood, and the Normal-Normal conjugacy, where a Normal prior pairs with a normal likelihood with known variance. These examples demonstrate how conjugate priors simplify posterior calculations and enhance analytical tractability.

Next, students delve into Bayesian supervised learning, covering linear, logistic, and nonparametric approaches, with an emphasis on applying Bayesian methods to real-world data and interpreting results.

The course also covers ways to perform posterior estimation, such as, Markov Chain Monte Carlo (MCMC) inference, including the Metropolis-Hastings algorithm and Gibbs sampling. We explore Bayesian high-dimensional regression techniques, such as the horseshoe prior, for handling models with many predictors. Additionally, students will learn about mixture models and Dirichlet processes, which are powerful tools for modelling heterogeneous data and uncovering latent structures.

We conclude with approximate inference methods, including variational inference and Approximate Bayesian Computation (ABC), essential for dealing with complex models and large datasets.

Arbeitsaufwand

Lecture with 2 SWS + 1 SWS exercise
 5 ECTS corresponds to approx. 150 hours
 approx. 30 hours lecture attendance,
 approx. 15 hours exercise attendance,
 approx. 80 hours post-processing (self-study)
 approx. 25 hours exam preparation (self-study)

Empfehlungen

- Knowledge in R or Python
- Mathematics-heavy lecture. The basics will be reviewed, but mathematical proficiency is helpful

M**4.3 Modul: Advanced Data Structures [M-INFO-107200]****Verantwortung:** Prof. Dr. Peter Sanders**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
5 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114223	Advanced Data Structures	4 LP	Sanders
T-INFO-114224	Advanced Data Structures Project/Experiment	1 LP	Sanders

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students acquire a systematic understanding of algorithmic issues and solution approaches in the area of advanced data structures, building on existing knowledge in the subject area of algorithms. They will also be able to apply learned techniques to related problems and interpret and comprehend current research topics in this area.

Upon successful completion of the course, students will be able to:

- explain terms, structures, basic problem definitions, and algorithms from the lecture;
- select which algorithms and data structures are suitable for solving a problem and, if necessary, adapt them to the requirements of a specific problem;
- use algorithms and data structures, analyze them mathematically, and prove the algorithmic properties.

Inhalt

In this lecture we deal with modern data structures for fundamental objects such as trees, graphs, integers, and strings. These data structures are the basis for many applications and an important part of efficient algorithms. We look at highlights from different research areas and learn techniques for solving a wide variety of problems.

In addition to the theoretical analysis of data structures, we also look at the practical performance of the various data structures and their applications.

Arbeitsaufwand

The lectures including the project/experiment with 5 CP corresponds to 150 working hours, which are divided approximately as follows:

- ca. 30 hours attending lectures
- ca. 60 hours preparing and following-up lectures
- ca. 30 hours working on the project/experiment
- ca. 30 hours preparing for the examination

M**4.4 Modul: Advanced Machine Learning and Data Science [M-WIWI-105659]**

Verantwortung: Prof. Dr. Maxim Ulrich
Einrichtung: KIT-Fakultät für Wirtschaftswissenschaften
Bestandteil von: [Minor Studies: Economics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
9 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	3

Pflichtbestandteile			
T-WIWI-111305	Advanced Machine Learning and Data Science	9 LP	Ulrich

Erfolgskontrolle(n)

Die Erfolgskontrolle erfolgt in Form einer Prüfungsleistung anderer Art. Die Note wird auf der Grundlage der Zwischenpräsentationen während des Projekts, der Qualität der Implementierung, der schriftlichen Abschlussarbeit und einer Endpräsentation bewertet.

Voraussetzungen

Keine

Qualifikationsziele

Nach einem erfolgreichen Projekt können die Studierenden:

- moderne Methoden des maschinellen Lernens zur Lösung eines datenwissenschaftlichen Problems auswählen und anwenden;
- sich in einem Team zielorientiert organisieren und ein umfangreiches Softwareprojekt im Bereich Data Science und Machine Learning zum Erfolg führen;
- ihre Data-Science- und Machine-Learning-Kenntnisse vertiefen
- ein finanzwirtschaftliches Problem mittels Data-Science und Machine-Learning-Algorithmen lösen.

Inhalt

Der Kurs richtet sich an Studenten mit einem Hauptfach in Data Science und/oder Machine Learning und/oder Quantitative Finance. Er bietet den Studierenden die Möglichkeit, praktische Kenntnisse über neue Entwicklungen im Spannungsfeld Finanzmärkte, Datenwissenschaft und des maschinellen Lernens zu erwerben. Das Ergebnis des Projekts soll nicht nur eine schriftliche Ausarbeitung sein, sondern die Implementierung von Methoden oder die Entwicklung eines Algorithmus im Bereich des maschinellen Lernens und der Datenwissenschaft. Typischerweise stammen Problemstellung und Daten aus Forschung und Innovation im Bereich des quantitativen Asset- und Risikomanagements.

Arbeitsaufwand

Gesamtaufwand für 9 Leistungspunkte: ca. 270 Stunden, die sich auf folgende Teile aufteilen: Kommunikation: Austausch während des Projekts: 30 h, Abschlusspräsentation: 10 h; Durchführung und Abschlussarbeit: Vorbereitung vor der Entwicklung (Problemanalyse und Lösungsentwurf): 70 h, Umsetzung der Lösung: 110 h, Tests und Qualitätssicherung: 50 h.

Empfehlungen

Keine

Lehr- und Lernformen

Projekt

M**4.5 Modul: Algorithm Engineering [M-INFO-100795]**

Verantwortung: Prof. Dr. Peter Sanders
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
5 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	4

Pflichtbestandteile			
T-INFO-101332	Algorithm Engineering	4 LP	Sanders
T-INFO-111856	Algorithm Engineering Pass	1 LP	Sanders

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

There are two partial achievements Algorithm Engineering and Algorithm Engineering Exercises. The partial achievement Algorithm Engineering Exercises must be started to be allowed to take the oral examination for Algorithm Engineering.

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The students acquire a systematic understanding of algorithmic problems and solution approaches in the field of Algorithm Engineering, building on existing knowledge in the subject area of algorithms. In addition, they will be able to apply learned techniques to related problems and interpret and comprehend current research topics in the field of Algorithm Engineering.

Upon successful completion of the course, the student will be able to

- Explain terms, structures, basic problem definitions, and algorithms from the lecture;
- select which algorithms and data structures are suitable for solving an algorithmic problem and, if necessary, adapt them to the requirements of a specific problem;
- Execute algorithms and data structures, analyze them mathematically precise and prove the algorithmic properties;
- Explain machine models from the lecture and analyze algorithms and data structures according to these models
- Analyze new problems from applications, reduce them to their algorithmic core and create a suitable abstract model; based on the concepts and techniques learned in the lecture, design and analyze own solutions in this model, and prove algorithmic properties in this model.

Inhalt

- What is Algorithm Engineering, Motivation etc.
- Realistic modeling of machines and applications
- practice-oriented algorithm design
- implementation techniques
- experimental techniques
- evaluation of measurements

The above skills are taught primarily using concrete examples. In the past these were for example the following topics from the area of basic algorithms and data structures:

- linked lists without special cases
- sorting: parallel, external, superscalar,...
- priority queues (cache efficient,...)
- search trees for integer keys
- Full text indexes
- graph algorithms: minimal spanning trees (external,...), route planning

In each of these cases, the focus is on the best known practical and theoretical methods. These usually differ considerably from the methods taught in beginners' lectures.

Arbeitsaufwand

Lecture and exercise with a combined 3 semester hours, 5 ECTS

5 ECTS correspond to about 150h of work, split into

about 45h visiting lectures and exercise or block seminar

about 25h preparation and follow-up on lectures

about 40h solving exercise tasks (programming, preparing presentation for mini seminar, etc)

about 40h exam preparation

M**4.6 Modul: Algorithmic Graph Theory [M-INFO-106960]****Verantwortung:** Dr. rer. nat. Torsten Ueckerdt**Einrichtung:** KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)
Leistungspunkte
5 LP**Notenskala**
Zehntelnoten**Turnus**
Unregelmäßig**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1

Pflichtbestandteile			
T-INFO-113918	Algorithmic Graph Theory	5 LP	Ueckerdt

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students know the basic concepts of algorithmic graph theory and the most important graph classes and their characterizations in this context, namely perfect graphs, chordal graphs, comparability graphs, as well as interval, split and permutation graphs. They will also be able to execute and analyze algorithms for recognizing these graphs and for solving basic algorithmic problems on these graphs. They are also able to identify subproblems in applied problems that can be expressed using these graph classes and to develop algorithms for new problems on these graph classes that are related to problems from the lectures.

Inhalt

Many basic problems that arise in many contexts, such as coloring problems or finding independent sets and maximal cliques, are NP-hard in general graphs. However, instances of these difficult problems that occur in applications are often much more structured and can therefore be solved efficiently. The lecture first introduces perfect graphs and their most important subclass, chordal graphs, and presents algorithms for various generally NP-hard problems on chordal graphs. Subsequently, in-depth concepts such as comparability graphs are discussed, with the help of which various other graph classes (interval, split and permutation graphs) can be characterized and recognized, and tools for the design of specialized algorithms for these are presented.

Arbeitsaufwand

Lecture with 3SWS, 5CP

5 CP corresponds to approx. 150 working hours, of which

approx. 45h lecture attendance

approx. 60 hours of follow-up work and completion of exercises

approx. 45h exam preparation

Empfehlungen

See partial achievements (Teilleistung)

M**4.7 Modul: Algorithms for Visualization of Graphs [M-INFO-106961]****Verantwortung:** Dr. rer. nat. Torsten Ueckerdt**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte
 5 LP

Notenskala
 Zehntelnoten

Turnus
 Unregelmäßig

Dauer
 1 Semester

Sprache
 Englisch

Level
 4

Version
 1

Pflichtbestandteile			
T-INFO-113919	Algorithms for Visualization of Graphs	5 LP	Ueckerdt

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students acquire a systematic understanding of algorithmic problems and solution approaches in the field of graph visualization, which builds on existing knowledge in the areas of graph theory and algorithmics.

After successfully completing the course, students will be able to

- explain concepts, structures and basic problem definitions from the lecture;
- execute layout algorithms for different graph classes, analyze them mathematically precisely and prove the algorithmic properties;
- explain complexity results from the lecture and independently perform similar reduction proofs for new layout problems;
- select which algorithms are suitable for solving a given layout problem and, if necessary, adapt them to the requirements of a concrete problem;
- select which algorithms are suitable for solving a given layout problem and, if necessary, adapt them to the requirements of a specific problem;
- analyze unknown visualization problems from graph drawing applications, reduce them to their algorithmic core and create an abstract model from this; design and analyze their own solutions in this model based on the concepts and techniques learned in the lecture and prove the algorithmic properties.

Inhalt

Networks are relationally structured data that are increasingly appearing in a wide variety of application areas. Examples range from physical networks, such as transportation and supply networks, to abstract networks, such as social networks. Network visualization is a fundamental tool for the investigation and understanding of networks.

Mathematically, networks can be modelled as graphs and the visualization problem can be reduced to the algorithmic core problem of determining a layout of the graph, i.e. suitable node and edge positions in the plane. Depending on the application and graph class, different requirements are placed on the type of drawing and the quality criteria to be optimized. The research field of graph drawing draws on approaches from classical algorithmics, graph theory and algorithmic geometry.

During the course, a representative selection of visualization algorithms will be presented and discussed in depth.

Arbeitsaufwand

Lecture and exercise with 3 SWS, 5 LP

5 LP corresponds to approx. 150 working hours, of which
 approx. 45 hours attendance of the lecture and exercise,
 approx. 25 hours preparation and follow-up,
 approx. 40 hours working on the exercise sheets
 approx. 40 hours exam preparation

M**4.8 Modul: Algorithms II [M-INFO-107201]****Verantwortung:** Prof. Dr. Peter Sanders**Einrichtung:** KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114225	Algorithms II	6 LP	Sanders

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The student has an in-depth insight into the theoretical and practical aspects of algorithms and is able to identify and formally formulate algorithmic problems in various application areas. Furthermore, they know advanced algorithms and data structures from the areas of graph algorithms, algorithmic geometry, string matching, algebraic algorithms, combinatorial optimization, and external memory algorithms. They are able to independently understand algorithms they are unfamiliar with, associate them with the above areas, apply them, determine their running time, evaluate them, and select appropriate algorithms for given applications. Furthermore, the student is able to adapt existing algorithms to related problems. In addition to algorithms for concrete problems, the student knows advanced techniques of algorithmic design. This includes parameterized algorithms, approximation algorithms, online algorithms, randomized algorithms, parallel algorithms, linear programming, and algorithm engineering techniques. For given algorithms, the student is able to identify techniques used to better understand these algorithms. In addition, they are able to select appropriate techniques for a given problem and use them to design their own algorithms.

Inhalt

This module is designed to provide students with the basic theoretical and practical aspects of algorithm design, analysis, and engineering. It teaches general methods for designing and analyzing algorithms for basic algorithmic problems, as well as the basic principles of general algorithmic methods such as approximation algorithms, linear programming, randomized algorithms, parallel algorithms, and parameterized algorithms.

Arbeitsaufwand

Lecture with 3 semester hours + 1 semester hour exercise

6 ECTS correspond to about 180 hours

about 45h visiting the lectures

about 15h visiting the exercises

about 90h follow-up of lectures and solving the exercise sheets

about 30h preparation for the exam

M**4.9 Modul: Artificial Intelligence & IT-Security [M-INFO-106810]**

Verantwortung: Prof. Dr. Christian Wressnegger
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113668	Artificial Intelligence & IT-Security	6 LP	Wressnegger

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students know basic concepts of applying artificial intelligence and machine learning in computer security, and are able to evaluate the performance, quality, and security of such systems.

- Students know and understand basic concepts of features and feature engineering in computer security as well as basic attacks against learning-based systems.
- Students know how to apply AI in computer security.
- Students are able differentiate attack vectors against AI.
- Students understand limits of learning-based security solutions.

Inhalt

The lecture is about combining the fields of artificial intelligence, machine learning and computer security in practice. Many tasks in the computer security landscape are based on manual labor, such as searching for vulnerabilities or analyzing malware. Here, machine learning can be used to establish a higher degree of automation, providing more "intelligent" security solutions (AI for Security). However, also these learning-based systems can be attacked and need to be secured (Security of AI). As an example, viciously crafted inputs can be exploited by an adversary to cause devastating damage in the application area. It thus is of utmost importance to investigate, research, and know about the security properties of AI methods.

The module introduces students to theoretic and practical aspects of AI in computer security as well as security of AI. We cover basics on features and feature engineering in the security domain, discuss fundamental learning settings in security and point out "Dos and Don'ts" of using AI/ML in computer security. Moreover, we put particular focus on "Explainable AI" (XAI) and its use in computer security, before we introduce attacks and defense against learning-based systems as discussed in the first half of the course. We cover input-manipulation attacks (e.g., adversarial examples), model-manipulation attacks (e.g., backdooring attacks), privacy attacks (e.g., model stealing and membership inference) and attacks against XAI.

Arbeitsaufwand

- 58h attendance time
- 56h preparation and follow-up time
- 66h exam preparation

Empfehlungen

The basics of IT security and artificial intelligence are a prerequisite.

M**4.10 Modul: Automated Planning and Scheduling [M-INFO-104447]****Verantwortung:** Prof. Dr. Peter Sanders**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: Area of Specialization: Theoretical Foundations
 Area of Specialization: Algorithm Engineering
 Elective Studies in Informatics

Leistungspunkte
 5 LP

Notenskala
 Zehntelnoten

Turnus
 Unregelmäßig

Dauer
 1 Semester

Sprache
 Englisch

Level
 4

Version
 1

Pflichtbestandteile			
T-INFO-109085	Automated Planning and Scheduling	5 LP	Sanders

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

- The students will be able to model various planning tasks in the PDDL language and solve them using off-the-shelf planners.
- The students will understand the approaches used in automated planning and scheduling algorithms, which will allow them to efficiently model and solve real world planning and scheduling problems by selecting the proper algorithms for the given task.

Inhalt

The course offers an introduction to the methods and techniques used in automated planning and scheduling. The course is focused on classical deterministic planning, i.e., planning in a fully observable deterministic environment. The students will learn how to use automated planners and schedulers and also how they work. The topics covered in the lecture include:

- applications of automated planning in artificial intelligence
- formalization of planning problems and the PDDL language
- computational complexity of planning and scheduling
- basic state space search algorithms (forwards/backwards search)
- heuristic search algorithms and planning heuristics
- plan space planning
- planning graph and the graph plan algorithm
- satisfiability based planning
- hierarchical task network planning
- classical scheduling approaches
- constraint-based scheduling
- planning for virtual agents in computer games

Arbeitsaufwand

2 SWS lecture + 1 SWS exercises

(Preparation and follow-up time: 4h/week for lecture plus 2h/week for exercises; exam preparation: 15h)

Total workload: (2 SWS + 1 SWS + 4 SWS + 2 SWS) x 15h + 15h exam preparation = 9x15h + 15h = 150h = 5 ECTS

M**4.11 Modul: Autonomous Learning for Intelligent Robot Perception [M-INFO-106608]**

Verantwortung: Prof. Dr. Rudolph Triebel
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113327	Autonomous Learning for Intelligent Robot Perception	4 LP	Triebel

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students are capable of describing the details of different methods for autonomous learning, and they can place them in the context of intelligent robot perception. They are able to derive mathematical principles of these algorithms and they can name and describe relevant applications.

Inhalt

This lecture conveys the main principles of Intelligent Robot Perception, where the major focus is on machine learning techniques that are particularly useful for robot vision applications. The most important design criteria for these methods are run-time and data efficiency, safety, and autonomy, where the latter refers to independence of human interactions and the ability to take decisions during learning (aka. active learning). In the lecture, we will analyse modern learning techniques that meet these criteria, and we will show concrete applications of these in robotic perception tasks such as object detection and pose estimation, grasp detection and semantic mapping.

Arbeitsaufwand

120h

Empfehlungen

A basic understanding of probability theory and linear algebra is required

M**4.12 Modul: Beating the Worst-Case in Practice: Unexpectedly Efficient Algorithms [M-INFO-107599]**

Verantwortung: TT-Prof. Dr. Thomas Bläsius
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte
6 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-114898	Beating the Worst-Case in Practice: Unexpectedly Efficient Algorithms	6 LP	Bläsius

Erfolgskontrolle(n)

See partial achievements. (Teilleistung)

Voraussetzungen

See partial achievements. (Teilleistung)

Qualifikationsziele

The students

- can practically apply the knowledge acquired in the basic modules on algorithm engineering,
- are able to independently produce efficient implementations of algorithmic methods,
- master the methodology for the practical evaluation of algorithms, including the processing, analysis, and interpretation of measurement data,
- possess the ability to communicate the results they obtain.

In addition, participants are able to analyze how various typical properties of instances influence the efficiency of algorithms.

Inhalt

In this lab, students implement and evaluate a set of given algorithmic approaches on their own. The emphasis is on methods that, on real-world instances, perform far better than worst-case theory would suggest. We investigate this gap in performance using empirical methods throughout the course.

Arbeitsaufwand

Practical course with 4 SWS, 6 LP

6 LP corresponds to approx. 180 working hours, of which

approx. 20 hours attendance time,

approx. 130 hours implementation and evaluation,

approx. 30 hours elaboration and preparation of the presentation

M**4.13 Modul: Biologically Inspired Robots [M-MACH-106903]**

Verantwortung: Prof. Dr.-Ing. Arne Rönna
Einrichtung: KIT-Fakultät für Maschinenbau
Bestandteil von: Minor Studies: Biologically Inspired Robotics

Leistungspunkte
3 LP

Notenskala
Zehntelnoten

Turnus
Jedes Sommersemester

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
2

Pflichtbestandteile			
T-MACH-113856	Biologically Inspired Robots	3 LP	Rönna

Erfolgskontrolle(n)

Siehe Teilleistung

Voraussetzungen

Keine

Qualifikationsziele

Studierende kennen verschiedenen Entwurfsprinzipien der Methode "Bionik" in der Robotik und können Modelle für Kinematik, Mechanik, Regelung und Steuerung, Perzeption und Kognition analysieren und bewerten.

Studierende verstehen die Leichtbaukonzepte und Materialeigenschaften natürlicher Vorbilder. Sie sind mit den Konzepten und Methoden der Leichtbaurobotik vertraut und können resultierenden Auswirkungen auf die Energieeffizienz mobiler Robotersysteme beschreiben.

Studierende können verschiedenen natürlichen Muskeltypen und ihre Funktionsweise unterscheiden. Außerdem kennen sie die korrespondierenden, künstlichen Muskelsysteme und können das zugrundeliegende Muskelmodell ableiten.

Studierende kennen die wichtigsten Sinne des Menschen, sowie die dazugehörige Reizverarbeitung und Informationskodierung. Sie können hierfür technologische Sensoren ableiten, die die gleiche Funktion in der Robotik übernehmen.

Studierende können die Funktionsweise eines Zentralen Mustergenerators (CPG) gegenüber einem Reflex abgrenzen. Sie können Neuro-Oszillatoren theoretisch herleiten und die Steuerung der Laufbewegung eines Roboters hiermit erläutern. Weiterhin können sie basierend auf den „Cruse Regeln“ Laufmuster für sechsbeinige Roboter erzeugen.

Studierende können verschiedenen Lokomotionsarten sowie passende Stabilitätskriterien für Laufbewegungen unterscheiden. Sie kennen die wichtigsten Laufmuster für mehrbeinige Laufroboter und können diese in einem Gait-Diagramm darstellen.

Studierende kennen die wichtigsten Algorithmen bei maschinellen Lernverfahren und können ihre Vor- und Nachteile in der Robotik erklären.

Studierende kennen die Subsumption System-Architektur und können die Vorteile einer reaktiven Systemarchitektur bewerten. Sie können „Verhalten“ für biologisch inspirierte Roboter zu Verhaltensnetzwerken zusammenfügen.

Studierende können die mendelschen Gesetze anwenden und die Unterschiede zwischen Meiose und Mitose erklären. Weiterhin können sie den genetische Grundalgorithmus erklären.

Studierende können die größten Herausforderungen bei der Entwicklung innovativer, humanoider Robotersysteme identifizieren und kennen Lösungsansätze sowie erfolgreiche Umsetzungen.

Inhalt

Die Vorlesung biologisch inspirierte Roboter beschäftigt sich intensiv mit Robotern, deren mechanische Konstruktion, Sensorkonzepte oder Steuerungsarchitektur von der Natur inspiriert wurden. Im Einzelnen wird jeweils auf Lösungsansätze aus der Natur geschaut (z.B. Leichtbaukonzepte durch Wabenstrukturen, menschliche Muskeln) und dann auf Robotertechnologien, die sich diese Prinzipien zu nutze machen um ähnliche Aufgaben zu lösen (leichte 3D Druckteile oder künstliche Muskeln in der Robotik).

Nachdem diese biologisch inspirierten Technologien diskutiert wurden, werden konkrete Robotersysteme und Anwendungen aus der aktuellen Forschung präsentiert, die diese Technologien erfolgreich einsetzen. Dabei werden vor allem mehrbeinige Laufroboter, schlangenartige und humanoide Roboter vorgestellt, und deren Sensor- und Antriebskonzepte diskutiert.

Der Schwerpunkt der Vorlesung behandelt die Konzepte der Steuerung und Systemarchitekturen (z.B. verhaltensbasierte Systeme) dieser Robotersysteme, wobei die Lokomotion im Mittelpunkt steht. Die Vorlesung endet mit einem Ausblick auf zukünftige Entwicklungen und dem Aufbau von kommerziellen Anwendungen mit diesen Robotern.

Arbeitsaufwand

90 Arbeitsstunden, davon ca.:

- 30h für Präsenzzeit in Vorlesungen
- 30h für Vor- und Nachbereitungszeiten
- 30h für Prüfungsvorbereitung und Teilnahme an der mündlichen Prüfung

M**4.14 Modul: Business Innovation in Optics and Photonics [M-ETIT-101834]**

Verantwortung: Prof. Dr. Werner Nahm
Einrichtung: KIT-Fakultät für Elektrotechnik und Informationstechnik
Bestandteil von: [Minor Studies: Electrical Engineering](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	2

Pflichtbestandteile			
T-ETIT-104572	Business Innovation in Optics and Photonics	4 LP	Nahm

Erfolgskontrolle(n)

Type of Examination: examination of another type

Duration of Examination: 4 group presentations à 20 minutes (approx.)

Modality of Exam: The exam consists of four group presentations. 2nd day: Technology Presentation. 3rd day: Development plan presentation. 4th day: Business Canvas presentation. Final presentation: Business pitch. The overall impression is rated.

Voraussetzungen

none

Qualifikationsziele

The student has an understanding how innovative concepts for optical and photonics products are transferred into a successful business development. The student knows about and makes first The student develops practical experience of how to analyse, develop, and pitch innovative business ideas based on optics & photonics technologies, specifically in the field of spectroscopy and hyperspectral imaging. Through a series of short lectures followed by extensive group work over the first week of the block course the student acquires hands-on experience of business development as lived in a technology start-up environment. The students are given a basic introduction to spectroscopy and are challenged to propose innovative businesses enabled by evolving technology and market needs. To assist in this process the students are introduced to the innovation methodology "Deep Tech Exploration" and primed on themes from intellectual property management to effective pitching.

The students organize themselves into teams, and work cooperatively within the team to develop and deliver the complete business pitch.

In summary the students:

- learn innovation methodologies to assess and optimize the desirability, viability, and feasibility of business ideas.
- are introduced to methods to assess and generate relevant intellectual property.
- get to apply the knowledge gained with extensive group work developing their own business idea and pitch.

Inhalt

This course is instructed by ZEISS employees, active within the ZEISS corporate innovation ecosystem.

Monday: Introduction into spectroscopy and the deep tech exploration method

- Technology introduction, state-of-the-art
- Scenario Introduction: Found your own start-up
- Overview Deep Tech Exploration methodology / Select your use case

Tuesday: Deep dive – deep tech exploration

- Methods and ideation techniques
- Fundamental principles in Innovation / Value Proposition
- Define your tech product

Wednesday: Business case development

- How to protect your business idea with IP?
- Develop your business idea with the help of the Business Model Canvas
- Writing a Business plan

Thursday: Storytelling and pitching

- Entrepreneurial finance: tech product problem sizing / ballparking
- Storytelling
- Preparing an elevator pitch

Friday-Thursday of second week: Final pitch preparation

- Self-organized by groups

Friday of second week: Final pitch

Zusammensetzung der Modulnote

The module grade results of the assessment of the presentations. Details will be given during the lecture.

Arbeitsaufwand

Total 120 h, thereof 34 h contact hours and 86 h preparation, homework, and self-studies

Empfehlungen

Good knowledge in optics & photonics. Personal motivation and interest for getting deeper into business development aspects, methods and tools. Commitment to active, regular and continuous participation in the group work.

M**4.15 Modul: CAD Engineering Project for Intelligent Systems [M-MACH-106905]**

Verantwortung: Prof. Dr.-Ing. Arne Rönnau
Einrichtung: KIT-Fakultät für Maschinenbau
Bestandteil von: [Minor Studies: Biologically Inspired Robotics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	3

Pflichtbestandteile			
T-MACH-113857	CAD Engineering Project for Intelligent Systems	6 LP	Rönnau

Erfolgskontrolle(n)

Siehe Teilleistung

Voraussetzungen

Keine

Inhalt

Das Modul *CAD Engineering Project for Intelligent Systems* ist ein projektbasiertes Mastermodul, das Studierenden die Möglichkeit bietet, ingenieurwissenschaftliche Kompetenzen im Bereich der Entwicklung intelligenter mechatronischer Systeme praxisnah zu erlangen und zu vertiefen. In interdisziplinären Teams konzipieren, entwerfen und realisieren die Studierenden ein robotischen Prototypen entlang eines vollständigen Entwicklungszyklus.

Ausgehend von einer definierten Problemstellung entwickeln die Studierenden ein technisches Konzept und setzen dieses mittels CAD-Software (Autodesk Fusion 360) in ein detailliertes mechanisches Design um. Ein besonderer Fokus liegt dabei auf Grundlagen des Rapid-Prototyping, funktionsgerechten Auslegung von Bauteilen, korrekter Parametrisierung, der Berücksichtigung von Fertigungsrestriktionen sowie der Integration mechanischer und elektronischer Komponenten, sowie deren softwaretechnische Anbindung.

Die entworfenen Bauteile werden überwiegend mittels additiver Fertigungsverfahren (insbesondere 3D-Druck) hergestellt und durch grundlegende konventionelle Fertigungstechniken ergänzt. Parallel dazu integrieren die Studierenden geeignete Sensoren, Aktoren und weitere elektronische Komponenten in das System und entwickeln die notwendige Software zur Steuerung und Regelung des robotischen Prototyp.

Im Anschluss wird das entwickelte System aufgebaut, in Betrieb genommen und unter realistischen Bedingungen getestet und evaluiert. Die Ergebnisse werden abschließend dokumentiert und präsentiert. Das Modul fördert neben fachlichen Kompetenzen insbesondere Projektmanagementfähigkeiten, Teamarbeit, interdisziplinäres Denken sowie eine systematische und lösungsorientierte Arbeitsweise, wie sie für die Entwicklung intelligenter technischer Systeme in Forschung und Industrie erforderlich sind.

Voraussetzungen:

Grundlegende Kenntnisse in gängigen CAD-Software Lösungen, bestenfalls Fusion 360.

Die Studierenden müssen eine kurze Bewerbung einreichen, in der sie ihre Motivation und Erfahrungen darlegen. Die Bewerbung erfolgt über Ilias: (ilias.studium.kit.edu) - **Deadline: 16.04.2026** .

Zusammensetzung der Modulnote

In die Modulnote gehen die Beurteilung des Konstruktionsprojekts, der schriftlichen Ausarbeitungen und des Vortrags ein. Nähere Angaben erfolgen zu Beginn der Veranstaltung.

Anmerkungen

Weitere Hinweise finden Sie hier teaching.imi.kit.edu

Arbeitsaufwand

1. Präsenzstudienzeit Praktikum: 60 h (4 SWS*15 h)
 2. Arbeit an dem Projekt, Selbststudienzeit inkl. Prüfungsvorbereitung: 120 h
- Summe: 180 h

Inhaltliche Aufteilung:

- Initialer Entwurf / Idee einer (biologisch inspirierten) mechatronische Komponente: 20h
- Konstruktion mittels CAD-System und Gesamtsystem-Design: 60h
- 3D Druck, Montage, Integration Elektronik und Funktionstests: 80h
- Dokumentation und Bericht: 10h
- Meetings: Kickoff, Zwischen- und Abschluss-Präsentation und Diskussionen sowie Treffen mit Betreuern: 10h

M**4.16 Modul: Channel Coding: Algebraic Methods for Communications and Storage [M-ETIT-105616]**

Verantwortung: Prof. Dr.-Ing. Laurent Schmalen
Einrichtung: KIT-Fakultät für Elektrotechnik und Informationstechnik
Bestandteil von: [Minor Studies: Electrical Engineering](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-ETIT-111244	Channel Coding: Algebraic Methods for Communications and Storage	3 LP	Schmalen

Erfolgskontrolle(n)

The exam is held as an oral exam of approx. 20 min.

Qualifikationsziele

The students are able to analyse and assess problems of algebraic channel coding. They can apply methods of algebraic coding theory in the context of communication systems for data transmission and data storage and are able to assess their implementation. Additionally, they will get knowledge to current research topics and research results.

Inhalt

This course focuses on the formal and mathematical basics for the design of coding schemes in digital communication systems. These include schemes for data transmission, data storage and networking. The course starts by introducing the necessary fundamentals of algebra which are then used to derive codes for different applications. Besides codes that are important for data transmission applications, e.g., BCH and Reed-Solomon-Codes, we also investigate codes for the efficient storage and reconstruction of data in distributed systems (locally repairable codes) and codes that increase the throughput in computer networks (network codes). Real applications are always given to discuss practical aspects and implementations of these coding schemes. Many of these applications are illustrated by example code in software (python/MATLAB).

Zusammensetzung der Modulnote

The module grade of the module corresponds to the grade of the oral exam.

Arbeitsaufwand

1. Attendance to the lecture: $15 * 2 \text{ h} = 30 \text{ h}$
2. Preparation and review: $15 * 4 \text{ h} = 60 \text{ h}$
3. Preparation for the exam: included in preparation and review
4. In total: $90 \text{ h} = 3 \text{ LP}$

Empfehlungen

Knowledge of basic engineering as well as basic knowledge of communications engineering.

Previous attendance of the lectures "Communication Engineering I" and "Probability Theory" is recommended.

M**4.17 Modul: Channel Coding: Graph-Based Codes [M-ETIT-105617]**

Verantwortung: Prof. Dr.-Ing. Laurent Schmalen
Einrichtung: KIT-Fakultät für Elektrotechnik und Informationstechnik
Bestandteil von: [Minor Studies: Electrical Engineering](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-ETIT-111245	Channel Coding: Graph-Based Codes	6 LP	Schmalen

Erfolgskontrolle(n)

The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 30 minutes in which preparatory tasks are solved.

Voraussetzungen

none

Qualifikationsziele

Students will be able to understand and apply advanced and modern methods of channel coding. They get to know various tools of modern coding theory for the analysis and optimization of coding schemes, conceptual design approaches of error correction building blocks as well as applications in digital communications (for example, 5G). Additionally, they will get knowledge to current research topics and research results.

Inhalt

The course expands on the topics dealt with in the lecture "Verfahren der Kanalcodierung". The focus is on modern methods that have been brought into practice in the past few years and that achieve the capacity limits postulated by Shannon. For this purpose, known techniques have to be extended and new methods have to be learnt additionally. The lecture introduces the theoretical limits very quickly and follows with a discussion on the basic concepts of channel coding, including block codes. Based on this, modern error correction methods like LDPC codes, spatially coupled codes, and Polar codes are treated in depth. The lecture ends with a view on the application of channel coding in classical and distributed storage scenarios and in computer networks. Many of the applications are illustrated with example implementations in software (python/MATLAB).

Zusammensetzung der Modulnote

The modul grade is the grade of the oral exam.

Arbeitsaufwand

- Lecture attendance time: $15 * 3 \text{ h} = 45 \text{ h}$
- Presence time Exercise: $15 * 1 \text{ h} = 15 \text{ h}$
- Lecture preparation / revision: $15 * 3 \text{ h} = 45 \text{ h}$
- Exercise: $15 * 1 \text{ h} = 15 \text{ h}$
- Exam preparation and attendance: 60 h

Total workload: approx. 180 h = 6 LP

Empfehlungen

Previous attendance of the lectures "Communication Engineering I" and "Probability Theory" is recommended. Knowledge from the lecture "Applied Information Theory" can be helpful. Previous attendance of the lecture "Verfahren der Kanalcodierung" can be helpful, but is not necessary.

M**4.18 Modul: Compiler Design [M-INFO-106966]****Verantwortung:** Prof. Dr. André Platzer**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
9 LP**Notenskala**
Zehntelnoten**Turnus**
Unregelmäßig**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
2

Pflichtbestandteile			
T-INFO-113925	Compiler Design	9 LP	Platzer

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

- Distinguish the main phases of a state-of-the-art compiler
- Understand static and dynamic semantics of an imperative language
- Develop parsers and lexers e.g. with parser generators, combinators
- Perform semantic analysis
- Translate abstract syntax trees to intermediate representations and static single assignment form
- Analyze the dataflow in an imperative language
- Perform standard compiler optimizations
- Generate assembly code
- Allocate registers using a graph-coloring algorithm
- Understand opportunities and limitations of compiler optimizations
- Appreciate design tradeoffs how representation affects optimizations
- Automatically manage memory using garbage collection
- Develop complex software following high-level specifications

Inhalt

This course covers the design and implementation of compiler and runtime systems for high-level programming languages, and examines the interaction between language design, compiler design, and runtime organization. Topics covered include lexical and syntactic analysis, semantic analysis, type-checking, program analysis, code generation and optimization, memory management, and runtime organization.

Compilers and principles of compiling are one fundamental core aspect of computer science. Compilers and several other parts of compiler technology (especially parsing, transformation, analysis, and optimization) play important roles in many systems built every day. The knowledge gained in this course should be broad enough that if you are confronted with the task of contributing to the implementation of a real compiler in the field or similar technology, you should be able to do so confidently and quickly.

Arbeitsaufwand

9 ECTS from 270h of coursework consisting of

- 60h=15*4h from 4SWS lectures
- 90h preparation, reading lecture notes, studying
- 100h developing a compiler
- 20h exam preparation

Empfehlungen

Students are expected to have significant experience in a high-level programming language. Students are also expected to follow the lecture notes.

M**4.19 Modul: Computational Geometry [M-INFO-107228]****Verantwortung:** TT-Prof. Dr. Thomas Bläsius**Einrichtung:** KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)
Leistungspunkte
6 LP**Notenskala**
Zehntelnoten**Turnus**
Unregelmäßig**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1

Pflichtbestandteile			
T-INFO-114251	Computational Geometry	6 LP	Bläsius
T-INFO-114252	Computational Geometry - Pass	0 LP	Bläsius

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students develop a systematic understanding of questions and solution approaches in the field of computational geometry, building on their existing knowledge of theoretical computer science and algorithms. Upon successful completion of the course, students will be able to:

- * explain concepts, structures, and fundamental problem definitions presented in the lectures
- * execute geometric algorithms, analyze them mathematically, and prove their properties
- * select appropriate algorithms and data structures for solving a given geometric problem and adapt them to specific problem scenarios if necessary
- * analyze unfamiliar geometric problems, reduce them to their algorithmic core, and create an abstract model; based on the concepts and techniques learned in the lecture, design their own solutions within this model, analyze them, and prove their properties

Inhalt

Spatial data is processed in a wide variety of areas in computer science, such as computer graphics and visualization, geographic information systems, robotics, and more. Computational geometry focuses on the design and analysis of geometric algorithms and data structures. This module introduces frequently used techniques and concepts in computational geometry, which are explored in depth using selected and application-related questions.

Arbeitsaufwand

Lecture with exercises, 4 hours per week (SWS), 6 ECTS corresponds to approximately 180 hours of work, including: ~60 hours attending lectures and exercises ~30 hours preparation and review ~60 hours working on exercise sheets ~30 hours exam preparation

Empfehlungen

Basic knowledge of algorithms and data structures (e.g., from the courses Algorithms 1 + 2) is expected.

M**4.20 Modul: Computational Imaging [M-INFO-106190]****Verantwortung:** Prof. Dr.-Ing. Jürgen Beyerer**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Robotics and Automation](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
5 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Wintersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1**Pflichtbestandteile**

T-INFO-112573	Computational Imaging	5 LP	Meyer
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Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Qualification goal: Students are able to model questions of machine vision optically and algorithmically and to process them using holistic optimization.

Learning objectives: Students know

- the essential components of machine vision, their optical modelling and suitable coding methods in the sense of computational imaging,
- methods for emitting, capturing and processing light fields for applications in photography and industrial image processing,
- the concept of light transport analysis, corresponding modelling, capturing and processing methods and
- approaches to holistic modelling and optimization of optical image capturing and processing systems.

Inhalt

Digital image acquisition and processing have revolutionized various fields of applications, e.g., medical imaging or automated visual inspection. Yet, the design of most such systems is still based on the separate and individual optimization of the employed illumination, image acquisition and image processing components. By following a holistic approach for system design, modelling and optimization, computational imaging methods yield superior performance with respect to the state of the art. After introducing the students into relevant basics of optics and signal theory, the lecture will thoroughly cover various topics of computational imaging. Accompanying practical exercises will complement the theoretical part of the lecture. The course will enable students to adequately model artificial vision problems in the sense of computational imaging in order to obtain holistically optimal solutions.

Arbeitsaufwand

Lecture with 2 SWS + 1 SWS exercise
 5 ECTS corresponds to approx. 150 hours
 approx. 30 hours lecture attendance,
 approx. 15 hours exercise attendance,
 approx. 90 hours post-processing and working on the exercises
 approx. 30 hours Exam preparation

Literatur

- Ayush Bhandari, Achuta Kadambi, Ramesh Raskar, Computational Imaging, MIT Press, 2022.
- Jürgen Beyerer, Fernando Puente León, Christian Frese, Machine Vision, Springer, 2015.
- Joseph. W. Goodman, Introduction to Fourier Optics. 4. Auflage W. H. Freeman, 2017.

M**4.21 Modul: Constructive Logic [M-INFO-106256]****Verantwortung:** Prof. Dr. André Platzer**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Theoretical Foundations](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
5 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Sommersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1

Pflichtbestandteile			
T-INFO-112704	Constructive Logic	5 LP	Platzer

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

- Understand the working principles of logic
- Understand how the meaning of a proposition comes from its verifications
- Distinguish propositions from judgments
- Use proof rules to conduct formal proofs
- Formalize informal problems into precise logical language
- Justify how proof rules fit to one another in sound and complete ways
- Assess the validity of a formal proof
- Understand propositions as types, proofs as programs, formulas as programs
- Relate constructive logic to computation and constructive proofs to functional programs
- Relate deductive proof search to computation in logic programming
- Relate induction to recursion and use induction to prove properties in and about logical systems
- Understand the principles and applications of logic programming

Inhalt

This course provides a thorough introduction to modern constructive logic, its roots in philosophy, its numerous applications in computer science, and its mathematical properties. The core topics of this course are intuitionistic logic, natural deduction, Curry-Howard isomorphism, propositions as types, proofs as programs, formulas as programs, functional programming, logic programming, Heyting arithmetic and primitive recursion, cut elimination, connections between classical and constructive logic, inductive definitions, sequent calculus, and decidable classes. Advanced topics may include type theory, proof search, linear logic, temporal logic, modal logic.

AnmerkungenCourse web page: <https://ifcps.org/course/constlog.html>**Arbeitsaufwand**

5 ECTS from 150h of coursework consisting of

45h = 15 * 3h from 3 SWS lectures

15h = 15 * 1h from 1 SWS exercises

90h preparation, reading lecture notes, studying

22h exam preparation

Empfehlungen

You will be expected to follow the lecture notes.

M**4.22 Modul: Cryptographic Protocols II: Universal Composability [M-INFO-107435]**

Verantwortung: Prof. Dr. Jörn Müller-Quade
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114602	Cryptographic Protocols II: Universal Composability	3 LP	Müller-Quade

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The student

- understands the problems that occur in protocol composition and can explain them using examples.
- knows and understands the frameworks, security concepts and their properties presented in the lecture and can explain, relate and prove these.
- knows and understands the protocols and proofs presented in the lecture and can reproduce them.
- can independently apply the techniques presented in the lecture, for example to prove or demonstrate the security or insecurity of simple protocols.

Inhalt

In the lecture "Kryptographische Protokolle", methods and building blocks for secure multi-party computation were presented. The security was shown for a single execution and under sequential composition.

In reality, however, protocols are executed concurrently - both multiple instances of the same protocol (concurrent composition) and multiple, different protocols (general composition) that were designed independently of each other.

This protocol composition is not adequately covered by classical security concepts: it may be that the execution of a single protocol instance can be proven to be secure. However, if the same protocol is executed several times simultaneously, all security may be lost.

In this lecture, we re-visit the topics of the "Kryptographische Protokolle" lecture with a focus on composability. The first part of the lecture therefore introduces the concept of universally composable (UC) security, which depicts the setting of general composition in which arbitrary protocols are executed concurrently. An important insight here is that UC security can only be achieved with the help of trust assumptions, so-called "setups". In the second part of the lecture, important protocols that fulfill this strong security are presented, for example for commitments or general secure multi-party computation. At the end of the lecture, further composing security notions and related protocols are considered, which are weaker than UC security, but without setups.

Arbeitsaufwand

90 h

Lectures: 24 h

Self-study during semester: 31 h

Preparation for the exam: 35 h

Empfehlungen

Students should be familiar with the contents of the modules "Theoretische Grundlagen der Kryptographie" (Theoretical Foundations of Cryptography) and „Kryptographische Protokolle“ (Cryptographic Protocols I).

M**4.23 Modul: Data Science and Artificial Intelligence for Energy Systems [M-INFO-106655]**

Verantwortung: TT-Prof. Dr. Benjamin Schäfer
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Telematics](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Area of Specialization: Data Science](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113402	Data Science and Artificial Intelligence for Energy Systems	6 LP	Schäfer

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

- Students obtained a foundational knowledge of data-driven methods in energy systems as an active research field. They can name some ongoing challenges.
- They can explain different data science methods and their applications in energy systems (including Langevin processes, superstatistics, (probabilistic) forecasts and explainable AI).
- Students can employ AI methods to solve problems in energy systems, including optimizing systems and forecasting time series.
- Students can exploit key properties of trained machine learning models and interpretability tools.
- Students can select suitable analysis tools, justify their choice and carry out data-driven analysis on power systems.

Inhalt

Artificial Intelligence (AI) is a key technology in many areas of society and research. Energy systems with the ongoing energy transition ("Energiewende") make it a fascinating field for deploying AI methods. AI and machine learning algorithms can play a crucial role in improving energy efficiency, optimizing power generation and distribution or enhancing system stability while facilitating additional renewable energy integration. In this lecture, we review some mechanics of energy systems, their design and optimization questions and how to solve these using data-driven approaches. We will discuss deterministic dynamics, as well as stochastic aspects of energy systems and will explore fundamental AI algorithms and their applications in energy systems. We will cover both classical time series methods as well as state-of-the-art AI techniques, e.g. for optimization or forecasting.

Arbeitsaufwand

Course workload:

1. Attendance time: 4 SWS x 15=60 (Course, exercise, etc.)
 2. Self-study: 6 h x 15 = 90 (independent review of course material, work on homework assignments)
 3. Preparation for the exam: 30h
- 60+90+30=180h= 6ECTS

Empfehlungen

Knowledge of AI basics is very helpful.

Previous participation in "Energieinformatik 1" and/or "Energieinformatik 2" is beneficiary but not mandatory.

Knowledge of Python is highly recommended.

M**4.24 Modul: Data Science for Finance [M-WIWI-105032]**

Verantwortung: Prof. Dr. Maxim Ulrich
Einrichtung: KIT-Fakultät für Wirtschaftswissenschaften
Bestandteil von: [Minor Studies: Economics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
9 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-WIWI-102878	Computational Risk and Asset Management	4,5 LP	Ulrich
T-WIWI-110213	Python for Computational Risk and Asset Management	4,5 LP	Ulrich

Erfolgskontrolle(n)

Die Modulprüfung erfolgt in Form einer Prüfungsleistung anderer Art. Die Prüfungsleistung anderer Art besteht aus einem Python-basierten "Takehome Exam". Am Ende der dritten Januarkalenderwoche bekommt der Student ein "Takehome Exam" ausgehändigt, welches er binnen 4 Stunden eigenständig und mittels Python bearbeitet und zurückschickt. Genaue Anweisungen werden zu Beginn der Lehrveranstaltung bekannt gegeben.

Die Prüfungsleistung anderer Art kann maximal einmal wiederholt werden. Eine fristgerechte Wiederholungsmöglichkeit findet am Ende der dritten Märzkalenderwoche des gleichen Jahres statt. Genauere Anweisungen werden zu Beginn der Lehrveranstaltung bekannt gegeben.

Voraussetzungen

Keine.

Qualifikationsziele

Das Ziel des Moduls ist mittels Data Science, Machine Learning und Finanzmarkttheorien bessere Investitions-, Risiko- und Assetmanagement-Entscheidungen zu generieren. Der Student lernt anwendungsorientiert und mittels echter Finanzmarktdaten Charakteristika verschiedener Assetklassen kennen. Wir verwenden Python und Webscraping Techniken um öffentlich zugängliche Finanzmarktdaten zu extrahieren, zu visualisieren und nach Mustern zu untersuchen. Interessante und nicht-öffentliche Finanzmarktdaten wie (Options- und Futuresdaten auf Aktien und Zinsen) werden für den Kurs zur Verfügung gestellt. Finanzmarkttheorien werden ebenfalls besprochen, um die Datenanalyse durch theoretische Kenntnisse zu verbessern. Studenten lernen durch die "Data Science-Brille" Aktien-, Zins-, Futures- und Optionsmärkte kennen. Durch die "Finanztheorie-Brille" verstehen Studenten, wie Muster mittels Finanztheorie kommuniziert und interpretiert werden können. Python ist das Bindeglied, durch welches wir Data Science und moderne Finanzmarktmodellierung zusammenbringen.

Inhalt

Das Modul umfasst unter anderem folgende Themen:

- Mustererkennung in Preis- und Ertragsdaten in Aktien-, Zinssatz-, Futures- und Optionsmärkten
- Quantitative Portfolio-Strategien
- Modellierung von Rücklaufdichten unter Verwendung von Instrumenten der Finanzökonomie, Datenwissenschaft und des maschinellen Lernens
- Bewertung von Aktien, festverzinslichen Wertpapieren, Futures und Optionen in einem kohärenten Rahmen, um möglicherweise Arbitragemöglichkeiten auszunutzen
- Neuronale Netze und Verarbeitung natürlicher Sprache

Arbeitsaufwand

Der Gesamtarbeitsaufwand für dieses Modul beträgt ca. 270 Stunden (9 Leistungspunkte). Die Gesamtstundenzahl ergibt sich aus dem Aufwand für das Studium von Onlinevideos, dem Bearbeiten von Quizfragen, dem Studium von Ipython- Notebooks, der Teilnahme an interaktiven "Python Data Sessions" und der Lektüre empfohlener Literatur.

Empfehlungen

Grundkenntnisse der Kapitalmarkttheorie.

Lehr- und Lernformen

Vorlesungen und Übungen

M**4.25 Modul: Decentralized Systems: Fundamentals, Modeling, and Applications [M-INFO-105334]**

Verantwortung: Prof. Dr. Hannes Hartenstein
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	4

Pflichtbestandteile			
T-INFO-110820	Decentralized Systems: Fundamentals, Modeling, and Applications	6 LP	Hartenstein

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele**Fundamentals & Modeling**

1. The student is able to recognize and distinguish distributed, federated, and decentralized systems.
2. The student understands the role of consensus, consistency, and coordination within the context of networked and decentralized systems.
3. The student understands the concept of Sybil attacks.
4. The student is familiar with decentralized algorithms like leader election for execution contexts with various guarantees.
5. The student understands the formally proven limits of fault tolerance and their underlying assumptions. This includes an understanding of synchronous and asynchronous network models which underpin the respective proofs. The student also understands several models for fault tolerance within the context of decentralized and distributed systems.
6. The student has a basic understanding of state machine replication.
7. The student knows how conflict-free replicated data types employ mathematics to provide eventually consistent data to collaborative peer-to-peer applications.

Applications

1. The student understands how decentralized storage, replication, and coordination is employed in practice.
2. The student has a fundamental understanding of blockchain-based systems and payment channels, as well as decentralized systems for storage, communication, and machine learning.
3. The student understands trust relations in distributed and decentralized systems applications.
4. The student is able to understand how theoretical foundations relate to networked and decentralized systems in practice.

Inhalt

Decentralized Systems (like blockchain-based systems) are distributed systems under the control of multiple, independent parties that act on their own, autonomous decisions. In this course, we cover fundamental theoretical aspects as well as up-to-date decentralized systems, and connect theory with current practice. We thereby address fault tolerance, security and trust, as well as performance aspects at the example of applications like Bitcoin, Ethereum, IPFS, Matrix, or the Fediverse. As a research-oriented lecture, we cover additional current topics like decentralizing artificial intelligence via decentralized, federated learning, or identity and access management in decentralized systems.

The lecture covers at least the following topics:

- Fundamentals
 - Peer-to-Peer Overlay Networks, Sybil and Eclipse Attacks
 - Formalization of decentralized systems, including models for their computation, communication, faults, and timing.
 - Leader election and broadcast in decentralized systems, based on different models for identities and timing.
 - Byzantine consensus in synchronous and asynchronous settings, including Bracha's fundamental algorithm for reliable broadcast, practical Byzantine fault tolerant consensus, and fundamental limits.
 - Consistency models and protocols, including conflict-free replicated data types.
- Applications
 - Decentralized storage systems, at the example of IPFS
 - Decentralized communication systems, at the example of Matrix for instant messaging
 - Distributed ledgers and blockchains, at the examples of Bitcoin and Ethereum, in particular Proof-of-Work and Proof-of-Stake consensus
 - Scaling techniques for distributed ledgers, at the example of payment channel networks and rollups
 - Decentralized, federated machine learning systems

Arbeitsaufwand

1. Attendance time (Course, exercise, etc.)

Lecture: 3 SWS: 3,0h x 15 = 45h

Exercise: 1 SWS: 1,0h x 15 = 15h

2. Self-study (e.g. independent review of course material, work on homework assignments)

Weekly preparation and follow-up of the lecture: 15 x 1h x 3 = 45h

Weekly preparation and follow-up of the exercise: 15 x 2h = 30h

3. Preparation for the exam: 45 h

$\Sigma = 180h = 6$ ECTS

Empfehlungen

Basics according to the lectures "Information Security" and "Introduction to Computer Networks" are recommended.

M**4.26 Modul: Deep Learning and Neural Networks [M-INFO-107197]****Verantwortung:** Prof. Dr. Jan Niehues**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
6 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Sommersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1**Pflichtbestandteile**

T-INFO-114219	Deep Learning and Neural Networks	6 LP	Niehues
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Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students will learn about the structure and function of different types of neural networks.

Students should learn the methods for training the various networks and their application to problems.

Students should learn the areas of application of the different types of networks.

Given a concrete scenario, students should be able to select the appropriate type of neural network.

Inhalt

This module introduces the use of neural networks for the solution of solving various problems in the field of machine learning, such as classification, prediction, control or inference. or inference. Different types of neural networks are covered and their areas of application are illustrated using examples.

Arbeitsaufwand

Lecture with 4 SWS, 6 CP.

6 LP corresponds to approx. 180 hours, of which

approx. 60 hours lecture attendance

approx. 90 hours post-processing (self-study)

approx. 30 hours exam preparation (self-study)

Empfehlungen

Prior successful completion of the core module "Cognitive Systems" is recommended.

M**4.27 Modul: Design and Architectures of Embedded Systems (ESII) [M-INFO-107230]****Verantwortung:** Prof. Dr.-Ing. Jörg Henkel**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114254	Design and Architectures of Embedded Systems (ESII)	3 LP	Henkel

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The student learns methods for mastering complexity and applies these methods to the design of embedded systems. He/she evaluates and selects specific architectures for embedded systems. Furthermore, the student receives an introduction to current research topics.

Inhalt

Nowadays, it is possible to integrate several billion transistors on a single chip and thus realize complete SoCs (systems-on-chip). The trend towards being able to use more and more transistors continues unabated, meaning that the complexity of such systems will also continue to increase. Computers will increasingly be ubiquitous, i.e. they will be integrated into the environment and will no longer be perceived as computers by humans. Examples include sensor networks, electronic textiles and many more. However, the physically possible complexity will not be readily achievable in practice, as there is currently a lack of powerful design processes capable of handling this high level of complexity. Powerful ESL tools ("Electronic System Level Design Tools") and novel architectures will be required. The focus of this lecture is therefore on high-level design methods and architectures for embedded systems. Since the power consumption of (mostly mobile) embedded systems is of crucial importance, one focus of the design methods will be on the design with regard to low power consumption.

Arbeitsaufwand

90 hours:

30 hours of lectures (15 weeks*2 hours)

30 hours of self study (15 weeks*2 hours)

15 hours of deepening (coding low level C and embedded C) to grasp the theory with practical

15 hours exam preparation

M**4.28 Modul: Digital Marketing [M-WIWI-106258]**

Verantwortung: Prof. Dr. Ann-Kristin Kupfer
Einrichtung: KIT-Fakultät für Wirtschaftswissenschaften
Bestandteil von: Minor Studies: Economics

Leistungspunkte
9 LP

Notenskala
Zehntelnoten

Turnus
Jedes Semester

Dauer
2 Semester

Sprache
Englisch

Level
4

Version
2

Pflichtbestandteile			
T-WIWI-112693	Digital Marketing	4,5 LP	Kupfer
Ergänzungsangebot (Wahl: 4,5 LP)			
T-WIWI-106981	Digital Marketing and Sales in B2B	1,5 LP	Klarmann, Konhäuser
T-WIWI-114174	Economic Decision Making	4,5 LP	Scheibehenne
T-WIWI-114893	International Entrepreneurship and Brand Development	6 LP	Kupfer, Terzidis
T-WIWI-107720	Market Research	4,5 LP	Klarmann
T-WIWI-112711	Media Management	4,5 LP	Kupfer
T-WIWI-111848	Online-Konzepte für Karlsruher Innenstadthändler	3 LP	Kupfer

Erfolgskontrolle(n)

The assessment is carried out as partial exams of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course, weighted by the credits and truncated after the first decimal.

Voraussetzungen

None

Qualifikationsziele

Students

- have an advanced knowledge about central marketing contents
- have a fundamental understanding of the marketing instruments
- know current fundamental principles and latest trends in the field of digital marketing
- know and understand several strategic concepts and how to implement them
- are able to implement their extensive marketing knowledge in a practical context
- are able to critically discuss and question theoretical concepts and current practices in marketing
- have theoretical knowledge that is fundamental for writing a master thesis in the field of marketing
- have gained insight into scientific research that prepares them to independently write a master's thesis
- have the theoretical knowledge and skills necessary to work in or collaborate with the marketing department of a company

Inhalt

The "Digital Marketing" module explores advanced marketing strategies within the context of digital transformation. Students develop a deep understanding of digital consumer behavior, the digital environment, and digital marketing strategies:

- **Principles of Digital Marketing:** In this mandatory course of the module, students learn about practices and strategies like search engine optimization, word-of-mouth, and influencer marketing.
- **Performance Analytics and Practical Applications:** Practical applications of SEO, SEA, and Content Management Systems. Students learn to monitor performance using Key Performance Indicators (KPIs), dashboards, and Return on Investment (RoI) calculations and apply their knowledge to real-world cases.
- **Contextualization:** Via course selections, students can learn about specific requirements for B2B digital strategies, media industries, and retailing.
- **Decision Psychology:** Students can decide to deepen their knowledge on consumer psychology, including heuristics, nudging, and risk perception.
- **Market Research & Methods:** Students can learn about advanced empirical techniques to interpret customer attitudes and prepare for scientific research.

By integrating these multidisciplinary perspectives, the module enables students to implement sophisticated marketing instruments and critically evaluate current practices in digital business environments.

Arbeitsaufwand

Total effort for 9 credit points: approx. 270 hours.

The exact distribution is done according to the credit points of the courses of the module.

M**4.29 Modul: Distributed Computing [M-INFO-107215]**

Verantwortung: Prof. Dr. Achim Streit
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114235	Distributed Computing	4 LP	Streit

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students understand the basic concepts of distributed systems, in particular Grid and Cloud Computing as well as the management and analysis of big and distributed data. They apply underlying paradigms and services to given examples. Students analyze methods and technologies of Grid and Cloud Computing as well as distributed data management, which are suitable for use in everyday and industrial application areas or which are used today by Google, Facebook, Amazon, etc. For this purpose, students will compare web/grid services, elementary grid functionalities, data lifecycles, metadata, archiving, cloud service types (IaaS, SaaS, PaaS) and public/private clouds using real-world examples.

Inhalt

The lecture introduces the world of distributed computing with a focus on fundamentals and technologies from Grid and Cloud Computing as well as the handling of Big Data. The lecture combines theory and application with the help of relevant examples from science and industry.

First, an introduction to the main characteristics of distributed systems is given. Then the topic of Grid Computing is discussed in more detail and the close relationship between Grid computing and distributed data management is illustrated using the example of the WLCG, the infrastructure for distributing, storing and analyzing data from the particle accelerator at CERN.

Subsequently, the topic of cloud computing is discussed and compared with the preceding. After the definition of basic terms and concepts, virtualization is introduced as one of the key technologies of Cloud Computing; finally, common architectures, services and components in the Cloud context are discussed using examples and in general.

Next, common methods for authorization and authentication in distributed environments will be discussed. The lecture includes the description of the basics of Authentication and Authorization Infrastructures (AAI) as well as different technologies, for example certificate- or token-based procedures.

In a further block of topics, concepts for the management and analysis of large or distributed data are presented. In this context tools and frameworks, as well as the lifecycle of data, its metadata and data storage are explained.

Arbeitsaufwand

2 SWS = 120 h per semester

- 30 h in the weekly lecture during the semester
- 90 h post-processing of lectures and self-learning of the content due to its complexity

M**4.30 Modul: Edge-AI in Software and Sensor Applications [M-INFO-107234]****Verantwortung:** Dr. Victor Pankratius**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114258	Edge-AI in Software and Sensor Applications	3 LP	Pankratius

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

QualifikationszieleQualification goals

After completing the module, students have the following skills. They...

- can name and explain the theoretical and practical aspects of software and sensor technology in the context of edge and fog computing
- can name and use techniques of software engineering and algorithm development for sensor applications
- can name and use methods of artificial intelligence in the context of resource constraints and fault tolerance
- can weigh the characteristic properties of the methods and tools presented, their advantages and disadvantages against each other and can select a suitable tool for a given application scenario.

Learning objectives

Students can name the relevant elements of a technical system and their tasks in edge/fog computing. Students are able to name resource constraints of different types (CPU, memory, communication, energy) and describe their effects on software and algorithm design. Students can describe functional principles of sensors of different types (e.g. microelectromechanical systems - MEMS), describe their functional principles in accelerators, gyroscopes, pressure/humidity sensors, particle detection, etc., explain applications and their context (e.g. gesture recognition in mobile phones/"wearables"/"hearables", localization & navigation, environmental measurements). Students are able to design software systems for edge and fog applications and to develop complex edge and fog software projects in an engineering manner. The problems and requirements of different application areas can be recognized, processed and transferred to a new context. Problems in recognizing patterns in sensor data, classification and prediction can be solved using model-based algorithms or machine learning approaches. Problems in deriving instructions for action can be solved using inference techniques.

Inhalt

Edge computing comprises applications, data and services that are relocated to the outer edges of networks. Such systems typically require local data processing with limited resources such as energy consumption, CPUs, memory or connectivity. Fog computing also combines these aspects with cloud architectures. The importance of these approaches is growing today for modern sensor applications and ranges from industrial applications to Internet-of-Things, ubiquitous computing, consumer applications in cell phones, wearables & hearables (e.g. health & fitness applications), drones or applications in augmented reality. At the same time, the proportion of hardware-related software is also growing in all sensor applications, which opens up new possibilities. In this context, artificial intelligence methods are becoming increasingly important in order to realize learning systems with improved autonomy and immediate feedback. This module presents the current status as well as research work and open problems.

Arbeitsaufwand

2 SWS: (2 SWS + 1.5 x 2 SWS) x 15 + 15 h exam preparation = 90 h = 3 ECTS

Empfehlungen

Knowledge of e.g. cognitive systems, software engineering, algorithms, computer networks & structures, low-power design is helpful.

M**4.31 Modul: Embedded Machine Learning Lab [M-INFO-105775]****Verantwortung:** Prof. Dr.-Ing. Jörg Henkel**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-111549	Embedded Machine Learning Lab	4 LP	Henkel

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The student will understand the main concept of machine learning (ML) on embedded systems, the constraints present on such platforms, and the design objectives for ML algorithms on such platforms. The student will be able to understand various concepts of compression of neural networks. The student will gain hands-on experience with current state-of-the-art ML frameworks, parameter tuning of algorithms, and will develop software programs for implementing the concepts. The student will be able to compare and analyze the current state-of-the-art algorithms regarding their flexibility and performance on embedded devices.

Inhalt

IoT devices more and more rely on ML models to perform their operations. They thereby also generate lots of data that should be used to improve these ML models through on-device learning. Devices need to perform the training with this data locally due to privacy constraints or communication limitations. However, the inference of neural networks, and especially the training, requires too many resources (computations, memory, energy, etc.) — unless the available resources are considered in the design.

This lab provides insights into deploying machine learning algorithms to embedded devices.

Since embedded devices operate with significantly lower resources than the commonly-employed high-end GPUs, making neural networks run fast without sacrificing much accuracy on embedded devices is a challenging task. The lab covers training and inference on resource-constrained devices, introducing state-of-the-art methodologies like pruning and quantization.

The students will learn about neural networks beyond theory, working with popular frameworks like TensorFlow, the effects of hyperparameters, and how they influence the network. Furthermore, the student will learn about resource and accuracy trade-offs in neural networks and design custom networks to achieve given resource or accuracy requirements.

This lab requires basic (theoretic) knowledge about neural networks and training. Further knowledge of Linux environments and Python is strongly advised since they will be intensively used in the lab and are the de-facto industry standard for machine learning research.

The students will meet every week. Exact dates and times will be fixed in the first kick-off meeting. Depending on the number of participants, students will work together in groups of 2-3 students.

Arbeitsaufwand

(2 SWS + 1.5*2 SWS)*10

+55 h final project

+15 h presentation & report

= 120 h = 4 ECTS

Empfehlungen

This lab requires a basic (theoretic) knowledge about neural networks and training. Further knowledge of Linux environments and Python is strongly advised since they will be intensively used in the lab and are the de-facto industry standard for machine learning research.

M**4.32 Modul: Empirical Software Engineering [M-INFO-107685]****Verantwortung:** Prof. Dr.-Ing. Anne Koziolk**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
4 LP**Notenskala**
Zehntelnoten**Turnus**
Unregelmäßig**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1

Pflichtbestandteile			
T-INFO-115054	Empirical Software Engineering	4 LP	Dehghani

Erfolgskontrolle(n)

See Partial Achievements (Teilleistung).

Voraussetzungen

See Partial Achievements (Teilleistung).

Qualifikationsziele

- Be able to report empirical studies in software engineering and identify potential sources of error as well as appropriate mitigation strategies.
- Be able to explain and apply statistical analysis methods.
- Analyze and evaluate empirical studies.
- Be able to plan and carry out empirical studies.

Inhalt

The lecture deals with the role of empiricism in software engineering. It introduces the most common empirical methods and points out common sources of error in empirical studies. The associated statistical methods for analyzing and presenting the data are taught. The lecture uses a number of scientific publications to illustrate the concepts and bring them to life.

Arbeitsaufwand

The lecture will take place once (one 1.5h session) per week. The workload will amount to 4.0 CP, as students will also complete exercises in preparation for the exam.

M**4.33 Modul: Engineering Self-Adaptive Systems [M-INFO-106626]****Verantwortung:** Prof. Dr. Raffaella Mirandola**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Wintersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1**Pflichtbestandteile**

T-INFO-113349	Engineering Self-Adaptive Systems	3 LP	Mirandola
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Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

- Understand the motivation for self-adaptation
- Get familiar with the basic principles and conceptual model of self-adaptation
- Understand how to engineer self-adaptive software systems from a software engineering perspective
- Understand the decision-making process using formal analysis at runtime for quality assurance
- Understand the notion of uncertainty in self-adaptive systems and how to tame it with formal verification at runtime
- Understand the level of adoption of self-adaptive systems in industry.

Inhalt

Self-adaptation is an important field of research and engineering that aims to address the challenging problem of how to engineer software systems that have to deal with uncertainties that can only be resolved at run time.

The course presents the basic principles of self-adaptation and introduces a conceptual feedback loop model of a self-adaptive system. It introduces quality models which can be used to estimate quality properties at runtime by a self-adaptive system to provide guarantees for the quality goals. The role played by the different types of uncertainties is then explored analyzing different possible approaches.

Arbeitsaufwand

Course workload:

30h in Class (lectures)

45h self-study during the semester

15h preparation for the exam

M**4.34 Modul: EU Data Protection Law [M-INFO-107030]**

Verantwortung: Gustavo Gil Gasiola
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Minor Studies: Law](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113887	EU Data Protection Law	3 LP	Gil Gasiola

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students are able to comprehend the EU data protection regulation, including the General Data Protection Regulation and related EU data regulations.

They know the foundations of data protection rules, including fundamental concepts (e.g., “personal data”, “processing”, “data subject”). They are also familiar with the principles of personal data processing (lawfulness, limited purpose, transparency, accountability) as well as the rights of the data subject.

They can identify the main obligations of the controller and the processor.

Students understand the conditions for the transfer of personal data to third countries.

They can identify the other regulations that govern data in the European Union.

Students are able to read and understand legal text related to data regulation.

They can understand and solve simple data protection cases.

Inhalt

The General Data Protection Regulation (GDPR) of the European Union is a milestone in protecting individuals from the unlawful use of their data. In a data-driven society, economy, and government, this protection has become essential to guarantee fundamental rights. In addition to its direct impact on the legal systems of all Member States, the GDPR has a major influence on third countries that have adopted similar regulations (e.g. Switzerland, Argentina, Brazil, South Africa, and many others). In this way, the EU Data Protection Regulation has established itself as the “gold standard” of data protection, providing guidance to address the challenges posed by new technologies and new ways of creating, using and sharing personal data. Understanding the structure of data protection in the EU is therefore essential to grasp its impact on individual rights, public administration, business models, and even technological development.

This lecture aims to provide a structured overview of the EU Data Protection Regulation, and to offer tools to understand the regulatory structure of the EU Data Regulation. The lecture will cover the following topics:

- Introduction to EU law
- Development of the EU data protection regulation
- Legal structure of data protection in the EU
- Role of national and sectoral laws
- Data protection as fundamental right
- Principles of data protection
- Lawfulness of personal data processing
- Anonymization and pseudonymization of personal data
- Special categories of personal data
- Rights of the data subject
- Transfer of personal data to third countries
- Responsibility of the controller and the processor
- Security of personal data and personal data breach
- Open Data Directive
- Data Governance Act
- Data Act

Arbeitsaufwand

- Attendance time to the lectures = 15 x 90 min = 22 h 30 min
- Self-study during the semester = 47 h 30 min
- Preparation for the exam = 20 h
- Total = 90 h

M**4.35 Modul: Explainable Artificial Intelligence [M-INFO-106302]**

Verantwortung: TT-Prof. Dr. Rudolf Lioutikov
Einrichtung: KIT-Fakultät für Elektrotechnik und Informationstechnik
 KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Human-centred Machine Intelligence](#)
[Area of Specialization: Data Science](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-112774	Explainable Artificial Intelligence	3 LP	Lioutikov

Erfolgskontrolle(n)

See Partial Achievements (Teilleistung).

Voraussetzungen

See Partial Achievements (Teilleistung).

Qualifikationsziele

- Students are able to understand problems and challenges of XAI
- Students can identify and differentiate different types and approaches of XAI
- Students can implement various XAI approaches
- Students understand current research questions and directions of XAI

Inhalt

Recent advances in Machine Learning and Deep Learning in particular have lead to the imminent introduction of AI agents into a wide variety of applications. However, the apparent “black-box” nature of these approaches hinders their application in both critical systems and close human-robot interactions. The sub-field of eXplainable Artificial Intelligence (XAI) aims to address this shortcoming. This lecture will introduce and discuss various concepts and methods of XAI and consider them from perspective of Robot Learning and Human-Robot Interaction.

The lecture will start with a (brief) introduction into relevant deep learning approaches, before discussing interpretable scene, task and behavior representations. Afterward the lecture will consider itself with Data-Driven and Goal-Driven AI. Finally, first approaches that incorporate XAI and XAI-based human feedback directly into the learning process itself will be discussed. An exemplary list of topics is given below:

- Introduction to XAI
 - Interpretable Machine Learning vs Explainable Machine Learning
- Primer / Introduction to relevant Deep Learning Concepts
 - MLPs and CNNs
 - Graph Neural Networks
 - Transformers
 - Diffusion Models
 - Score Based Methods
- Interpretable Structures
 - Scene Representations
 - Task Representations
 - Behavior Representations
- Data-Driven Explainable AI: XAI Methods for
 - Shapley Values
 - Saliency Maps
 - Concept Activation Vectors
 - Linguistic Neuron Annotation
- Goal-Driven Explainable AI: XAI Methods for
 - Generative Explaining Models
 - Behavior Verbalization
 - Behavior Visualization
- Interactive Learning
 - Integrating Human Feedback
 - Explanatory Interactive Learning

Arbeitsaufwand

Workload = 90h = 3 ECTS

- ca 30h lecture attendance
- ca 30h post-processing
- ca 30h exam preparation

Empfehlungen

- Experience in Machine Learning is recommended, e.g. through prior coursework.
 - The Computer Science Department offers several great lectures e.g., “Maschinelles Lernen - Grundlagen und Algorithmen” and “Deep Learning”
- A good mathematical background will be beneficial
- Python / PyTorch experience could be beneficial when we discuss practical examples/implementations.

M**4.36 Modul: Fine-Grained Complexity Theory & Algorithms [M-INFO-106644]**

Verantwortung: Prof. Dr.-Ing. Marvin Künnemann
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte
6 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-113391	Fine-Grained Complexity Theory & Algorithms	6 LP	Künnemann

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students know the foundations of fundamental algorithmic barriers in the polynomial-time and exponential-time regimes. They are able to use fine-grained reductions to relate the time complexity of different problems. They can derive conditional lower bounds from such reductions, based on established hardness assumptions. Furthermore, they know about the techniques underlying the fastest known algorithms for central problems in the field. 

Inhalt

- fine-grained reductions:
 - conditional lower bounds
 - main techniques for obtaining such reductions
- central hardness assumptions and their applications:
 - (Strong) Exponential Time Hypothesis
 - Orthogonal Vectors Hypothesis
 - 3SUM Hypothesis
 - APSP Hypothesis
- conditional lower bounds for string problems, algorithmic graph theory, geometry
- algorithmic techniques:
 - fastest known algorithms for central problems (SAT, Orthogonal Vectors, 3SUM, APSP)
 - polynomial method
 - applications of fast matrix multiplication
 - Fast Fourier Transform/polynomial multiplication

Arbeitsaufwand

Lecture with exercises, 4 SWS, 6 CP

6 CP amounts to 180 h, distributed as follows:

- about 60 h attendance of lectures and exercise sessions
- about 30 h of preparation and reviewing course material
- about 60 h solving exercise sheets
- about 30 h exam preparation

Empfehlungen

Basic knowledge of theoretical computer science and algorithm design is recommended.

M**4.37 Modul: Geometric Deep Learning [M-INFO-106237]****Verantwortung:** Jun.-Prof. Dr. Jan Stühmer**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
6 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Wintersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
2**Pflichtbestandteile**

T-INFO-112662	Geometric Deep Learning	6 LP	Stühmer
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Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students gain a theoretical and methodical approach to modern Deep Learning as well as knowledge and experience about the application of Deep Learning methods on networks and graphs

Students are able to apply this knowledge for understanding existing state-of-the-art Deep Learning architectures and for deriving novel architectures from first principles

Inhalt

- This module provides students with both theoretical and practical insights into modern Deep Learning
- In particular, we focus on a novel approach for understanding deep neural networks with mathematical tools from geometry and group theory
- This enables a methodical approach to Deep Learning: starting from first principles of symmetry and invariance, we derive different network architectures for analyzing unstructured sets, grids, graphs, and manifolds
- Topics of the course include: group theory, graph neural networks, convolutional neural networks, applications of geometric deep learning in diverse fields such as geometry processing, molecular dynamics, social networks, game playing (computer Go), processing of text and speech, as well as applications in medicine

Arbeitsaufwand

Lecture with 2 SWS + 2 SWS Tutorial, 6 LP
6 LP corresponds to 180 hours, including

- 1) Attendance of the course: 22.5h (15x90min slots)
- 2) Attendance of the exercise sessions: 22.5h (15x90min slots)
- 3) Self-study of course material and work on homework assignments: 60h (4h/week)
- 4) Preparation for the exam: 75h

Empfehlungen

Knowledge about the foundations of machine learning, group theory and linear algebra useful but not required.

M**4.38 Modul: Graph Partitioning and Graph Clustering in Theory and Practice [M-INFO-107211]**

Verantwortung: Prof. Dr. Peter Sanders
Dr. rer. nat. Torsten Ueckerdt

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
5 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114232	Graph Partitioning and Graph Clustering in Theory and Practice	4 LP	Sanders, Ueckerdt
T-INFO-114233	Graph Partitioning and Graph Clustering in Theory and Practice - Practical	1 LP	Sanders, Ueckerdt

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The aim of the lecture is to provide students with an initial insight into the problems of graph partitioning and graph clustering and to apply knowledge from graph theory and algorithmics.

On the one hand, the problems that arise are reduced to their algorithmic core and then solved efficiently. On the other hand, various modelling methods and their interpretations are discussed. After successfully completing the course, students will be able to apply the methods and techniques presented autonomously to related problems.

Inhalt

Many applications in computer science involve the clustering and partitioning of graphs, e.g. the finite element method in scientific simulations, digital circuit design, route planning, web graph analysis or the analysis of social networks.

A well-known example where good partitioning of unstructured graphs is needed is parallel processing, where graphs must be partitioned to distribute computations evenly over a given number of processors and minimise communication between them. k processors, the graph must be divided into k blocks of approximately equal size so that the number of edges between the blocks is minimal.

Since many partitioning and clustering problems occur in practice, the problems discussed are introduced and motivated, and both the theoretical and practical aspects of graph partitioning and graph clustering are taught, including heuristics, meta-heuristics, evolutionary and genetic algorithms as well as approximation and streaming algorithms.

Arbeitsaufwand

Lecture with project/experiment with 3 SWS, 5 CP correspond to approx. 150 working hours, of which
approx. 30 hours attending the lecture
approx. 60 hours of preparation and follow-up work
approx. 30 hours working on the project/experiment
approx. 30 hours exam preparation

M**4.39 Modul: Graphentheorie [M-MATH-101336]**

Verantwortung: Prof. Dr. Maria Aksenovich
Einrichtung: KIT-Fakultät für Mathematik
Bestandteil von: [Minor Studies: Mathematics](#)

Leistungspunkte
9 LP

Notenskala
Zehntelnoten

Turnus
siehe Anmerkungen

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
3

Pflichtbestandteile			
T-MATH-102273	Graphentheorie	9 LP	Aksenovich

Erfolgskontrolle(n)

Die Modulprüfung erfolgt in Form einer schriftlichen Prüfung (3h).

Durch die erfolgreiche Teilnahme am Übungsbetrieb kann ein Bonus erworben werden. Um einen Bonus zu bekommen, muss man jeweils 50% der Punkte für die Lösungen der Übungsblätter 1-6 sowie der Übungsblätter 7-12 erwerben. Liegt die Note der schriftlichen Prüfung zwischen 4,0 und 1,3, so verbessert der Bonus die Note um eine Notenstufe (0,3 oder 0,4).

Voraussetzungen

Keine.

Qualifikationsziele

Die Studierenden können grundlegende Begriffe und Techniken der Graphentheorie nennen, erörtern und anwenden. Sie können geeignete diskrete Probleme als Graphen modellieren und Resultate wie Menger's Satz, Kuratowski's Satz oder Turán's Satz, sowie die in den Beweisen entwickelten Ideen, auf Graphenprobleme anwenden. Insbesondere können die Studierenden Graphen hinsichtlich ihrer Kennzahlen wie Zusammenhang, Planarität, Färbbarkeit und Kantenzahl untersuchen. Sie sind in der Lage, Methoden aus dem Bereich der Graphentheorie zu verstehen und kritisch zu beurteilen. Desweiteren können die Studierenden in englischer Fachsprache kommunizieren.

Inhalt

Der Kurs über Graphentheorie spannt den Bogen von den grundlegenden Grapheneigenschaften, die auf Euler zurückgehen, bis hin zu modernen Resultaten und Techniken in der extremalen Graphentheorie. Insbesondere werden die folgenden Themen behandelt: Struktur von Bäumen, Pfaden, Zykeln, Wegen in Graphen, unvermeidliche Teilgraphen in dichten Graphen, planare Graphen, Graphenfärbung, Ramsey-Theorie, Regularität in Graphen.

Zusammensetzung der Modulnote

Die Modulnote ist Note der Prüfung.

Anmerkungen

- Unterrichtssprache: Englisch
- Dieses Modul ist eines der neun Kernmodule im Bereich Algebra und Geometrie von welchen mindestens sechs innerhalb aller zwei Jahre angeboten werden (mindestens vier verschiedene).

Arbeitsaufwand

Gesamter Arbeitsaufwand: 240 Stunden

Präsenzzeit: 90 Stunden

- Lehrveranstaltung einschließlich studienbegleitender Modulprüfung
- Selbststudium: 150 Stunden
 - Vertiefung der Studieninhalte durch häusliche Nachbearbeitung des Vorlesungsinhaltes
 - Bearbeitung von Übungsaufgaben
 - Vertiefung der Studieninhalte anhand geeigneter Literatur und Internetrecherche
 - Vorbereitung auf die studienbegleitende Modulprüfung

M**4.40 Modul: Hands-on Bioinformatics Practical [M-INFO-101573]**

Verantwortung: Prof. Dr. Alexandros Stamatakis
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Algorithm Engineering](#)

Leistungspunkte
3 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-103009	Hands-on Bioinformatics Practical	3 LP	Stamatakis

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The participants develop and document an open-source tool or pipeline for sequence-based data analysis of biological data. The tool is likely to cover one or more of the main topics of the corresponding lecture and shall be useful to and usable for the biological user community. If possible, the tool should be published in a peer-reviewed scientific journal. Participants learn to work in teams of 2-3 programmers, to use version management tools such as github, to analyse and optimise the runtime behaviour of programs using appropriate tools, to test C/C++ programs for memory leaks (e.g., using valgrind), and to improve the quality of their code using SoftWipe (<https://www.nature.com/articles/s41598-021-89495-8>). Participants will be able to independently carry out and document larger software projects in the field of bioinformatics and evaluate as well as improve code quality. They are able to write a scientific paper in a team.

Inhalt

In the practical course, we jointly develop an open-source tool (algorithms, analysis pipelines, parallelisation) with the aim of providing a new tool that is useful for biology and can be used by biologists at the end of the semester.

Arbeitsaufwand

WWeekly meetings with the supervisor 15 hours + internal team meetings 15 hours + programming time 45 hours + 15 hours writing paper or final report = 90 hours = 3 ECTS

M**4.41 Modul: Hardware Modeling and Simulation [M-ETIT-100449]**

Verantwortung: Dr.-Ing. Jens Becker
Prof. Dr.-Ing. Jürgen Becker

Einrichtung: KIT-Fakultät für Elektrotechnik und Informationstechnik

Bestandteil von: [Minor Studies: Electrical Engineering](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	2

Pflichtbestandteile			
T-ETIT-100672	Hardware Modeling and Simulation	4 LP	Becker, Becker

Erfolgskontrolle(n)

Achievement is examined in the form of a written examination lasting 120 minutes.

Voraussetzungen

none

Qualifikationsziele

After completing this module, students will be familiar with different hardware description languages and their applications in various abstraction levels. They will gain knowledge of the SPICE Hardware Description Language and become proficient in building and deriving the analog matrix for spice simulation. In the realm of digital design, they will develop a comprehensive understanding of the hardware description language VHDL, encompassing the VHDL Standard and its extensions, such as VHDL 2008, the 9-valued logic, and the VHDL-AMS standard. Furthermore, students will achieve a profound comprehension of simulator principles, particularly the delta cycle model. They will also grasp the fundamentals of fault simulations for testing fabricated circuits and learn to derive test vectors. Additionally, students will acquire an understanding of higher-level hardware construction languages like Chisel and SystemC.

Inhalt

In order to address the complexity of modern chips during development, it is essential to utilize modern hardware description languages. This course offers insights into the various levels of abstraction in these languages. It starts by covering the fundamentals of analog description using SPICE and then progresses through VHDL, VHDL-AMS, and Verilog. Additionally, the course introduces more abstract languages like Chisel and SystemC.

Topics covered in the course are:

- Design Process
- Basics of Modeling and Simulation
- Low Level Modeling
- VHDL
 - VHDL-AMS
 - 9-valued logic
 - Delta cycle simulation
 - Fault simulation
- Verilog
- Chisel
- SystemC

Zusammensetzung der Modulnote

The module grade results from the grade of the written examination.

Anmerkungen

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Arbeitsaufwand

The workload is covered by:

1. Participating in lectures and tutorials: 33h
2. Preparing and wrap up of the above named units: 66h
3. Exam preparation and presence: 21h

Sum: 120h = 4 LP

M**4.42 Modul: Hardware Synthesis and Optimization [M-ETIT-106963]**

Verantwortung: Prof. Dr.-Ing. Jürgen Becker
Einrichtung: KIT-Fakultät für Elektrotechnik und Informationstechnik
Bestandteil von: [Minor Studies: Electrical Engineering](#)

Leistungspunkte 6 LP	Notenskala Zehntelnoten	Turnus Jedes Sommersemester	Dauer 1 Semester	Sprache Englisch	Level 4	Version 1
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Pflichtbestandteile			
T-ETIT-113922	Hardware Synthesis and Optimization	6 LP	Becker

Erfolgskontrolle(n)

The examination takes place within the framework of an oral overall examination (approx. 30 minutes).

Voraussetzungen

none

Qualifikationsziele

Students know the basic steps required for the automated design of optimized digital circuits. They are able to classify them in the Y-chart and assess their complexity.

They will be able to name and explain the most important approaches for these design steps and evaluate them with regard to optimality and computational effort. This includes the ability to use algorithms for these approaches, e.g. selected graph algorithms, metaheuristics such as simulated annealing. The students are also able to determine their respective runtime complexities.

In addition, they can solve given problems from the field of design automation by selecting a suitable approach based on certain optimization criteria and applying it to the respective problem.

Inhalt

The module focuses on teaching the formal and methodological foundations for the automated design of optimized electronic systems. The relevant scientific and methodological properties of the methods used are discussed and their implementation in industrial practice is also taught.

The following topics are covered:

- Graph Algorithms and Complexity
- High-Level Synthesis
- Algorithms for Scheduling, Allocation and Binding Problems
- Register-Transfer-Level Synthesis
- Retiming Algorithms
- Logic Optimization
- Technology Mapping for Standard Cells and FPGAs
- Physical Design
- Placement of Standard Cells with ILP and Simulated Annealing
- Global and Detailed Routing

Zusammensetzung der Modulnote

The module grade is the grade of the oral exam.

Arbeitsaufwand

The workload includes (4 SWS):

1. 1. attendance in lectures and exercises: 60 h
 2. 2. preparation / follow-up: 60 h
 3. 3. preparation of and attendance in examination: 60 h
- A total of 180 h = 6 CR

Empfehlungen

Basic knowledge in the field of digital circuits, e.g. as taught in the course "Digital Technology" (2311615) is helpful.

M**4.43 Modul: Hardware-efficient Artificial Intelligence [M-INFO-107448]****Verantwortung:** Prof. Dr.-Ing. Jörg Henkel**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Wintersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
2

Pflichtbestandteile			
T-INFO-114628	Hardware-efficient Artificial Intelligence	3 LP	Henkel, Rapp

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Upon successful completion of the course, participants will be able to:

1. Understand and analyze the challenges of efficiently deploying deep learning models on hardware platforms.
2. Apply optimization techniques to improve the hardware efficiency of deep learning models, specifically to reduce latency, energy consumption, and memory usage.
3. Explain and implement the following methods in practical scenarios:
 - Knowledge Distillation
 - Neural Architecture Search (NAS)
 - Pruning
 - Quantization
4. Evaluate and adapt the use of these techniques for both cloud-based and embedded systems.
5. Perform model optimizations while maintaining high task performance.

Inhalt

Deep learning models are ubiquitous in modern applications, from computer vision to natural language processing. However, deploying these models efficiently on hardware platforms remains a significant challenge, leading to increased latency, energy consumption, memory usage, and cost. As deep learning models become larger and more complex, the need for optimization techniques to improve their performance on hardware is more critical than ever. This course discusses the principles and techniques for optimizing deep learning models for hardware deployment, focusing on both cloud and embedded systems.

The course covers various optimization techniques to improve model efficiency, reduce latency, and minimize memory usage while maintaining a high task performance. Precisely, the covered topics include

- Knowledge distillation
- Neural architecture search (NAS)
- Pruning
- Quantization

A basic understanding of deep learning concepts and neural networks is recommended. Furthermore, familiarity with hardware architectures and embedded systems is beneficial but not mandatory.

Arbeitsaufwand

lecture time: 24h

self study: 24h

exam preparation: 42h

total: 90h = 3 ECTS

Empfehlungen

A basic understanding of deep learning concepts and neural networks is expected; familiarity with hardware architectures and embedded systems is helpful but not required.

M**4.44 Modul: HRI and Social Robotics [M-INFO-106650]****Verantwortung:** TT-Prof. Dr. Barbara Bruno**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113396	HRI and Social Robotics	4 LP	Bruno
T-INFO-113397	HRI and Social Robotics - Pass	2 LP	Bruno

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students know the foundations of Human-Robot Interaction (HRI) and Social Robotics, including: design principles and methodologies, human factors influencing HRI (anthropomorphization), sensors, actuators and software architecture for social robotics, challenges and solutions for key abilities of social robots (spatial interaction, engagement detection, non-verbal interaction, verbal interaction, emotion generation and detection), research methods (study design principles, statistical tools for analyses) and have seen state-of-the-art research topics in the field including social learning, theory of mind, trust and ethical considerations in HRI.

Thanks to the exercise sessions and assignments, students gain first-hand knowledge and can independently apply techniques related to the above theory items, including for collecting stakeholders' feedback for a robot design, programming the robot's social behaviour along multiple modalities, extracting relevant user information from available sensors, designing and analysing HRI experiments.

Inhalt

The lectures cover all foundational topics in HRI (design principles and methodologies, human factors influencing HRI, sensors, actuators and software architecture for social robotics), challenges and solutions for key abilities of social robots (spatial interaction, engagement detection, non-verbal interaction, verbal interaction, emotion generation and detection), research methods (study design principles, statistical tools for analyses) and state-of-the-art topics including social learning, theory of mind and ethical considerations in HRI.

In the exercise sessions and related assignments students can experience first-hand how the theoretical concepts seen in the lectures can be applied in practice and learn how to collect stakeholders' feedback for a robot design, program the robot's social behaviour along multiple modalities, extract relevant user information from available sensors, design and analyse HRI experiments. At the end of the course, the students have a solid understanding of HRI, its principles, challenges and solutions and can autonomously apply such knowledge in practical contexts.

Arbeitsaufwand

Course workload:

- 1) Attendance of the course: 22.5h (15x90min slots)
- 2) Attendance of the exercise sessions: 22.5h (15x90min slots)
- 3) Self-study of course material and work on homework assignments: 60h (4h/week)
- 4) Preparation for the exam: 80h

Empfehlungen

Knowledge of the content of modules Robotics I - Introduction to Robotics is helpful.

M**4.45 Modul: Human Computer Interaction [M-INFO-107166]****Verantwortung:** Prof. Dr.-Ing. Michael Beigl**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Telematics](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114192	Human-Computer-Interaction	6 LP	Beigl
T-INFO-114193	Human-Computer-Interaction Pass	0 LP	Beigl

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

After completing the course, students will be able to

- reproduce basic knowledge about the field of human-machine interaction
- name and apply basic techniques for analysing user interfaces
- apply basic rules and techniques for designing user interfaces
- analyse and evaluate existing user interfaces and their function

Inhalt

Topics are:

1. human information processing (models, physiological and psychological principles, human senses, action processes),
2. design principles and design methods, input and output units for computers, embedded systems and mobile devices,
3. principles, guidelines and standards for the design of user interfaces
4. technical basics and examples for the design of user interfaces (text dialogues and forms, menu systems, graphical interfaces, interfaces in the WWW, audio dialogue systems, haptic interaction, gestures),
5. methods for modelling user interfaces (abstract description of interaction, embedding in requirements analysis and the software design process),
6. evaluation of systems for human-machine interaction (tools, evaluation methods, performance measurement, checklists).
7. practising the above basics using practical examples and developing independent, new and alternative user interfaces.

Arbeitsaufwand

The total workload for this course unit is approx. 180 hours (6.0 credits).

Attendance time: Attendance of the lecture 15 x 90 min = 22 h 30 min

Attendance time: Attendance of the exercise 8 x 90 min = 12 h 00 min

Preparation / follow-up of the lecture 15 x 150 min = 37 h 30 min

Preparation / follow-up of the exercise 8x 360min =48h 00min

Go through slides/script 2x 2 x 12 h =24 h 00 min

Prepare exam = 36 h 00 min

SUM = 180h 00 min

M**4.46 Modul: Humanoid Robots - Locomotion and Whole-Body Control [M-INFO-106649]**

Verantwortung: Prof. Dr. Katja Mombaur
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	2

Pflichtbestandteile			
T-INFO-113395	Humanoid Robots - Locomotion and Whole-Body Control	6 LP	Mombaur
T-INFO-114282	Humanoid Robots - Locomotion and Whole-Body Control -Pass	0 LP	Mombaur

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

By the end of the course, students will be able to:

- Develop kinematic and dynamic models of humanoid robots
- Understand basic principles of human whole-body movement
- Control gaits and other whole-body motions for humanoid robots and maintain balance
- Explain advanced methods for humanoid motion generation, optimization, and learning
- Give an overview of the state of the art in locomotion and whole-body control of humanoid robotics
- Complete a graduate level research project on humanoid robots including simulation and real-robot implementation

Inhalt

This course introduces fundamentals and recent developments in the field of humanoid robotics with a focus on locomotion and whole-body motions. We will cover kinematic and dynamic modeling of anthropomorphic systems, basic concepts of bipedal walking control, stability aspects, gait generation in different terrains, humanoid balance and push recovery, motion primitives and optimal control-based approaches, motion imitation and learning. The course will also give some insights in basic principles of passive dynamic walking, human motion generation and control and human motion modeling. Students will work with different robotics tools and perform a graduate level research project related to a whole-body humanoid robot.

This module is complementary to the course “4.290 Robotik II - Humanoide Robotik” which focuses on upper body motions and cognitive architectures while this course focuses on the specific aspects of legged humanoids and whole-body motions. The modules can be taken at the same time.

Zusammensetzung der Modulnote

See partial achievements (Teilleistung)

Anmerkungen

For capacity reasons (supervision of computational projects), the lecture is limited to 30 students. If necessary, a selection process will be carried out. Places will be allocated taking into account the students' subject-specific skills.

Arbeitsaufwand

Estimated effort for this module is 180 hours:

60h - Lecture and exercises (2+2 SWS)

40h - Repetition of lecture contents, preparation of assignments

80h – Work on final project, documentation and presentation

Empfehlungen

Attendance of the lectures Robotics I - Introduction to Robotics and Mechano-Informatics in Robotics is required.

M**4.47 Modul: Humanoid Robots - Seminar [M-INFO-107152]****Verantwortung:** Prof. Dr.-Ing. Tamim Asfour**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114170	Humanoid Robots - Seminar	3 LP	Asfour

Erfolgskontrolle(n)

See partial Achievements (Teilleistung)

Voraussetzungen

See partial Achievements (Teilleistung)

Qualifikationsziele

The students choose a topic from the field of humanoid robotics, e.g. robot design, motion generation, perception or learning. They conduct a literature research on this topic under the guidance of a scientific supervisor. At the end of the semester, they present the results and write a term paper in English in the form of a scientific publication.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

The student gained experience with literature research on a current research topic. He/she explored, understood and compared different approaches to a selected scientific problem. The student is able to write a summary of their literature research in the form of a scientific publication in English and to give a scientific talk on it.

Arbeitsaufwand

Seminar with 2 SWS, 3 LP
 3 LP corresponds to 90 hours, including
 45 hours literature research
 25 hours manuscript preparation
 10 hours preparation of the presentation
 10 hours attendance time

Empfehlungen

Attending the lectures Robotics I – Introduction to Robotics, Robotics II: Humanoid Robotics, Robotics III – Sensors and Perception in Robotics, Mechano-Informatics and Robotics and Wearable Robotic Technologies is recommended.

M**4.48 Modul: Interdisciplinary Qualifications [M-INFO-107254]**

Verantwortung: Prof. Dr. Bernhard Beckert
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Interdisciplinary Qualifications](#)

Leistungspunkte
6 LP

Notenskala
best./nicht best.

Turnus
Jedes Semester

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

SQ- Master (Wahl: zwischen 1 und 6 LP)			
T-INFO-102051	Lesegruppe Softwaretechnik	1 LP	Reussner
T-INFO-111474	Selbstverbuchung-HOC-SPZ-FORUM-PEBA-benotet	1 LP	Coerdts
T-INFO-111475	Selbstverbuchung-HOC-SPZ-FORUM-PEBA-benotet	2 LP	Coerdts
T-INFO-111476	Selbstverbuchung-HOC-SPZ-FORUM-PEBA-benotet	3 LP	Coerdts
T-INFO-111477	Selbstverbuchung-HOC-SPZ-FORUM-PEBA-unbenotet	1 LP	Coerdts
T-INFO-111478	Selbstverbuchung-HOC-SPZ-FORUM-PEBA-unbenotet	2 LP	Coerdts
T-INFO-111479	Selbstverbuchung-HOC-SPZ-FORUM-PEBA-unbenotet	3 LP	Coerdts
T-INFO-111839	Information, Wissenschaft und Verantwortung – aktuelle ethische Herausforderungen der IT	1 LP	Kaplan
T-INFO-112148	Information, Wissenschaft und Verantwortung – aktuelle ethische Herausforderungen der IT	2 LP	Kaplan
T-INFO-115027	Selbstverbuchung-HOC-SPZ-FORUM-PEBA-benotet	2 LP	Coerdts
T-INFO-115028	Selbstverbuchung-HOC-SPZ-FORUM-PEBA-benotet	1 LP	Coerdts
T-INFO-115029	Selbstverbuchung-HOC-SPZ-FORUM-PEBA-unbenotet	2 LP	Coerdts
T-INFO-115030	Selbstverbuchung-HOC-SPZ-FORUM-PEBA-unbenotet	1 LP	Coerdts
T-INFO-115077	Selbstverbuchung-HOC-SPZ-FORUM-PEBA-unbenotet	4 LP	Coerdts

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Learning objectives can be divided into three main categories, which complement each other:

1. Orientation knowledge

- Students become aware of the cultural imprint of their position and are able to consider the views and interests of others (across disciplinary, cultural and linguistic boundaries).
- They expand their ability to participate in academic or public discussions in an appropriate and relevant manner.

2. practical orientation

- Students gain insight into the routines of professional action.
- They further develop their learning skills.
- They expand their ability to act by improving their foreign language skills.
- They can link basic business and legal issues to their field of experience.

3. basic skills

- Students can acquire new knowledge independently in a planned, targeted and methodologically sound manner and apply this knowledge to solve tasks and problems.
- They can evaluate their own work.
- They have efficient working techniques, can set priorities, make decisions and take responsibility.

Inhalt

The House of Competence (HoC) is the central, research-based institution in the field of interdisciplinary competence development at KIT and offers students of all disciplines a broad learning portfolio. The HoC seminar program is divided into focus areas that aim to develop interdisciplinary skills for studies and careers. The three HoC laboratories are mainly responsible for the focus areas: the Methods LAB, Learning LAB and Writing LAB.

The courses in the HoC program can be credited in the areas of "Key Qualifications" (SQ), "Additional Professional Qualifications" (BOZ) and in the "Personal Competence Module" for student teachers (MPK). The requirements for the respective degree programs can be found in the applicable examination and study regulations. The current seminar program, which is published each semester, can be found on the HoC homepage at www.hoc.kit.edu.

Anmerkungen

German courses and/or language courses in the native language are not recognized as key qualifications.

Only those examination and study achievements can be credited, which cannot be taken in the computer science or supplementary subjects. Certificates of attendance are not accepted.

Arbeitsaufwand

Each credit point corresponds to approx. 25-30 hours of work (by the student). This is based on the average student who achieves an average performance. The workload includes (for a lecture)

1. Attendance time in lectures, exercises
2. Preparation and follow-up of the same
3. Exam preparation and attendance in the same.

M**4.49 Modul: Internet of Everything [M-INFO-100800]**

Verantwortung: Prof. Dr. Martina Zitterbart
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-101337	Internet of Everything	4 LP	Zitterbart

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students

- know the challenges of the Internet of Everything (IoE) from both a technical and legal perspective
- know and understand the risks to user privacy in the IoE as well as basic mechanisms and protocols to protect it
- are familiar with the basic architectures and protocols in the field of wireless sensor networks and the Internet of Things.

Students know the platforms and applications of the Internet of Everything. Students have an understanding the challenges of designing protocols and applications for the IoE.

Students know and understand the risks to the privacy of users of the future IoE. They know protocols and mechanisms to enable future applications, such as smart metering and smart traffic, while protecting the privacy of users.

Students know and understand classic sensor network protocols and applications, such as media access procedures, routing protocols, transport protocols and mechanisms for topology control. Students know and understand the interaction of individual communication layers and the influence on, for example, the energy requirements of the systems.

Students know protocols for the Internet of Things such as 6LoWPAN, RPL, CoAP and DICE. Students understand the challenges and assumptions that have led to the standardization of protocols.

Students have a basic understanding of security technologies in IoE. They know typical protection goals and attacks, as well as building blocks and protocols to implement the protection goals.

Inhalt

The lecture deals with selected protocols, architectures, procedures and algorithms that are essential for IoE. In addition to classic topics from the field of wireless sensor-actuator networks, such as media access and routing, this also includes new challenges and solutions for the security and privacy of transmitted data in IoE. Socially and legally relevant aspects are also addressed.

Arbeitsaufwand

Lecture with 2 SWS plus follow-up/exam preparation, 4 CP.

4 CP corresponds to approx. 120 working hours, of which

approx. 30 hours lecture attendance

approx. 60 hours preparation/follow-up work

approx. 30 hours exam preparation

Empfehlungen

See partial achievements (Teilleistung)

M**4.50 Modul: Introduction to Bioinformatics for Computer Scientists [M-INFO-100749]****Verantwortung:** Prof. Dr. Alexandros Stamatakis**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)**Voraussetzung für:** [T-INFO-101287 - Seminar: Hot Topics in Bioinformatics](#)
[T-INFO-103009 - Hands-on Bioinformatics Practical](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Wintersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1

Pflichtbestandteile			
T-INFO-101286	Introduction to Bioinformatics for Computer Scientists	3 LP	Stamatakis

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students attain comprehensive knowledge of standard methods, algorithms, theoretical principles and open problems in the field of sequence-based bioinformatics (biological principles, sequence assembly, pairwise sequence alignment, multiple sequence alignment, phylogenetic tree reconstruction under parsimony, likelihood and Bayesian models, coalescent inference in population genetics). They develop the ability to categorise and evaluate algorithms and problems. They can select suitable models and methods for a given biological data analysis problem and can justify their choice. Students will be able to design analysis pipelines for biological data analysis.

Inhalt

Initially, some basic concepts and mechanisms of biology are introduced. Subsequently, algorithms and models from the fields of sequence analysis (sequence alignment, dynamic programming, sequence assembly), population genetics, and discrete as well as numerical algorithms for inferring molecular phylogenetic trees (parsimony, likelihood, Bayesian inference) are discussed. Furthermore, discrete operations on trees are treated (e.g., topological distances between trees, consensus tree algorithms). A substantial part of the lectures will cover the practical implementation, the optimisation, and the parallelisation of the respective methods.

Arbeitsaufwand

2 SWS lecture + 1.5 * 2 SWS follow-up) * 15 + 15 hours exam preparation = 90 hours = 3 ECTS

M**4.51 Modul: IT Security [M-INFO-106998]**

Verantwortung: Prof. Dr. Jörn Müller-Quade
Prof. Dr. Christian Wressnegger

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	2

Pflichtbestandteile			
T-INFO-113960	IT Security	6 LP	Müller-Quade, Wressnegger

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students

- have in-depth knowledge of cryptography and IT security
- know and understands sophisticated techniques and security primitives to achieve the protection goals
- know and understand scientific evaluation and analysis methods of IT security (game-based formalization of confidentiality and integrity, security and anonymity notions)
- have a good understanding of types of data, personal data, legal and technical fundamentals of privacy protection
- know and understand the fundamentals of system security (buffer overflow, return-oriented programming, ...)
- know different mechanisms for anonymous communication (TOR, Nym, ANON) and can assess their effectivity

Inhalt

This advanced mandatory module deepens different topics of IT security. These include in particular:

- Elliptic curve cryptography
- Threshold cryptography
- Zero-knowledge proofs
- Secret sharing
- Secure multi-party computation and homomorphic encryption
- Methods of IT security (game-based analysis and the UC model)
- Crypto-currencies and consensus through proof-of-work/stake
- Anonymity on the Internet, anonymity with online payments
- Privacy-preserving machine learning
- Security of machine learning
- System security and exploits
- Threat modeling and quantification of IT security

Arbeitsaufwand

Course workload:

1. Attendance time: 56 h
2. Self-study: 56 h
3. Preparation for the exam: 68 h

Empfehlungen

Attendance of the lecture Information Security is recommended.

Literatur

Literature:

- Katz/Lindell: Introduction to Modern Cryptography (Chapman & Hall)
- Schäfer/Roßberg: Netzsicherheit (dpunkt)
- Anderson: Security Engineering (Wiley, and online)
- Stallings/Brown: Computer Security (Pearson)
- Pfleeger, Pfleeger, Margulies: Security in Computing (Prentice Hall)

M**4.52 Modul: Kombinatorik [M-MATH-102950]**

Verantwortung: Prof. Dr. Maria Aksenovich
Einrichtung: KIT-Fakultät für Mathematik
Bestandteil von: [Minor Studies: Mathematics](#)

Leistungspunkte
9 LP

Notenskala
Zehntelnoten

Turnus
siehe Anmerkungen

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
4

Pflichtbestandteile			
T-MATH-105916	Kombinatorik	9 LP	Aksenovich

Erfolgskontrolle(n)

Die Modulprüfung erfolgt in Form einer schriftlichen Prüfung (2h).

Durch die erfolgreiche Teilnahme am Übungsbetrieb kann ein Bonus erworben werden. Um einen Bonus zu bekommen, muss man jeweils 50% der Punkte für die Lösungen der Übungsblätter 1-6 sowie der Übungsblätter 7-12 erwerben. Liegt die Note der schriftlichen Prüfung zwischen 4,0 und 1,3, so verbessert der Bonus die Note um eine Notenstufe (0,3 oder 0,4).

Voraussetzungen

Keine

Qualifikationsziele

Die Studierenden können grundlegende Begriffe und Techniken der Kombinatorik nennen, erörtern und anwenden. Sie können kombinatorische Probleme analysieren, strukturieren und formal beschreiben. Die Studierenden können Resultate und Methoden, wie das Inklusions-Exklusions-Prinzip, Erzeugendenfunktionen oder Young Tableaux, sowie die in den Beweisen entwickelten Ideen, auf kombinatorische Probleme anwenden. Insbesondere sind sie in der Lage, die Anzahl der geordneten und ungeordneten Arrangements gegebener Größe zu bestimmen oder die Existenz solcher Arrangements zu beweisen oder zu widerlegen. Die Studierenden sind fähig, Methoden aus dem Bereich der Kombinatorik zu verstehen und kritisch zu beurteilen. Desweiteren können die Studierenden in englischer Fachsprache kommunizieren.

Inhalt

Die Vorlesung bietet eine Einführung in die Kombinatorik. Angefangen mit Problemen des Abzählens und Bijektionen, werden die klassischen Methoden des Inklusion-Exklusions-Prinzip und der erzeugenden Funktionen behandelt. Weitere Themengebiete beinhalten Catalan-Familien, Permutationen, Partitionen, Young Tableaux, partielle Ordnungen und kombinatorische Designs.

Zusammensetzung der Modulnote

Die Modulnote ist Note der schriftlichen Prüfung.

Anmerkungen

- Unterrichtssprache: Englisch
- Dieses Modul ist eines der neun Kernmodule im Bereich Algebra und Geometrie von welchen mindestens sechs innerhalb aller zwei Jahre angeboten werden (mindestens vier verschiedene).

Arbeitsaufwand

Gesamter Arbeitsaufwand: 240 Stunden

Präsenzzeit: 90 Stunden

- Lehrveranstaltung einschließlich studienbegleitender Modulprüfung

Selbststudium: 150 Stunden

- Vertiefung der Studieninhalte durch häusliche Nachbearbeitung des Vorlesungsinhaltes
- Bearbeitung von Übungsaufgaben
- Vertiefung der Studieninhalte anhand geeigneter Literatur und Internetrecherche
- Vorbereitung auf die studienbegleitende Modulprüfung

Empfehlungen

Kenntnisse aus den Vorlesungen Lineare Algebra 1 und 2 sowie Analysis 1 und 2 sind empfohlen.

M**4.53 Modul: Lab Project: Speech Translation [M-INFO-107176]**

Verantwortung: Prof. Dr. Jan Niehues
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114205	Lab Project: Speech Translation	6 LP	Niehues

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The student

- is able to develop a language translation system using state-of-the-art methods.
- can evaluate language translation systems.
- can present his/her findings in a scientific lecture.

Inhalt

The use of deep learning technologies has significantly improved the quality of machine translation of text and speech in recent years. In this internship, students will develop a language translation system for a new language pair using state-of-the-art methods.

In the first part of the internship, students are introduced step-by-step to the development of a translation system and its evaluation. To this end, the various subtasks must be solved. In the second part of the internship, the students are asked to independently analyse various improvements to the system.

Arbeitsaufwand

180h

Approx. 15h presence

Approx. 15h pre/post processing

Approx. 140h self-study

Approx. 10h Preparation of scientific presentation

Empfehlungen

Students should have understood the theoretical principles as introduced in the lectures Deep Learning or Machine Translation.

M**4.54 Modul: Localization of Mobile Agents [M-INFO-107663]****Verantwortung:** Prof. Dr.-Ing. Uwe Hanebeck**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-115014	Localization of Mobile Agents	6 LP	Hanebeck
T-INFO-115015	Localization of Mobile Agents - Pass	0 LP	Hanebeck

Erfolgskontrolle(n)

See Partial Achievements (Teilleistung).

Voraussetzungen

See Partial Achievements (Teilleistung).

Qualifikationsziele

Participants in this course will gain a broad overview of probabilistic machine learning tools and learn how to creatively combine them to solve a wide variety of problems.

Inhalt

This module provides a systematic introduction to the field of localization methods. To facilitate entry, the module is divided into four central topics. Dead reckoning (1) deals with the step-by-step determination of a vehicle's position from dynamic parameters such as speed or steering angle, as well as inertial measurement unit (IMU) data. Localization with the aid of measurements to external, known landmarks is part of static localization (2). Specifically, signal times of arrival (multilateration and GPS), distance measurements, and angle measurements (bearings) are covered. In addition to closed-form solutions for special measurements, the method of least squares for fusing arbitrary measurements is introduced. This also includes nonlinear optimization methods and certifiable global optimization. Dynamic localization (3) deals with the combination of dead reckoning and static localization using the Kalman filter, which is used in virtually all object tracking scenarios and other state estimation problems. General estimator properties are also discussed. The course concludes with simultaneous localization and mapping (SLAM) (4), which allows localization even when the location of landmarks is partially unknown. Thereby the measurement association problem is also addressed.

Arbeitsaufwand

Per week: 📅

2 SWS Presence

2h Follow-up

6h Digital exercise with computation tasks

2h exam preparation

= 12h/week und 180h/semester

Empfehlungen

Knowledge of a higher-level programming language with mature libraries for scientific and numerical computing (e.g., Julia, Matlab, Python) is an advantage.

M**4.55 Modul: Logical Foundations of Cyber-Physical Systems [M-INFO-106102]****Verantwortung:** Prof. Dr. André Platzer**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)

Leistungspunkte
6 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Wintersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
3

Pflichtbestandteile			
T-INFO-112360	Logical Foundations of Cyber-Physical Systems	6 LP	Platzer

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele**In modeling and control**, successful students will

- understand core principles behind CPS. A solid understanding of these principles is important for anyone who wants to integrate cyber and physical components to solve problems that no part could solve alone.
- develop models and controls. In order to understand, design, and analyze CPS, it is important to be able to develop models for the relevant aspects of a CPS design and to design controllers for the intended functionalities based on appropriate specifications, including modeling with differential equations.
- identify relevant dynamical aspects. It is important to be able to identify which types of phenomena influence a property of a system. These allow us to judge, for example, where it is important to manage adversarial effects, or where a nondeterministic model is sufficient.

In computational thinking, successful students should be able to

- identify safety specifications and critical properties. In order to develop correct CPS designs, it is important to identify what “correctness” means, how a design may fail to be correct, and how to make it correct.
- understand abstraction in system designs. The power of abstraction is essential for the modular organization of CPS, and the ability to reason about separate parts of a system independently.
- express pre- and post-conditions and invariants for CPS models. Pre- and post-conditions allow us to capture under which circumstance it is safe to run a CPS or a part of a CPS design, and what safety entails. They allow us to achieve what abstraction and hierarchies achieve at the system level: decompose correctness of a full CPS into correctness of smaller pieces. Invariants achieve a similar decomposition by establishing which relations of variables remain true no matter how long and how often the CPS runs.
- reason rigorously about CPS models. Reasoning is required to ensure correctness and find flaws in CPS designs. Both informal and formal reasoning in a logic are important objectives for being able to establish correctness, which includes rigorous reasoning about differential equations.

In CPS skills, successful students will be able to

- understand the semantics of a CPS model. What may be easy in a classical isolated program becomes very demanding when that program interfaces with effects in the physical world.
- develop an intuition for operational effects. Intuition for the joint operational effect of a CPS is crucial, e.g., about what the effect of a particular discrete computer control algorithm on a continuous plant will be.
- understand opportunities and challenges in CPS and verification. While the beneficial prospects of CPS for society are substantial, it is crucial to also develop an understanding of their inherent challenges and of approaches for minimizing the impact of potential safety hazards. Likewise, it is important to understand the ways in which formal verification can best help improve the safety of system designs.

Inhalt

Cyber-physical systems (CPSs) combine cyber capabilities (computation and/or communication) with physical capabilities (motion or other physical processes). Cars, aircraft, and robots are prime examples, because they move physically in space in a way that is determined by discrete computerized control algorithms. Designing these algorithms to control CPSs is challenging due to their tight coupling with physical behavior. At the same time, it is vital that these algorithms be correct, since we rely on CPSs for safety-critical tasks like keeping aircraft from colliding. In this course we will strive to answer the fundamental question posed by Jeannette Wing:

“How can we provide people with cyber-physical systems they can bet their lives on?”

The cornerstone of this course design are hybrid programs (HPs), which capture relevant dynamical aspects of CPSs in a simple programming language with a simple semantics. One important aspect of HPs is that they directly allow the programmer to refer to real-valued variables representing real quantities and specify their dynamics as part of the HP.

This course will give you the required skills to formally analyze the CPSs that are all around us—from power plants to pacemakers and everything in between—so that when you contribute to the design of a CPS, you are able to understand important safety-critical aspects and feel confident designing and analyzing system models. It will provide an excellent foundation for students who seek industry positions and for students interested in pursuing research.

Anmerkungen

Course web page: <https://lfcps.org/course/lfcps.html>

Arbeitsaufwand

6 ECTS from 180h of coursework consisting of

45h = $15 * 3$ from 3 SWS lectures

15h = $15 * 1$ from 1 SWS exercises

68h preparation, reading textbook, studying

40h solving exercises

12h exam preparation

Empfehlungen

The course assumes prior exposure to basic computer programming and mathematical reasoning. This course covers the basic required mathematical and logical background of cyber-physical systems. You will be expected to follow the textbook as needed: André Platzer. Logical Foundations of Cyber-Physical Systems. Springer 2018. DOI:10.1007/978-3-319-63588-0

M**4.56 Modul: Low Power Design [M-INFO-100807]****Verantwortung:** Prof. Dr.-Ing. Jörg Henkel**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Sommersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1**Pflichtbestandteile**

T-INFO-101344	Low Power Design	3 LP	Henkel
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Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students are made aware of various low power design optimizations employed in state-of-the-art embedded devices. This involves optimization techniques that incorporate embedded machine learning algorithms to enhance system performance. At the end of the lecture, the students will be able to recognize the challenges involved in crafting efficient low power designs and how to tackle them.

Inhalt

The lecture provides an overview of design methods, synthesis tools, estimation models, software techniques, operating system strategies, scheduling algorithms, embedded machine learning methods, etc., with the aim of minimizing the power consumption of embedded devices without compromising their performance. Both the research-relevant and industry-prevalent topics at different level of abstractions (from circuit to system) are discussed in this lecture.

Arbeitsaufwand

Attendance time: 30 hours (2 SWS × 15 weeks)

Self-study: 45 hours (1.5 × 2 SWS × 15 weeks)

Exam preparation: 15 hours

Total: 90 hours (3 ECTS)

M**4.57 Modul: Machine Learning - Foundations and Algorithms [M-INFO-107169]**

Verantwortung: Prof. Dr. Gerhard Neumann
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	2

Pflichtbestandteile			
T-INFO-114197	Machine Learning - Foundations and Algorithms	6 LP	Neumann

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

- Students acquire knowledge of the basic methods of Machine Learning
- Students acquire the mathematical knowledge to understand the theoretical foundations of Machine Learning
- Students can categorize, formally describe and evaluate methods of Machine Learning
- Students can apply their knowledge to select appropriate models and methods for selected problems in the field of Machine Learning.

Inhalt

The field of Machine Learning has made enormous progress in recent years and good knowledge of Machine Learning is becoming increasingly in demand on the job market. Machine Learning describes the acquisition of knowledge by an artificial system based on experience or data. Rules or certain calculations no longer have to be manually coded but can be extracted from data by intelligent systems.

This lecture provides an overview of essential and current methods of Machine Learning. After reviewing the necessary mathematical background, the lecture primarily deals with algorithms for classification, regression, and density estimation, with a focus on the mathematical understanding of probabilistic methods and neural networks.

Examples of topics include:

- Basics in Linear Algebra, Probability Theory, Optimization and Constraint Optimization
- Linear Regression
- Linear Classification
- Model Selection, Overfitting, and Regularization
- Support Vector Machines
- Kernel Methods
- Bayesian Learning and Gaussian Processes
- Neural Networks
- Dimensionality Reduction
- Density estimation
- Clustering
- Expectation Maximization
- Graphical Models

Arbeitsaufwand

180h, divided into:

- ca 45h lecture attendance
- approx. 15h attending exercises
- approx. 90h post-processing and working on the exercise sheets
- ca 30h exam preparation

M**4.58 Modul: Machine Learning for Natural Sciences [M-INFO-106959]****Verantwortung:** Prof. Dr. Pascal Friederich**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Area of Specialization: Data Science](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113916	Machine Learning for Natural Sciences	3 LP	Friederich
T-INFO-113917	Machine Learning for Natural Sciences - Pass	3 LP	Friederich

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Qualification Objectives

- Students are able to name relevant machine learning methods, describe them, as well as develop independent proposals on how questions in the natural sciences and material sciences can be answered using machine learning methods.

Learning Objectives

- Necessary knowledge for the selection and, if necessary, the adaptation of suitable machine learning models.
- Knowledge about data availability and, if necessary, planning of training data generation
- Knowledge of practical implementation, training, and systematic evaluation of machine learning models in python using common libraries (sklearn, TensorFlow, Keras, PyTorch, etc.)
- Knowledge of ways and means to systematically analyze and interpret results.

Inhalt

This module covers the theoretical and practical aspects of machine learning methods and their application to problems in natural sciences, especially in materials science and chemistry. Students gain insight into machine learning fundamentals as well as current research topics of this still young interdisciplinary field. Topics covered include the application of machine learning methods for medical image analysis, sequence analysis and generation, the prediction of material and molecular properties, generative models for materials design, Bayesian methods for decision making in autonomous experiments, as well as interpretation possibilities of all methods for gaining scientific understanding.

A practical exercise based on jupyter notebooks gives students insight into the practical aspects of machine learning for natural sciences and supports the learning process.

Arbeitsaufwand

4 SWS: (2 SWS Lecture + 2 SWS Exercise + 1,5 x 4 SWS Preparation) x 15 + 30 h exam preparation
 = 180 h

Empfehlungen

- Knowledge of the basics of machine learning is helpful but not required
- Interest in natural science topics is required
- Basic knowledge of python is recommended. It has to be acquired during the semester through self-study

M**4.59 Modul: Machine Learning in Climate and Environmental Sciences [M-INFO-106470]**

Verantwortung: TT-Prof. Dr. Peer Nowack
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Human-centred Machine Intelligence](#)
[Area of Specialization: Data Science](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113083	Machine Learning in Climate and Environmental Sciences	6 LP	Nowack
T-INFO-113085	Machine Learning in Climate and Environmental Sciences - Pass	0 LP	Nowack

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Learning objectives:

Students will be able to effectively address complex data science challenges. They can design and use robust strategies/modelling pipelines for machine learning applications in the climate and environmental sciences, which are transferable to other disciplines.

Their acquired knowledge will include major classes of machine learning techniques, how to choose and differentiate among algorithms in a variety of problem settings, ways of assessing important data properties that could for example help or interfere with modelling goals, and methods to combine data-driven modelling with prior scientific system understanding to increase performance and trustworthiness of machine learning.

Students will learn how to implement these approaches in Python, using major machine learning software packages.

Inhalt

This module covers key concepts for real-world applications of machine learning, focusing on environmental data science. These include:

- foundations of machine learning (e.g., curse of dimensionality, cross-validation, cost functions, feature engineering)
- several widely applied regression, classification, and unsupervised learning algorithms (e.g., LASSO, random forests, Gaussian processes, neural networks, LSTMs, transformers, self-organizing maps)
- time series forecasting and causal inference.
- explainable AI (e.g., SHAP value analyses, feature permutation methods, intrinsically interpretable methods).

These concepts will be discussed in applied contexts, using current research examples from the climate and environmental sciences, including: climate change modelling, machine learning emulation of numerical models, forecasting air pollution and wildfires, understanding coupled dynamical systems such as global teleconnections in climate science, challenges in modelling non-stationary systems (e.g., predicting extreme weather events under global warming), and anomaly detection in measurement data.

The lectures are accompanied by computer exercises in which students learn how to implement and modify machine learning modelling pipelines first-hand.

Arbeitsaufwand

Concerning in-person events, this is a 4 SWS module: 2 SWS for lectures, 2 SWS for exercises

Overall:

(2 SWS lectures + 2 SWS exercises + 1.5 x 4 SWS preparation and homework) x 15 +30 h preparation for the exam = 180 h = 6 ECTS

Empfehlungen

- Previous programming experience, e.g. in scientific contexts or in computer science, is required.
- Knowledge of fundamentals about machine learning is an advantage.
- Knowledge of the Python programming language is an advantage.
- Good knowledge of mathematical concepts such as linear algebra is an advantage.
- An interest in scientific questions important for the climate- and environmental sciences.

M**4.60 Modul: Master's Thesis [M-INFO-106828]****Verantwortung:** Prof. Dr. Gerhard Neumann**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Master's Thesis](#)**Leistungspunkte**
30 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Semester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1

Pflichtbestandteile			
T-INFO-113697	Master's Thesis	30 LP	Neumann

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

- In den folgenden Bereichen müssen in Summe mindestens 60 Leistungspunkte erbracht worden sein:
 - Area of Specialization: Algorithm Engineering
 - Area of Specialization: Computer Graphics and Geometry Processing
 - Area of Specialization: Cryptography and Security
 - Area of Specialization: Data Science
 - Area of Specialization: Design of Embedded Systems and Computer Architectures
 - Area of Specialization: Human-centred Machine Intelligence
 - Area of Specialization: Parallel Computing
 - Area of Specialization: Robotics and Automation
 - Area of Specialization: Software Engineering and Compiler Construction
 - Area of Specialization: System Architecture
 - Area of Specialization: Telematics
 - Area of Specialization: Theoretical Foundations
 - Elective Studies in Informatics
 - Interdisciplinary Qualifications
 - Minor Studies: Biologically Inspired Robotics
 - Minor Studies: Economics
 - Minor Studies: Electrical Engineering
 - Minor Studies: Law
 - Minor Studies: Mathematics
 - Minor Studies: Mechanical Engineering
- In den folgenden Bereichen müssen in Summe mindestens 15 Leistungspunkte erbracht worden sein:
 - Area of Specialization: Algorithm Engineering
 - Area of Specialization: Computer Graphics and Geometry Processing
 - Area of Specialization: Cryptography and Security
 - Area of Specialization: Data Science
 - Area of Specialization: Design of Embedded Systems and Computer Architectures
 - Area of Specialization: Human-centred Machine Intelligence
 - Area of Specialization: Parallel Computing
 - Area of Specialization: Robotics and Automation
 - Area of Specialization: Software Engineering and Compiler Construction
 - Area of Specialization: System Architecture
 - Area of Specialization: Telematics
 - Area of Specialization: Theoretical Foundations

Qualifikationsziele

- In the Master's thesis, students work independently on a topic in computer science in line with the current state of research.
- Students demonstrate a comprehensive understanding of the scientific methods and procedures relating to the topic.
- Students select suitable methods and apply them correctly. If necessary, they adapt them accordingly or develop them further.
- Students critically compare their results with other approaches and evaluate their results.
- Students form a scientific opinion and are able to present and defend this and their results in discussions.

Inhalt

- The Master's thesis should demonstrate that students are able to work on a problem from their subject independently and within a limited period of time using scientific methods that reflect the current state of research.
- The completion period is six months. Upon justified request by the student, the Examination Board may extend the completion time by a maximum of three months. The Master's thesis can also be written in English in agreement with the supervisor.
- If the Master's thesis is to be written outside the faculty, this requires the approval of the Examination Board.
- The Master's thesis may also be approved in the form of a group thesis if the contribution of the individual students to be assessed as examination performance is clearly distinguishable.
- When submitting the Master's thesis, students must confirm in writing that they have written the thesis independently and have not used any sources or aids other than those specified by them, that they have identified the passages taken verbatim or in terms of content as such and that they have observed the statutes of the Karlsruhe Institute of Technology (KIT) for ensuring good scientific practice in the currently valid version.
- The date of issue of the topic and the date of submission of the Master's thesis must be recorded.

Arbeitsaufwand

The workload for the module is approximately 900 hours.

M**4.61 Modul: Mobile Communication [M-INFO-107245]**

Verantwortung: Prof. Dr. Oliver Waldhorst
Prof. Dr. Martina Zitterbart

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114271	Mobile Communication	4 LP	Waldhorst, Zitterbart

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students

- know the basic concepts of mobile communication and can evaluate basic methods and influencing factors of wireless communication
- are familiar with the structure and functionality of prominent, practically relevant mobile communication systems (e.g. GSM, UMTS, WLAN)
- know typical problems in mobile communication systems and can evaluate, select and apply suitable methods to solve them

Students are familiar with typical problems in wireless transmission (e.g. signal propagation, attenuation) and can explain these using examples and relate them to each other. They can also recognize where these problems typically occur when designing different communication systems.

Students are familiar with a portfolio of methods for modulating digital data, multiplexing, coordinating competing media access and mobility management. They will be able to explain these in their own words, evaluate them and select suitable candidates when designing mobile communication systems.

Students master the basic concepts of wireless local networks according to IEEE 802.11 and wireless personal networks with Bluetooth. They can explain these and compare the respective variants with each other. They will also be able to analyze and evaluate media access in detail.

Students master the structure of digital telecommunications systems such as GSM, UMTS and LTE as well as the individual tasks of the respective components and their detailed interaction in the overall system. They are familiar with the conceptual differences between the systems presented and can explain in their own words why certain methods from the portfolio are used in the respective systems.

Students will be familiar with basic routing methods in self-organizing wireless ad hoc networks and will be able to analyse these comprehensively and evaluate their use depending on the application scenario. Furthermore, they master the basic concepts of mobility support on the Internet (Mobile IP and Mobile IPv6).

Inhalt

The lecture first discusses typical problems in wireless transmission, such as signal propagation, attenuation, reflections and interference. Based on this, it develops a portfolio of methods for modulation of digital data, multiplexing, coordination of competing media accesses and mobility management. To illustrate where and how these methods are used in practice, typical mobile communication systems of great practical relevance are presented in detail. These include wireless local area networks according to IEEE 802.11, wireless personal networks with Bluetooth as well as wireless telecommunication systems such as GSM, UMTS with HSPA and LTE. Discussions of mechanisms at the network layer (mobile ad-hoc networks and MobileIP) and transport layer round off the lecture.

Arbeitsaufwand

Lecture with 2 SWS plus follow-up/exam preparation, 4 CP.

4 CP corresponds to approx. 120 working hours, of which

approx. 30 hours lecture attendance

approx. 60 hours preparation/follow-up work

approx. 30 hours exam preparation

M**4.62 Modul: Mobile Communications [M-ETIT-105971]**

Verantwortung: Prof. Dr.-Ing. Peter Rost
Einrichtung: KIT-Fakultät für Elektrotechnik und Informationstechnik
Bestandteil von: [Minor Studies: Electrical Engineering](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-ETIT-112127	Mobile Communications	4 LP	Rost

Erfolgskontrolle(n)

The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 15 minutes in which preparatory tasks are solved.

Voraussetzungen

none

Qualifikationsziele

Students are enabled to analyze and assess functionalities of mobile communication systems. They learn how to apply and implement fundamental methods of the lecture "Communications Engineering I" in mobile radio networks. Furthermore, students will be enabled to understand requirements and limitations of mobile applications.

Inhalt

At the beginning, this course describes exemplary applications of mobile communications and elaborates on resulting requirements. Based on a solid understanding of those requirements, selected approaches and techniques will be presented that are solving the respective challenges in mobile communication systems. To this end, algorithms as well as system architectures are discussed in order to acquire solid knowledge on the radio network, the core network and the integration with applications and services.

Zusammensetzung der Modulnote

Grade of the module corresponds to the grade of the oral exam.

Arbeitsaufwand

1. Attendance time in lectures: $15 \cdot 2 \text{ h} = 30 \text{ h}$
2. Preparation and follow-up of lectures: $15 \cdot 2 \text{ h} = 30 \text{ h}$
3. Attendance time in exercises: $15 \cdot 1 \text{ h} = 15 \text{ h}$
4. Preparation and follow-up of exercises: $15 \cdot 1 \text{ h} = 15 \text{ h}$
5. Preparation for the oral exam: 30 h

In total: 120 h = 4 LP

Empfehlungen

Knowledge of basic engineering as well as basic knowledge of communications engineering and Previous attendance of the lecture "Communication Engineering I" is recommended. Sound English language skills are required.

M**4.63 Modul: Model-Driven Software Development [M-INFO-106931]****Verantwortung:** Prof. Dr. Ralf Reussner**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	2

Pflichtbestandteile			
T-INFO-113896	Model-Driven Software Development	3 LP	Burger, Reussner

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

- * Students understand model-driven approaches for software and systems development, and they can use and assess them.
- * Students can create metamodels and transformations using established model-driven development processes and standards of the OMG (MOF, QVT, XMI, UML, etc.), as well as state-of-the-art languages and tool} (Xtext, Xtend, Xpand, etc.)
- * Students know the theoretical background of model transformation languages.
- * Students know practical applications of model-driven technologies.
- * Students can assess standards and technologies and can estimate their respective advantages and disadvantages.

Inhalt

Model-driven software development pursues the development of software systems on the basis of models. The models are not only used to document, design and analyse an initial system, as is usual in conventional software development, but rather serve as primary development artefacts from which the final system can be generated in its entirety if possible. This focus on models offers a number of advantages, such as an increase in the level of abstraction at which the system is specified, improved communication options that can extend to the end customer through domain-specific languages (DSL), and an increase in the efficiency of software development through automated transformations of the created models to the source code of the system. However, there are still some unresolved challenges in the use of model-driven software development, such as model versioning, evolution of DSLs, maintenance of transformations or the combination of teamwork and MDSD. Although MDSD is already used in practice due to the advantages mentioned, the challenges mentioned also offer opportunities for current research.

The lecture introduces concepts and techniques that are part of MDSD. As a basis, the systematic creation of meta-models and DSLs including all necessary components (concrete and abstract syntax, static and dynamic semantics) is introduced. This is followed by a general discussion of the concepts of transformation languages and an introduction to some selected transformation languages. The embedding of MDSD in the software development process provides the necessary foundations for their practical use. The remaining lectures deal with further issues such as model versioning, model coupling, MDSD standards, teamwork based on models, testing of model-driven software, as well as the maintenance and further development of models, meta-models and transformations. Finally, model-driven methods for analysing software architecture models are covered as an advanced unit. The lecture deepens concepts from existing courses such as software engineering or compiler construction or transfers and extends them to model-driven approaches. Furthermore, formal techniques are applied in transformation languages, such as graph grammars, logical calculi or relational algebras.

Arbeitsaufwand

(2 SWS + 1.5 x 2 SWS) x 15 + 15 h exam preparation = 90 h

M**4.64 Modul: Moderne Methoden der Kombinatorik [M-MATH-106957]**

Verantwortung: Prof. Dr. Maria Aksenovich
Einrichtung: KIT-Fakultät für Mathematik
Bestandteil von: [Minor Studies: Mathematics](#)

Leistungspunkte
6 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-MATH-113911	Moderne Methoden der Kombinatorik	6 LP	Aksenovich

Erfolgskontrolle(n)

Die Modulprüfung erfolgt in Form einer mündlichen Gesamtprüfung (ca. 30 min).

Voraussetzungen

Keine

Qualifikationsziele

Die Studierenden verstehen leistungsfähige moderne Methoden der Kombinatorik und können diese anwenden.

Inhalt

Der Kurs befasst sich mit modernen Methoden der Kombinatorik, einschließlich probabilistischer und algebraischer Methoden. Jede vorgestellte Methode wird mit mehreren Anwendungen illustriert.

Der probabilistische Teil umfasst die folgenden Themen: Zufallsgraphen, Linearität der Erwartung, Methode des zweiten Moments und Lovasz-Lemma. Der algebraische Teil umfasst: polynomiale Methoden, spektrale Methoden und lineare algebraische Techniken.

Zusammensetzung der Modulnote

Die Modulnote ist die Note der mündlichen Prüfung.

Arbeitsaufwand

Gesamter Arbeitsaufwand: 180 Stunden

Präsenzzeit: 60 Stunden

- Lehrveranstaltung einschließlich studienbegleitender Modulprüfung

Selbststudium: 120 Stunden

- Vertiefung der Studieninhalte durch häusliche Nachbearbeitung des Vorlesungsinhaltes
- Bearbeitung von Übungsaufgaben
- Vertiefung der Studieninhalte anhand geeigneter Literatur- und Internetrecherche
- Vorbereitung auf die studienbegleitende Modulprüfung

Empfehlungen

Kenntnisse in linearer Algebra und Stochastik werden dringend empfohlen. Die Kurse *Graphentheorie* und *Kombinatorik* werden empfohlen, sind aber nicht erforderlich.

M**4.65 Modul: Motion in Human and Machine - Seminar [M-INFO-102555]****Verantwortung:** Prof. Dr.-Ing. Tamim Asfour**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Human-centred Machine Intelligence](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Sommersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
3

Pflichtbestandteile			
T-INFO-105140	Motion in Human and Machine - Seminar	3 LP	Asfour

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students understand methods for modeling, processing, and analyzing human motions. They can explain approaches for capturing biological motion and biomechanical simulation.

Students can describe methods for learning motion primitives from human demonstrations and understand how human motion is mapped to robots with different kinematics and dynamics. They can apply these methods in new contexts.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

This interdisciplinary block seminar addresses methods for modeling, generating, and controlling movements in human and robot systems. Students gain insight into this interdisciplinary field, learning fundamentals of capturing biological motion, biomechanical simulation, robotics, and machine learning.

The seminar begins with the generation of human movement as an effect of muscle contraction. It covers how observed movement patterns can be identified and categorized. Methods for learning movement primitives from human demonstrations are presented, including approaches for reproducing these patterns on humanoid robots. The seminar addresses the challenge of mapping human motion to robots with different kinematics and dynamics. Students learn how motion primitives enable transfer of human movements to robotic systems and their application to motion generation in humanoid robots.

Students work in small groups on collaborative projects that apply the learned methods. These projects allow students to deepen their understanding through practical application of the methods.

Anmerkungen

This interdisciplinary seminar is conducted in cooperation with the University of Stuttgart and the University of Tübingen.

Arbeitsaufwand

Seminar with 3 SWS, 3 LP

3 LP corresponds to 90 hours, including

30 hours attendance time

15 hours group work

20 hours literature research

20 hours manuscript preparation

5 hours video creation

Empfehlungen

Programming experience in C++, Python or Matlab is recommended.

Attending the lectures Robotics I – Introduction to Robotics, Robotics II: Humanoid Robotics, Robotics III - Sensors and Perception in Robotics, Mechano-Informatics and Robotics and Wearable Robotic Technologies is recommended.

M**4.66 Modul: Multimodal Artificial Intelligence [M-INFO-107676]**

Verantwortung: Prof. Dr. Jan Niehues
Prof. Dr.-Ing. Rainer Stiefelhagen

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-115041	Multimodal Artificial Intelligence	6 LP	Niehues, Stiefelhagen

Erfolgskontrolle(n)

See Partial Achievements (Teilleistung).

Voraussetzungen

See Partial Achievements (Teilleistung).

Qualifikationsziele

- The students know the relevant elements of a multimodal AI .
- The students understand the algorithms and methods of multimodal AI.
- The students are able to understand the different sub-components to develop and analyze a system .
- The students can transfer this knowledge to new applications, as well as analyze and compare different methods.

Inhalt

Due to the successes in research, multimodal AI systems are increasingly integrated into our everyday lives. These are, for example, systems that can understand and generate language and speech or analyze images and videos. Furthermore, advanced video and image generation AI shows leads to new applications

Based on the knowledge of the lecture "Introduction to AI", the students learn to understand, develop and evaluate multimodal AI Systems.

In order to bring this knowledge closer to the students, the lecture is divided into 5 parts. First, the lecture investigates method of speech and language perception followed by a part about visual perception. The second part deals speech and language generation followed by a part about video processing. Finally, multimodal AI methods are presented.

Arbeitsaufwand

Lecture with 3 SWS + 1 SWS exercise , 6 CP.

6 LP corresponds to approx. 180 hours, of which

approx. 45 hours lecture attendance

approx. 15 hours exercise visit

approx. 90 hours post-processing and processing of the exercise sheets (self-study)

approx. 30 hours exam preparation (self-study)

Empfehlungen

See Partial Achievements (Teilleistung).

M**4.67 Modul: Nano- and Quantum Electronics [M-ETIT-105604]**

Verantwortung: Prof. Dr. Sebastian Kempf
Einrichtung: KIT-Fakultät für Elektrotechnik und Informationstechnik
Bestandteil von: [Minor Studies: Electrical Engineering](#)

Leistungspunkte
6 LP

Notenskala
Zehntelnoten

Turnus
Jedes Sommersemester

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-ETIT-111232	Nano- and Quantum Electronics	6 LP	Kempf

Erfolgskontrolle(n)

The assessment of success takes place in the form of a written examination lasting 120min. The grade corresponds to the result of the written examination.

Voraussetzungen

none

Qualifikationsziele

Students will understand the physical limits of CMOS scaling and will be able to analyze the function of conventional nanoelectronic devices. Students will also understand the operation of novel nanoelectronic and quantum electronic devices and will be able to design this kind of devices that are based on quantum mechanical effects. They develop the ability to design nanoelectronic sensors and devices and can understand and analyze the fabrication methods for nano- and quantum electronic devices.

Inhalt

Nanoelectronics deals with integrated circuits whose typical length scale is well below 100nm. In this regime, physical effects, in particular of quantum mechanical origin, occur and strongly influence the scaling of classical microelectronic devices. This ultimately leads to a new form of electronic components as well as novel operation principles. A special form of nanoelectronics is quantum electronics in which quantum mechanical effects are exploited on purpose to build an entirely new class of devices whose performance reaches far beyond any other microelectronics devices. Well-known examples are superconducting digital electronics which enables to build, for example, microprocessors with clock rates exceeding several 100GHz, or the quantum computer, which will lead to a change of paradigms in the field of information processing.

Within this context, the module "Nano- and quantum electronics" intends to give students an overview of the theoretical and practical aspects of nano- and quantum electronics. In particular, it discusses the following topics:

- Limitations of conventional CMOS technology
- Quantum mechanical effects in the field of nano- and quantum electronics (quantized conductance, Coulomb blockade, tunnel effect, etc.)
- Hot-electron effect
- Nano- and quantum-technological manufacturing and analysis methods
- Nanostructure field-effect transistors
- Quantum dots
- Carbon nanotube field-effect transistor
- Resonant tunnel diodes
- Unipolar resonant tunnel transistor
- Single Electron Transistor (SET)
- Josephson junction based analog and digital electronics
- Quantum bits, quantum computers and quantum computing

The tutorial is closely linked to the lecture and deals with special aspects concerning the development of nano- and quantum electronics. In particular, the development and system integration of such devices for various applications is discussed by means of exercises.

Zusammensetzung der Modulnote

The module grade is the grade of the written examination.

Arbeitsaufwand

A workload of approx. 175h is required for the successful completion of the module. This is composed as follows:

- Attendance time in lectures and exercises: $18 \cdot 1.5h + 6 \cdot 1.5h = 36h$
- Preparation and follow-up of lectures: $21 \cdot 3h = 54h$
- Preparation and follow-up of tutorials: $7 \cdot 5h = 35h$
- Preparation for the exam: 50h

Empfehlungen

Successful completion of the modules "Superconductivity for Engineers" and „Einführung in die Quantentheorie für Elektrotechniker“ is recommended.

M**4.68 Modul: Natural Language Processing [M-INFO-107178]**

Verantwortung: Prof. Dr. Jan Niehues
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114207	Natural Language Processing	6 LP	Niehues

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

- To familiarise the student with the problems that exist in natural language processing
- The student should be introduced to the basic techniques for solving the problems.
- The student should gain an insight into current research in the field of natural language processing and be able to use the language processing and can use the acquired knowledge to work on current research topics

Inhalt

Summarise today's lecture? When were neural networks invented? Artificial intelligence that can answer these questions has long been a dream of mankind. And today we are seeing the first programmes that can solve these problems. In this lecture we will provide the skills and knowledge to develop solutions to these problems of natural language processing using state-of-the-art methods.

After an introduction to the challenges of natural language processing, the different tasks in natural language processing are discussed. One focus of the course is on methods from the field of deep learning. Firstly, sequence classification tasks such as sentiment analysis are covered. Next, methods of sequence labelling are discussed, such as those used in the recognition of proper names or the determination of part-of-speech tags. The lecture will then discuss sequence-to-sequence methods. These models are used in many natural language processing tasks, such as machine translation, automatic summarisation and automatic question answering.

In this course, the important challenges in the development of systems will be addressed: The representation of words, neural architectures to model language, methods to train complex models, and finding the most likely output.

Arbeitsaufwand

Lecture with 4 SWS, 6 CP.

6 LP corresponds to approx. 180 hours, of which

approx. 60 hours lecture attendance

approx. 90 hours post-processing (self-study)

approx. 30 hours exam preparation (self-study)

M**4.69 Modul: Natural Language Processing and Software Engineering [M-INFO-107233]**

Verantwortung: Prof. Dr.-Ing. Anne Koziolk
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114257	Natural Language Processing and Software Engineering	3 LP	Koziolk

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students know basic concepts of linguistics such as syntax, semantics and pragmatics, and can explain and compare them. They are familiar with lexical relations such as polysemy, homonymy, and troponymy and can identify relevant examples. Furthermore, they can identify and compare connections between the relations.

Students are familiar with basic concepts of computational linguistics. Basic techniques such as part-of-speech tagging, lemmatization, word similarities and disambiguation can be explained. Associated methods (lexical, rule-based, or probabilistic) can be described and their respective strengths and weaknesses assessed. Different parsing methods can be named, explained and conceptually reproduced.

Students can describe and compare the structure, content and benefits of different knowledge bases. In addition to the overarching concepts of ontology, lexical databases and other knowledge representations, they are also familiar with specific representatives, such as WordNet, DBpedia and similar, and can use them.

Students understand the connection between the functionality of basic computational linguistics techniques and their applicability in software engineering. In addition, they can break down tool chains into individual components and evaluate them. In particular, students will be able to analyze and evaluate different applications. These include automated modeling, improving requirements specifications, and traceability link recovery. In addition, students can explain the concept of large language models (LLMs) and their application and use in the field of language processing. Students can identify application scenarios in software engineering for text analysis systems and design their own solutions.

Inhalt

This lecture provides the basics for the automated processing of natural language texts. Language processing is becoming increasingly important.

Linguistic input plays a critical role in interactive systems, such as voice commands, assistance systems, and query interfaces. Additionally, the analysis and processing of text-based software artifacts represents an important field of research. Computational linguistics is therefore not only of great importance for software applications, but also for software engineering itself.

The aim of this lecture is to provide basic knowledge of natural language processing (NLP) and its potential applications in the development of software systems. Key topics include the automated analysis of texts, the challenges posed by the inherent ambiguity of natural language, the translation of natural language texts into software models, and the use of large language models (LLMs) in software engineering. The lecture will also explore current research developments and trends in the field.

Arbeitsaufwand

3 ECTS correspond to approximately 90 hours of work, including:

approx. 30 hours of attending lectures

approx. 45 hours of preparation and follow-up work

approx. 15 hours of exam preparation

M**4.70 Modul: Network Security: Architectures and Protocols [M-INFO-107218]**

Verantwortung: Prof. Dr. Martina Zitterbart
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114238	Network Security: Architectures and Protocols	4 LP	Zitterbart

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students

- know basic challenges, protection goals and cryptographic building blocks that are relevant for the design of secure communication systems
- are proficient in security-relevant communication protocols (e.g. Kerberos, TLS, IPSec) and can identify and explain basic security mechanisms
- have the ability to analyze and evaluate communication protocols from a security perspective
- have the ability to assess and evaluate the quality of security mechanisms in relation to the required security objectives

In particular, students are familiar with typical attack techniques such as eavesdropping, interception or replaying and can explain these using examples. In addition, students are familiar with cryptographic primitives such as symmetric and asymmetric encryption, digital signatures, message authentication codes and can apply these in particular for the design of secure communication services.

Students are familiar with the Kerberos distributed authentication service and can explain the protocol flow in their own words and name basic concepts (e.g. tickets). In addition, students are familiar with relevant communication protocols for protecting communication on the Internet (e.g. IPsec, TLS) and can explain these and analyze and evaluate their security properties.

Students know different methods for network access protection and can explain and compare common authentication methods (e.g. CHAP, PAP, EAP). Furthermore, students are proficient in methods for protecting wireless access networks and can analyze and evaluate methods such as WEP, WPA and WPA2.

Students master different trust models and can explain and apply basic technical concepts (e.g. digital certificates, PKI) in their own words. In addition, students develop an understanding of data protection aspects in communication networks and can explain and apply technical procedures to protect privacy.

Inhalt

The lecture "Network Security: Architectures and Protocols" looks at challenges and techniques in the design of secure communication protocols as well as data protection and privacy issues. Complex systems such as Kerberos are examined in detail and their design decisions with regard to security aspects are highlighted. Special focus is placed on PKI fundamentals, infrastructures and specific PKI formats. Further emphasis is placed on the common security protocols IPsec and TLS/SSL as well as protocols for infrastructure protection.

Arbeitsaufwand

Lecture with 2 SWS plus follow-up/exam preparation, 4 CP.

4 CP corresponds to approx. 120 working hours, of which

approx. 30 hours lecture attendance

approx. 60 hours preparation/follow-up work

approx. 30 hours exam preparation

M**4.71 Modul: Next Generation Internet [M-INFO-100784]**

Verantwortung: Dr.-Ing. Roland Bless
Prof. Dr. Martina Zitterbart

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte 4 LP	Notenskala Zehntelnoten	Turnus Jedes Sommersemester	Dauer 1 Semester	Sprache Englisch	Level 4	Version 1
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Pflichtbestandteile			
T-INFO-101321	Next Generation Internet	4 LP	Bless, Zitterbart

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students

- know the basic properties and architectural concepts of the Internet as well as its limitations. know newer transport protocols and current approaches to increase the flexibility of Internet-based communication and can apply this knowledge in practice. are familiar with concepts for quality of service support and group communications and can apply mechanisms for their implementation on the Internet. have the ability to analyze and evaluate peer-to-peer systems and advanced routing protocols. are familiar with concepts of satellite networking and quantum Internet. In particular,
- know important architectural concepts and design principles that are used on the Internet and can explain these using examples or apply them themselves when designing systems. In addition, students know the concept of quality of service and important quality of service parameters, are familiar with basic mechanisms for supporting quality of service (e.g. classifiers, traffic shapers, queuing and scheduling strategies, resource reservation), can analyze and evaluate them and can apply them to the design of communication systems.
- Moreover, students know the requirements and challenges for today's transport protocols and newer congestion control algorithms and can analyze and assess trade-offs of the presented approaches.
- know the characteristics of peer-to-peer systems, can explain them and can compare different forms of organization. Furthermore, students master routing procedures in such decentrally organized peer-to-peer systems and can explain how they work in detail in their own words.
- Similarly, students know inherent trade-offs for routing in the Internet and can explain newer approaches in their own words.
- In addition, students develop an understanding of the functioning of newer approaches to increase the flexibility of communication networks (e.g. network virtualization, software-defined networking, service function chaining) and can analyze, explain, and apply technical procedures for their implementation.
- Moreover, students know properties of satellite and quantum networks and their corresponding challenges.

Inhalt

The lecture focuses on current developments in Internet-based network technologies. First, architectural principles of today's Internet are presented and discussed, subsequently nowadays and future challenges are motivated. The lecture also discusses approaches and paradigms beyond the current Internet architecture, methods for quality-of-service support, newer transport protocols and congestion control approaches as well as group communication support. Deployments of the presented technologies in IP-based networks are discussed. The lecture presents advanced approaches such as programmable networks, network virtualization as well as newer approaches and protocols for routing, satellite networking, and peer-to-peer networks. A brief introduction to the technology of a future quantum Internet is provided as well.

Arbeitsaufwand

Lecture with 2 SWS plus follow-up/exam preparation, 4 CP.

4 CP corresponds to approx. 120 working hours, of which

approx. 30 hours lecture attendance

approx. 60 hours preparation/follow-up work

approx. 30 hours exam preparation

Literatur

J.F. Kurose, K.W. Ross; Computer Networking: A Top-Down Approach. Pearson, 2022, 8th Edition, ISBN 978-1292405469

M**4.72 Modul: Optical Engineering [M-ETIT-100456]**

Verantwortung: Prof. Dr. Wilhelm Stork
Einrichtung: KIT-Fakultät für Elektrotechnik und Informationstechnik
Bestandteil von: [Minor Studies: Electrical Engineering](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-ETIT-100676	Optical Engineering	4 LP	Stork

Erfolgskontrolle(n)

Achievement will be examined in an oral examination (approx. 20 minutes).

Voraussetzungen

none

Qualifikationsziele

The students from different backgrounds refresh and elaborate their knowledge of engineering optics and photonics. They will get to know the basic principles of optical designs. They will connect these principles with real-world applications and learn about their problems and how to solve them. The students will know about the human view ability and the eye system. After the module they will be able to judge the basic qualities of an optical system by its quantitative data.

After the course, students will:

- understand fundamental optical phenomena and apply it to solve optical engineering problems;
- work with the basic tools of optical engineering, i.e. ray-tracing by abcd-matrices;
- get a broad knowledge on real-world applications of optical engineering;
- learn about the potential of optical design for industrial, medical and day-to-day applications;
- know up-to-date optical engineering problems and its solutions.

Inhalt

The course "Optical Engineering" teaches the practical aspects of designing optical components and instruments such as lenses, microscopes, optical sensors and measurement systems, and optical disc systems (e.g. CD, DVD, HVD). The course explains the layout of modern optical systems and gives an overview over available technology, materials, costs, design methods, as well as optical design software. The lectures will be given in the form of presentations and accompanied by individual and group exercises. The topics of the lectures include:

- I. Introduction (Optical Phenomena)
- II. Ray Optics (thin/thick lenses, principal planes, ABCD-matrices, chief rays, examples: Eye, IOL)
- III. Popular Applications (Magnifying glass, microscope, telescope, Time-of-flight)
- IV. Wave Optics (Interference, Diffraction, Spectrometers, LDV)
- V. Aberrations I (Coma, defocus, astigmatism, spherical aberration)
- VI. Fourier Optics (Periodical patterns, FFT spectrum, airy-patterns)
- VII. Aberration II (Seidel and Zernike Aberrations, MTF, PSF, Example: Eye)
- VIII. Fourier Optics II (Kirchhoff + Fresnel, contrast, example: Hubble-telescope)
- IX. Diffractive Optics Applications (Gratings, holography, IOL, CD/DVD/Blu-Ray-Player)
- X. Interference (Coherence, OCT)
- XI. Filters and Mirrors (Filters, antireflection, polarization, micro mirrors, DLPs)
- XII. Laser and Laser Safety (Laser principle, laser types, laser safety aspects)
- XIII. Displays (Pico projectors, LCD, LED, OLED, properties of displays)

Zusammensetzung der Modulnote

The module grade is the grade of the oral exam.

Arbeitsaufwand

total 120 h, hereof 45 h contact hours (30 h lecture, 15 h problem class), and 75 h homework and selfstudies

Empfehlungen

Solid mathematical background.

Literatur

E. Hecht: Optics

J.W. Goodman: Introduction to Fourier optics

K.K. Sharma: Optics - Principles and Applications

M**4.73 Modul: Optimization and Synthesis of Embedded Systems (ESI) [M-INFO-107229]**

Verantwortung: Prof. Dr.-Ing. Jörg Henkel
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114253	Optimization and Synthesis of Embedded Systems (ESI)	3 LP	Henkel

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The student can develop embedded systems. They can specify, synthesize and optimize their own hardware. They learn the hardware description language and are familiar with the special boundary conditions of the design of embedded systems.

Inhalt

The cost-effective and error-free development of embedded systems represents a challenge that should not be underestimated and which is having an ever greater influence on the added value of the overall system. In Europe in particular, the design of embedded systems is playing an increasingly important economic role in many sectors of the economy, such as the automotive industry, so that a number of well-known companies are already involved in the development of embedded systems.

The lecture deals comprehensively with all aspects of the development of embedded systems at hardware, software and system level. This includes diverse areas such as modelling, optimization and synthesis of systems.

Arbeitsaufwand

90 hours:

30 hours of lectures (15 weeks*2 hours)

30 hours of self study (15 weeks*2 hours)

15 hours of deepening (coding RTL) to grasp the theory with practical

15 hours exam preparation

M**4.74 Modul: Optimization Methods for Machine Learning and Engineering [M-INFO-105329]**

Verantwortung: Prof. Dr.-Ing. Jürgen Beyerer
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
5 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-110809	Optimization Methods for Machine Learning and Engineering	5 LP	Beyerer, Pfrommer

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students are able to formulate practical tasks as optimisation problems and solve them using suitable algorithmic methods.

Learning objectives: The students know

- The most important categories of (convex) optimisation problems and their mathematical foundations
- The associated algorithmic solution methods and their runtime complexity
- Techniques for modelling practical tasks as optimisation problems (machine learning, engineering, finance)
- Methods for transforming and approximating optimisation problems for the use of resource-efficient methods

Inhalt

The term optimization refers to techniques for the identification of the best solution in a complex problem setting. Many applications from machine learning and engineering are based on solving an optimization problem. This lecture introduces the major theoretical and algorithmic tools for solving of convex optimization problems. Practical problems for machine learning, engineering and further application domains are used as illustration. The students apply their knowledge to practical optimization problems in tutorial exercises.

Arbeitsaufwand

Lecture with 2 SWS + 1 SWS exercise
 5 ECTS corresponds to approx. 150 hours
 approx. 30 hours lecture attendance,
 approx. 15 hours attending exercises,
 approx. 90 hours of post-processing and working on the exercise sheets
 approx. 30 hours exam preparation

Literatur

- Boyd, Stephen, and Lieven Vandenberghe. Convex optimization. Cambridge university press, 2004.
- Luenberger, David G. Optimization by vector space methods. John Wiley & Sons, 1969.
- Sra, Suvrit, Sebastian Nowozin, and Stephen J. Wright, eds. Optimization for machine learning. MIT Press, 2012.

M**4.75 Modul: Parallel Algorithms [M-INFO-107199]**

Verantwortung: Prof. Dr. Peter Sanders
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
5 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114221	Parallel Algorithms	4 LP	Sanders
T-INFO-114222	Parallel Algorithms Pass	1 LP	Sanders

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The students acquire a systematic understanding for algorithmic problems and their solutions in the field of parallel algorithms, building on existing knowledge in algorithmics. Additionally, they are able to apply learned techniques to related problems and to interpret and comprehend current research topics.

After successful attendance of the course, the students are able to

- explain terms, structures, basic problem definitions and algorithms from the lecture;
- decide which algorithms and data structures are suitable for solving a given problem and, if necessary, adapt them to the requirements of a specific problem;
- execute algorithms and data structures, conduct a mathematically precise analysis, and prove their algorithmic properties;
- explain machine models from the lecture and analyze algorithms and data structures in them;
- analyze new problems from application contexts, reduce them to their algorithmic core and design an abstract model; design own solutions in this model using concepts and techniques from the lecture, analyze them and prove the algorithmic properties.

Inhalt

Models and their relation to real machines:

- shared memory - PRAM
- message passing - BSP
- circuits

Analysis: speedup, efficiency, scalability

Basic techniques:

- SPMD
- parallel divide-and-conquer
- collective communication
- load balancing

Concrete algorithms (examples):

- collective communication (including large data volumes): broadcast,
- reduce, prefix sums, all-to-all exchange
- matrix computations
- sorting
- list ranking
- minimum spanning trees
- load balancing: master worker with adaptive problem size, random
- polling, random distribution

Arbeitsaufwand

Lecture and exercise with 3 semester hours per week, 5 ECTS correspond to approx. 150 working hours, consisting of

- approx. 30 h attendance of the lecture and exercise session / block seminar
- approx. 60 h preparation and follow-up work
- approx. 30 h working on exercise sheets / preparation of seminar presentation
- approx. 30 h exam preparation

Empfehlungen

See partial achievements (Teilleistung)

M**4.76 Modul: Parameterized Algorithms [M-INFO-107167]****Verantwortung:** TT-Prof. Dr. Thomas Bläsius**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte
 6 LP

Notenskala
 Zehntelnoten

Turnus
 Unregelmäßig

Dauer
 1 Semester

Sprache
 Englisch

Level
 4

Version
 1

Pflichtbestandteile			
T-INFO-114194	Parameterized Algorithms	6 LP	Bläsius
T-INFO-114195	Parameterized Algorithms - Pass	0 LP	Bläsius

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students acquire a systematic understanding of the parameterised approach in the runtime analysis of algorithms, as well as the associated techniques for algorithm design, which build on existing knowledge in theoretical computer science and algorithmics. After successfully completing the course, students will be able to

- reproduce and explain basic algorithmic techniques and analysis techniques in the field of parameterised algorithms,
- execute parameterised algorithms by way of example, analyse them with mathematical precision and prove their properties,
- select which algorithms or algorithmic techniques are suitable for a given parameterised problem,
- analyse unknown problems with regard to their parameterised complexity.

Inhalt

Many problems that arise in practice are NP-hard and therefore generally (presumably) cannot be solved in polynomial time. Nevertheless, these problems can often be solved efficiently because the inputs are "benign". One way to formally capture this benignity of the instances is to consider the parameterised complexity. This involves associating a parameter k with each instance, which represents a measure of the complexity of the input. The aim is then to find an algorithm whose runtime depends only polynomially on the input size n but possibly exponentially on the parameter k . Compared to the rough classification of a problem as polynomially solvable or NP-hard, the parameterised approach offers a much more differentiated view of hard problems.

Arbeitsaufwand

Lecture with tutorial with 4 SWS, 6 CP

6 CP corresponds to approx. 180 working hours, of which

approx. 60 hours attending the lecture and tutorial

approx. 30 hours of preparation and follow-up work

approx. 60 hours working on the exercise sheets

approx. 30 hours exam preparation

Empfehlungen

Basic knowledge of algorithms and data structures (e.g. from the lectures Algorithms 1 + 2) is helpful.

M**4.77 Modul: Participatory Technology Design [M-INFO-107170]****Verantwortung:** Prof. Dr. Kathrin Gerling**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
6 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Wintersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1

Pflichtbestandteile			
T-INFO-114199	Participatory Technology Design	6 LP	Gerling
T-INFO-114200	Participatory Technology Design - Pass	0 LP	Gerling

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

After completing the course, students will be able to reproduce basic and advanced theoretical concepts from human-machine interaction and participatory technology design. Furthermore, they will be able to apply relevant methods for participatory design and evaluation to given problems, taking into account the needs of users and ethical aspects, and derive concrete design recommendations from the results. Finally, students are able to recognise and critically reflect on the connections between participation, design, implementation and evaluation of technologies.

Inhalt

In human-machine interaction, the participatory development of new technologies - i.e. the direct and equal involvement of users in the development process - is becoming increasingly important. It is used, for example, in the development of body-centred and wearable systems, or contributes to the design of solutions in the field of smart and assisted living and personal robotics. Participation is often realised through interviews, focus groups and design workshops; new technologies are also regularly evaluated as part of user studies. The direct involvement of users is intended to ensure that the resulting technologies are better adapted to people's needs and that their benefits and relevance for individuals and society can be increased as a result.

The lecture deals with current research approaches to the participatory design of new technologies and covers the following topics in particular, while continuously taking ethical aspects into account:

- Design approaches, in particular theory-driven design, ethical approaches such as value-sensitive design, and future-oriented approaches such as speculative design and design fiction
- Typical methods of participation in the design and development of technologies, and reflection on the associated challenges and opportunities
- Participatory and user-centred evaluation of technologies and implications for society, research and development

In the associated exercise, students actively develop relevant concepts and reflect on theoretical aspects in their application in the context of practical examples. In addition, current research publications are analysed as part of the exercise.

Arbeitsaufwand

The total workload for this course is approx. 180 hours (6 credits).

Approximately...

28h for attending the lecture,

24 hours for attending the exercises,

40h for preparation and follow-up of the lecture,

40h for preparation and follow-up of the exercise,

48h for exam preparation.

Empfehlungen

Knowledge of the basics of human-machine interaction is helpful.

M**4.78 Modul: Post-quantum Cryptography from Lattices [M-INFO-107642]**

Verantwortung: Prof. Dr. Jörn Müller-Quade
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Elective Studies in Informatics](#)

Leistungspunkte
3 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-114987	Post-quantum Cryptography from Lattices	3 LP	Müller-Quade

Erfolgskontrolle(n)

See Partial Achievements (Teilleistung).

Voraussetzungen

See Partial Achievements (Teilleistung).

Qualifikationsziele

The student

- knows the assumptions and framework underlying lattice-based cryptography.
- understands and is able to explain the common tools of the area.
- knows and understands the protocols and proofs presented in the lecture and can reproduce them.
- can apply the contents of the lecture to construct new (simple) schemes or analyze the security of given (simple) schemes.

Inhalt

The lecture "Post-Quantum (Lattice-based) Cryptography" introduces the framework of lattice-based cryptography, its common tools and concepts, and important constructions of advanced cryptographic primitives. The central hardness assumptions underlying lattice-based cryptography are believed to be intractable even for quantum algorithms, and as such provide a foundation for post-quantum security. Beyond that, lattice-based cryptography enables a wealth of constructions of theoretical and practical interest.

Arbeitsaufwand

90 h

Lectures: 24 h

Self-study during semester: 31 h

Preparation for the exam: 35 h

Empfehlungen

See Partial Achievements (Teilleistung).

M**4.79 Modul: Practical Course on Network Security Research [M-INFO-107244]**

Verantwortung: Prof. Dr. Martina Zitterbart
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte
3 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-114270	Practical Course on Network Security Research	3 LP	Hock, Zitterbart

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students are able to understand, justify, evaluate and classify the selected topic or the selected focus from the field of network security.

They know the basic principles relevant to the selected topic and can apply these in practice. Students are also able to derive concrete work steps from a task description and to document, summarize and present the results obtained.

Inhalt

The research practical course on network security is offered alongside the module Network Security: Architectures and Protocols [M-INFO-100782]. The practical course gives students the opportunity to selectively deepen a specific topic from the above-mentioned lecture with current research relevance. The topic may vary and will be announced when registering for the practical course (example: "Attacks and Anomalies in the context of the Border Gateway Protocol").

The practical course consists of five sections:

- Familiarization with the topic
- Selection of a suitable practical focus in coordination with the supervising chair
- Practical implementation of the focus
- Presentation of the results in a colloquium (lecture)
- Preparation of a research report (3-5 pages)

Arbeitsaufwand

3 ETCS:

Attendance time / meetings in large and small groups: 15h

Selection of the focus: 10h

Conception + specification of the focus: 10h

Implementation of the focus: 20h

Research report and colloquium: 20h

Empfehlungen

The module Network Security: Architectures and Protocols [M-INFO-100782] should have been started or completed.

M**4.80 Modul: Practical Course on Telematics Research [M-INFO-107220]**

Verantwortung: Prof. Dr. Martina Zitterbart
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte
3 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-114239	Practical Course on Telematics Research	3 LP	Zitterbart

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students are able to understand, justify, evaluate and classify the selected topic or focus from the field of telematics.

They know the basic principles relevant to the selected topic and can apply these in practice. Students are also able to derive concrete work steps from a task description and to document, summarize and present the results obtained.

Inhalt

The telematics research internship is offered alongside the telematics module [M-INFO-100801]. The internship gives students the opportunity to selectively deepen a specific topic from the above-mentioned lecture with current research relevance. The topic may vary and will be announced when registering for the practical course (example: "Visualization and anomaly detection in the context of the Border Gateway Protocol").

The practical course consists of the following sections:

- Familiarization with the topic
- Selection of a suitable practical focus in coordination with the supervising chair
- Practical implementation of the focus
- Presentation of the results (colloquium, research report)

Arbeitsaufwand

Attendance time / meetings in large and small groups: 15h

Selection of the focus: 10h

Conception + specification of the focus: 10h

Implementation of the focus: 20h

Research report / colloquium: 20h

Empfehlungen

A pronounced scientific interest in the topics of network security is a prerequisite: no prefabricated exercises are worked on, instead the internship requires a high degree of personal initiative.

M**4.81 Modul: Practical Course: Internet of Things (IoT) [M-INFO-103706]****Verantwortung:** Prof. Dr.-Ing. Jörg Henkel**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-107493	Practical Course: Internet of Things (IoT)	4 LP	Henkel

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

- Students will understand the core concepts of IoT systems, including design objectives, application domains, and requirements.
- They will gain skills in developing software programs for IoT embedded devices, debugging, and testing software on hardware.
- They will be capable of integrating and evaluating IoT systems comprising sensors, processors, wireless communication modules, and data storage.

Inhalt

- The lab provides hands-on experience in IoT system design and development, focusing on the following aspects:
- Embedded, machine learning algorithms, intelligence, and connectivity in IoT systems.
- Challenges and optimization techniques for embedded software in resource-constrained IoT devices.
- Practical design and implementation of IoT devices with applications in smart environments, healthcare, and more.
- Integration of IoT components: sensors, processors, wireless radios, and storage systems.

Arbeitsaufwand

Attendance time: 45 hours

Final project: 55 hours

Final presentation & Report: 20 hours

Total = 120 hours = 4 ECTS

Empfehlungen

- Familiarity with other (than C) languages like Python could be helpful as well.
- Basic knowledge from the modules "Design and Architectures of Embedded Systems (ESII)" and "Optimization and Synthesis of Embedded Systems (ESI)" are helpful but not essential for understanding the lab.

M**4.82 Modul: Practical Course: Advanced Topics in High Performance Computing, Data Management and Analytics [M-INFO-105870]**

Verantwortung: Prof. Dr. Achim Streit
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-111803	Practical Course: Advanced Topics in High Performance Computing, Data Management and Analytics	6 LP	Streit

Erfolgskontrolle(n)

See partial achievements (Teilleitung)

Voraussetzungen

See partial achievements (Teilleitung)

Qualifikationsziele

Students know and can apply tools and techniques in the fields of high-performance computing, data management and data analysis. They acquire the possibility to analyze complex scenarios and develop solutions for this. Besides working on the content, students improve their competences in communication and presentation.

Inhalt

Participants will have the chance to deepen their knowledge of high-performance computing, data management and data analysis and to apply it in a practical way. The tasks to be worked on come from the subfields:

- HPC simulations (e.g., parallelization, MPI, performance engineering)
- HPC systems and operating environment (e.g., On Demand File Systems, Infiniband Networks, Job Scheduling)
- Machine Learning and Data Mining (e.g., RapidMiner, scikit)
- Data-Intensive Computing (e.g., Hadoop, Spark).
- HPC and data analysis with Python (e.g., Numpy, Scipy, Pandas, Dask, Parsl)
- Distributed & Parallel File Systems (e.g., glusterFS, BeeGFS)
- Object Storage (e.g., S3, CEPH)
- Data Management System (e.g., dCache, iRods)
- Databases (e.g., SQL, NoSQL)
- Workflow management systems for HPC and data analysis (e.g., FireWorks, AiiDA, SimStack)
- Opportunistic resource integration and utilization (e.g., using COBalD/TARDIS)
- Authentication and authorization infrastructure (e.g., OpenID, SAML)

Students are individually supervised by scientific staff of the Scientific Centre for Computing and can apply their skills in a practical and research-oriented way by being involved in current research tasks (e.g., Helmholtz program, BMBF and EU projects).

Arbeitsaufwand

3 SWS = 150 h per semester

- 12 h in meetings during the semester (kick-off, regular meetings with the supervisor, final meeting including presentation)
- 18 h preparation of meetings
- 120 h working on the topic and preparation of the exam

Empfehlungen

Knowledge in the area of databases, data management, data analytics, parallel computing is helpful.

M**4.83 Modul: Practical Course: AI for Climate and Weather Predictions [M-INFO-106800]****Verantwortung:** TT-Prof. Dr. Peer Nowack**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Elective Studies in Informatics](#)**Leistungspunkte**
6 LP**Notenskala**
Zehntelnoten**Turnus**
Unregelmäßig**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1**Pflichtbestandteile**

T-INFO-113659	Practical Course: AI for Climate and Weather Predictions	6 LP	Nowack
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Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students will be able to

- define current opportunities and challenges in building advanced AI models for climate and weather predictions.
- explain advanced AI model architectures.
- generate and critically assess output of state-of-the-art AI models.
- professionally present their results both orally and in a concise scientific paper.

Inhalt

Students will learn how to work with state-of-the-art AI models for climate science and weather forecasting.

For example, typical AI models will include recent releases of

- Foundation models for climate science and weather forecasting.
- Generative AI models for tasks such as ensemble generation of weather forecasts and of climate change simulations for uncertainty quantification.
- Transformer and graph neural network models for weather forecasting.
- Climate model emulators.

Each student will be able to select from a variety of topics to explore in their practical experiments. These could include, but are not limited to:

- The representation of physical concepts in data-driven AI models (e.g., does the model indirectly learn to “understand physics”?).
- Detecting and understanding failure modes of AI models.
- Forecast accuracy and uncertainty quantification for AI-generated ensembles of simulations.
- Effective solutions to post-processing AI results and/or to modifying AI model architectures.
- Assessing if certain AI architectures perform significantly better for specific tasks.

Arbeitsaufwand

In-person introductory session, individual and group meetings, final presentation sessions: 30h

Practical tasks – getting started, implementation, experiments, analysis: 100h

Write up results in the style of a scientific paper and preparation of final presentation: 50h

Empfehlungen

- Knowledge of the Python programming language.
- Good knowledge of mathematical concepts such as linear algebra is an advantage.
- An interest in scientific questions around climate science and weather forecasting.

M**4.84 Modul: Practical Course: Application Security [M-INFO-106996]**

Verantwortung: Prof. Dr. Jörn Müller-Quade
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113958	Practical Course: Application Security	4 LP	Müller-Quade

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele**Qualification objective:**

Students are able to identify security-relevant weaknesses and errors in a program analysis and suggest corrections.

Learning objectives:

- Students know and understand the programming model of x86 processors and their assembly language and can apply it.
- Students know and understand common error types, attack techniques and countermeasures and can reproduce these independently.
- Students are able to read and analyze a compiled program and examine it for vulnerabilities.
- Students are able to independently carry out attacks in simple scenarios to prove the relevance of the programming error.

Inhalt

This module is dedicated to techniques for exploiting programming errors and common countermeasures, such as:

- Buffer overflows
- Shellcode Injection
- Return Oriented Programming
- Address Space Layout Randomization
- Stack Canaries

Arbeitsaufwand

Attendance time: 15 h

Solving the tasks: 75

Preparation for exam: 30

(1 SWS + 5 SWS) x 15 + 30 h exam preparation = 120 h

Empfehlungen

- The basics of IT security are assumed.
- The content of the lectures "Computer Organization" and "Operating Systems" should be known.

M**4.85 Modul: Practical Course: Artificial Intelligence & Security Lab (AISEC-Lab) [M-INFO-106867]**

Verantwortung: Prof. Dr. Christian Wressnegger
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113760	Practical Course: Artificial Intelligence & Security Lab (AISEC-Lab)	4 LP	Wressnegger

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students know how to apply basic concepts of artificial intelligence and machine learning, and are able to evaluate the performance of such systems on real-world data from computer security research.

- Students know and understand concepts of machine learning for computer security.
- Students are able independently design, implement, and evaluate learning-based systems.
- Students understand limits of learning-based approaches.

Inhalt

In this practical course, the students develop learning-based systems for different computer security tasks, thereby intensifying their knowledge gained in the corresponding lectures. The students have the unique opportunity to design, implement, and evaluate systems based on real-world data used in computer security research.

The module is composed of multiple units with several individual tasks/challenges covering different topics from classical computer security research to security of artificial intelligence. In each unit, the students develop an approach, train and validate it on known data, and submit their solution to the course platform, where the approach is tested against unknown data.

Arbeitsaufwand

- 30h attendance time
- 70h Time to complete the exercises
- 15h Preparation of final presentation
- 5h attendance time (final event)

Empfehlungen

The basics of IT security and artificial intelligence are a prerequisite.

M**4.86 Modul: Practical Course: Building PETs [M-INFO-107412]**

Verantwortung: Prof. Dr. Thorsten Strufe
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114550	Practical Course: Building PETs	4 LP	Schadt, Strufe

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students are able to implement basic methods to handle data in a privacy-preserving manner.

Inhalt

This practical course teaches students the basics of Privacy Enhancing Technology implementations. The methods consist of data privacy notions (k-anonymity and Differential Privacy), private information retrieval and anonymous communication. Students must implement those methods in small example programs in a programming language of their choice.

Anmerkungen

Participation slots are limited. The sign-up modality can be found on the institute website.

Arbeitsaufwand

Attendance time: 9h

Practical work: 100h

Lecture preparations/follow-up: 11h

Empfehlungen

Familiarity with a programming language and a HTTP library

M**4.87 Modul: Practical Course: Chip Design I [M-INFO-107265]****Verantwortung:** Prof. Dr. Mehdi Baradaran Tahoori**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114298	Practical Course: Chip Design I	3 LP	Tahoori

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Hands-on, practical learning by working on hardware-related projects

Inhalt

The Project Lab is a unique opportunity for students to engage in hands-on, practical learning by working on hardware-related projects from various fields, including

- Hardware-based Neural-Networks implementation
- Neuromorphic computing
- Printed Electronics
- Computation in Memory
- Open-source electronic design automation (EDA) tools extension
- Field Programmable Gate Arrays (FPGA)
- Risc-V architecture
- Hardware Security
- Reliability and Test
- other Emerging Technologies

Students can work individually or in groups of 2-4, collaborating to tackle challenges based on a selected topic. The lab accepts a limited number of participants each term based on the number of offered projects, ensuring a focused and immersive experience for everyone involved.

Project topics are carefully defined and curated each term to align with active research initiatives within the Chair of Dependable Nano Computing (CDNC). Participants not only contribute to these projects but also have the potential to co-author research papers and make tangible contributions to the academic community.

The lab emphasizes practical skills, especially in hardware-related fields, offering students access to state-of-the-art tools and technologies. It provides an invaluable opportunity to bridge the gap between theory and practice while preparing for a future in research, development, or industry.

Arbeitsaufwand

4 SWS of practical workload including meetings with the supervisor. 90h.

M**4.88 Modul: Practical Course: Chip Design II [M-INFO-107266]**

Verantwortung: Prof. Dr. Mehdi Baradaran Tahoori
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114299	Practical Course: Chip Design II	3 LP	Tahoori

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Hands-on, practical learning by working on hardware-related projects

Inhalt

The Project Lab is a unique opportunity for students to engage in hands-on, practical learning by working on hardware-related projects from various fields, including

- Hardware-based Neural-Networks implementation
- Neuromorphic computing
- Printed Electronics
- Computation in Memory
- Open-source electronic design automation (EDA) tools extension
- Field Programmable Gate Arrays (FPGA)
- Risc-V architecture
- Hardware Security
- Reliability and Test
- other Emerging Technologies

Students can work individually or in groups of 2-4, collaborating to tackle challenges based on a selected topic. The lab accepts a limited number of participants each term based on the number of offered projects, ensuring a focused and immersive experience for everyone involved.

Project topics are carefully defined and curated each term to align with active research initiatives within the Chair of Dependable Nano Computing (CDNC). Participants not only contribute to these projects but also have the potential to co-author research papers and make tangible contributions to the academic community.

The lab emphasizes practical skills, especially in hardware-related fields, offering students access to state-of-the-art tools and technologies. It provides an invaluable opportunity to bridge the gap between theory and practice while preparing for a future in research, development, or industry.

Arbeitsaufwand

4 SWS of practical workload including meetings with the supervisor. 90h.

M**4.89 Modul: Practical Course: Continuous Software Engineering [M-INFO-107677]**

Verantwortung: Prof. Dr.-Ing. Anne Koziolk
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-115043	Practical Course: Continuous Software Engineering	6 LP	Koziolk

Erfolgskontrolle(n)

See Partial Achievements (Teilleistung)

Voraussetzungen

See Partial Achievements (Teilleistung)

Qualifikationsziele

Students are able to

- understand scientifically motivated problem statements and, in cooperation with supervisors, derive requirements for the software to be developed or extended.
- develop medium-sized programs using current development environments, or familiarize themselves with medium-to large-scale software systems and further develop them.
- present project progress to supervisors in regular meetings and identify and communicate potential obstacles.
- evaluate third-party code in the context of code reviews, identify potential weaknesses, and discuss them.
- present a (further) developed software system in a short presentation.
- document a (further) developed software system

Inhalt

Agile software development methods refer to a class of iterative and incremental development approaches that place particular emphasis on executable software, early feedback from stakeholders, and embracing necessary changes. In these approaches, efforts that are not directly related to producing running software are viewed critically.

In this lab course, participants use and extend state-of-the-art tools from both practice and research to support agile software development. The following types of tools are considered:

- tools that leverage large language models (LLMs) to support agile software development methods, e.g., creating trace links from existing software engineering artifacts and models
- tools for semi-automatically deriving models from source code and runtime information, in particular performance data
- tools for simpler, domain-specific modeling
- tools for automatically maintaining consistency between development artifacts and models

Depending on the chosen topic, different technologies and programming languages are used, including:

- LLMs
- Java, Python, Java/TypeScript
- Spring Boot, React, Maven, EMF (Eclipse Modeling Framework)
- development of VS Code extensions or language servers
- Vitruvius
- Palladio Simulator
- ARDoCo
- JPlag

Course assessment is graded individually; group work is intended. The lab is embedded in the chair's ongoing research and offers plenty room for creativity. The assignments are practice-oriented and prepare students for realistic tasks in both research and industry.

The lab can be completed entirely in English. If preferred, supervision and presentations in German are also possible.

Anmerkungen

In earlier semesters, this course was named „Praktikum: Werkzeuge für agile Modellierung“.

Arbeitsaufwand

Course workload:

20 hours of work for onboarding,

120 hours of work for development,

20 hours of work for weekly meetings and follow-up,

10 hours for preparing and conducting a code review,

10 hours for preparing and delivering the final presentation.

In total, this amounts to 180 hours of work.

Empfehlungen

See Partial Achievements (Teilleistung)

M**4.90 Modul: Practical Course: Digital Design & Test Automation Flow [M-INFO-102570]**

Verantwortung: Prof. Dr. Mehdi Baradaran Tahoori
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-105565	Practical Course: Digital Design & Test Automation Flow	3 LP	Tahoori

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The objective of this lab is to have a hands-on practice on major steps in digital design and test automation flow, from system-level specification to physical design and verification.

Inhalt

Electronic Design Automation (EDA) is used to develop nearly all novel electronic systems that we use in our daily lives, such as smartphones or laptops. In order to manage the high complexity of these systems, all steps in the design and verification phases are done automatically with the help of EDA tools.

The objective of this lab is to have a hands-on practice on major steps in digital design and test automation flow, from system-level specification to physical design and verification, using industrial EDA toolsets which are predominantly used in the industry and academia.

The students will work on some sample designs and go through all major design and test steps, one by one, in different sessions of the lab. So, by the end of this lab, they become familiar with the steps and tool chain in the digital design and test automation flow. The topics include system-level specification and simulation; high-level synthesis; logic-level synthesis and simulation; design for testability; test pattern generation and fault simulation; physical design and verification; timing analysis and closure; area, delay, and power estimation and analysis.

Arbeitsaufwand

4 SWS / 3 CP = 90 h/week

Empfehlungen

Knowledge of "Dependable Computing" and "Fault Tolerant Computing" and Computer Architecture is helpful.

M**4.91 Modul: Practical Course: Efficient Parallel C++ [M-INFO-107203]****Verantwortung:** Prof. Dr. Peter Sanders**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Algorithm Engineering](#)
[Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)

Leistungspunkte
 6 LP

Notenskala
 Zehntelnoten

Turnus
 Unregelmäßig

Dauer
 1 Semester

Sprache
 Englisch

Level
 4

Version
 1

Pflichtbestandteile			
T-INFO-114228	Practical Course: Efficient Parallel C++	6 LP	Sanders

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The students

- can use the methods of algorithm engineering in order to implement and evaluate given algorithmic problems and data structures in C++.
- recognize factors that lead to inefficient code and can, if possible, replace them with more efficient constructions.
- understand how to use the presented techniques for parallelization and to generate thread-safe codes with the given means.
- know the features of the standard library and are able to use them selectively.
- can test the codes generated by them for correctness and performance, furthermore they can represent and analyze the obtained results.

Inhalt

In this practical course students solve multiple programming tasks in C++. The main focus is on the efficient implementation and their evaluation through extensive experiments. The programming tasks are motivated by scientific work in the field of algorithm engineering.

They cover complex algorithms as well as advanced data structures, furthermore advanced programming techniques and parallelization (thread management capabilities of the standard library).

Arbeitsaufwand

- ~ 15h attendance time
- ~ 25h presentation/discussion/assessment of the solutions (with preparation)
- ~ 140h working on the tasks (implementation and evaluation)

M**4.92 Modul: Practical Course: Engineering Approaches to Software Development [M-INFO-107678]****Verantwortung:** Prof. Dr. Ralf Reussner**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
6 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Semester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1

Pflichtbestandteile			
T-INFO-115044	Practical Course: Engineering Approaches to Software Development	6 LP	Reussner

Erfolgskontrolle(n)

See Partial Achievements (Teilleistung)

.

Voraussetzungen

See Partial Achievements (Teilleistung).

Qualifikationsziele

Students are able to

- understand scientifically motivated problem statements and, in cooperation with supervisors, derive requirements for the software to be developed or extended.
- develop medium-sized programs using current development environments, or familiarize themselves with medium-to large-scale software systems and further develop them.
- present project progress to supervisors in regular meetings and identify and communicate potential obstacles.
- evaluate third-party code in the context of code reviews, identify potential weaknesses, and discuss them.
- present a (further) developed software system in a short presentation.

Inhalt

The engineering-oriented development of software is an indispensable prerequisite for building large systems. Accordingly, software engineers must systematically analyze the system's quality already during software design and, where possible, also predict it.

In this practical course, participants use and extend state-of-the-art tools from both practice and research to evaluate and predict the performance of software systems. These tools provide solutions for the following tasks:

- assessing the scalability of software depending on the execution environment
- maintaining consistency across different development artifacts

The development tasks are drawn from the following areas:

- MDSD (Model-Driven Software Development)
- plugin development
- benchmarking
- reverse engineering

The technologies used include:

- Java, Python, Java/TypeScript
- Spring Boot, React, Maven, EMF (Eclipse Modeling Framework)
- Development of VS Code extensions or language servers
- Vitruvius
- Palladio Simulator

Course assessment is graded individually; group work is intended. The lab is embedded in the chair's ongoing research and offers plenty room for creativity. The assignments are practice-oriented and prepare students for realistic tasks in both research and industry.

The lab can be completed entirely in English. If preferred, supervision and presentations in German are also possible.

Anmerkungen

In earlier semesters, this course was named „Praktikum Ingenieursmäßige Software-Entwicklung“.

Arbeitsaufwand

Course workload:

20 hours of work for onboarding,

120 hours of work for development,

20 hours of work for weekly meetings and follow-up,

10 hours for preparing and conducting a code review,

10 hours for preparing and delivering the final presentation.

In total, this amounts to 180 hours of work.

Empfehlungen

See Partial Achievements (Teilleistung).

M**4.93 Modul: Practical Course: Fine-grained Algorithm Design and Engineering [M-INFO-106784]**

Verantwortung: Prof. Dr.-Ing. Marvin Künnemann
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte
6 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-113635	Practical Course: Fine-grained Algorithm Design and Engineering	6 LP	Künnemann

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students should be able to apply knowledge in the specializations “Algorithmtechnik” and “Theoretische Grundlagen” to derive fast algorithms and their implementations for a given algorithmic problem.

This includes:

- modeling a given problem of interest as a well-defined algorithmic problem as well as identifying reasonable relaxations
- performing a literature search to identify algorithmic ideas previously proposed for a given problem
- researching a given algorithmic or conditional lower bound technique and investigating its applicability on a given problem
- implementing resulting algorithms efficiently
- creating reasonable benchmark data sets (generated randomly, via reductions or from real-world data sources)
- evaluating an implementation on benchmark data and gaining insights on possible improvements of the model, algorithm or implementation.

Furthermore, the students can constructively engage in a team setting and are able to clearly communicate their ideas and results.

Inhalt

Each group of students will receive a topic among a list of possible algorithmic problems with relevance for fine-grained and parameterized complexity (usually from the fields of graph theory, computational geometry or string problems). In some cases, the proposed topic is the subject of an ongoing algorithmic contest (e.g., the PACE challenge), providing an opportunity of participation as part of the practical course.

Under supervision, each group will:

- research the theoretical state-of-the-art for their algorithmic problem and/or design a novel algorithm,
- implement one or more algorithmic approaches
- evaluate and improve them using appropriate benchmark data sets.

The course aims to investigate the connections between worst-case upper & conditional lower bounds and fast practical implementations.

Arbeitsaufwand

6 CP correspond to ~ 180 h, distributed roughly as follows:

- ~ 40 h meetings, literature review, etc.
- ~ 100 h implementation and evaluation
- ~ 40 h preparation of presentation and report

Empfehlungen

- Basic knowledge of algorithms and data structures is assumed.
- Knowledge of fine-grained complexity is helpful, but not required.

M**4.94 Modul: Practical Course: FPGA Programming [M-INFO-102661]**

Verantwortung: Prof. Dr. Mehdi Baradaran Tahoori
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-105576	Practical Course: FPGA Programming	3 LP	Tahoori

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students will learn to design and to simulate digital circuits with FPGA.

Inhalt

This lab emphasizes on the practical aspects of Field Programmable Gate Arrays (FPGAs). In the beginning, a short background introduction on FPGAs is given, followed by a tutorial on the workflow of configuring and programming an FPGA. This lab includes FPGA design using schematic layouts as well as several example of VHDL/Verilog programming to implement some sample digital circuits. Students will learn to design and to simulate digital circuits with FPGA. The design will be compiled on run a FPGA. The lab is designed around the DE2-115 prototyping board, which provides a programmer, program memory, and array of switches, buttons, LEDs, an LCD, and several I/O ports.

Arbeitsaufwand

4 SWS / 3 CP = 90 h/week

Empfehlungen

Knowledge of "Dependable Computing" and "Fault Tolerant Computing" and Computer Architecture is helpful.

M**4.95 Modul: Practical Course: General-Purpose Computation on Graphics Processing Units [M-INFO-100724]**

Verantwortung: Prof. Dr.-Ing. Carsten Dachsbacher
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	3

Pflichtbestandteile			
T-INFO-109914	Practical Course: General-Purpose Computation on Graphics Processing Units	3 LP	Dachsbacher

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students should acquire the ability to use programmable graphics hardware using suitable interfaces (e.g. OpenCL, CUDA) to solve scientific and technical calculations. The students should thereby acquire the practical ability to systematically develop a parallel, efficient programme on the basis of suitable algorithms. Students learn basic algorithms for parallel architectures, are able to analyse and evaluate them, and practice their use in practical applications.

Inhalt

The practical course covers basic concepts for the use of modern graphics hardware for technical and scientific calculations and simulations. Starting with basic algorithms, e.g. parallel reduction or matrix multiplication, the practical course imparts knowledge about the properties and capabilities of modern graphics processors (GPUs). As part of the practical course, students work on smaller sub-projects in which they acquire knowledge about the algorithms used and apply them to a specific problem; OpenCL or CUDA, for example, is used as a programming interface.

Arbeitsaufwand

Attendance time = 12h

Preparation/post-processing = 78h

M**4.96 Modul: Practical Course: Human-Centred Robotics [M-INFO-106646]****Verantwortung:** Prof. Dr. Katja Mombaur**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113393	Practical Course: Human-Centred Robotics	6 LP	Mombaur

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students learn to understand and scrutinize complex scientific topics and to reproduce and verify published results. They gain in-depth knowledge and practical experience in the field of motion generation and control of human-centered robots while working on a specific project task. They also learn how to plan, work together and communicate in a team. Students will be able to present their project results in a scientific presentation, demonstrate the practical results and answer detailed questions. They can also summarize their project results in writing using Latex in the style of a scientific paper and place them in a scientific context.

Inhalt

Human-centered robots are robots that directly interact with humans or support humans in their motions. This includes humanoid robots, but also wearable robots (exoskeletons and prostheses) or external physical assistive devices. In this practical course, students learn how to implement theoretical knowledge about human-centered robots and use it to solve a given task based on a individual project with robot hardware. Projects can either focus on code development for a given hardware or on the development or modification of robot hardware along with the basic code. Students learn about the challenges of working with real robot hardware vs model computations, and about working principles and practical implementation of sensors and actuators.

Anmerkungen

For capacity reasons (supervision of practical projects and workspace) , the practical course is limited in terms of projects and number of students. Specific project topics will be different each term and will be announced in a presentation during the first semester week.

Subsequently, a selection and topic assignment will be carried out, taking into account the students' subject and topic specific skills.

Arbeitsaufwand

Estimated effort for this module is 180 hours:

20h – In person events (kickoff meeting, individual meetings with supervisor, presentations)

130h – Individual project work

30h - Writing report and preparing presentation

Empfehlungen

Knowledge in Robotics (e.g. from the class Robotics 1 and follow-ups) are very helpful.

M**4.97 Modul: Practical Course: Interactive Theorem Proving [M-INFO-107494]****Verantwortung:** Prof. Dr. Bernhard Beckert**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Theoretical Foundations](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Unregelmäßig**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1

Pflichtbestandteile			
T-INFO-114691	Practical Course: Interactive Theorem Proving	3 LP	Beckert, Stübinger

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students are able to use an interactive theorem prover such as Isabelle/HOL and can formulate and prove formal statements in the field of programming language technology (e.g. related to type safety, compilers, and semantics).

Students can name and describe the foundational features of the theorem prover, including its meta-logic, unification, resolution and rule application as well as the axiomatisations of higher-order logic.

Furthermore, students are able to write proofs themselves in the proof language. They can assess which proof methods (simplification, case distinction, induction, etc.) are appropriate for a given statement. They are also able to formulate statements which fit a given proof method. Students can apply advanced methods for structuring proofs (e.g. using Isar). They know the automatic and manual tactics of the theorem prover and can decide in which situations these should be applied.

Students are proficient in defining (recursive) data types, functions and inductive predicates and can prove statements about them. Students can implement simple algorithms from the field of program analysis (e.g. constant propagation) and verify its correctness with the help of the theorem prover.

Inhalt

In this lab course students are familiarized with the use of the interactive theorem prover Isabelle/HOL and use it to formalise and verify a project. In the first half of the course, important principles of theorem proving (e.g. simplification, recursion, inductive definitions and proofs) are learned and practiced via exercise sheets. In the second half of the course, students will work independently in teams formalising and verifying a topic in the area of formal methods, e.g. in programming language theory, type systems, or compilers. Project topics are set at the end of the first half of the course.

Arbeitsaufwand

3 CP equate to approximately 90 SWS. These are distributed as follows:

- approx. 15 h attendance time
- approx. 30 h work on the exercise sheets
- approx. 43 h work on the independent project
- approx. 2 h work on preparing the project presentation

Empfehlungen

- Knowledge of functional programming is assumed.
- Some knowledge of logic (e.g. via courses such as Formale Systeme and/or Constructive Logic) will be helpful, but is not required

M**4.98 Modul: Practical Course: Low Power Design and Embedded Systems [M-INFO-104031]****Verantwortung:** Prof. Dr.-Ing. Jörg Henkel**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	2

Pflichtbestandteile			
T-INFO-108323	Practical Course: Low Power Design and Embedded Systems	4 LP	Henkel

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Upon completion of this lab, students will:

- Apply and evaluate different hardware/software optimization techniques for low power and energy consumption under given constraints (e.g., performance) on embedded systems.
- Develop a deep understanding of system-level resource management techniques in modern systems. They will learn to apply machine learning methods to automate and optimize complex resource allocation decisions, thereby acquiring practical skills in data collection, model training, and iterative system tuning.
- Collaborate effectively in a team to practically solve technical problems related to power, temperature and energy optimizations on a real hardware platform.

Inhalt

This lab explores different software and hardware approaches for power reduction on modern embedded systems, considering other relevant metrics and constraints such as performance, power, temperature, chip area, among others, both on simulation and real-hardware platforms.

The course is divided in two main topics:

- Smart resource management (RM) for multi-/many-core computing systems: This topic covers system level techniques based on RM control knobs such as application mapping, dynamic task migration and dynamic voltage and frequency scaling (DVFS), using both from the heuristic and machine-learning methods.
- Hardware/Software design for low power systems: This topic includes the application of several design principles among the computing stack to optimize different metrics such as performance, power and energy in embedded systems. Techniques include compiler optimizations, HW/SW co-simulation, design-space exploration and high-level synthesis.

Arbeitsaufwand

Attendance time:

40 hours

Project work:

60 hours

Final Report preparation:

20 hours

Total: 120 hours (4 ECTS)

Empfehlungen

Students should be familiar with software development practices under Linux-based systems. Practical knowledge in C/C++ as well as Python is required.

M**4.99 Modul: Practical Course: Mathematical and Computational Methods in Robotics and AI [M-INFO-107609]**

Verantwortung: Prof. Dr. Katja Mombaur
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114911	Practical Course: Mathematical and Computational Methods in Robotics and AI	6 LP	Mombaur

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students learn to understand scientific topics. They will learn how to address specific project tasks and solve them in a structured way. They will obtain deepened knowledge about fundamental mathematical and computational methods, including their potential, strengths and limitations and in which areas they are suited to be applied. Students learn to present their results to a an audience, as well as suitably documenting their results in form or in addition to a scientific paper and being capable of assessing the advantages and disadvantages of the methods used.

Inhalt

Efficient mathematical and computational methods are an irreplaceable cornerstone of modern research and industrial applications in forms such as numerical simulation, optimization or inference, ranging from the generation of task-specific trajectories for robotic systems over stabilization of existing robots to data-driven prediction and inference as well as AI-applications.

In this practical course, students learn, on the basis of an individual programming project, how to implement and apply fundamental mathematical and computational methods to solve an existing task in a robotics application or related fields and gain experience leveraging methods from this area in said real-world context.

Anmerkungen

Limited number of projects and participants. Specific project topics will be different each term and will be announced in a presentation during the first semester week or can be individually agreed upon with people supervising the practical. Students can propose topics on their own as well.

Arbeitsaufwand

Estimated effort for this module is 180 hours:

20h – In person events (kickoff meeting, individual meetings with supervisor, presentations)

130h – Individual project work

30h - Writing report and preparing presentation

Empfehlungen

Basic mathematical knowledge will be helpful, depending on topic. For Projects in the context of robotics, knowledge of robotics is recommended, but not strictly necessary.

M**4.100 Modul: Practical Course: Model-Driven Software Development [M-INFO-106932]****Verantwortung:** Prof. Dr. Ralf Reussner**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
6 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Sommersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1**Pflichtbestandteile**

T-INFO-113897	Practical Course: Model-Driven Software Development	6 LP	Burger, Reussner
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Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students can

- understand and apply the model-driven development process
- Express facts as a metamodel and create a suitable domain-specific language (DSL)
- Formulate restrictions in the OCL language
- Create and apply model-to-model transformations
- Create model-to-text transformations
- Create graphical editors for metamodels
- Develop textual syntaxes for metamodels and DSLs
- use current tools in the field of model-driven software development

Inhalt

Model-driven development methods have become particularly popular thanks to the Eclipse Modelling Framework (EMF) and the OMG standards MOF, UML and QVT. Advanced software development concepts such as product lines, generative programming and model transformations now make it possible to develop software more flexibly and quickly and to use it on different platforms. Domain-specific languages (DSL) and the graphical and textual editors generated from them can be easily created.

This practical course deals with current techniques of model-driven software development (MDSD). Students work with current frameworks and languages such as EMF, QVT, ATL and XText and create a domain-specific language and model transformations.

Arbeitsaufwand

96 working hours for exercises, 48 working hours for project work, 16 working hours for preparing the final presentation, 20 working hours for weekly meetings and final presentation. This results in a total of 180 working hours.

M**4.101 Modul: Practical Course: Movement and Technology [M-INFO-106648]**

Verantwortung: Prof. Dr. Katja Mombaur
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113394	Practical Course: Movement and Technology	6 LP	Mombaur

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students learn to analyze and understand complex scientific topics in the area of human motion capture and motion analysis. They gain in-depth knowledge and practical experience with motion capture technology, experiment planning, and analysis. They also learn how to plan, work together and communicate in an interdisciplinary team. Students will be able to present their project results in a scientific presentation, demonstrate the practical results and answer detailed questions. They can also summarize their project results in writing using Latex and place them in a scientific context.

Inhalt

In this joint course between Informatics and Sports Science, and in the sense of research-oriented teaching, students learn about current research projects of the BioRobotics Lab (Informatics) and the BioMotion Center (Sports Science) at the interface of motor control and biomechanics of human movement. This research involves the use of latest motion capture technology, advanced analysis tools, and partly also assistive robotics technology. Students work in teams (interdisciplinary teams between students from different study programs are highly encouraged) to carry out motion capture experiments, analyze the data and present the results in written and oral form. Depending on the specific project, these motion capture studies are either stand-alone studies just for this course or part of a larger research project at one of the organizing research groups.

Anmerkungen

Limited number of projects and participants. Specific project topics will be different each term and will be announced in a presentation during the first semester week.

Arbeitsaufwand

Estimated effort for this module is 180 hours:

20h – In person events (kickoff meeting, individual meetings with supervisor, presentations)

120h – Individual project work

40h - Writing report and preparing presentation

Empfehlungen

Knowledge in Robotics (e.g. from the class Robotics 1 and follow-ups) are very helpful.

Programming skills.

M**4.102 Modul: Practical Course: Natural Language Dialog Systems [M-INFO-107177]**

Verantwortung: Prof. Dr. Jan Niehues
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte
6 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-114206	Practical Course: Natural Language Dialog Systems	6 LP	Niehues

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The student

- is able to develop a dialogue system using state-of-the-art methods.
- develop a dialogue system.
- can evaluate dialogue systems.
- can present his/her findings in a scientific lecture.

Inhalt

Thanks to major advances in the field of deep learning and, in particular, large language models, it is now possible to develop dialogue systems and chatbots that can support people in many situations.

As part of this internship, students will develop a personal assistant for various application scenarios. To do this, students must first deal with data collection and data preparation. This data should then be used to develop a chatbot for the addressed application using freely available pre-trained models. In addition, the students will investigate various options for evaluating the systems.

In the final part of the internship, students can independently choose a focus to improve their initial system. The final systems will be presented in a final presentation.

Arbeitsaufwand

180 h

M**4.103 Modul: Practical Course: Real-world Vulnerability Discovery and Exploits [M-INFO-106627]**

Verantwortung: Prof. Dr. Christian Wressnegger
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113350	Practical Course: Real-world Vulnerability Discovery and Exploits	4 LP	Wressnegger

Erfolgskontrolle(n)

Siehe Teilleistung.

Voraussetzungen

Siehe Teilleistung.

Qualifikationsziele

- ← Students know and understand exploitation techniques.
- ← Students are able to independently research software vulnerabilities.
- ← Students are comfortable engaging with software vendors in vulnerability disclosure.

Inhalt

Students understand modern exploitation techniques and can apply them. Moreover, they get familiar with the vulnerability disclosure process of prominent software vendors, reporting their findings.

Arbeitsaufwand

- 2h Präsenzzeit/ Woche (Vorträge)
- 5h Projektarbeit/ Woche
- 10h Vorbereitung Abschlusspräsentation
- 5h Präsenzzeit (Abschlussveranstaltung)

Insgesamt 120h

Empfehlungen

Praktikum Anwendungssicherheit

M**4.104 Modul: Practical Course: Route Planning [M-INFO-107600]**

Verantwortung: TT-Prof. Dr. Thomas Bläsius
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte
6 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-114899	Practical Course: Route Planning	6 LP	Bläsius, Ueckerdt

Erfolgskontrolle(n)

See partial achievements. (Teilleistung)

Voraussetzungen

See partial achievements. (Teilleistung)

Qualifikationsziele

The students

- can practically apply the knowledge from the the module "Algorithmen für Routenplanung",
- are capable of efficiently implementing given topics in route planning (e.g., separator computation, shortest path acceleration techniques, ...),
- master the steps from design to implementation and evaluation in the practical application of algorithmic methods,
- possess the ability to act in a results-oriented manner within a team, critically assess their own actions, and demonstrate strong communication skills.

The participants are furthermore able to efficiently implement algorithms with consideration for modern computer architectures, as well as to plan and conduct meaningful experimental evaluations.

Inhalt

In the practical course route planning, various topics from route planning are assigned, which are to be implemented independently by small student groups.

A main focus is placed on efficient programming in C++ or Rust.

Arbeitsaufwand

Practical course with 4 SWS, 6 LP

6 LP corresponds to approx. 180 working hours, of which
 approx. 10 hours attendance time,
 approx. 12 hours working on the exercises,
 approx. 128 hours implementation phase,
 approx. 30 hours elaboration and preparation of the presentation

M**4.105 Modul: Practical Course: Security, Usability and Society [M-INFO-105453]**

Verantwortung: Prof. Dr. Thorsten Strufe
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte
4 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-110990	Practical Course: Security, Usability and Society	4 LP	Strufe

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students know established security and data protection programs, can implement them in apps and can carry out user studies.

Learning objectives:

- Students know and understand the methods for developing privacy-friendly apps and can apply them.
- Students are able to implement various applicable security measures in programs.
- Students can set up and conduct user studies.
- Students are able to prepare and present a report of their work.

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Inhalt

The internship "Security, Usability and Society" covers topics such as usable security and privacy programs as well as conducting user studies.

Topics include:

- Privacy-friendly apps
- Programming usable security measures
- Conducting usable security user studies

Arbeitsaufwand

Attendance time: 15 h

Solving the tasks: 75

Preparation of presentation and report: 30

M**4.106 Modul: Practical Course: Smart Energy System [M-INFO-105955]**

Verantwortung: Prof. Dr. Veit Hagenmeyer
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-112030	Practical Course: Smart Energy System Lab	6 LP	Jumar

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

After successfully completing the course, students will be able to

- explain and categorize a set of current research issues in modern energy systems
- explain structure and objectives of the KIT Energy Lab and its role in smart grid research
- conduct selected research activities in the field of innovative, application-oriented information, automation and system technology for sustainable energy systems
- isolate a research activity / hypothesis suitable for evaluation in the Smart Energy System Simulation and Control Center
- plan, organize and manage their laboratory / experimental work to fit a given time-frame
- concisely present and defend their research activity and results in front of a peer audience
- structure and writeup their work in research-paper-style
- work in team

Inhalt

As part of the two-week lab course, students work in teams of two on a project topic derived from current research questions of the Smart Energy System Simulation and Control Centre of the Energy Lab (<https://www.iai.kit.edu/RPE.php>). A list of topics is made available in advance via the Campus Plus Portal. Interested students should indicate their preferences based on which the topics will be assigned.

The group of students typically goes through the following phases: Concept development/experiment preparation, implementation/experiment execution, evaluation/experiment analysis, presentation of results and write-up. The teams will work independently on the assigned project - supervision is provided by research assistants.

The course begins with a joint kick-off event. This includes an organizational introduction, a tour of the Energy Lab and a brief presentation of all project topics. For the following preparation phase students are provided with literature and material to thoroughly familiarize themselves with the topic, research background and literature, plan and prepare experiments etc. In the two lab weeks the experiments are conducted and a scientific writing workshop is held. At the end of the lab course each team has to defend their work in a 10 min presentation followed by 5 minutes of discussion in front of peers and examiners. Additionally, the work needs to be documented in the style of a scientific paper.

Arbeitsaufwand

6 credit points corresponds to approx. 180 working hours, of which

- Attendance time / meetings in large and small groups: 10h
- Select and carry out project work: 140h
- Writing a research report and preparing a presentation: 30 hours

Empfehlungen

- Knowledge of the fundamentals of energy informatics is a prerequisite.
- Knowledge of the fundamentals of electrical engineering and energy technology is required.
- Knowledge of the basics of mechatronics, data analysis and signal processing is helpful.
- Knowledge of power systems or power electronics is helpful.

M**4.107 Modul: Practical Course: Software Defined Networking [M-INFO-107221]**

Verantwortung: Prof. Dr. Martina Zitterbart
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114240	Practical Course: Software Defined Networking	6 LP	Zitterbart

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The student understands the concepts behind the SDN approach and applies this knowledge to design solutions for new problems. He/she is able to develop an application in group work that implements a specific functionality in an SDN network. From the outset, the student plans his/her solution approaches from the point of view of interoperability with the solutions of the other groups. The participants jointly decide on compromise solutions, if these are necessary, in order to be able to operate the applications of the different groups together without disruption.

Inhalt

The internship deals with the realization of a software project in the field of Software-Defined Networking (SDN). With SDN, the control and monitoring of a network is outsourced to a controller. The actual forwarding hardware can then be programmed via the OpenFlow interface.

As part of the internship, we want to find out together to what extent this technology can also be used within our own four walls. To this end, we will design and develop an SDN home router that enables users to monitor and control their network using SDN applications. In small groups, we will build or recreate various functions from the home network sector, e.g. a firewall or parental control. A monitoring system that breaks down the Internet consumption of all connected computers is also conceivable. Or a traffic engineering mechanism that ensures that you can still enjoy YouTube even when your younger brother is downloading a 100 GB game. Many other variants are conceivable. We decide together in the internship what will be implemented in the end. Your own ideas are very welcome!

Arbeitsaufwand

180h

M**4.108 Modul: Practical Course: Software Development and Application of Mobile, Bio-Inspired Robots [M-MACH-106904]**

Verantwortung: Prof. Dr.-Ing. Arne Rönnau
Einrichtung: KIT-Fakultät für Maschinenbau
Bestandteil von: [Minor Studies: Biologically Inspired Robotics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-MACH-113854	Practical Course: Software Development and Application of Mobile, Bio-Inspired Robots	6 LP	Rönnau

Erfolgskontrolle(n)

Siehe Teilleistung

Voraussetzungen

Keine

Qualifikationsziele

Studierende sind in der Lage, biologisch inspirierte, mobile Roboter zu verstehen und deren Software zu erweitern.

Inhalt

Dieses Modul vermittelt Studierenden den Umgang mit und die Erweiterung biologisch inspirierter, mobiler Robotik. Dabei werden die Themenbereiche Regelungstechnik, Computer-Vision, 3D Kartierung, Navigation und Mensch-Maschine-Interaktion behandelt.

Die Studierenden arbeiten in Gruppen und erstellen einen gemeinsamen Abschlussbericht und eine gemeinsame Präsentation.

Arbeitsaufwand

180h

- 30h wöchentliches Regeltreffen
- 120h Vor- und Nachbereitungszeiten
- 30h Präsentations- und Berichtvorbereitung

Empfehlungen

Siehe Teilleistung

M**4.109 Modul: Practical Course: Visual Computing [M-INFO-101567]****Verantwortung:** Prof. Dr.-Ing. Carsten Dachsbacher**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Elective Studies in Informatics](#)**Leistungspunkte**
6 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Semester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1

Pflichtbestandteile			
T-INFO-103000	Practical Course: Visual Computing	6 LP	Dachsbacher

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

In this course, practical problems from the core area of computer graphics and the broader field of visual computing are solved where graphics hardware is used. In individual sub-projects, or self-defined larger projects, the application of various computer graphics techniques and the use of modern graphics hardware are practised. In addition, students can work together in a team to solve the tasks of the work placement.

Inhalt

The practical course deals with specific topics, some of which were addressed in corresponding lectures on the specialisation subject of computer graphics, and explores these in greater depth. Previous attendance of the respective lecture is helpful, but not a prerequisite for attendance.

Arbeitsaufwand

Attendance time = 30h

Preparation/follow-up = 150h

Empfehlungen

Programming skills in C/C++ are recommended.

M**4.110 Modul: Practical Introduction to Hardware Security [M-INFO-107241]****Verantwortung:** Prof. Dr. Mehdi Baradaran Tahoori**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114267	Practical Introduction to Hardware Security	6 LP	Tahoori

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The goal of this course, which is a combination of lectures and lab assignments, is to have a hands-on experience on basic concepts and new developments in hardware security, by combining both theory and practice in a coherent course. The theoretical concepts for each topic will be presented to the students in form of lectures, followed by a set of lab assignments on both hardware and software platforms to be performed by the students for each topic.

Inhalt

1. Hardware security primitives (PUF, TRNG)
2. Hardware Implementation of encryption modules (AES)
3. Passive Attack with side channel (on AES)
4. Active fault attack (on AES)

Arbeitsaufwand

4 SWS / 6 ECTS = 180h

2 SWS lecture (1,5h) + 2 SWS practical course (1,5h) / week

Empfehlungen

Knowledge of Digital Design (lecture TI)

Practical Course "FPGA Programming"

M**4.111 Modul: Practical SAT Solving [M-INFO-107238]****Verantwortung:** Prof. Dr. Peter Sanders**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Theoretical Foundations](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
5 LP**Notenskala**
Zehntelnoten**Turnus**
Unregelmäßig**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
2

Pflichtbestandteile			
T-INFO-114262	Practical SAT Solving	5 LP	Balyo, Iser, Sanders, Schreiber

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students are able to evaluate combinatorial problems, assess their complexity, and solve them using computers.

Students learn how to solve combinatorial problems efficiently using SAT Solving. Students are able to assess the practical complexity of decision and optimization problems, encode problems as SAT problems, and implement efficient solution procedures for combinatorial problems.

Students gain insight into state-of-the-art solution methods for SAT and related problems and their implementations in SAT solvers.

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Inhalt

The problem of propositional satisfiability (SAT) is an outstanding problem of computer science from a theoretical as well as practical perspective. Being the first problem proven to be NP-complete, it serves as a fundamental tool for research in complexity theory. Moreover, SAT solving has been established as one of the most important fundamental methods in hardware and software verification, and is used to solve hard combinatorial problems in industrial practice as well. This module aims to provide students with the theoretical and practical aspects of SAT-Solving. Covered are:

1. basics, historical development
2. encodings, e.g. cardinality constraints
3. phase transitions in random problems
4. local search (GSAT, WalkSAT, ..., ProbSAT)
5. resolution, Davis-Putnam algorithm, DPLL algorithm, look-ahead algorithm
6. efficient implementations, data structures
7. heuristics in the DPLL algorithm
8. CDCL algorithm, clause learning, implication graphs
9. restarts and heuristics in the CDCL algorithm
10. preprocessing, inprocessing
11. generation of proofs and their checking
12. parallel SAT solving (guiding paths, portfolios, cube-and-conquer)
13. related problems: MaxSAT, MUS, #SAT, QBF
14. advanced applications: Bounded model checking, planning, satisfiability-modulo-theories

Anmerkungen**Arbeitsaufwand**

Lecture (2 SWS) + exercise (1 SWS)

(Preparation and follow-up: 4h/week, exercises: 2h/week, preparation for exam: 15h)

Total workload: (2 SWS + 1 SWS + 4 SWS + 2 SWS) x 15 h + 15h preparation = 9x15h + 15h = 150h = 5 ECTS

M**4.112 Modul: Privacy Enhancing Technologies [M-INFO-105452]**

Verantwortung: Prof. Dr. Thorsten Strufe
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-110989	Privacy Enhancing Technologies	6 LP	Strufe

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

This course will provide students with a basic understanding of privacy risks, the most common technologies to tackle them and the human factors shaping their design. The course will analyze the adversary models and evaluation metrics underlying the design of privacy-enhancing technologies

- The students have a critical reasoning about privacy,
- have knowledge in the evaluation of privacy risks,
- understand the design aspects of privacy-enhancing technologies,
- are familiar with the latest research in the field
- are able to analyze and discuss the space of solutions to a given privacy problem

Inhalt

The following topics will be covered

- Freedom of information, the surveillance economy, and other motivations for privacy
- Privacy metrics and adversary models
- Anonymous communications
- Data-perturbative privacy-enhancing technologies
- Anonymization algorithms for databases
- Homomorphic encryption and zero knowledge proofs
- Selective disclosure for identity management
- Usable privacy
- Applying privacy principles and case studies

Arbeitsaufwand

Attendance time in lectures: 45 h

Preparation and follow-up of the same: 90 h

Exam preparation and attendance in the same: 45 h

M**4.113 Modul: Probability and Computing [M-INFO-107168]**

Verantwortung: TT-Prof. Dr. Thomas Bläsius
Prof. Dr. Peter Sanders

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
5 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114196	Probability and Computing	5 LP	Bläsius, Katzmann, Sanders

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The students

- understand when and why randomisation is useful or necessary for solving an algorithmic problem,
- can explain central design methods and analysis tools of randomised algorithms,
- can design and explain simple randomised algorithms and data structures for solving a problem,
- can decide which tools are suitable for the analysis of given randomised algorithms and data structures and apply them.

Inhalt

Randomised algorithms and data structures make their approach dependent on random experiments. While the design of deterministic algorithms is often driven by a pessimistic view of worst-case behaviour, randomised algorithms rely on approaches that occasionally fail but usually perform much better.

The runtime of such algorithms as well as the solution quality (in the case of optimisation problems) and sometimes also the correctness (in the case of computational problems) are then subject to chance. A formal analysis therefore focusses on expected values and probabilities of success. We will look at classical examples as well as current research topics from the field of hashing and graph theory. Specific design methods (such as probability amplification) and advanced analysis tools of probability theory (such as coupling, Poissonisation and concentration bounds) will be applied. It will often turn out that randomised approaches are more efficient or simpler than all (or at least all known) deterministic approaches.

We will also briefly consider on the theoretical side how randomised complexity classes relate to known classes such as P and NP, and on the practical side we will clarify how randomised algorithms can be implemented on common (essentially deterministic) computers with pseudorandomness.

Arbeitsaufwand

Lecture with exercise with 3 SWS, 5 LP
approx. 45h attendance of the lecture and exercise
approx. 30h preparation and follow-up work
approx. 45 hours working on the exercise sheets
approx. 30h exam preparation

Empfehlungen

Basic knowledge of algorithms and data structures (e.g. from the lectures Algorithms 1 + 2) as well as basic knowledge of probability theory (e.g. from the lecture Introduction to Stochastics) are helpful.

M**4.114 Modul: Public International Law with an Economic Law Focus [M-INFO-107029]**

Verantwortung: Dr. Sebastian Kasper-Schöniger
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Minor Studies: Law](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113381	Public International Law	3 LP	Zufall

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Competency Goals:

- Participating students will be able to navigate the plethora of multilateral treaties to detect relevant international law for specific cases.
- They can develop solutions for legal problems based on case law of international courts and tribunals.
- Students will be able to read and comprehend international treaties and case law.
- They will have a fundamental understand of the interplay between various subfields of public international law.
- Students can identify and explain current issues in public international law.

Inhalt

The lecture is designed to provide participating students with a general understanding of the foundations, subjects, and sources of public international law, its interplay with national legal regimes, and more detailed knowledge of particular subfields of public international law.

Since the lecture targets students of information systems, particular focus will be given to economic topics in international law, such as investment and trade law aspects. Due to the general importance of climate change for today's (economic) law, international climate change law and environmental law will form further focus areas.

In addition, a concise overview on human rights law, the law on State responsibility, and the peaceful settlement of disputes will be provided.

Throughout the lecture, important case law will be referenced and students are expected to read relevant cases in part to facilitate a discussion of such cases and their relevance for a subject field. Although the United Nations, including its principal judicial organ, the International Court of Justice, is one of the, if not the, key international organization in public international law, further international organizations (eg, Council of Europe, World Trade Organization) and their respective law(s) will also be touched.

Students are advised to have a statute book at hand that includes the most important international treaties and conventions (eg, Evans, Blackstone's International Law Documents, currently 15th ed 2021).

Conducting the lecture in English intends to facilitate students to link their ideas and arguments to current debates in international law.

Arbeitsaufwand

90h

Empfehlungen

- General knowledge of (public) law (eg, through participating in public law or EU law modules) is helpful but not necessary.
- Interest in international affairs and politics is welcomed.

M**4.115 Modul: Reinforcement Learning [M-INFO-105623]**

Verantwortung: TT-Prof. Dr. Rudolf Lioutikov
Prof. Dr. Gerhard Neumann

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	2

Pflichtbestandteile			
T-INFO-111255	Reinforcement Learning	6 LP	Lioutikov, Neumann

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

- Students are able to understand the RL problem and challenges.
- Students can differentiate between different RL algorithm and understand their underlying theory
- Students will know the mathematical tools necessary to understand RL algorithms
- Students can implement RL algorithms for various tasks
- Students understand current research questions in RL

Inhalt

Reinforcement Learning (RL) is a sub-field of machine learning in which an artificial agent has to interact with its environment and learn how to improve its behaviour by trial and error. For doing so, the agent is provided with an evaluative feedback signal, called reward, that he perceives for each action performed in its environment. RL is one of the hardest machine learning problems, as, in contrast to standard supervised learning, we do not know the targets (i.e. the optimal actions) for our inputs (i.e. the state of the environment) and we also need to consider the long-term effects of the agent's actions on the state of the environment. Due to recent successes, RL has gained a lot of popularity with applications in robotics, automation, health care, trading and finance, natural language processing, autonomous driving and computer games. This lecture will introduce the concepts and theory of RL and review current state of the art methods with a particular focus on RL applications in robotics. An exemplary list of topics is given below:

- Primer in Machine Learning and Deep Learning
- Supervised Learning of Behaviour
- Introduction in Reinforcement Learning
- Dynamic Programming
- Value Based Methods
- Policy Optimization and Trust Regions
- Episodic Reinforcement Learning and Skill Learning
- Bayesian Optimization
- Variational Inference, Max-Entropy RL and Versatility
- Model-based Reinforcement Learning
- Offline Reinforcement Learning
- Inverse Reinforcement Learning
- Hierarchical Reinforcement Learning
- Exploration and Artificial Curiosity
- Meta Reinforcement Learning

Arbeitsaufwand

Approximately 180 hours, divided into:

- 45 hours of lecture attendance
- 15 hours of exercise attendance
- 90 hours of post-processing and working on exercise sheets
- 30 hours of exam preparation.

Empfehlungen

- It is recommended that students are familiar with the content of the "Foundations of Artificial Intelligence" lecture.
- Good Python knowledge is recommended.
- Good mathematical background knowledge is recommended.

M**4.116 Modul: Reliable Computing I [M-INFO-100850]****Verantwortung:** Prof. Dr. Mehdi Baradaran Tahoori**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Unregelmäßig**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1**Pflichtbestandteile**

T-INFO-101387	Reliable Computing I	3 LP	Tahoori
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Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The objective of this course is to become familiar with general and state of the art techniques used in design and analysis of fault-tolerant digital systems.

Inhalt

The objective of this course is to become familiar with general and state of the art techniques used in design and analysis of fault-tolerant digital systems. The students will study and investigate existing fault-tolerant systems. Both Hardware and software methods will be studied and new research topics will be investigated.

This course overviews reliable (fault-tolerant) computing and the design and evaluation of dependable systems, and provides a base for research in reliable systems. Models and methods are used in the analysis and design of fault-tolerant and highly reliable computer systems will be taught in this course. Topics include faults and their manifestations, fault/error modeling, reliability, availability and maintainability analysis, system evaluation, performance-reliability trade-offs, system level fault diagnosis, hardware and software redundancy techniques, and fault-tolerant system design methods.

Arbeitsaufwand

2 SWS: (2 SWS + 1,5 x 2 SWS) x 15 + 15 h preparation for the exam = 90 h = 3 ECTS 

M**4.117 Modul: Research Focus Class: Blockchain & Cryptocurrencies [M-INFO-106654]**

Verantwortung: Prof. Dr. Hannes Hartenstein
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte
6 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-113400	Research Focus Class: Blockchain & Cryptocurrencies	3 LP	Hartenstein
T-INFO-113401	Research Focus Class: Blockchain & Cryptocurrencies - Seminar	3 LP	Hartenstein

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

- Students are familiar with current issues in the field of blockchain and cryptocurrencies and can identify specific research questions.
- Students have the necessary basic knowledge to identify, discuss and scientifically address current issues in the subject area.
- Students are able to independently develop a research topic and find and process related literature.
- Students are familiar with research methods in the field of decentralized systems and have gained initial experience in a specific research topic.
- Students can write a paper according to scientific standards.
- Students can present and discuss a research topic in a colloquium.

Inhalt

Blockchains such as Ethereum are decentralized systems that are currently receiving a lot of attention both in practice and in research. These systems can not only be used to carry out payment transactions in a decentralized manner, but also to programmatically record and enforce processes between mutually distrustful parties in so-called smart contracts. In particular, security and fairness properties as well as scalability in terms of transaction throughput play a key role.

This course begins with a lecture in which the basics of blockchains and Ethereum in particular are taught and current problems are introduced. After an introduction to the structure and functionality of Ethereum, advanced aspects that are necessary to address current research questions will be covered. The basics of scientific methodology in dealing with decentralized systems are also covered. The basic knowledge imparted in the lecture will be applied and consolidated in the seminar - the second part of the course - through the students' own research work.

The seminar offers the opportunity to work on a self-chosen topic in the field of blockchains and cryptocurrencies, which is facilitated by the previous lecture and direct consultation. The students' task is to find and process literature on the chosen topic and to work on the chosen topic. The results are documented in a paper according to scientific standards and presented in a colloquium.

Anmerkungen

Places are limited. Information about the registration process is given in the first lecture. Registration is usually carried out via CampusPlus or Wiwi-Portal. A listing in one of them indicates that the module is offered in the current term.

Arbeitsaufwand

6 ECTS = 180 hours

- Lecture attendance and discussion (20 hours)
- Lecture preparation and follow-up (20 hours)
- Literature research (20 hours)
- Implementation of self-chosen project (60 hours)
- Writing a scientific report (60 hours)

Empfehlungen

Knowledge from 'Decentralized Systems: Fundamentals, Modeling, and Applications' [M-INFO-105334] and skills from 'Scientific Methods to Design and Analyze Secure Decentralized Systems' [M-INFO-105780] are of advantage.

M**4.118 Modul: Research Practical Course on Behavior Foundation Models [M-INFO-107728]**

Verantwortung: TT-Prof. Dr. Rudolf Lioutikov
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-115118	Research Practical Course on Behavior Foundation Models	6 LP	Lioutikov

Erfolgskontrolle(n)

See Partial achievements (Teilleistung).

Voraussetzungen

See Partial achievements (Teilleistung).

Qualifikationsziele

Students learn to understand and scrutinise complex scientific topics and to reproduce and check published results. and to reproduce and verify published results. Students gain in-depth knowledge in the field of interactive learning and experience with the use of novel learning methods.

Inhalt

Each student will select a topic in the field of Behavior Foundation Models. The organizers will suggest topics but the students are welcome suggest relevant topics. The students will then implement and evaluate several algorithms corresponding to the chosen topic. The experimental evaluation will be documented in a report and presented to their peers.

It is highly recommended to take this research project in combination with the Seminar on Behavior Foundation Models, where the students get the chance to acquire the required background on the literature.

Arbeitsaufwand

Workload = 180h = 6 ECTS

- Attendance time: 15h
- Project work: 135h
- Writing scientific report + preparing presentation: 30h

Empfehlungen

We highly recommend to take this research project in combination with the Seminar on Behavior Foundation Models.

It is highly recommended to attend the "Explainable Artificial Intelligence" lecture in parallel or prior to this project.

- Experience in Machine Learning is recommended, e.g. through prior coursework.
 - The Computer Science Department offers several great lectures e.g., "Maschinelles Lernen - Grundlagen und Algorithmen" and "Deep Learning"
- A good mathematical background will be beneficial
- Python experience is recommended
- We might use the PyTorch deep learning library In the exercises. Some prior knowledge in this is helpful but not necessary.

M**4.119 Modul: Research Practical Course: Artificial Intelligence & Security [M-INFO-106866]**

Verantwortung: Prof. Dr. Christian Wressnegger
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113759	Research Practical Course: Artificial Intelligence & Security	6 LP	Wressnegger

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Qualifikationsziele: Students understand how to interpret results from state-of-the-art research and are able to actively contribute to timely research.

Lernziele:

- ← Students know and understand concepts of recent research at the intersection of artificial intelligence and computer security.
- ← Students are able independently research topics and methods in this field of research.
- ← Students understand limits of current approach in computer security research.

Inhalt

In this practical course, the students work on a project at the intersection of artificial intelligence, machine learning, and computer security. They come in contact with and participate in timely and state-of-the-art research in this exciting field. In this scope, the students read up on a sub-field, design and implement a learning-based system, and conduct evaluations on real-world data.

Topics include but are not limited to adversarial machine learning, explainability of machine learning in computer security, learning-based attack detection, and vulnerability discovery.

Arbeitsaufwand

- 140h Projektarbeit
- 20h Abschlussbericht
- 15h Vorbereitung Abschlusspräsentation
- 5h Präsenzzeit (Abschlussveranstaltung)

M**4.120 Modul: Research Practical Course: Explainable Artificial Intelligence [M-INFO-106495]**

Verantwortung: TT-Prof. Dr. Rudolf Lioutikov
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113114	Research Practical Course: Explainable Artificial Intelligence	6 LP	Lioutikov

Erfolgskontrolle(n)

See partial achievements (Teilleistung).

Voraussetzungen

See partial achievements (Teilleistung).

Qualifikationsziele

Students learn to understand and question complex scientific topics and to reproduce and verify published results. to reproduce and review published results. Students gain in-depth knowledge of interactive learning and experience with the use of novel learning methods.

Inhalt

Each student will select a topic in the field of Explainable Artificial Intelligence and study and analyze it in the context of Robot Learning. The organizers will suggest topics but the students are welcome suggest relevant topics. The students will then implement and evaluate several algorithms corresponding to the chosen topic. The experimental evaluation will be documented in a report and presented to their peers.

It is highly recommended to take this research project in combination with the "Explainable Artificial Intelligence" Seminar, where the students get the chance to acquire the required background on the literature.

Anmerkungen

It is only possible to resign within two weeks after assignment of the topic.

Arbeitsaufwand

Workload = 180h = 6 ECTS

- Attendance time: 15h
- Project work: 135h
- Write scientific report + prepare presentation: 30h

Empfehlungen

We recommend taking this Praktikum after attending the "Explainable Artificial Intelligence" lecture in the summer semester.

We highly recommend to take this research project in combination with the "Explainable Artificial Intelligence" seminar.

- Experience in Machine Learning is recommended, e.g. through prior coursework.
 - The Computer Science Department offers several great lectures e.g., "Maschinelles Lernen - Grundlagen und Algorithmen" and "Deep Learning"
- A good mathematical background will be beneficial
- Python experience is recommended
- We might use the PyTorch deep learning library In the exercises. Some prior knowledge in this is helpful but not necessary.

M**4.121 Modul: Research Project Deep Learning for Robotics [M-INFO-107174]**

Verantwortung: Prof. Dr. Gerhard Neumann
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114203	Research Project Deep Learning for Robotics	6 LP	Neumann

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students learn to understand and scrutinise complex scientific topics and to reproduce and verify published results. Students gain in-depth knowledge in the field of learning with robots and experience with the use of novel learning methods.

Inhalt

Each student has to choose one of the offered topics from the area of deep learning / robot learning / deep reinforcement learning / deep imitation learning. The students need to implement one or several algorithms and evaluate them against available baselines on standard benchmark tasks as well as on (custom-made) physically realistic simulations and/or a real robot platform. The experiments have to be documented in a report. Students will work in teams of 2. It is recommended to take this course together with the seminar "Deep learning for robotics" where the students will acquire the required background on the literature.

Arbeitsaufwand

Workload: 180h

Attendance time: 15h

Project work: 135h

Writing a report + preparing a presentation: 30h

Empfehlungen

- Experience in Machine Learning is recommended.
- Python experience is recommended
- We will use the PyTorch deep learning library. Some prior knowledge in this is helpful but not necessary.

M**4.122 Modul: Research Project: Generative AI for Autonomous Agents [M-INFO-107163]**

Verantwortung: Prof. Dr. Gerhard Neumann
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114189	Research Project: Generative AI for Autonomous Agents	6 LP	Neumann

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students will learn to understand, question, and reproduce complex scientific topics and published results. They will gain in-depth knowledge in the field of learning for decision making and experience with the application of novel learning methods.

Inhalt

This practical research course explores advanced machine learning methods and generative AI such as diffusion models to empower autonomous agents with intelligent decision-making capabilities. Students will delve into:

- Generative Models for Decision Making
- Reinforcement Learning (RL)
- Imitation Learning
- Multi-Agent Systems
- Uncertainty Quantification
- Learning Prediction Models of Physical Processes
- Time-Series Modeling
- Discovery and Inference of Latent Variables

Each student will choose one of the offered topics, implement one or several algorithms, and evaluate them against available baselines using standard benchmark tasks. The course emphasizes hands-on experimentation, requiring students to document their findings in a detailed report. Students will work in teams of two, closely collaborating with their supervisor with the aim of achieving publishable results. This course provides students with their first experience in running a research project in machine learning, including algorithm design, evaluation, benchmarking, deploying algorithms on HPC hardware, and paper writing.

Arbeitsaufwand

Workload: 180h

Attendance time: 15h

Project work: 135h

Writing a report + preparing a presentation: 30h

Empfehlungen

- Experience in Machine Learning is recommended.
- Python experience is recommended
- We will use the PyTorch deep learning library. Some prior knowledge in this is helpful but not necessary.

M**4.123 Modul: Resilient Networking [M-INFO-105591]**

Verantwortung: Prof. Dr. Thorsten Strufe
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-111209	Resilient Networking	6 LP	Strufe

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

This course will provide students with a basic understanding of threats to the Internet, and the most common technologies to tackle them. The course will analyze the adversary models and evaluation metrics underlying their design.

- The students have a critical reasoning about network resilience,
- have knowledge in the evaluation of threats to network operation,
- understand the design aspects of protection measures,
- are familiar with the latest research in the field
- are able to analyze and discuss the space of solutions to a given challenge to Internet security and robustness

Inhalt

The lecture resilient networking provides an overview on the basics of secure networks as well as on current threats and respective countermeasures. Especially bandwidth-depleting Denial of Service attacks represent a serious threat. Moreover, over the last years the number of targeted and highly sophisticated attacks on company and governmental networks increased. To make it worse, as a new trend at the moment, the interconnection of the Internet with cyber physical systems takes place. Such systems, e.g., the energy network (smart grid), transportation systems and large industrial facilities, are critical infrastructures with severe results in case of their failure. Thus, the Internet that interconnects these systems has evolved to a critical infrastructure as well.

The lecture introduces the current state-of-the-art in the research towards resilient networks. Resilience-enhancing techniques can be generally classified in proactive and reactive methods. Proactive techniques are redundancy and compartmentalization. Redundancy allows to tolerate attacks to a certain extent, while compartmentalization attempts to restrict the attack locally and preventing its expansion across the whole system. Reactive techniques follow a three step approach by comprising the phases of detecting an attack, mitigate its impacts, and finally restore a system's usual operation.

Based upon this categorisation of resilience strategies the lecture will give an excursus to graph theory and will introduce generic strategies to increase the resilience of networks, e.g., proactively establishing backup routes and fast restoration strategies. Furthermore, the lecture will provide an overview on BGP routing and the Domain Name Service, as two essential Internet services. Both services are presented and current attacks as well as corresponding countermeasures are described. Moreover, Denial of Service attacks and their mitigation are observed in detail as well as mechanism for increasing the resilience of P2P networks. Finally, Intrusion Detection systems are covered as mechanisms to mitigate the impacts of successful attacks.

Arbeitsaufwand

1. Attendance time in lectures: 45 h
2. Preparation and follow-up of the same: 90 h
3. Exam preparation and attendance in the same: 45 h

Empfehlungen

Knowledge of the basics of cryptography and computer networks is helpful.

M**4.124 Modul: Robotic Intelligence: Navigation and Mobile Manipulation [M-MACH-107621]**

Verantwortung: Prof. Dr.-Ing. Arne Rönnau
Einrichtung: KIT-Fakultät für Maschinenbau
Bestandteil von: [Minor Studies: Biologically Inspired Robotics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-MACH-114952	Robotic Intelligence: Navigation and Mobile Manipulation	6 LP	Rönnau

Erfolgskontrolle(n)

Siehe Teilleistung

Voraussetzungen

Siehe Teilleistung

Qualifikationsziele

Die Studierenden erwerben ein fundiertes Verständnis der grundlegenden Prinzipien sowie aktueller Methoden der Robotik in den Bereichen Navigation, mehrbeinige Lokomotion und Manipulation mit mobilen Systemen. Sie sind in der Lage, geeignete Verfahren des maschinellen Lernens, der simultanen Lokalisierung und Kartierung (SLAM), der Entscheidungsfindung sowie der kollisionsfreien Pfadplanung anzuwenden, um komplexe reale Umgebungen autonom, beispielsweise mit einem Laufroboter, zu durchqueren. Ein besonderer Schwerpunkt liegt auch auf der direkten physischen Interaktion von Laufrobotern mit dem Untergrund und Hindernissen. Die Studierenden lernen, geeignete Machine Learning Ansätze zur Lokomotion zu entwerfen und in dynamischen Simulationsumgebungen zu trainieren, um Bewegungen auf einfachem, ebenem Gelände ebenso wie das Überwinden komplexer Hindernisse, etwa Treppen, zu erlernen.

Darüber hinaus erwerben die Studierenden Kenntnisse über verschiedene Verfahren der Manipulation, dem robotischen Greifen und Handhaben von Objekten. Sie sind in der Lage, sowohl klassische Manipulationsansätze als auch moderne, auf maschinellem Lernen basierende Ansätze zu analysieren, zu vergleichen und hinsichtlich ihrer Leistungsfähigkeit zu bewerten. In Kombination können mobile Roboter so einen Weg durch schwieriges Gelände finden und am Ziel autonom Objekte manipulieren.

Inhalt

Die Vorlesung *Robotic Intelligence: Navigation and Mobile Manipulation* vermittelt grundlegende und fortgeschrittene Konzepte zur Entwicklung intelligenter, autonomer Roboter für den Einsatz in komplexen Umgebungen. Im Fokus stehen Methoden zur Navigation, mehrbeinigen Lokomotion und mobilen Manipulation sowie deren Integration in reale Robotersysteme. Behandelt werden Verfahren der Sensorfusion, Lokalisierung und Kartierung (SLAM), Entscheidungsfindung sowie der kollisionsfreien Bewegungs- und Pfadplanung. Weiter werden maschinelle Lernverfahren zur Steuerung von Laufrobotern in schwierigem Gelände sowie zur direkten physischen Interaktion mit der Umwelt vorgestellt. Ergänzend werden Konzepte der mobilen Manipulation behandelt, die es Robotern ermöglichen, Objekte sicher zu greifen, zu handhaben oder mit ihrer Umgebung zu interagieren. Dabei werden sowohl klassische als auch moderne, lernbasierte Ansätze betrachtet. Durch die Kombination dieser Ansätze können Robotersysteme ein hohes Maß an Autonomie erreichen. Die Vorlesung verbindet theoretische Grundlagen mit anwendungsorientierten Methoden und befähigt die Studierenden, autonome robotische Systeme und deren Methoden für reale Einsatzszenarien zu entwickeln und kritisch zu bewerten.

Ein wichtiger Teil des Kurses ist die praktische Gruppenarbeit im *Machine Intelligence and Robotics Lab (MaiRo)*, in dem die Studierenden Aufgaben mit echten Sensoren und Robotern lösen und dabei sowohl technische Kompetenzen als auch Team-Fähigkeiten für die Lösung komplexer Navigations- und Manipulationsaufgaben mobiler Systeme entwickeln.

Die Themen umfassen:

- Grundlagen des maschinellen Lernens und Sensorfusion
- Simultane Lokalisierung und Kartierung (SLAM) in unbekannten Umgebungen
- Ansätze zur Bewegungsplanung und Kollisionsvermeidung
- Intelligente Entscheidungsfindung und Aufgabenplanung
- Reinforcement Learning für die Lokomotion von Laufrobotern
- Machine Learning Ansätze zum Passieren oder Überwinden komplexer Hindernisse
- Manipulation von Objekten in statischen Szenarien
- Mobile Manipulation in komplexen Umgebungen

Am Ende des Kurses sind Studierende in der Lage, Methoden zur Navigation und mobilen Manipulation zu entwerfen, zu implementieren und zu integrieren. Diese Ansätze ermöglichen es mobilen Systemen, durch komplexe, dynamische Umgebungen autonom zu navigieren, hierfür gelernte Lokomotionskonzepte einzusetzen sowie verschiedene Manipulationsaufgaben zu lösen.

Zusammensetzung der Modulnote

Schriftliche Klausur: Dauer 60 Minuten. Durch erfolgreiche Bearbeitung der Übungen kann ein Notenbonus erreicht werden.

Anmerkungen

Siehe weitere Details zu der Organisation der Veranstaltung: lehre.imi.kit.edu sowie in ILIAS.

Arbeitsaufwand

180 Arbeitsstunden, davon ca.:

- 30h für Präsenzzeit in Vorlesungen
- 110h für Übungen und Gruppenarbeiten
- 40h für Prüfungsvorbereitung und Teilnahme an der schriftlichen Prüfung

Lehr- und Lernformen

Vorlesung, Übung und Gruppenarbeit

M**4.125 Modul: Robotic Intelligence: Perception and Scene Understanding [M-MACH-107558]**

Verantwortung: Prof. Dr.-Ing. Arne Rönnau
Einrichtung: KIT-Fakultät für Maschinenbau
Bestandteil von: [Minor Studies: Biologically Inspired Robotics](#)

Leistungspunkte 6 LP	Notenskala Zehntelnoten	Turnus Jedes Wintersemester	Dauer 1 Semester	Sprache Englisch	Level 4	Version 1
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Pflichtbestandteile			
T-MACH-114813	Robotic Intelligence: Perception and Scene Understanding	6 LP	Rönnau

Erfolgskontrolle(n)

Siehe Teilleistung

Voraussetzungen

Siehe Teilleistung

Qualifikationsziele

Die Studierenden verstehen die grundlegenden Prinzipien und aktuelle Methoden aus dem Bereich Robotik für die Wahrnehmung und das Verstehen von Szenen. Sie erlangen die Fähigkeit, geeignete Ansätze des maschinellen Lernens und der Bildverarbeitung auszuwählen und anzuwenden, um komplexe reale Umgebungen zu interpretieren, von sensorischen Rohdaten bis hin zu strukturierten Szenendarstellungen. Die Studierenden sind in der Lage, Systeme zu entwerfen, die mehrere Wahrnehmungsmodalitäten kombinieren und eine robuste Objekterkennung, Verfolgung, Segmentierung und 3D-Szenenanalyse ermöglichen. Darüber hinaus können sie diese Methoden kritisch bewerten und für Robotikanwendungen anpassen, die ein Situationsbewusstsein und/oder fundierte Entscheidungen erfordern.

Inhalt

Die Vorlesung *Robotic Intelligence: Perception and Scene Understanding* kombiniert moderne KI-basierte Methoden zur maschinellen Wahrnehmung mit den technischen Fähigkeiten, die für deren reale Anwendung in der Robotik erforderlich sind. Sie deckt die gesamte Wahrnehmungskette ab, von der Erfassung und Kalibrierung von Sensordaten bis hin zur Interpretation von Szenen und der Integration in Robotersysteme, wie z.B. mobile Laufroboter.

Die Studierenden werden zunächst die Grundlagen in maschinellern Lernen und Bildverarbeitung lernen, die z.B. die Bildklassifikation, Objekterkennung, Tracking und Segmentierung von Objekten in 2D und 3D abdeckt. Anschließend werden fortgeschrittene Deep-Learning-Architekturen betrachtet, darunter Vision Transformers, multimodale Wahrnehmungssysteme und End-To-End multimodale ML-Ansätze. Ein besonderer Schwerpunkt liegt auf Techniken zur Integration von Bildverarbeitung, Tiefenwahrnehmung und anderen Wahrnehmungsmodalitäten, die eine robuste Wahrnehmung in anspruchsvollen realen Umgebungen ermöglichen.

Ein wichtiger Teil des Kurses ist die praktische Gruppenarbeit im *Machine Intelligence and Robotics Lab (MaiRo)*, in dem die Studierenden praktische Aufgaben mit echten Sensoren und Robotern lösen und dabei sowohl technische Kompetenzen als auch Team-Fähigkeiten zum Lösen komplexen Wahrnehmungsaufgaben für mobile Systeme entwickeln.

Die Themen umfassen:

- Grundlagen des maschinellen Lernens und der Bildverarbeitung für die Robotik
- Vision Transformer und aktuelle, komplexe Deep-Learning-Architekturen
- Objekterkennung, -verfolgung und -segmentierung (2D und 3D)
- Instanz-, semantische und panoptische Segmentierung
- Multimodale Erfassung, Fusion und End-to-End multimodale ML-Ansätze
- World Models und ihre Bedeutung und Anwendung in der Robotik
- Sensor-Modellierung, Kalibrierung und Integration in das Gesamtsystem

Am Ende des Kurses sind Studierende in der Lage, intelligente robotische Systeme zur Wahrnehmung zu entwerfen, zu implementieren und zu integrieren, die es mobilen Systemen ermöglichen, komplexe, dynamische Umgebungen zu verstehen und darin möglichst autonom zu agieren.

Zusammensetzung der Modulnote

Schriftliche Klausur: Dauer 60 Minuten. Durch erfolgreiche Bearbeitung der Übungen kann ein Notenbonus erreicht werden.

Arbeitsaufwand

180 Arbeitsstunden, davon ca.:

- 30h für Präsenzzeit in Vorlesungen
- 110h für Übungen und Gruppenarbeiten
- 40h für Prüfungsvorbereitung und Teilnahme an der schriftlichen Prüfung

Lehr- und Lernformen

Vorlesung, Übung und Gruppenarbeit

M**4.126 Modul: Robotics - Practical Course [M-INFO-107155]****Verantwortung:** Prof. Dr.-Ing. Tamim Asfour**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114172	Practical Course: Robotics	6 LP	Asfour

Erfolgskontrolle(n)

See partial Achievements (Teilleistung)

Voraussetzungen

See partial Achievements (Teilleistung)

Qualifikationsziele

The student knows concrete solutions for different problems in robotics. He/she uses methods of inverse kinematics, grasp and motion planning, and visual perception. The student can implement solutions in the programming languages C++ and Python with the help of suitable software frameworks.

Inhalt

The practical course is offered as an accompanying course to the lectures Robotics I-III. Every week, a small team of students will work on solving a given robotics problem. The list of topics includes robot modeling and simulation, inverse kinematics, robot programming via state charts, collision-free motion planning, grasp planning, robot vision and robot learning.

Arbeitsaufwand

Practical course with 4 SWS, 6 LP

6 LP corresponds to 180 hours, including

2 hours introductory event

18 hours initial familiarization with the software framework

120 hours group work

40 hours attendance time

Empfehlungen

Attending the lectures Robotics I – Introduction to Robotics, Robotics II: Humanoid Robotics, Robotics III - Sensors and Perception in Robotics and Mechano-Informatics and Robotics is recommended.

M**4.127 Modul: Robotics I - Introduction to Robotics [M-INFO-107162]**

Verantwortung: Prof. Dr.-Ing. Tamim Asfour
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114190	Robotics I - Introduction to Robotics	6 LP	Asfour

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The students are able to apply the presented concepts to simple and realistic tasks from robotics. This includes mastering and deriving the mathematical concepts relevant for robot modeling. Furthermore, the students master the kinematic and dynamic modeling of robot systems, as well as the modeling and design of simple controllers. The students know the algorithmic basics of motion and grasp planning and can apply these algorithms to problems in robotics. They know algorithms from the field of image processing and are able to apply them to problems in robotics. They are able to model and solve tasks as a symbolic planning problem. The students have knowledge about intuitive programming procedures for robots and know procedures for programming and learning by demonstration.

Inhalt

The lecture provides an overview of the fundamentals of robotics using the examples of industrial robots, service robots and autonomous humanoid robots. An insight into all relevant topics is given. This includes methods and algorithms for robot modeling, control and motion planning, image processing and robot programming. First, mathematical basics and methods for kinematic and dynamic robot modeling, trajectory planning and control as well as algorithms for collision-free motion planning and grasp planning are covered. Subsequently, basics of image processing, intuitive robot programming especially by human demonstration and symbolic planning are presented.

In the exercise, the theoretical contents of the lecture are further illustrated with examples. Students deepen their knowledge of the methods and algorithms by independently working on problems and discussing them in the exercise. In particular, students can gain practical programming experience with tools and software libraries commonly used in robotics.

Arbeitsaufwand

Lecture with 3 SWS + 1 SWS Tutorial, 6 LP
 6 LP corresponds to 180 hours, including
 $15 \cdot 3 = 45$ hours attendance time (lecture)
 $15 \cdot 1 = 15$ hours attendance time (tutorial)
 $15 \cdot 6 = 90$ hours self-study and exercise sheets
 30 hours preparation for the exam

M**4.128 Modul: Robotics II - Humanoid Robotics [M-INFO-107123]****Verantwortung:** Prof. Dr.-Ing. Tamim Asfour**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114152	Robotics II - Humanoid Robotics	3 LP	Asfour

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students can explain the challenges and goals of humanoid robotics research, particularly regarding the implementation of complex sensorimotor and cognitive abilities in humanoid robots. They are familiar with the history of the field and understand how biomechanical models of the human body inform humanoid robot design.

Students have comprehensive knowledge of human and robotic grasping and manipulation. They can analyze human grasping and manipulation strategies, understand taxonomies, and evaluate different computational approaches for grasping and manipulation. They understand the challenges in transferring concepts from human studies to humanoid robots.

Students understand fundamentals of learning from human demonstration and imitation learning. They can explain the learning from demonstration cycle and methods for learning generalized task representation, especially task constraints from demonstrations, and how learned behaviors are reproduced on robots.

Students can describe cognitive and AI-based architectures for humanoid robots, including approaches to address the signal-to-symbol gap and human-inspired memory architectures that enable intelligent robot behavior.

Inhalt

This lecture addresses the implementation of complex sensorimotor and cognitive abilities in humanoid robots, with humans serving as inspiration. The lecture begins by addressing the motivation for humanoid robotics. The history of humanoid robotics and biomechanical models of the human body that inform robot design are covered, along with some mechatronic principles underlying humanoid robot systems.

The first main topic comprehensively addresses grasping and manipulation. After covering fundamentals, the lecture discusses neuroscientific insights and concepts like grasp phases and grasping synergies to reduce control complexity. Taxonomies are presented as structured frameworks for understanding the space of possible constraints in grasping and manipulation. The lecture addresses computational approaches for grasping known, familiar, and unknown objects, covering both classical methods and modern learning-based approaches using deep learning and vision-language-action models.

The second main topic covers learning from demonstration and imitation learning. After introducing the topic, fundamentals are presented including the learning from demonstration cycle. The lecture emphasizes learning task models on both symbolic/semantic and subsymbolic/sensorimotor levels. While methods for capturing human demonstrations are briefly covered, the focus is on semantic segmentation of human demonstrations and learning of task constraints that capture the task and allow generalization. Movement primitives are presented as efficient representations that allow robots to generalize learned behaviors to new situations.

The lecture concludes with cognitive and AI-based architectures for humanoid robots, discussing state-of-the-art approaches, methods to address the signal-to-symbol gap, and human-inspired memory architectures that enable intelligent robot behavior.

Arbeitsaufwand

Lecture with 2 SWS, 3 ECTS.

3 ECTS corresponds to approx. 90 hours, thereof:

approx. $15 * 2h = 30h$ attendance time

approx. $15 * 2h = 30h$ self-study prior/after the lecture

approx. 30h preparation for the exam and exam itself

Empfehlungen

Having visited the lecture on Robotics I – Introduction to Robotics is recommended

M**4.129 Modul: Robotics III - Sensors and Perception in Robotics [M-INFO-107130]**

Verantwortung: Prof. Dr.-Ing. Tamim Asfour
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114155	Robotics III - Sensors and Perception in Robotics	3 LP	Asfour

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students can explain the main sensor principles used in robotics and distinguish between proprioceptive sensors (encoders, force/torque sensors, IMUs) and exteroceptive sensors (proximity, range, visual, and tactile sensors). They understand and can characterize sensor properties such as resolution, range, accuracy, and bandwidth, and can explain the principles of analog-to-digital conversion. Students are able to propose, analyze, and justify suitable sensor concepts for specific robotic tasks, taking into account trade-offs between different sensor modalities.

Students can apply fundamental perception methods for robotic applications. This includes robot vision techniques such as feature detection and matching, object detection and recognition, and semantic segmentation, including deep learning-based approaches. They can describe and analyze perception methods for manipulation tasks, including object pose estimation, grasp detection, active vision strategies, and haptic exploration. In addition, students can explain point cloud processing methods for registration and surface reconstruction, as well as principles of simultaneous localization and mapping (SLAM).

Inhalt

This lecture complements the lecture "Robotics I" and provides a comprehensive overview of sensors and perception methods in robotics. It is divided into two main parts.

The first part introduces fundamental concepts of perception in robotics, including the distinction between sensation and perception within the perception-cognition-action loop. It covers sensor fundamentals, such as sensor characteristics (resolution, range, accuracy, bandwidth), analog-to-digital conversion, and common sensor classification schemes. Proprioceptive sensors for measuring the robot's internal state are discussed, including encoders (optical, magnetic, absolute, and incremental), force/torque sensors, and inertial measurement units (IMUs). The lecture then covers exteroceptive sensors, including proximity sensors, range sensors (LiDAR, time-of-flight cameras, ultrasonic sensors), visual sensors (monocular, stereo, RGB-D cameras), and tactile sensing technologies (capacitive, resistive, and optical).

The second part focuses on processing and interpreting sensor data. Topics in robot vision include image representation, feature detection and matching, object detection and recognition, semantic segmentation, and corresponding deep learning approaches. Point cloud processing is covered with emphasis on data structures, registration, surface reconstruction, and semantic segmentation. The lecture also addresses perception for manipulation, covering object pose estimation, grasp detection, visual servoing, active vision, and haptic exploration. It concludes with the simultaneous localization and mapping (SLAM) problem, including EKF SLAM, Graph SLAM, and FastSLAM.

Arbeitsaufwand

Lecture with 2 SWS, 3 ECTS.
 3 ECTS corresponds to approx. 90 hours, thereof:
 approx. 15 * 2h = 30h attendance time
 approx. 15 * 2h = 30h self-study prior/after the lecture
 approx. 30h preparation for the exam and exam itself

Empfehlungen

Having visited the lectures on Robotics I – Introduction to Robotics is recommended.

M**4.130 Modul: Robotics Institute Germany - Lecture Series [M-INFO-107598]**

Verantwortung: Prof. Dr.-Ing. Tamim Asfour
 TT-Prof. Dr. Barbara Bruno
 Prof. Dr. Jan Niehues

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114895	Robotics Institute Germany - Lecture Series	3 LP	Asfour, Bruno, Niehues

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students can discuss the foundations of various open research directions related to AI-powered robotics, as well as their open challenges, relevant findings/breakthroughs and state-of-the-art approaches.

Inhalt

The course is structured as an inter-institutional lecture series, concurrently offered by various German institutions affiliated with the Robotics Institute Germany (RIG). In the lecture series students are offered an overview of significant open research directions related to AI-powered robotics by leading figures in the field. The lecturers are affiliated with various RIG partner institutions and students from various RIG partner institutions attend the lectures remotely. The lectures cover topics such as cognitive robotics, robot learning, robot perception, manipulation, natural interaction and communication with robots, human-robot interaction, soft robotics and responsible robotics. The exposure to top researchers in the field from different German institutions and the possibility of exchanges with students from other German universities are unique opportunities offered by this lecture series.

Arbeitsaufwand

Lecture series with 2 SWS, 3 LP. 3 LP corresponds to approx. 90 hours, of which: approx. 22.5 hrs. of compulsory attendance approx. 45 hours of literature research approx. 22.5 hrs. exam preparation

Empfehlungen

Knowledge of the content of module Robotics I - Introduction to Robotics is helpful.

M**4.131 Modul: Sampling Methods for Machine Learning [M-INFO-107090]****Verantwortung:** Prof. Dr.-Ing. Uwe Hanebeck**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114133	Sampling Methods for Machine Learning	6 LP	Hanebeck
T-INFO-114134	Sampling Methods for Machine Learning - Pass	0 LP	Hanebeck

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students will understand and be able to implement various sampling techniques, from basic random number generation to advanced methods like normalizing flows. They will develop the ability to evaluate sampling quality, optimize procedures, and select appropriate methods for specific machine learning tasks. Graduates will be capable of independently developing sampling solutions for complex problems and critically assessing different approaches. Their comprehensive understanding will enable them to engage with current developments in the field and apply their knowledge effectively in both research and practical applications. This will be supported via a digital exercise.

Inhalt

Sample-based inference is the de-facto standard for solving otherwise infeasible problems in machine learning, estimation, and control under (unavoidable) uncertainties. Thus, it is an important foundation for further studies. This lecture gives a thorough overview of state-of-the-art sampling methods and discusses current developments from the research frontier.

The first part shows how to efficiently sample large numbers of random samples from given densities starting with the special cases of uniform and Gaussian distributions. For sampling from arbitrary densities, important techniques such as inverse transform sampling, Knothe-Rosenblatt maps, Markov chain Monte Carlo, normalizing flows, and Langevin equations are introduced.

The second part is concerned with deterministic or low-discrepancy sampling, where the goal is to find a set of representative samples of a given density. These are usually obtained by optimization, which, in contrast to random samples, leads to good coverage, high homogeneity, and reproducible results. To analyze and synthesize such samples, various statistical tests and discrepancy measures are presented. This includes scalar tests such as the Cramér-von Mises test, Kolmogorov-Smirnov test, and multivariate generalizations based on Localized Cumulative Distributions and Stein discrepancy.

Finally, advanced topics such as importance sampling and sampling from the posterior density in a Bayesian update are discussed. Typical applications of sample-based inference include Bayesian neural networks, information fusion, and reinforcement learning.

Arbeitsaufwand

Per week:

2 SWS Presence

2h Follow-up

6h Digital exercise with programming tasks

2h Exam preparation

= 12h/week und 180h/semester

Empfehlungen

Knowledge of a higher programming language with sophisticated libraries for scientific-numerical computing (e.g. Julia, Matlab, Python) is advantageous.

M**4.132 Modul: Scientific Methods to Design and Analyze Secure Decentralized Systems [M-INFO-105780]**

Verantwortung: Prof. Dr. Hannes Hartenstein
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
5 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-111568	Scientific Methods to Design and Analyze Secure Decentralized Systems	5 LP	Hartenstein

Erfolgskontrolle(n)

Siehe Teilleistung.

Voraussetzungen

Siehe Teilleistung.

Qualifikationsziele

1. Philosophy of Science: The student understands epistemological principles like the scientific and mathematical process, within the context of networked and decentralized systems. The student knows about the current limits of scientific research, especially in regards to the security of a given decentralized system.
2. Empirical Methods: Observation / Monitoring: The student is able to construct setups to monitor system properties related to performance or security. The student knows how to observe a decentralized system like an overlay network without interference, i.e., without impact on the behavior to measure as well as the overall system functionality.
3. Combined Empirical / Formal Methods: The student has a fundamental understanding of Discrete Event Simulations, as well as stochastic modelling and random number generation. The student is able to conduct a simulation study consisting of observation, modelling, simulation, validation, and result analysis.
4. Formal Methods: The student knows how to apply formal methods like formal verification / model checking and model comparison / simulation-based proofs to decentralized systems. The student understands tradeoffs between empirical and formal methods, and can choose suitable methods for given research tasks.
5. Applications in Research: The student understands how the methods of this lecture are applied to practical examples, and knows how to apply the methods on problems of a researcher's everyday life.

Inhalt

Decentralized Systems (like peer-to-peer- or blockchain-based systems) are systems controlled by multiple parties who make their own independent decisions to reach a common goal. However, not knowing which parties are trustworthy and which are betrayers requires a radically different way of thinking. Based on the lecture "Decentralized Systems: Fundamentals, Modeling, and Applications", in this lecture, we cover the necessary scientific methods to analyze existing and to create new decentralized systems. We treat both, selected empirical and formal methods and their tradeoffs, as well as the overarching philosophy of science behind the research process. Together with its practical parts, this lecture provides the foundational scientific toolbox to work on the decentralized systems of the future.

Arbeitsaufwand

1. Attendance time (Course, exercise,): 3 SWS: 15 x 3h = 45h
 2. Self-study (e.g. independent review of course material, work on homework assignments)
Weekly preparation and follow-up of the lecture/exercise: 15 x 3 SWS x 1,5h = 67,5h
 3. Preparation for the exam: 37,5h
- $\Sigma = 150h = 5$ ECTS

Empfehlungen

Prior knowledge on the abstract concepts as well as concrete use cases of decentralized systems is strongly recommended. The "Decentralized Systems: Fundamentals, Modeling, and Applications" lecture covers all necessary aspects, but equivalent lectures and / or self-study can also be sufficient.

M**4.133 Modul: Seminar Dependable Computing [M-INFO-102662]****Verantwortung:** Prof. Dr. Mehdi Baradaran Tahoori**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-105577	Seminar Dependable Computing	3 LP	Tahoori

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The objective of this seminar is to become familiar with general and state of the art techniques used in design and analysis of fault-tolerant digital systems.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

Reliability plays a major role in design of contemporary and next generation electronics. In many safety-critical application domains, reliability is considered as the main design criteria. With nanoscale technologies, the reliability of individual devices is decreasing, therefore, reliable computing must be considered in the design flow in order to ensure correctness of computing.

The objective of this seminar is to become familiar with general and state of the art techniques used in design and analysis of fault-tolerant digital systems. This seminar overviews reliable (fault-tolerant) computing and the design and evaluation of dependable systems, and provides a base for research in reliable systems.

The topics include study and investigation of existing and classical fault-tolerant systems as well as current trend in the research of reliable computing. Since reliability spans from hardware to software, and from device-level to system-level, various topics can be envisioned in the scope of this seminar and the prospective students can choose specific topic from a wide range of areas based on their interests and background.

Arbeitsaufwand**4 SWS / 3CP = 90h/week****Empfehlungen**

Knowledge of "Fault Tolerant Computing" and Computer Architecture is helpful.

M**4.134 Modul: Seminar in Privacy [M-INFO-107242]**

Verantwortung: Prof. Dr. Thorsten Strufe
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114268	Seminar in Privacy	4 LP	Strufe

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The seminar deals with current topics from the research field of technical data protection.

These include, for example:

- Attacks on private information in behavioral data
- Anonymous communication
- Publication of anonymized usage data (semantic/syntactic privacy)
- Understanding and supporting the use of online media
- Security in networks

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

The student is able to

- carry out a literature search based on a given topic, identify and evaluate the relevant literature;
- independently compile research results from IT security and technical data protection;
- analyze and discuss scientific studies and place them in their context;
- carry out their own classifications and evaluations of scientific studies, report on them in writing and present the results in a short scientific presentation.

Arbeitsaufwand

Workload attendance time in the seminar: 10h

Research and preparation of a paper: 75h

Reviewing and commenting on the preliminary papers of fellow students: 5h

Preparing the presentation: 30h

Empfehlungen

Fundamentals of IT security, computer networks and distributed systems are required

M**4.135 Modul: Seminar on Behavior Foundation Models [M-INFO-107729]**

Verantwortung: TT-Prof. Dr. Rudolf Lioutikov
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-115119	Seminar on Behavior Foundation Models	3 LP	Lioutikov

Erfolgskontrolle(n)

See Partial Achievements (Teilleistung).

Voraussetzungen

See Partial Achievements (Teilleistung)..

Qualifikationsziele

Students are able to independently understand a complex research topic, present the content in a concise and understandable way and prepare a scientific report summarizing the topic.

Students are able to independently understand a complex research topic, present the content in a concise and understandable way and prepare a scientific report summarizing the topic. Students get a deeper understanding of state-of-the-art learning algorithms and get to know current research challenges.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

Each student will select several related papers in the field of Behavior Foundation Models. The organizers will suggest several papers but the students will be encouraged to identify and research additional relevant papers during the semester. The students will then prepare a presentation and a basic scientific research paper.

It is highly recommended to take this seminar in combination with the Research Practical Course on Behavior Foundation Models (Forschungspraktikum), where the students get the chance to deepen their understanding, implement and evaluate their presented work.

Arbeitsaufwand

Workload = 90 h = 3 ECTS

- Attendance time: 15hr
- Self-study: 45h
- Writing a scientific report: 20h
- Prepare presentation: 10h

Empfehlungen

We highly recommend to take this seminar in combination with the Research Practical Course on Behavior Foundation Models (Forschungspraktikum).

It is highly recommended to attend the "Explainable Artificial Intelligence" lecture in parallel or prior to this seminar.

- Experience in Machine Learning is recommended, e.g. through prior coursework.
 - The Computer Science Department offers several great lectures e.g., "Maschinelles Lernen - Grundlagen und Algorithmen" and "Deep Learning"
- A good mathematical background will be beneficial
- Python experience is recommended
- We might use the PyTorch deep learning library in the exercises. Some prior knowledge in this is helpful but not necessary.

M**4.136 Modul: Seminar: Advanced Topics in Natural Language Processing [M-INFO-107353]**

Verantwortung: Prof. Dr. Jan Niehues
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114433	Seminar: Advanced Topics in Natural Language Processing	3 LP	Niehues

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students learn to familiarise themselves independently with topics based on scientific literature and prepare them for presentations.

From the other presentations, students gain in-depth knowledge in sub-areas of machine translation and learn to critically analyse the work presented.

Inhalt

Machine translation now makes it possible to automatically translate both written texts and spoken language into another language. In statistical approaches to machine translation, methods from machine learning are primarily used to train statistical models for the translation process.

In the seminar, current research results on various aspects of the systems will be discussed. Selected publications from the fields will be presented by the participants. Possible topics include improvement of word order and grammar of the target language, adaptation to topic or genre, treatment of spoken language phenomena, error correction, ...

Students are familiar with the DFG Code of Good Scientific Practice and successfully apply these guidelines when writing their scientific work.

Arbeitsaufwand

90h

Empfehlungen

Previous knowledge from the lecture "Machine Translation" is an advantage.

M**4.137 Modul: Seminar: Advanced Topics on SAT Solving [M-INFO-107209]****Verantwortung:** Prof. Dr. Peter Sanders**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114231	Seminar: Advanced Topics on SAT Solving	3 LP	Iser, Sanders

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

With selected high-influence papers from the field of SAT solving, we take a close look at how SAT solvers evolved in the past decade and learn about the major cornerstones of modern and efficient large scale SAT solving systems.

Arbeitsaufwand

Attendance time (3-4 dates): 4.5 - 6h

Reading, summarising and relating (2-3 papers): 30 - 40h

Preparation of the presentation: 16 - 24h

Total 90h

Empfehlungen

Knowledge of the basics from "SAT Solving in Practice" is helpful.

M**4.138 Modul: Seminar: Algorithm Engineering [M-INFO-106086]****Verantwortung:** Prof. Dr. Peter Sanders**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte
 4 LP

Notenskala
 Zehntelnoten

Turnus
 Unregelmäßig

Dauer
 1 Semester

Sprache
 Englisch

Level
 4

Version
 1

Pflichtbestandteile			
T-INFO-112312	Seminar: Algorithm Engineering	4 LP	Sanders

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students can

- carry out a literature search based on a given topic, identify, locate, evaluate and finally analyse the relevant literature.
- prepare presentations in a scientific context. To this end, students master techniques that enable them to prepare and present the content to be presented to the audience.
- prepare their written seminar paper (as required later for further academic work) in accordance with the requirements and quality standards of academic writing, taking into account the format requirements specified by academic publishers for the publication of documents.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

This seminar covers various topics from the field of algorithm engineering. The focus can be on scalability, parallelism, efficiency or theoretical guarantees of algorithms. Example topics may include graph algorithms, sorting algorithms, string algorithms, SAT solvers, data structures or other algorithms. The exact focus of the seminar for the current semester will be announced in advance on the institute website by the chair of Prof. Sanders.

Participants in the seminar carry out their own literature research, present their results to their fellow students and prepare a paper.

The exact formalities will be announced at a kick-off event at the beginning of the semester, which will also be announced on the institute's website.

Arbeitsaufwand

4 LP corresponds to approx. 120 working hours, of which

- 10h seminar attendance
- 45h Literature research, assessment and evaluation of relevant literature
- 25h preparation of own presentation
- 25h Preparation of the written paper
- 15h preparation and follow-up work

Empfehlungen

Knowledge of algorithms is an advantage. Exemplary lectures are Algorithms I, Algorithms II, Algorithm Engineering and Parallel Algorithms.

M**4.139 Modul: Seminar: Applications and Extensions of Timed Systems [M-INFO-106512]**

Verantwortung: Jun.-Prof. Dr. Maike Schwammberger
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)

Leistungspunkte
4 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-113132	Seminar: Applications and Extensions of Timed Systems	4 LP	Schwammberger

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The students can understand, model and analyse time-critical systems. Further on, they can apply the learned topics to real-world problems. They can independently work on a given topic in a team of two students and present the topic adequately within a paper and in front of an audience. The students can also critically discuss the works of the other students in plenum discussions.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

Many of the (embedded) software systems we are confronted with in everyday life have time-critical functionalities. For example, in the event of an accident, an airbag should be activated within a specific, very short, period of time. As another example: we expect fast response times from our smartphones so that we can use them conveniently and purposefully.

When modeling software systems, "time" is therefore a decisive factor. In this seminar, various mechanisms to formalise and analyse so-called real-time systems are discussed. The lecture also focuses on applications of timed systems. For instance, the following topics are dealt with:

- Timed Games
- Applications of Timed Automata (e.g. UPPAAL in Space)
- Synthesising Strategies using UPPAAL Stratego
- Duration Calculus (e.g. Satisfiability, Calculus)
- Interval Temporal Logic vs Duration Calculus

The module will consist of an introductory lecture part, where some basic topics around timed systems are introduced. For the second half of the module, the students will prepare papers and topic talks each in teams of two students. Additionally, a conference-style peer-review process for the papers is planned amongst the students. It is also expected that the students actively discuss their topics with their fellow students.

Arbeitsaufwand

4 ECTS correspond to 120 working hours, of which
 approx. 10 hours attendance of an introductory lecture incl. preparation and wrap-up
 approx. 60 hours independent examination of a given topic + writing a paper
 approx. 30 hours preparation of a lecture
 approx. 20 hours block seminar, incl. preparation and follow-up (e.g. review)

Empfehlungen

Knowledge in areas of theoretical computer science and modeling of (embedded) software systems is helpful (e.g. CTL, finite automata, first order logic). It is also helpful, but not at all necessary, to have knowledge of the topics of the summer term lecture "Timed Systems". Necessary topics from that lecture will also be introduced in the beginning of the winter term, if necessary.

M**4.140 Modul: Seminar: Artificial Intelligence for Energy Systems [M-INFO-106490]**

Verantwortung: TT-Prof. Dr. Benjamin Schäfer
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Telematics](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113110	Seminar: Artificial Intelligence for Energy Systems	4 LP	Schäfer

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

- Students obtained a foundational knowledge of AI in energy systems as an active research field and can name some ongoing challenges
- Students are able to independently conduct a literature review on a given topic.
- Students are able to present their knowledge in a written and structured report
- Students are able to orally present results and discuss topics of the seminar in the broader context of the field

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

Artificial Intelligence (AI) is a key technology in many areas of society and research. Energy systems with the ongoing energy transition ("Energiewende") make it a fascinating field for deploying AI methods. AI and machine learning algorithms can play a crucial role in improving energy efficiency, optimizing power generation and distribution or enhancing system stability while facilitating additional renewable energy integration. This seminar will explore fundamental AI algorithms and their applications in energy systems. Examples may include forecasting of energy demand or renewable generation, explainability of algorithms as well as optimization via AI.

Arbeitsaufwand

20h attendance time (kick-off and talks by other students)
 20h literature review
 40h writing of own contribution
 10h per-review for other students
 30h preparation of the final presentation
 120h=4ECTS

Empfehlungen

Previous participation in "Energieinformatik 1" and/or "Energieinformatik 2" is beneficiary but not mandatory.

M**4.141 Modul: Seminar: Bionic Algorithms and Robot Technologies [M-MACH-106902]**

Verantwortung: Prof. Dr.-Ing. Arne Rönnau
Einrichtung: KIT-Fakultät für Maschinenbau
Bestandteil von: [Minor Studies: Biologically Inspired Robotics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	2

Pflichtbestandteile			
T-MACH-113842	Seminar: Bionic Algorithms and Robot Technologies	3 LP	Rönnau

Erfolgskontrolle(n)

Siehe Teilleistung

Voraussetzungen

Keine

Qualifikationsziele

Ziel ist das selbständige Erarbeiten eines wissenschaftlichen Themas im Themenfeld der biologisch inspirierten Algorithmen und Robotertechnologien.

Die Studierenden sind in der Lage selbständig eine Literaturrecherche über den Stand der Forschung durchzuführen, fremde Arbeiten treffend zusammenzufassen, untereinander in Bezug zu setzen und zu bewerten.

Die Ergebnisse und Inhalte können in englischer Fachsprache schriftlich ausgearbeitet und im Rahmen eines Vortrags in Englisch präsentiert werden.

Die Studierenden sind mit dem DFG-Kodex „Leitlinien zur Sicherung guter wissenschaftlicher Praxis“ vertraut und wenden diese Leitlinien erfolgreich bei der Erstellung Ihrer wissenschaftlichen Arbeit an.

Inhalt

Biologisch inspirierte Roboter und deren Methoden und Technologien übertragen Konzepte zur Problemlösung aus der Natur unter anderem in die mechanische Konstruktion, Sensorik, Navigation, Steuerung oder Interpretation. Dabei werden diese Lösungsansätze durch technische Systeme angenähert. Die Bandbreite der von der Biologie inspirierten Robotik reicht von mehrbeinigen Laufrobotern, verteilten Sensorkonzepten und dem Leichtbau über Methodiken des maschinellen Lernens bis hin zu neuromorpher Hardware.

Arbeitsaufwand

90h Arbeitsaufwand

- Literaturrecherche: 35h
- Ausarbeitung der Seminararbeit: 35h
- Vorbereitung der Abschlusspräsentation: 10h
- Präsenzzeit: Kickoff, Präsentation und Diskussion sowie Treffen mit Betreuern: 10h

Empfehlungen

Der Besuch der Vorlesung "Biologically Inspired Robots" ist hilfreich.

M**4.142 Modul: Seminar: Continuous Software Engineering [M-INFO-105309]**

Verantwortung: Prof. Dr.-Ing. Anne Koziolk
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	2

Pflichtbestandteile			
T-INFO-110794	Seminar: Continuous Software Engineering	4 LP	Koziolk

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students can

- carry out a literature search based on a given topic, identify, locate, evaluate and finally analyze the relevant literature.
- prepare their seminar paper (and later their Bachelor's/Master's thesis) with a minimum of training, taking into account the format requirements specified by all publishers for the publication of documents.
- Preparing presentations in a scientific context. To this end, techniques are introduced that make it possible to prepare and present the content to be presented in an auditorium-appropriate manner.

present the results of their research in written form, as is generally the case in scientific publications.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and apply these guidelines successfully in the preparation of their scientific work.

Inhalt

Modern software engineering takes place in short cycles that enable rapid feedback Technologies such as build servers and containerization enable fast, frequent and automatic deployment of software in productive operation and rapid feedback into development (DevOps).

The term "Continuous Software Engineering" summarizes the interlocking of the various activities.

The seminar will examine various current challenges in the field of continuous software engineering, including the engineering of applications with machine learning components.

Arbeitsaufwand

25 working hours for the literature research

55 working hours for the preparation of the thesis and peer reviews

20 working hours for the preparation of the final presentation

20 working hours for the final block event and meeting with the supervisor.

This results in a total of 120 working hours

M**4.143 Modul: Seminar: Critical Topics in AI [M-INFO-106958]**

Verantwortung: Prof. Dr. Pascal Friederich
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113915	Seminar: Critical Topics in AI	3 LP	Friederich

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Qualification objectives:

- Students are able to work independently on literature about a topic of current research and to critically evaluate it, to find and understand relevant publications, and to classify and process their content accordingly in order to be able to present the chosen topic area in the form of a presentation and a written paper.

Learning Objectives:

- Overview of positive as well as negative impact of AI technology
- Overview of scientific and related ethical issues in current AI research

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

This seminar covers the technical as well as ethical aspects of critical issues in AI. Topics covered include bias in machine learning methods, ethically and socially critical applications of AI, and the impact of AI on society. The exact topics will be determined in each semester.

Students will work independently on an advanced topic and critically engage with it, presenting and discussing their findings in a lecture and summarizing them in a seminar paper.

Arbeitsaufwand

Total 90 h, of which:

- Introductory courses: 4 h
- Literature research: 30 h
- Writing the report (10-15 pages) and preparing the presentation (30+15 minutes): 50 h
- Presentation of the results: 6 h

Empfehlungen

Interest in social topics and research questions is required

M**4.144 Modul: Seminar: Current Trends in Theoretical Computer Science [M-INFO-107027]**

Verantwortung: Prof. Dr.-Ing. Marvin Künnemann
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte
4 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-114091	Seminar: Current Trends in Theoretical Computer Science	4 LP	Künnemann

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students are able to:

- perform a literature review on the basis of a given topic/scientific paper, to read and understand relevant scientific works in theoretical computer science and to identify the scientific context.
- present a scientific paper and its context.
- lead the discussion on the merits of the paper.
- create a written report of their topic in accordance to usual quality standards for scientific writing

Inhalt

This seminar discusses current trends and topics in theoretical computer science, with a focus on algorithms & complexity theory. This includes short deep-dives into hot topics of research, as well as possibly required background material. Topics covered in this class may well inspire subsequent research projects/thesis topics.

The seminar is organized in a reading group format: Each student is expected to perform a basic reading of the topic for each session. A designated session leader (either a student or non-student participant) has prepared the material more thoroughly and leads the discussion, usually including a basic presentation. Every student participant is expected to lead a session at least once.

The sessions usually take place in blocked format (i.e., on a small number of dates).

The specific contents vary, but often focus on 2-3 main themes of current research in the field.

Arbeitsaufwand

4 CP amounts to 120 h, distributed as follows:

- about 15 h attendance in class
- about 45 h reviewing the literature
- about 40 h preparation of presentation/discussion
- about 20 h writing of scientific report

Empfehlungen

Basic knowledge of theoretical computer science and algorithm design is recommended.

M**4.145 Modul: Seminar: Deep Learning for Robotics [M-INFO-107175]****Verantwortung:** Prof. Dr. Gerhard Neumann**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114204	Seminar: Deep Learning for Robotics	3 LP	Neumann

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students are able to independently understand a complex research topic, present the content in a concise and understandable way and prepare a scientific report summarizing the topic.

Students are able to independently understand a complex research topic, present the content in a concise and understandable way and prepare a scientific report summarizing the topic. Students get a deeper understanding of state-of-the-art learning algorithms and get to know current research challenges.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

Each student has to choose one of the offered topics from the area of deep learning / robot learning / deep reinforcement learning / deep imitation learning. Each topic consists of several research papers for which the students have to prepare a presentation as well as a report in form of a scientific research paper. It is recommended to take the seminar together with the "Research Project Deep Learning for Robotics", where the presented algorithms will be implemented and evaluated. Students will work in teams of 2.

Arbeitsaufwand

Workload = 90 h = 3 ECTS

Attendance time: 15h

Self-study: 45h

Writing a scientific report: 20h

Prepare presentation: 10h

Empfehlungen

Attendance of the lecture "Machine Learning - Fundamentals and Algorithms" is recommended.

M**4.146 Modul: Seminar: Embedded Systems I [M-INFO-107231]****Verantwortung:** Prof. Dr.-Ing. Jörg Henkel**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Semester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
2**Pflichtbestandteile**

T-INFO-114255	Seminar: Embedded Systems I	3 LP	Henkel
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Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students learn the basics of scientific work in the form of literature research, writing a scientific paper and giving a presentation to a specialist audience.

Learning objectives:

Students learn to read conference papers, articles in specialist journals and standard literature. Furthermore, they interpret these texts in order to give an overview of the topic in their own words in a paper. Finally, they also present an overview of the topic to other computer scientists. In doing so, they are trained in scientific writing in the form of expression, text structure and reduction to the essentials.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines when writing their scientific work.

Inhalt

This module bundles the seminars at the Chair of Embedded Systems:

Internet of Things

Machine Learning

Embedded Security and Architectures

For current information, please check the course catalog and the Chair of Embedded Systems homepage at <https://ces.itec.kit.edu>.**Arbeitsaufwand**

90 h

M**4.147 Modul: Seminar: Embedded Systems II [M-INFO-107232]****Verantwortung:** Prof. Dr.-Ing. Jörg Henkel**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** Area of Specialization: Design of Embedded Systems and Computer Architectures
Elective Studies in Informatics

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114256	Seminar: Embedded Systems II	3 LP	Henkel

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele**Qualification objective:** Students learn the basics of scientific work in the form of literature research, writing a scientific paper and giving a presentation to a specialist audience.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines when writing their scientific work.

Learning objectives: Students learn how to read conference papers, articles in specialist journals and standard literature. Furthermore, they interpret these texts in order to give an overview of the topic in their own words in a paper. Finally, they also present an overview of the topic to other computer scientists. In doing so, they are trained in scientific writing in the form of expression, text structure and reduction to the essentials.**Inhalt**

This module bundles the seminars at the Chair of Embedded Systems:

Internet of Things (IoT) for Healthcare

Internet of Things (IoT) in Embedded Systems

Approximate Computing

Thermal-aware Embedded Systems

Dependability in Internet of Things (IoT)

Performance Optimization for Multicore Chips

Power Efficient Reliability

Distributed Decision Making

Low Power Design for Embedded Systems

Reconfigurable Embedded Systems

Mixed Criticality Systems

Security in Internet of Things (IoT)

For current information, please refer to the course catalog and the Chair of Embedded Systems homepage at <http://ces.itec.kit.edu>.**Anmerkungen**

This is identical to the module 'Seminars: Embedded Systems I' and enables participation in a second seminar at the CES Chair..

Arbeitsaufwand

90 h

M**4.148 Modul: Seminar: Exoskeletons & Motion Capture [M-INFO-106927]**

Verantwortung: Prof. Dr. Katja Mombaur
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113892	Seminar: Exoskeletons & Motion Capture	3 LP	Mombaur

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

- The students know the state of the art of exoskeletons and assistive robots and current medical applications.
- Students are familiar with the most important methods and devices for measuring human movements with and without robotic mobility assistance systems
- Students are able to independently research, understand, critically evaluate and summarize scientific literature on a given topic (usually in English).
- Students will be able to prepare and deliver a scientific presentation, taking into account the level of knowledge of the other seminar participants, and to answer detailed questions on the topic
- Students will be able to ask questions about scientific presentations of other students and make active contributions to scientific discussions
- Students will be able to use Latex to create a scientific text (in English) incorporating the sources they have read.

Inhalt

This seminar provides an insight into the current state of research and the practical use of exoskeletons in medicine and their potential to improve the quality of life of people with and without physical disabilities. It also gives an overview of different motion capture technologies, which play an important role in exoskeleton research but have many other applications. The proseminar is offered by the research group Optimization and Biomechanics for Human-Centred Robotics at the IAR.

Exoskeletons are used to improve mobility and are worn by people directly on the body to support or completely replace their muscle strength. One class of exoskeletons assists people with mobility impairments to walk, stand and perform other physical activities so that they can regain their independence and participate in the activities of daily living. Other types of exoskeletons are used by healthy people to prevent injuries in difficult working conditions.

The applications of motion capture include many disciplines in which human behavior and movements are of interest, such as sports science, biomechanics, medicine, psychology, etc., but they also play an important role in the creation of animated films and computer games. In robotics, and especially in the development of exoskeletons, they are important for analyzing human-robot interaction and improving robot design and control. Various motion capture technologies exist for recording movement kinematics, ground reaction forces and muscle activity.

The proseminar presentation topics cover the spectrum of different exoskeleton types and motion capture technologies.

Schedule of the proseminar:

In an event in the first week of the semester, the topic of the seminar and the individual lecture topics are presented in detail and distributed to the students according to their preferences. In a second session in week 3, an introduction to academic literature research, writing and presentations will take place. Afterwards, the students carry out a literature search under supervision and prepare a presentation and a corresponding paper. As the scientific literature is mostly in English, the paper should also be written in English. Presentations can be held in either English or German and take place in one or more block courses during the last third of the semester.

Anmerkungen

Limited number of participants

Arbeitsaufwand

Estimated effort for this module is 90 hours:

20h – In person events (kickoff meeting, individual preparatory meetings, seminar block)

20h – Literature research

20h – Preparation of presentation

30h - Paper writing

Empfehlungen

Knowledge in Robotics (e.g. from the class Robotics 1 and follow-ups) is helpful

M**4.149 Modul: Seminar: Explainable Artificial Intelligence [M-INFO-106497]****Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113115	Seminar: Explainable Artificial Intelligence	3 LP	Lioutikov

Erfolgskontrolle(n)

See partial achievements (Teilleistung).

Voraussetzungen

See partial achievements (Teilleistung).

Qualifikationsziele

Students are able to independently understand a complex research topic, present the content in a concise and understandable way and prepare a scientific report summarizing the topic.

Students are able to independently understand a complex research topic, present the content in a concise and understandable way and prepare a scientific report summarizing the topic. Students get a deeper understanding of state-of-the-art learning algorithms and get to know current research challenges.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

Each student will select several related papers in the field of Explainable Artificial Intelligence and study and analyze it in the context of Robot Learning. The organizers will suggest several papers but the students will be encouraged to identify and research additional relevant papers during the semester. The students will then prepare a presentation and a basic scientific research paper.

It is highly recommended to take this seminar in combination with the "Explainable Robot Intelligence" Research Project (Forschungspraktikum), where the students get the chance to deepen their understanding, implement and evaluate their presented work.

Anmerkungen

It is only possible to resign within two weeks after assignment of the topic.

Arbeitsaufwand

Workload = 90 h = 3 ECTS

- Attendance time: 15h
- Self-study: 45h
- Writing a scientific report: 20h
- Preparing a presentation: 10h

Empfehlungen

We recommend taking this research project after attending the "Explainable Artificial Intelligence" lecture in the summer semester.

We highly recommend to take this seminar in combination with the "Explainable Artificial Intelligence" research project (Forschungspraktikum).

- Experience in Machine Learning is recommended, e.g. through prior coursework.
 - The Computer Science Department offers several great lectures e.g., "Maschinelles Lernen - Grundlagen und Algorithmen" and "Deep Learning"
- A good mathematical background will be beneficial
- Python experience is recommended
- We will use the PyTorch deep learning library. Some prior knowledge in this is helpful but not necessary.

M**4.150 Modul: Seminar: Exploring Robotics - Insights from Science Fiction, Research and Society [M-INFO-106651]**

Verantwortung: TT-Prof. Dr. Barbara Bruno
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113398	Seminar: Exploring Robotics - Insights from Science Fiction, Research and Society	3 LP	Bruno

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The students gain experience with literature research on a current research topic. They explore, understand and compare different approaches to a selected scientific problem. The students are able to write a summary of their literature research in the form of a scientific publication in English and give a scientific talk on it.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

The students choose a topic from the field of robotics (e.g. remote control, behavior-based robotics, human-robot interaction, the "uncanny valley," natural language understanding, machine learning) and conduct a research on it that, building on literature findings, also includes and addresses the perspectives of society and the general media (as given by science fiction books, movies and games, as well as media and news outlets) and technology assessment (including social/societal expectations and needs, ethical implications, and risks/benefits analyses).

Students work under the guidance of a scientific supervisor. At the end of the semester, they present the results and write an elaboration in English in the form of a scientific publication.

Arbeitsaufwand

Seminar with 2 SWS, 3 LP.

3 LP corresponds to approx. 90 hours, of which

approx. 45 hours of literature research

approx. 25 hrs. elaboration

approx. 10 hrs. preparation of presentation

approx. 10 hrs. compulsory attendance

Empfehlungen

Knowledge of the content of modules Robotics I - Introduction to Robotics, Robotics II: Humanoid Robotics, Robotics III - Sensors and Perception in Robotics is helpful.

M**4.151 Modul: Seminar: Failing Forward: Robotics, Critical Reading and the Making of Good Research [M-INFO-107436]**

Verantwortung: TT-Prof. Dr. Barbara Bruno
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114604	Seminar: Failing Forward: Robotics, Critical Reading and the Making of Good Research	3 LP	Bruno

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The students refine their critical reading skills and gain experience with literature research on a current research topic. They can identify and propose good research practices. The students are able to present their findings and reflections in the form of a scientific publication in English and give a scientific talk on it.

Inhalt

The students choose a topic from the field of human-robot interaction (e.g. about robots in public spaces, or robots and diversity, or companion robots for older adults). They then conduct a literature research under the guidance of a scientific supervisor, aimed at identifying what makes good research and good research reporting on the chosen topic. At the end of the semester, they present the results and write an elaboration in English in the form of a short scientific position paper.

Arbeitsaufwand

Seminar with 2 SWS, 3 LP.
 3 LP corresponds to approx. 90 hours, of which
 approx. 45 hours of literature research
 approx. 25 hrs. elaboration
 approx. 10 hrs. preparation of presentation
 approx. 10 hrs. compulsory attendance

Empfehlungen

Knowledge of the content of modules Robotics I - Introduction to Robotics, Robotics II: Humanoid Robotics, Robotics III - Sensors and Perception in Robotics is helpful.

M**4.152 Modul: Seminar: Fine-Grained Complexity Theory & Algorithms [M-INFO-106645]**

Verantwortung: Prof. Dr.-Ing. Marvin Künnemann
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte
4 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-113392	Seminar: Fine-Grained Complexity Theory & Algorithms	4 LP	Künnemann

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students are able to:

- perform a literature review on the basis of a given topic/scientific paper, to read and understand relevant scientific works in algorithms & complexity theory and to identify the scientific context.
- present a scientific paper and its context. This includes competency in tools and techniques for making the content accessible for a target audience.
- create a written report of their topic in accordance to usual quality standards for scientific writing

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

Selected topics from the field of fine-grained complexity theory & algorithm design. This consists of recent papers on fine-grained hardness assumptions, conditional lower bounds and algorithmic results for important problems from various sub-areas.

Each student will present a topic and summarize it in a scientific report.

Arbeitsaufwand

4 CP amounts to 120 h, distributed as follows:

- about 10 h attendance in class
- about 40 h literature search and review
- about 40 h preparation of presentation
- about 30 h writing of scientific report

Empfehlungen

Basic knowledge of theoretical computer science and algorithm design is recommended.

Concurrent or previous attendance of the lecture "Fine-Grained Complexity Theory & Algorithms" is helpful, but not required. This seminar can be attended independently.

M**4.153 Modul: Seminar: Hot Topics in Artificial Intelligence & Security 1 [M-INFO-106868]**

Verantwortung: Prof. Dr. Christian Wressnegger
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113761	Seminar: Hot Topics in Artificial Intelligence & Security 1	4 LP	Wressnegger

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students know basic concepts artificial intelligence and machine learning in computer security, and are able to understand/interpret results from state-of-the-art research.

- Students know and understand basic concepts of combining artificial intelligence and computer security.
- Students are able independently research topics and methods.
- Students understand limits of current methods and applications

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work

Inhalt

This seminar is concerned with the combination of artificial intelligence, machine learning and computer security in practice. Many tasks in the security landscape are based on manual labor, such as searching for vulnerabilities or analyzing malware. Here, machine learning can be used to establish a higher degree of automation, providing more "intelligent" security solutions (AI for Security). However, also these learning-based systems can be attacked and need to be secured (Security of AI).

This module is part of a seminar series to intensifies the contents of the AISEC lecture. It can be attended individually and in no particular order. The module puts focus on timely topics from recent research and teaches students to work up results from state-of-the-art research. To this end, the they will read up on a sub-field, prepare a seminar report, and present their work at the end of the term to their colleagues.

Arbeitsaufwand

- 30h Literature research
- 60h Elaboration of the seminar paper
- 20h Preparation of final presentation
- 10h attendance time

Empfehlungen

The basics of IT security and artificial intelligence are a prerequisite.

M**4.154 Modul: Seminar: Hot Topics in Artificial Intelligence & Security 2 [M-INFO-106869]**

Verantwortung: Prof. Dr. Christian Wressnegger
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-113762	Seminar: Hot Topics in Artificial Intelligence & Security 2	4 LP	Wressnegger

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students know basic concepts artificial intelligence and machine learning in computer security, and are able to understand/interpret results from state-of-the-art research.

- Students know and understand basic concepts of combining artificial intelligence and computer security.
- Students are able independently research topics and methods.
- Students understand limits of current methods and applications

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work

Inhalt

This seminar is concerned with the combination of artificial intelligence, machine learning and computer security in practice. Many tasks in the security landscape are based on manual labor, such as searching for vulnerabilities or analyzing malware. Here, machine learning can be used to establish a higher degree of automation, providing more "intelligent" security solutions (AI for Security). However, also these learning-based systems can be attacked and need to be secured (Security of AI).

This module is part of a seminar series to intensifies the contents of the AISEC lecture. It can be attended individually and in no particular order. The module puts focus on timely topics from recent research and teaches students to work up results from state-of-the-art research. To this end, the they will read up on a sub-field, prepare a seminar report, and present their work at the end of the term to their colleagues.

Arbeitsaufwand

- 30h Literature research
- 60h Elaboration of the seminar paper
- 20h Preparation of final presentation
- 10h attendance time

Empfehlungen

The basics of IT security and artificial intelligence are a prerequisite.

M**4.155 Modul: Seminar: Hot Topics in Bioinformatics [M-INFO-100750]**

Verantwortung: Prof. Dr. Alexandros Stamatakis
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-101287	Seminar: Hot Topics in Bioinformatics	3 LP	Stamatakis

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Participants will be able to understand, critically evaluate and compare current scientific publications in the field of sequence-based bioinformatics. They are able to present, understand, and critically assess the algorithms and models from current publications orally and in writing at a level that corresponds to the quality of scientific publications and conference presentations. They are able to suggest possible extensions to existing work and assess if the results are reproducible.

Inhalt

The field of Bioinformatics is by now established as an independent application area of computer science. One of the main objectives of classical bioinformatics is to generate biological knowledge (usually from molecular data, e.g., DNA data sets) using appropriate models and algorithms. The so-called molecular data flood, which is being driven by increasingly faster and cheaper methods for extracting DNA, presents bioinformatics with new challenges regarding data storage and processing. These challenges range from discrete algorithms on strings and trees to parallel processing of data and large numerical simulations on supercomputers. The aim of the module is to provide an insight into the many facets of current bioinformatics research.

Arbeitsaufwand

10 hours of topic selection + 10 hours of attending the seminar lectures + 30 hours of reading and understanding the paper(s) + 10 hours of lecture preparation + 30 hours for writing the report = 90 hours = 3 ECTS

M**4.156 Modul: Seminar: Hot Topics in Decentralized Systems [M-INFO-104891]**

Verantwortung: Prof. Dr. Hannes Hartenstein
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte
3 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-109922	Seminar: Hot Topics in Decentralized Systems	3 LP	Hartenstein

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The student is familiar with the current state of research in the field of decentralised systems.

The student is able to familiarise him/herself independently with a current research topic and the associated fundamentals by identifying relevant literature and processing it in a structured manner.

The student is able to write a paper according to scientific standards.

The student is able to present and discuss a scientific topic in a colloquium.

The student is able to consider the challenges of a specific technical problem in the context of decentralised systems and transfer existing solution approaches to the given problem and evaluate them with regard to performance and security.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

The seminar deals with current work in the field of decentralised systems. Based on current research work, challenges and approaches are identified. Corresponding solutions are analysed and compared. Finally, the reference to related domains is established.

Arbeitsaufwand

Kick-off events: 4h

Meeting with the supervisor: 4h

Presentation dates: 8h

Literature research: 25h

Writing the paper and preparing the presentation: 50h

Total: 91h = 3 ECTS points

Empfehlungen

Knowledge of the basics of IT security management for networked systems and the basic security module is helpful

M**4.157 Modul: Seminar: Hot Topics in Explainable Artificial Intelligence (XAI) [M-INFO-106392]**

Verantwortung: Prof. Dr. Christian Wressnegger
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-112917	Seminar: Hot Topics in Explainable Artificial Intelligence (XAI)	4 LP	Wressnegger

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students know concepts of explainable machine learning and are able to understand/interpret results from state-of-the-art research.

- Students know and understand concepts of methods for explaining machine learning algorithms.
- Students are able independently research topics and methods in the field of explainable machine learning.
- Students understand limits of current approaches for explaining machine learning.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

This seminar is concerned with explainable machine learning in computer security. Learning-based systems often are difficult to interpret, and their decisions are opaque to practitioners. This lack of transparency is a considerable problem in computer security, as black-box learning systems are hard to audit and protect from attacks.

The module introduces students to the emerging field of explainable machine learning and teaches them to work up results from recent research. To this end, the students will read up on a sub-field, prepare a seminar report, and present their work at the end of the term to their colleagues.

Topics cover different aspects of the explainability of machine learning methods for the application in computer security in particular.

Arbeitsaufwand

- 24h literature research
- 48h Elaboration of the seminar paper
- 24h Review of preliminary work by fellow students
- 16h preparation of final presentation
- 8h attendance time

In total 120h

M**4.158 Modul: Seminar: Information Systems [M-INFO-107686]**

Verantwortung: Prof. Dr.-Ing. Klemens Böhm
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Semester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-115055	Seminar: Information Systems	3 LP	Böhm

Erfolgskontrolle(n)

See Partial Achievements (Teilleistung).

Voraussetzungen

See Partial Achievements (Teilleistung).

Qualifikationsziele

- Students receive an initial introduction to scientific work in a specific specialist area in the field of information systems.
- Working on the seminar paper also prepares students for writing their Master's thesis.
- By attending the seminars, students are taught not only techniques of scientific work but also key qualifications.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines when writing their scientific work.

Inhalt

The Chair of Information Management Systems offers at least one seminar per semester on a selected information systems topic (every seminar at the "Chair of Information Management Systems" that is not a proseminar counts as an "Information Systems Seminar").

These are usually current research topics, for example from the fields of databases, data mining or workflow management.

Details are announced each semester (notices and homepage of the Chair of Information Management Systems).

Arbeitsaufwand

- General introduction 2h
- Introduction to presentations 2h
- Reading and understanding articles and publications 6x4,5h=27h
- Creating a paper 6x4,5h=27h
- Creating a presentation 6x4,5h=27h
- Lectures 10x0.5h=5h

Total = 90h (3 ECTS)

Empfehlungen

Lectures of the Chair of Information Management Systems that match the topic of the seminar are strongly recommended.

M**4.159 Modul: Seminar: Interpretability and Causality in Machine Learning [M-INFO-107217]**

Verantwortung: Jun.-Prof. Dr. Jan Stühmer
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114237	Seminar: Interpretability and Causality in Machine Learning	3 LP	Stühmer

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Qualification target:

Students acquire the foundations of scientific literature research, writing of a scientific report, and presenting their results in front of an audience.

Learning objectives:

Students independently acquire an understanding of their research topic from scientific literature such as conference papers, journal papers and textbooks.

They are able to independently present the content in a concise and understandable way in a written report and in a presentation in front of an audience.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

Topic of this Masterseminar are machine learning approaches and deep learning methods for learning of interpretable representations. These methods enable to reconstruct underlying principles from data, for example the reconstruction of generative factors of a dataset.

Starting from these methods for interpretable representations, we will discuss further methods for causal discovery, that enable the inference of causal dependencies in data.

Methods and algorithms covered include for example variational inference, contrastive learning, as well as statistical methods for factor analysis.

There will be a kick-off meeting at the beginning of the semester and 2-3 block seminars towards the end of the term.

Dates for both will still be determined.

The Masterseminar will be held in English language.

Arbeitsaufwand

90h

Empfehlungen

Attendance of the lecture "Machine Learning - Fundamentals and Algorithms" is recommended.

M**4.160 Modul: Seminar: Law and Legal Studies [M-INFO-107028]****Verantwortung:** TT-Prof. Dr. Frederike Zufall**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Minor Studies: Law](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Semester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1

Pflichtbestandteile			
T-INFO-114094	Seminar: Law and Legal Studies	3 LP	Gil Gasiola, Pathak

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The student

- deals with a defined problem in the field of law,
- analyzes and discusses problems within the framework of the courses and in the final seminar papers,
- discusses, presents and defends subject-specific arguments within a given task,
- organizes the preparation of the final seminar papers largely independently.

Accompanied by the relevant examiners, the student practises independent scientific work when writing the final seminar papers and presenting them.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines when writing their scientific work.

Inhalt

The module consists of a seminar that is thematically related to law. A list of approved courses will be published on the Internet.

Arbeitsaufwand

The total workload for this module is approx. 90 hours (3 credits) for attendance time, preparation and follow-up work as well as the course examination.

The actual workload varies depending on the specific seminar chosen and is described for the individual course.

M**4.161 Modul: Seminar: Machine Learning in Climate and Environmental Sciences [M-INFO-106719]****Verantwortung:** TT-Prof. Dr. Peer Nowack**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Unregelmäßig**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1

Pflichtbestandteile			
T-INFO-113519	Seminar: Machine Learning in Climate and Environmental Sciences	3 LP	Nowack

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The students will learn to:

- independently discuss current research topics on machine learning in climate and environmental sciences.
- summarize published research in a structured way and explain it in their own words, in group discussions and in the form of a presentation.
- contrast modern problem-solving approaches and methods and propose suitable solutions for a variety of subject-relevant issues.
- Optional: students are invited to develop their own research ideas on the basis of what they have learned and to refine them in discussion with their supervisors. Such ideas could be pursued as a project internship, in the "Practical Research course" or in the form of a master's thesis.

Inhalt

Machine learning (ML) methods are already ubiquitous in many areas of society and research. This is especially true for climate and environmental sciences, where ML algorithms help e.g. to improve predictions of climate change and weather, or to optimize energy supply systems. In this session, we will discuss cutting-edge publications on ML applications in climate and environmental sciences, as well as the underlying theory behind the classes of algorithms. While organizers will suggest initial papers, students will be encouraged to seek out additional relevant literature throughout the semester.

The seminar will cover both the in-depth study of the climate/environmental sciences topic as well as of the specific machine learning method(s) employed in the literature. It will include two short and one longer final presentation from each student. The first presentation will focus solely on the chosen climate or environmental event or phenomenon, while the second presentation will cover the machine learning methods employed in studying it. Next to suggested reading by the module organizers, students will be encouraged to seek out additional relevant literature throughout the semester.

Towards the end, students will compile their findings into the final presentation accompanied by a scientific report, presenting the results in the form of a lecture.

Arbeitsaufwand

Total 90 h, consisting of:

Seminar attendance and personal meetings with the supervisors: 10 h

Literature research: 30 h

Writing the seminar paper and preparing the final presentation: 50 h

Empfehlungen

- An interest in climate and environmental sciences topics is a prerequisite.

M**4.162 Modul: Seminar: Operating Systems [M-INFO-107205]**

Verantwortung: Prof. Dr.-Ing. Frank Bellosa
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Elective Studies in Informatics](#)

Leistungspunkte
3 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-114230	Seminar: Operating Systems	3 LP	Bellosa

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students analyse and present scientific work in the field of operating systems.

In addition to techniques of scientific work, key qualifications are also taught in an integrative manner by attending the seminars.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines when writing their scientific work.

Inhalt

The seminar is dedicated to a current area of operating system research.

Arbeitsaufwand

30 h = 2 SWS * 15 attendance

30 h preparation

10 h Presentation

20 h elaboration

90 h = 3 ECTS

M**4.163 Modul: Seminar: Post-Quantum Cryptography [M-INFO-105585]**

Verantwortung: Prof. Dr. Jörn Müller-Quade
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Elective Studies in Informatics](#)

Leistungspunkte
3 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-111200	Seminar: Post-Quantum Cryptography	3 LP	Müller-Quade

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The student

- can describe the mathematical foundations of different quantum-hard problems
- knows basic post-quantum cryptosystems
- analyzes and discusses specific problems within the scope of his seminar paper
- presents technical arguments and is able to defend them within the scope of a certain assignment
- manages and prepares his seminar paper and his presentation largely independent

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

The Seminar deals with the foundations of post-quantum cryptography and quantum hard problems.

First, the mathematical basics describing several quantum-hard problems are introduced in introductory lectures. Subsequently, different post-quantum cryptosystems and common cryptographic notions will be introduced. Furthermore the seminar covers related topics, such as provability in the event of quantum adversaries.

Arbeitsaufwand

Attendance time in seminar: 15 h

Writing the paper: 30 h

Designing and preparing the presentation: 45 h

Empfehlungen

Basic knowledge of IT-Security and cryptography are recommended.

M**4.164 Modul: Seminar: Practical Graph Algorithms [M-INFO-107264]**

Verantwortung: Prof. Dr. Henning Meyerhenke
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte
4 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-114297	Seminar: Practical Graph Algorithms	4 LP	Meyerhenke

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students are able to:

- perform a literature review on the basis of a given topic/scientific paper, to read and understand relevant scientific works in algorithm engineering for graph problem and to identify the scientific context.
- present a scientific paper and its context. This includes competency in tools and techniques for making the content accessible for a target audience.
- create a written report of their topic in accordance to usual quality standards for scientific writing
- critically assess the work of other participants and make constructive suggestions for improvement.
- Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

This seminar covers various topics from the field of practical graph algorithms such as small subgraph detection, graph robustness, centrality computations, and related ones. The exact focus of the seminar for the current semester will be announced in advance on the website of Prof. Meyerhenke's chair. Participants in the seminar carry out their own literature research, present their results to their fellow students and prepare a paper.

The seminar will be held in several blocks, partially online, partially on-site. The exact formalities will be announced at an online kick-off event at the beginning of the semester, which will also be announced on the course website mentioned above.

Arbeitsaufwand

4 LP corresponds to approx. 120 working hours, of which

- 15h seminar attendance
- 35h Literature research, assessment and evaluation of relevant literature
- 35h preparation of own presentation
- 35h preparation of the scientific report

Empfehlungen

Knowledge of algorithms, in particular graph algorithms, is a clear advantage. Exemplary lectures are Algorithms I and Algorithms II.

M**4.165 Modul: Seminar: Privacy and Security [M-INFO-107216]**

Verantwortung: Prof. Dr. Thorsten Strufe
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114236	Seminar: Privacy and Security	4 LP	Strufe

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The student is able to

- conduct a literature search based on a given topic, identify and evaluate the relevant literature;
- independently compile research results from IT security and technical data protection;
- analyze and discuss scientific studies and place them in their context;
- conduct their own classifications and evaluations of scientific studies, report on them in writing and present the results in a short scientific presentation.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

The seminar deals with current topics from the research field of data protection and security.

These include, for example:

- Privacy attacks on communication
- Network security
- Anonymized online services
- Evaluation of the anonymity of online services
- Anonymized publication of data (differential privacy, k-anonymity)
- Transparency/awareness-enhancing systems
- Behavioral analysis of media use
- Biometric authentication

Arbeitsaufwand

Seminar attendance time: 10h

Researching and writing a paper: 75h

Reviewing and commenting on the preliminary papers of fellow students: 5h

Preparing the presentation: 30h

Empfehlungen

Fundamentals of IT security, computer networks and distributed systems are required

M**4.166 Modul: Seminar: Proofs from THE BOOK [M-INFO-103306]****Verantwortung:** Prof. Dr. Peter Sanders**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Theoretical Foundations](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Unregelmäßig**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1**Pflichtbestandteile**

T-INFO-106604	Seminar: Proofs from THE BOOK	3 LP	Sanders
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Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The students learn to follow and understand complex mathematical proofs on their own. They learn to represent these proofs in an appealing manner and present the proofs to the other participants using a blackboard.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

According to the Hungarian mathematician Paul Erdős, God is keeping a book – the BOOK – under wraps that contains the most elegant mathematical proofs. Erdős' loftiest goal was to find such proofs from the BOOK.

After Erdős' death in 1996, Martin Aigner and Günter Ziegler published the book "Proofs from THE BOOK" in 1998. The book has also been published in German with the title "Das BUCH der Beweise". In Aigner and Ziegler's collection, there are some 40 of the most elegant proofs which are handled as candidates for BOOK-proofs.

In this seminar, the participants will present and discuss proofs from "Proofs from THE BOOK" and other well known and well studied proofs in the area of mathematics and informatics.

Arbeitsaufwand

Seminar with 2 SWS, 3 LP

3 LP correspond to about 90h of work, split into

about 20h attendance

about 60h preparation for seminar

about 10h follow-up

Empfehlungen

The German version "Das Buch der Beweise" is available online at the KIT library within the KIT network. The English version "Proofs from THE BOOK" is available as a physical copy at the KIT library. We recommend having a look inside either version before registering for this seminar.

M**4.167 Modul: Seminar: Quantum Information Theory [M-INFO-105408]**

Verantwortung: Prof. Dr. Jörn Müller-Quade
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Elective Studies in Informatics](#)

Leistungspunkte
3 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-110904	Seminar: Quantum Information Theory	3 LP	Müller-Quade

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The student...

- understands the basics of Quantum Information Theory.
- understands formalizing quantum states via state vectors and is able to autonomously use the state vector formalism to design and analyze quantum algorithms.
- knows and understands the quantum gates introduced in the seminar.
- knows the visual quantum circuit tool "Quirk" and is able to autonomously apply it to design and analyze quantum algorithms.
- knows and understands the quantum problems and algorithms discussed in the seminar and is able to explain them and relate them to one another.
- knows and understands the impact quantum algorithms have on classic cryptography.
- knows and understands the basics of and presented protocols for quantum key distribution.
- is able to autonomously apply the techniques presented in the seminar, e.g. to prove correctness of simple quantum algorithms.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

- Basics of Quantum Information Theory
- Formalism for dealing with quantum systems
- "Quirk"
- Important quantum problems and algorithms
- Quantum key distribution
- Quantum walks

Arbeitsaufwand

Seminar attendance time: 18h

Preparation and follow-up work: 12h

Preparation of a presentation: 30h

Preparation of a written examination: 30h

Empfehlungen

Students should be familiar with the contents of the module "Linear Algebra 1 and 2", as well as the basics of IT security.

M**4.168 Modul: Seminar: Recent Highlights in Algorithms [M-INFO-107172]**

Verantwortung: TT-Prof. Dr. Thomas Bläsius
Prof. Dr. Peter Sanders

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte
4 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-114201	Seminar: Recent Highlights in Algorithms	4 LP	Bläsius, Sanders, Ueckerdt

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students can,

- carry out a literature search based on a given topic, identify, locate, evaluate and finally analyse the relevant literature.
- prepare presentations in a scientific context. To this end, students master techniques that enable them to prepare and present the content to be presented in a manner suitable for an audience.
- prepare their written seminar paper (as required later for further academic work) in accordance with the requirements and quality standards of academic writing, taking into account the format requirements specified by academic publishers for the publication of documents.
- critically assess the work of other participants and make constructive suggestions for improvement.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

The seminars offered as part of this seminar module deal with current topics in algorithm technology and explore them in depth. As a rule, the prerequisite for passing the module is the preparation of a written paper of max. 15 pages and an oral presentation of at least 45 minutes.

Arbeitsaufwand

Seminar with 2SWS, 4LP

4 LP corresponds to approx. 120 working hours, of which

approx. 10h seminar attendance

approx. 40 hours of literature research, assessment and evaluation of relevant literature

approx. 30h preparation of own presentation

approx. 30 hours writing the paper

approx. 10h Reading two papers and formulating constructive criticism and suggestions for improvement in writing

M**4.169 Modul: Seminar: Secure Multiparty Computation [M-INFO-105761]**

Verantwortung: Prof. Dr. Jörn Müller-Quade
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Elective Studies in Informatics](#)

Leistungspunkte
3 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-111501	Seminar: Secure Multiparty Computation	3 LP	Müller-Quade

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students learn to familiarize themselves thoroughly with scientific papers, to present them to other students and to deal with questions on their topic in a subsequent discussion round.

Students will be able to differentiate between different protocols for secure multiparty computation and weigh up their advantages and weaknesses.

Students will be able to present academic publications from the research field of secure multiparty computation in a suitable manner, place them in the historical context of the research field and critically examine the results and findings presented.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and apply these guidelines successfully in the preparation of their academic work.

Inhalt

In the setting of secure multiparty computation, two or more parties with private inputs wish to compute some joint function of their inputs. The security requirements of such a computation are privacy (meaning that the parties learn the output and nothing more), correctness (meaning that the output is correctly distributed), independence of inputs, and more. Due to its generality, secure computation is a central tool in cryptography.

In this seminar, we examine modern protocols for secure multiparty computation of arbitrary functions.

Arbeitsaufwand

Attendance time in seminar: 15 h

Meeting with supervisors: 5 h

Independent work in relation to the individual seminar topic: 70 h

Empfehlungen

Knowledge of the content of the lecture Cryptographic Protocols is assumed.

M**4.170 Modul: Seminar: Software Architecture, Security and Privacy [M-INFO-107236]**

Verantwortung: Prof. Dr. Ralf Reussner
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Unregelmäßig	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114260	Seminar: Software Architecture, Security and Privacy	4 LP	Reussner

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students can,

- carry out a literature search based on a given topic, identify, locate, evaluate and finally analyze the relevant literature.
- prepare their seminar paper (and later their Bachelor's/Master's thesis) with a minimum of training, taking into account the format requirements specified by all publishers for the publication of documents.
- Preparing presentations in a scientific context. To this end, techniques are introduced that make it possible to prepare and present the content to be presented in an auditorium-appropriate manner.
- present the results of their research in written form, as is generally the case in scientific publications.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and apply these guidelines successfully in the preparation of their scientific work.

Inhalt

Anyone who processes personal data automatically must effectively protect this data from unauthorized access in order to act in accordance with data protection laws, but also to prevent damage to reputation and trustworthiness should data protection violations become public. Protecting personal data from unauthorized access and complying with other data protection obligations is therefore actually one of the most important goals in software design and operation.

However, looking at data protection in isolation does not do justice to reality. If an attacker gains access to personal data, voluntary commitments and internal data protection regulations no longer apply. In case of doubt, the operator of the software is liable for severe fines. Effective security precautions are therefore indispensable as a mainstay for protecting personal data.

Security-critical vulnerabilities must be identified at an early stage, ideally before the vulnerability is introduced. Such quality assessments are performed by software architecture-based analyses. How security can be described and analyzed at the software architecture level is the subject of ongoing research, as is the question of whether - and how - security can be expressed in figures.

In this seminar, students deal with these questions and the state of research at the interface between data protection, security and software architecture. Possible topics are located in one or more of these areas.

Arbeitsaufwand

25 working hours for literature research

55 working hours for writing the thesis and preparing peer reviews

20 working hours for preparing the final presentation

20 working hours for the final block event and meeting with the supervisor.

This results in a total of 120 working hours

M**4.171 Modul: Seminar: Speech-to-Speech Translation [M-INFO-107179]****Verantwortung:** Prof. Dr. Jan Niehues**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Sommersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1**Pflichtbestandteile**

T-INFO-114208	Seminar: Speech-to-Speech Translation	3 LP	Niehues
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Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students learn to familiarise themselves independently with topics based on academic literature and prepare them for presentations.

From the other presentations, students gain in-depth knowledge in sub-areas of language-to-language translation

By evaluating the presentations of their fellow students, students improve their social skills.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Inhalt

Speech-to-speech translation is a popular application that combines automatic speech recognition and machine translation. However, a user-friendly combination requires more than just a linear connection of the individual techniques.

In this seminar, students work independently on individual topics from the fields of automatic speech recognition, machine translation and their combination into speech-to-speech translation systems using the literature provided and present the summarised findings to the other participants in the seminar in the form of a slide-based presentation.

Arbeitsaufwand

90 h

M**4.172 Modul: Seminar: Type Systems and Static Program Analysis [M-INFO-107625]**

Verantwortung: Prof. Dr. Sebastian Erdweg
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114955	Seminar: Type Systems and Static Program Analysis	4 LP	Erdweg

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

None.

Qualifikationsziele

Students can:

- carry out a literature search based on a given topic. They can identify, locate, evaluate and comprehend the relevant literature.
- prepare and deliver a scientific presentation on a given topic, using adequate presentation techniques.
- write a seminar paper that meets the requirements and quality standards of academic writing.
- participate in academic discussions with their peers.

Students are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and apply these guidelines successfully in the preparation of their scientific work.

Inhalt

Type systems and static program analyses predict the behavior of programs through source-code analysis, but without executing the program. They are fundamental for isolating ill-defined program behavior (e.g., from type errors), but also can provide guarantees (e.g., memory safety). Topics covered in this seminar include the design, definition, formalization, and application of type systems and static program analyses, including type inference, dependent typing, control flow, data flow, abstract interpretation, soundness, and precision.

Arbeitsaufwand

10 working hours for independent study of the seminar topic
 25 working hours for literature research
 20 working hours for preparing the presentation
 55 working hours for writing the thesis and preparing peer reviews
 10 working hours for the final block event and meetings with the supervisor

This results in a total of 120 working hours

Empfehlungen

Students should be familiar with the foundational knowledge about type systems (e.g., typing rules, type checking, polymorphism) or static program analysis (e.g., data flow, abstract interpretation).

M**4.173 Modul: Service Design Thinking [M-WIWI-101503]**

Verantwortung: Prof. Dr. Gerhard Satzger
Prof. Dr. Orestis Terzidis

Einrichtung: KIT-Fakultät für Wirtschaftswissenschaften

Bestandteil von: [Minor Studies: Economics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
9 LP	Zehntelnoten	Jedes Wintersemester	2 Semester	Englisch	4	1

Pflichtbestandteile			
T-WIWI-102849	Service Design Thinking	9 LP	Satzger, Terzidis

Erfolgskontrolle(n)

Die Modulprüfung erfolgt in Form einer Gesamtprüfung (nach §4(2), 3 SPO). Die Gesamtnote des Moduls entspricht der (Drittel-)Note der Prüfung (nach §4(2), 3 SPO).

Voraussetzungen

Keine

Qualifikationsziele

Studierende

- gewinnen ein umfassendes Verständnis der weltweit anerkannten Innovationsmethodik "Design Thinking", wie sie an der Stanford University gelehrt wird
- wenden die erlernte Methodik im Rahmen eines echten Innovationsprojekts an, das von einem Praxispartner gestellt wird
- konzeptionieren neue, kreative Lösungen durch umfassende Analyse der Bedürfnisse von Service-Nutzern
- entwickeln frühzeitig und eigenständig Prototypen, testen diese und verbessern sie iterativ, um damit die von der Partnerorganisationen gestellten Herausforderung zu lösen
- lernen, in einem interdisziplinären und internationalen Umfeld zu kommunizieren, zu präsentieren und sich zu vernetzen

Inhalt

Kursphasen (jeweils ca. 4 Wochen):

Design Space Exploration:

- Erkundung des Problemraums durch Hinterfragen der gestellten Aufgabe.
- Einarbeiten in den Themenbereich der jeweiligen Innovationsherausforderung aus der Praxis.
- Erheben erster Eindrücke, Anforderungen und Bedürfnisse der Personen, die mit der Problemstellung in Zusammenhang stehen.

Critical Function Prototype:

- Aufbau eines intensiven Verständnisses von den Bedürfnissen der Zielgruppe der jeweiligen Herausforderung.
- Ableiten von kritischen Funktionen aus Sicht der Kunden, die zur Lösung des Gesamtproblems beitragen könnten.
- Bau von Prototypen für die kritischen Funktion und Testen dieser in realen Kundensituationen.

Dark Horse Prototype:

- Umkehrung von bislang getroffenen Annahmen und Erfahrungen. Das Ziel ist die Entwicklung von radikal neuen und unkonventionellen Ideen.
- Umsetzung der Ideen in einfache Prototypen und anschließender Test.

Funky Prototype:

- Integration der einzelnen erfolgreich getesteten Funktionen aus der Critical Function und Dark Horse Phase zu Lösungskonzepten. Diese werden ebenso getestet und weiterentwickelt.

Functional Prototype:

- Selektion erfolgreicher Funky Prototypen und Entwicklung dieser in Richtung hoch aufgelöster Prototypen. Der endgültige Lösungsansatz für das Projekt wird detailliert niedergelegt und Feedback dazu eingeholt.

Final Prototype:

- Umsetzung des finalen Prototyps und Präsentation vor dem Partnerunternehmen sowie dem SUGAR Netzwerk.

Anmerkungen

Aufgrund der Projektarbeit ist die Zahl der Teilnehmer beschränkt. Das Modul (und auch die Teilleistung) geht über zwei Semester. Es startet jedes Jahr Ende September und läuft bis Ende Juni des darauffolgenden Jahres. Ein Einstieg ist nur zu Programmstart im September (Bewerbung im Mai/Juni) möglich. Weitergehende Informationen zum Bewerbungsprozess und dem Programm selbst finden Sie in der Teilleistungsbeschreibung sowie über die Website des Programms (<https://sdtkarlsruhe.de/>). Ferner führen die Dozenten jedes Jahr im Mai eine Informationsveranstaltung zum Programm durch. Dieses Modul ist Teil des KSRI-Lehrprofils.

Arbeitsaufwand

Der Aufwand für dieses Modul beträgt ca. 2 Tage pro Woche über einen Zeitraum von 9 Monaten. Der Aufwand für dieses praxisnahe Modul ist somit vergleichsweise hoch. Die Ursache dafür ist, dass die Teilnehmenden in internationalen Teams mit Studierenden anderer Universitäten sowie Partnerorganisationen zusammenarbeiten und echte Innovationsherausforderungen lösen.

Der Arbeitsaufwand in Höhe von ca. 270 Stunden verteilt sich dabei auf ca. 105 Stunden (3,5 LP) im ersten und 165 Stunden (5,5 LP) im zweiten Semester.

Empfehlungen

Diese Veranstaltung findet in englischer Sprache statt – Teilnehmer sollten sicher in Schrift und Sprache sein.

Unsere bisherigen Teilnehmer fanden es empfehlenswert, das Modul zu Beginn des Master-Programms zu belegen.

M**4.174 Modul: Simulation and Optimization in Robotics and Biomechanics [M-INFO-106504]**

Verantwortung: Prof. Dr. Katja Mombaur
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Elective Studies in Informatics](#)

Leistungspunkte 6 LP	Notenskala Zehntelnoten	Turnus Jedes Wintersemester	Dauer 1 Semester	Sprache Englisch	Level 4	Version 2
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Pflichtbestandteile			
T-INFO-113123	Simulation and Optimization in Robotics and Biomechanics	6 LP	Mombaur
T-INFO-114838	Simulation and Optimization in Robotics and Biomechanics - Pass	0 LP	Mombaur

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The students

- can explain advanced principles of modeling, optimization and control of dynamic processes, in particular mechanical systems and can apply them
 - can model, classify and analyze complex motions in robotics or biomechanics, and investigate specific properties such as stability.
 - can apply nonlinear optimization and optimal control methods and can compare and evaluate different mathematical approaches.
 - know how to use software tools based on C++ and Lua for modeling, simulation, optimization and visualization of humanoid and robotic systems
- are capable of solving optimal control problems numerically and to evaluate the quality of the solution.

Inhalt

The goal of this course is to give a practical introduction into simulation and optimization of motions in robotics and biomechanics. Simulation and optimization play an important role in generating and controlling motions in complex robotics systems and in predicting and analyzing motions of humans. Theory and methods will be covered, but the focus is on the use software tools for modeling, simulation, optimization and visualization of multibody systems. Topics covered include:

- Dynamic process modeling
- Transforming real world problems into mathematical models
- Modeling of complex robotics and biomechanics systems (e.g. humanoids), based on previous modeling knowledge
- Common template models for bipedal walking and running in robotics and biomechanics
- Simulation of mechanical / robotics systems (Integrators and Initial value problems)
- Boundary value problems
- Nonlinear optimization problems
- Optimal control problems
- Direct and indirect methods for optimal control problems, focus on direct methods, especially direct multiple shooting
- Stability of dynamical systems, stability in biomechanics and robotics

Anmerkungen

Limitation to 30 participants

Arbeitsaufwand

Estimated effort for this module is 180 hours:

60h - Lecture and exercises (2+2 SWS)

80h - Independent work (repetition of lecture contents, preparation of assignments)

40h - Exam preparation

M**4.175 Modul: Software Architecture and Quality [M-INFO-107237]****Verantwortung:** Prof. Dr. Ralf Reussner**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Wintersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1

Pflichtbestandteile			
T-INFO-114261	Software Architecture and Quality	3 LP	Reussner

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students can explain the role of components and explicit software architecture descriptions for engineering software development.

They can also explain the basic concepts of component-based software development.

Students are familiar with advanced concepts of view-based metamodeling and can apply these to the scenarios of the software development domain.

In addition, they can use procedures for the documentation, evaluation and reuse of software architectures, such as architecture patterns or architecture styles.

Furthermore, they can differentiate between and use different software development processes.

Students can design models for software quality characteristics such as performance.

The effects of architecture design decisions on software quality characteristics such as performance can also be analyzed.

Inhalt

In many software development projects, the software architecture is the main determining factor for software quality. Runtime properties such as performance or reliability, as well as maintainability, essentially depend on the architecture of a software system.

In the lecture, students learn about and apply modern approaches to software architecture modeling and analysis, which can be used to predict the quality characteristics of the system at design time. The lecture thus lays the scientific foundations for software design as an engineering discipline, as the methods learned enable an understanding of the effects of architectural design decisions on software quality. In particular, software qualities such as performance, reliability and maintainability are discussed.

In connection with software architecture, software components are also introduced as "software building blocks". In particular, techniques for the reuse of architectural knowledge such as patterns, styles and reference architectures and product lines are discussed.

The lecture deals with the Palladio component model as a description language for software components and architectures.

Using the Palladio component model, role models for the design and development of component-based software are presented in addition to quality prediction.

Its use is demonstrated using industry-related case studies and techniques for evaluating the quality of your software architecture are illustrated.

The lecture covers technologies such as MOF, OCL and architecture-centered, model-driven software development (AC-MDSD). Modern middleware from practice such as Java EE / EJB is also presented.

Arbeitsaufwand

(2 SWS + 1.5 x 2 SWS) x 15 + 15 h exam preparation = 90 h

M**4.176 Modul: Software Engineering II [M-INFO-107235]**

Verantwortung: Prof. Dr.-Ing. Anne Koziolk
 Prof. Dr. Raffaella Mirandola
 Prof. Dr. Ralf Reussner

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)

Leistungspunkte 6 LP	Notenskala Zehntelnoten	Turnus Jedes Sommersemester	Dauer 1 Semester	Sprache Englisch	Level 4	Version 1
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Pflichtbestandteile			
T-INFO-114259	Software Engineering II	6 LP	Koziolk, Mirandola, Reussner

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Software processes: Students understand evolutionary and incremental development and can describe the advantages over the sequential approach. They can describe the phases and disciplines of the unified process.

Requirements engineering: Students can describe the terms of requirements engineering and name activities in the requirements engineering process. They can classify and assess requirements according to the facets of type and representation. They can apply basic guidelines for specifying natural language requirements and describe prioritization procedures for requirements. Describe the purpose and elements of use case models. You can classify use cases according to their granularity and objectives. You can create use case diagrams and use cases. They can derive system sequence diagrams and operation contracts from use cases and can describe their role in the software development process.

Software architecture: Students can reproduce and explain the definition of software architecture and software components. They can explain the difference between software architecture and software architecture documentation. They can describe the advantages of explicit architecture and the factors influencing architecture decisions. You can assign design decisions and elements to the layers of an architecture. You will be able to describe what component models define. They can describe the components of the Palladio component model and discuss some of the design decisions made.

Enterprise Software Patterns: Students can characterize enterprise applications and decide for a described application which properties it fulfills. They know patterns for structuring domain logic, architectural patterns for data access and object-relational structure patterns. They can select a suitable pattern for a design problem and justify the selection based on the advantages and disadvantages of the patterns.

Software design: Students can assign the responsibilities resulting from system operations to classes or objects in object-oriented design using the GRASP patterns and thus design object-oriented software.

Software quality: Students know the principles for readable program code, can identify violations of these principles and develop proposals for solutions.

Model-driven software development: Students can describe the goals and the idealized division of labor of model-driven software development (MDSD) and reproduce and explain the definitions for model and metamodel. They can discuss the goals of modeling. You will be able to describe the model-driven architecture and express constraints in the Object Constraint Language. You can express simple transformation fragments of model-to-text transformations in a template language. You can weigh up the advantages and disadvantages of MDSD.

Embedded systems: Students will be able to explain the principle of a real-time system and why they are usually implemented as parallel processes. They can describe a rough design process for real-time systems. They can describe the role of a real-time operating system. They can distinguish between different classes of real-time systems.

Reliability: Students can describe the various dimensions of reliability and categorize a given requirement. They can illustrate that unit tests are not sufficient to evaluate software reliability and can describe how usage profile and realistic error data have an influence.

Domain-driven design (DDD): Students are familiar with the design metaphor of ubiquitous language, Closed Contexts, and Strategic Design. They can describe a domain using the DDD concepts, entity, value objects, services, and improve the resulting domain model using the patterns of aggregates, factories, and depots. They know the different types of interactions between Closed Contexts and can apply them.

Security (in the sense of security): Students can describe the basic ideas and challenges of security assessment. They can recognize common security problems and propose solutions.

Inhalt

Requirements engineering, software development processes, software quality, software architectures, MDD, Enterprise Software Patterns software maintainability, software security, dependability, embedded software, middleware, domain-driven design

Anmerkungen

The Software Engineering II module is a basic module.

Arbeitsaufwand

Preparation and follow-up time 1.5 h / 1 SWS

Total workload:

$(4 \text{ SWS} + 1.5 \times 4 \text{ SWS}) \times 15 + 30 \text{ h exam preparation} = 180 \text{ h} = 6 \text{ ECTS}$

M**4.177 Modul: Software Product Line Engineering [M-INFO-107212]****Verantwortung:** Prof. Dr.-Ing. Ina Schaefer**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Sommersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1**Pflichtbestandteile**

T-INFO-114234	Software Product Line Engineering	3 LP	Schaefer
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Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students understand the essential concepts (such as modularity, variation point, feature model, feature mapping, configuration, product generator, and product) and techniques (such as feature-oriented domain analysis, variant extraction, delta modelling, variant space analyses, product generation, testing of software product lines) of the development of software product lines, their relationships and their assignment to problem and solution spaces. They are able to understand and apply the different methods for designing software product lines, such as feature-orientated domain analysis or variant extraction. Students are familiar with various product generation strategies and know their advantages and disadvantages in practical use. Students are familiar with techniques for the maintenance of software product lines, such as variant space analysis, the generation of product samples and the testing of software product lines, and are able to apply these. In addition, students are familiar with current results and issues from the research field of software product lines and understand their significance, e.g. results from the field of language product lines.

Learning objectives: Students are able to independently design, implement and maintain a software product line. Students can apply feature-orientated domain analysis to a given domain and design a software product line based on a domain description and implement it in practice with tool support. Students can use variant extraction independently and with tool support to design a software product line from a series of product variants of a software system and implement it by refactoring. Students can select a suitable product generation strategy for a given domain and implement it with tool support. Students can analyse and improve the variant space of a given software product line. Students know different techniques to maintain a software product line and can analyse the variant space, generate product samples and develop tests for a given software product line.

Inhalt

This module teaches students the procedures and techniques for the development and maintenance of multi-variant software systems using software product lines. The lecture will provide an overview of the basic goals, processes, concepts and techniques in the development and maintenance of software product lines. It is subdivided into the subject areas of the problem space and the solution space. In the first topic area, topics such as feature-oriented domain analysis, feature models and analyses of the variant space are dealt with, whereas in the second topic area, different techniques for product generation and testing of product lines are discussed and demonstrated in practice. In addition, current results and questions from software product line research are presented and discussed.

Arbeitsaufwand

(2 SWS + 1.5 x 2 SWS) x 15 + 15 h exam preparation = 90 h

Empfehlungen

Basic knowledge from the lectures Software Engineering II [T-INFO-101370] and Formal Systems [T-INFO-101336] is helpful.

M**4.178 Modul: Software Security Engineering [M-INFO-106344]****Verantwortung:** Prof. Dr. Ralf Reussner**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Cryptography and Security](#)
[Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
3 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-112862	Software Security Engineering	3 LP	Gerking, Reussner

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Qualification target: Participants will be able to apply measures to detect or avoid vulnerabilities in different development phases.

Learning objectives:

- Participants acquire the ability to name criteria from security standards and evaluate their fulfillment.
- They master central security principles and their application to specific use cases.
- They can formalize security policies (based on security models) and recognize violations of policies.
- They are familiar with the handling and processing of security incidents.

Inhalt

The course deals with the engineering of cyber security along the development cycle of software systems. This includes constructive and analytical development measures to achieve protection goals through systematic prevention and detection of vulnerabilities. The course familiarizes participants with the adoption and implementation of security measures in various development phases. Relevant fundamentals from the field of formal security models are introduced.

Arbeitsaufwand

(2 SWS + 1.5 x 2 SWS) x 15 + 15 h exam preparation = 90 h

Empfehlungen

Knowledge of Software Engineering I and Software Engineering II is recommended.

M**4.179 Modul: Software-Evolution [M-INFO-100719]****Verantwortung:** Prof. Dr. Ralf Reussner**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Wintersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1**Pflichtbestandteile**

T-INFO-101256	Software-Evolution	3 LP	Reussner
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Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students learn about the particular challenges of long-lived software systems and the possibilities of influencing the future development of a software system through targeted software evolution. Students will learn which tools and concepts they can use in the context of software evolution and which factors have an impact on the software development process. In addition to the theoretical basics, students will gain insight into practical examples and suitable tools that simplify the handling of software evolution. A cross-section of implementation aspects, techniques, management and concepts will be conveyed to the participants of the lecture. Students are enabled to analyze, evaluate and improve software systems.

Inhalt

The lecture Software Evolution covers: Software development processes, special features of long-lived software systems, evolution scenarios for software systems, software architecture development, software refurbishment, implementation techniques, architecture patterns, traceability, software evaluation methods, maintainability analyses and tools to support software evolution.

Arbeitsaufwand

(2 SWS + 1.5 x 2 SWS) x 15 + 15 h exam preparation = 90 h

M**4.180 Modul: Softwaretest und Quality Management (SQM) [M-INFO-107239]****Verantwortung:** Prof. Dr.-Ing. Ina Schaefer**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
5 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Wintersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1**Pflichtbestandteile**

T-INFO-114263	Softwaretest und Quality Management (SQM)	5 LP	Schaefer
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Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

After completing the module, participants will be familiar with the basic principles of software testing. They will be able to apply the general testing process and master the activities and techniques to support it. Participants will be able to specify test cases in all phases of the software life cycle. They know test procedures and methods with which they can prepare and carry out software tests efficiently. They are familiar with common methods of test management methods and test tools for automating test activities.

Inhalt

1. Basics (introduction, definition of terms, principles of software testing, fundamental test process, psychology of testing)
2. Testing in the software life cycle (general V-model, component test, integration test, system test, acceptance test, testing of new product versions, overview of test types)
3. Static testing (structured group tests, static analyses, metrics)
4. Dynamic testing (black-box procedure, white-box procedure, experience-based test case determination)
5. Test management (test organisation and planning, economic aspects, test strategy, management of test work, error management, requirements for configuration management).
6. Testing tools (types, selection, introduction)
7. Modern test procedures (model-based testing, regression testing, testing of variant-rich systems)
8. Debugging

Anmerkungen

At the end of the course there is also the opportunity to be certified as an "ISTQB - Certified Tester - Foundation Level". A date and the modalities for the exam will be agreed on in the lecture.

Arbeitsaufwand

150h

M**4.181 Modul: Student Innovation Lab [M-ETIT-105073]**

Verantwortung: Prof. Dr.-Ing. Sören Hohmann
 Prof. Dr. Werner Nahm
 Prof. Dr.-Ing. Eric Sax
 Prof. Dr. Wilhelm Stork
 Prof. Dr. Orestis Terzidis
 Prof. Dr.-Ing. Thomas Zwick

Einrichtung: KIT-Fakultät für Elektrotechnik und Informationstechnik

Bestandteil von: [Minor Studies: Electrical Engineering](#)

Leistungspunkte
15 LP

Notenskala
Zehntelnoten

Turnus
Jedes Semester

Dauer
2 Semester

Sprache
Englisch

Level
4

Version
2

Pflichtbestandteile			
T-ETIT-110291	Innovation Lab	9 LP	Hohmann, Nahm, Sax, Stork, Zwick
T-WIWI-102864	Entrepreneurship	3 LP	Terzidis
T-WIWI-110166	SIL Entrepreneurship Projekt	3 LP	Terzidis

Erfolgskontrolle(n)

This module consists of an 60-minute written exam on the contents of the Entrepreneurship lectures, as well as 5 other types of exams on the contents of the seminar Entrepreneurship and Innovation Lab in the form of term papers and presentations. All exams results are graded. Details will be given during the course.

Voraussetzungen

none

Qualifikationsziele**Personal competence**

- Reflection faculty:
The students are able to analyze, evaluate and develop an alternative for action for certain elements of action in social interaction
- Decision-making ability:
The students are able to prepare a decision template in time and to provide the necessary arguments for alternative decisions and therefore are able to decide in time.
- Interdisciplinary teamwork
Students are able to detect their limits of competence in one domain and to adjust to a the non-specialist domain. The students are able to detect a lack in competence and to compensate this lack via competences of other team members. The students are able to communicate their domain-specific knowledge and develop a basic understanding of other domains.
- Value-based action:
The students are able to use selected psychological tools to determine their own values. They are able to match these values with team members and reflect if their offer fits these values.

Social competence

- Ability to cooperate:
The students are able to analyze and judge their cooperative behavior in a group.
- Communication competence:
The students are able to present their information in persuasive, focused and target group oriented way.
- Ability to deal with conflicts:
The students are able to detect conflicts in advance, analyze them and name solution concepts.

Innovation and entrepreneurship competence

- Agile product development:
The students are able to apply methods of agile product development e.g. Scrum.
- Methodical innovation retrieval:
The students are able to conduct processes for user- and technology-centered innovation to develop sustainable value propositions for certain target groups (e.g. Design Thinking (DT), Technology Application Selection (TAS)-process).
- Orientation on management of new technology-based firms (NTBF):
The students are able to name central concepts of intellectual property and legal structures. The students are able to name the most important tasks of entrepreneurial leadership. They are able to name the most common form of business modeling and to setup a business plan. The students know important approaches to establish an organization. The students are able to determine the ownership structure in an investment situation. The students are able to name marketing concepts and setup a business model.
- Generate investment readiness:
The students are able to setup rudimentary revenue and cost plan. Furthermore, they are able to establish a project plan for a company in order to derive an investment plan. The students are able to present their business proposal to investors and develop empathy for the investors.
- Competence to develop a business model:
The students are able to apply respective tools for business modeling e.g. Business Model Canvas. The students are able to develop and assess alternative business models.
- Risk handling:
The students are able to name basic risks w.r.t. requirements, technical limitations and profitability. The students are able to apply methods of customer interaction for evaluation of requirements and willingness to pay. The students are able to setup a rudimentary competitors analyze. The students are able to name and identify risks and present potential reactions.

Systemic technical competence

- Problem solution competence:
The students are able to analyze, assess and structurally solve a technical problem.
- Agile methodology of system development:
The students are able to name and apply different system development processes.
- Validation in volatile environment:
The students are able to conduct technical and economical validation under volatile constraints. For this, they are able to name the constraints and interpret the results of the validation.
- Functional decomposition:
The students are able to identify, interpret and derive functional requirements from complex customer needs.
- Architecture development:
The students are able to recognize coherences from the functional requirements and derive a suitable system architecture.

Inhalt

This module strives to combine technical, social and personal competences from the technical and entrepreneurial domain. The objective is to prepare students as best as possible for entrepreneurial activity within or outside of an established organization. Our teaching methods are research-based with a practical orientation.

The lecture Entrepreneurship as the essential component offers the theoretical basis and provides insight in important theoretical concepts and empirical evidence. Currently released case studies and practical experiences of successful founders support the theoretical and empirical content. In order to run a company for the long term additional knowledge is important. That's why the lecture also teaches basic principles for opportunity recognition, business modeling, an introduction to entrepreneurial marketing and leadership. Customer-based design methods from the lean startup approach as well as methods of technology-centered innovation are presented. Future founders have to be able to develop and handle resources such as financial and human capital, infrastructure and intellectual property. Further aspects tackle the establishment of an organization and funding of the own project.

The knowledge taught in the lecture Entrepreneurship will be applied in an application-oriented seminar and the labs. Hence we use an action learning approach to extend the taught knowledge by practical skills and reflection capabilities. In an team of five, the students will experience their way from the ideation process to the final pitch in front of investors.

The students are able to choose between the following options concerning the labs:

- The Automation Innovation Lab offers drones as an innovation platform for cooperative swarm solutions.
- The Industry 4.0 Innovation Lab enables innovation in the context of the next industrial revolution via mobile robot platforms.
- In the Interconnected Intelligent Systems Lab innovations in the context of Assisted Living and Smart Housing are enabled by providing a rich assembly set of mobile robots, actuators and sensors.
- The Computer Vision for Health Lab offers a selection of state-of-the-art imaging devices and powerful computing hardware for innovative image-based applications for medicine and healthcare.

The module also presents methods of agile system development (Scrum) along with associated validation methods as well as methods for functional prototyping. Gate plans are used within the module to determine the progress of the project. Methods for single person work and teamwork are presented and applied. Additionally group-specific knowledge of the different roles of team members, solutions to conflict situations and interdisciplinary teams are presented.

Zusammensetzung der Modulnote

The module grade consists of the written exam of the Lecture Entrepreneurship (40%), of the submissions and presentation of the Innovation Lab (40%) and of the submissions and presentation of the SIL Entrepreneurship Project (20%).

Anmerkungen

An application is required to participate in this module. Information about the application: www.kit-student-innovation-lab.de.

Arbeitsaufwand

Lecture Entrepreneurship: 32h attendance time, 48h preparation and follow-up time, 10h preparation time for assessment

Seminar Entrepreneurship: 34h attendance time, 3h preparation and follow-up time, 53h preparation time for assessment.

Innovation Lab: 8h attendance time, 213h preparation and follow-up time, 49h preparation time for assessment.

This results in a total of 450 hours and a total of 15 LPs for both semesters ($15 \cdot 30 / 2 = 225$).

Empfehlungen

It is recommended to attend the lecture Entrepreneurship at the same time as the seminar Entrepreneurship Project and the Innovation Lab in the winter semester.

Lehr- und Lernformen**Related courses:**

Lecture Entrepreneurship
Seminar Entrepreneurship Project
Innovation Labs

Please note that the courses must be booked in parallel.

Related exams:

Written exams covering the content of lecture Entrepreneurship
Presentation of the Value Profile (seminar Entrepreneurship)
Submission of the Business Plan (seminar Entrepreneurship)
Submission of a Technical Report with requirements list and system architecture (Innovation Lab)
Submission of the reflection of the Gate Plans (Innovation Lab)
Presentation of the High-fidelity (Innovation Lab)

M**4.182 Modul: Systems and Software Engineering [M-ETIT-100537]**

Verantwortung: Prof. Dr.-Ing. Eric Sax
Einrichtung: KIT-Fakultät für Elektrotechnik und Informationstechnik
Bestandteil von: [Minor Studies: Electrical Engineering](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	3

Pflichtbestandteile			
T-ETIT-100675	Systems and Software Engineering	6 LP	Sax

Erfolgskontrolle(n)

Written exam of 90 minutes.

Students are given the opportunity to earn a grade bonus through separate task assignments. If the grade of the written exam is between 4.0 and 1.3, the bonus improves the grade by a maximum of one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the lecture. Bonus points do not expire and remain valid for exams taken at a later date.

Voraussetzungen

none

Qualifikationsziele

Students are able to analyse and explain the functional principles and applications of embedded systems.

- Students are able to evaluate and apply maturity models as well as Software Development Life Cycle models including the waterfall model, V-model, prototyping model, agile models, and DevOps.
- Students are able to apply various creativity techniques to develop innovative solutions to problems. They will be able to derive and analyse requirements.
- Students are familiar with diagram formats software modelling languages; they can evaluate and create these based on problem descriptions of an application area. They will be able to create and evaluate functional, data-oriented, algorithmic, state-oriented, and object-oriented views.
- Students are able to understand and apply various aspects of the realization of embedded systems. They will be able to consider implementation alternatives: hardware, co-design and scheduling aspects.
- Students are familiar with the various testing phases in a project and can explain them. They can assess the reliability of a system and understand the concept of functional safety.

Inhalt

The focus of the course is on processes and methods for the design of systems composed of electrical, electronic and electronically programmable systems that contain software, hardware and mechanical components. The desired competencies of the course include the knowledge and goal-oriented use of modeling techniques, design processes, description and representation tools as well as specification languages that correspond to the current state of the art.

Zusammensetzung der Modulnote

The grade is determined by the written exam and the bonus points.

Arbeitsaufwand

- 15 weeks of 1.5h attendance in lecture and of 1.5h attendance in each exercise = 45 h
- 15 weeks of 2h preparation and follow-up per lecture and at least 3h preparation for each exercise (includes processing of exercise sheets and additional learning material, the processing of tasks for the acquisition of bonus points) = 75h
- Preparation for the exam = 60h

In total: 180h

Empfehlungen

Knowledge in Digital Technology and Information and Automation Technology (e.g. module M-ETIT-102102 and M-ETIT-106336)

M**4.183 Modul: Systems Engineering for Automotive Electronics [M-ETIT-100462]**

Verantwortung: Prof. Dr.-Ing. Jürgen Bortolazzi
Einrichtung: KIT-Fakultät für Elektrotechnik und Informationstechnik
Bestandteil von: [Minor Studies: Electrical Engineering](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-ETIT-100677	Systems Engineering for Automotive Electronics	4 LP	Bortolazzi

Erfolgskontrolle(n)

Die Art und Weise (schriftliche oder mündliche Prüfung) der Erfolgskontrolle wird zu Beginn der Veranstaltung bekanntgegeben.

Die Prüfung findet ohne Hilfsmittel statt.

Voraussetzungen

keine

Qualifikationsziele

Die Studierenden kennen den systematischen Entwicklungsprozess von elektrischen und elektronischen Systemen und Architekturen im Umfeld der Fahrzeugtechnik sowie der Automobilindustrie. Sie sind in der Lage die systematische Entwicklung unterstützenden Werkzeuge anzuwenden sowie Elektrik- und Elektronikarchitekturen modellbasiert zu beschreiben. Sie können in den Domänen funktionale und physikalische Modellierung Systeme analysieren und beurteilen.

Inhalt

Die Vorlesung vermittelt Kenntnisse bezüglich Methoden, Techniken und Vorgehensweisen die in den Phasen der Entwicklung von elektrischen und elektronischen Systemen für Fahrzeuge zum Einsatz kommen.

Zusammensetzung der Modulnote

Die Modulnote ist die Note der schriftlichen bzw. mündlichen Prüfung. Der Besuch von Labor / Übung zur Vorlesung ist Voraussetzung für die Zulassung zur Prüfung.

Anmerkungen

Die Art und Weise (schriftliche oder mündliche Prüfung) der Erfolgskontrolle wird zu Beginn der Veranstaltung bekanntgegeben.

Die Vorlesung wird im Haupttermin schriftlich geprüft, für den Nachholtermin kann die Prüfung auch mündlich erfolgen.

Die Prüfung findet ohne Hilfsmittel statt.

Der Besuch von Labor / Übung zur Vorlesung ist Voraussetzung für die Zulassung zur Prüfung.

Arbeitsaufwand

1. Präsenzzeit in Vorlesung und Übung: 60 h
2. Vor-/Nachbereitung derselbigen: 30 h
3. Klausurvorbereitung: 30 h

Empfehlungen

Empfohlen wird der Besuch der Vorlesung SE (23611)

M**4.184 Modul: Telematics [M-INFO-107243]**

Verantwortung: Prof. Dr. Martina Zitterbart
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114269	Telematics	6 LP	Zitterbart

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students

- master protocols, architectures, and methods and algorithms that are used on the Internet for routing and for establishing a reliable end-to-end connection, as well as various media allocation procedures in local networks.
- have an understanding of the systems and the problems that appear in a global, dynamic network as well as the mechanisms used to remedy them.
- are familiar with current developments such as SDN and data center networking.
- know methods to manage and administrate networks.

Students master the basic protocol mechanisms for establishing reliable end-to-end communication. Students have detailed knowledge of the mechanisms used in TCP for congestion and flow control and can discuss the issue of fairness with multiple parallel transport streams. Students can analytically determine the performance of transport protocols and know methods that fulfill special requirements of TCP, such as high data rates and short latencies. Students are familiar with current topics such as problems introduced by utilization of middle boxes in the Internet, the use of TCP in data centers and multipath TCP. Students can use transport protocols in practice.

Students know the functions of routers in the Internet and can reproduce and apply common routing algorithms. Students can reproduce the architecture of a router and know different approaches to buffer placement as well as their advantages and disadvantages.

Students understand the distinction of routing protocols into interior and exterior gateway protocols and have detailed knowledge of the functionality and properties of common protocols such as RIP, OSPF and BGP. The students are familiar with current topics such as SDN.

Students know the function of media allocation and can classify and analytically evaluate media allocation processes. Students have in-depth knowledge of Ethernet and are familiar with various Ethernet forms and their differences, especially current developments such as real-time Ethernet and data center Ethernet. Students can reproduce and apply the spanning tree protocol.

Students can reproduce the technical characteristics of DSL. Students are familiar with the concept of label switching and can compare existing approaches such as MPLS.

Inhalt

- Introduction
- End-to-end data transport
- Routing protocols and architectures
- Media allocation
- Bridges
- Data transmission
- Further selected examples
- Network management

Arbeitsaufwand

Lecture with 3 SWS plus follow-up/exam preparation, 6 CP.

6 CP corresponds to approx. 180 working hours, of which

approx. 60 hours lecture attendance

approx. 60 hours preparation/follow-up work

approx. 60 hours exam preparation

M**4.185 Modul: Testing Digital Systems I [M-INFO-100851]****Verantwortung:** Prof. Dr. Mehdi Baradaran Tahoori**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Unregelmäßig**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1

Pflichtbestandteile			
T-INFO-101388	Testing Digital Systems I	3 LP	Tahoori

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The course provides the basic techniques for testing digital circuits

Inhalt

Testing of digital circuits plays a critical role during the design and manufacturing cycles. It also ensures the quality of parts shipped to the customers. Test generation and design for testability are integral parts of automated design flow of all electronics products. The objective of this course is to provide the foundations for developing test methods for digital systems and provides the techniques necessary to practice design for testability.

This course encompasses the theoretical and practical aspects of digital systems testing and the design of easily testable circuits. Topics include Introduction to Testing (testing definition, types of test, automatic test equipment, test economics, and quality models), Failures and Errors (definitions, failure modes, failure mechanisms, reliability defects), Faults (fault models, stuck-at faults, bridging faults, timing faults, transistor-level faults, functional-level faults, effectiveness of different fault models based on real data), Logic and Fault Simulation (fault equivalence and fault collapsing, true-value simulation, fault simulation algorithms, statistical methods), Test Generation for Combinational Circuits (algebraic methods, path-tracing (D-alg, PODEM, FAN), testability metrics, test file compression), Digital Design-For-Testability and Internal Scan Design (ad-hoc methods, scan architectures, scan-based test methodology).

Arbeitsaufwand

2 SWS: (2 SWS + 1,5 x 2 SWS) x 15 + 15 h preparation for the exam = 90 h = 3 ECTS

M**4.186 Modul: Testing Digital Systems II [M-INFO-102962]****Verantwortung:** Prof. Dr. Mehdi Baradaran Tahoori**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [Area of Specialization: Design of Embedded Systems and Computer Architectures](#)
[Elective Studies in Informatics](#)**Leistungspunkte**
3 LP**Notenskala**
Zehntelnoten**Turnus**
Jedes Sommersemester**Dauer**
1 Semester**Sprache**
Englisch**Level**
4**Version**
1**Pflichtbestandteile**

T-INFO-105936	Testing Digital Systems II	3 LP	Tahoori
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Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The objective of this course is to provide more advanced topics on testing of digital systems and complement the foundation covered in Testing Digital Systems I.

Inhalt

Testing of digital circuits plays a critical role during the design and manufacturing cycles. It also ensures the quality of parts shipped to the customers. Test generation and design for testability are integral parts of automated design flow of all electronic products. The objective of this course is to provide more advanced topics on testing of digital systems and complement the foundation covered in Testing Digital Systems I.

Topics include Functional and Structural Testing (design verification vectors, exhaustive test, pseudo-exhaustive test, pseudo-random testing), Essentials of Test Generation for Sequential Circuits (state-machine initialization, time-frame expansion method), Built-in Self Test (test economics of BIST, test pattern generation, output response analysis, BIST architectures), Boundary Scan (Boundary scan architectures, BS test methodology), Delay Testing (path delay test, hazard-free, robust, and non-robust delay tests), transition faults, delay test schemes), Current-Based Testing (motivation, test vectors for IDDQ, variations of IDDQ), Memory Test (memory test algorithm, memory BIST, memory repair), and DFT for System-on-Chip.

Arbeitsaufwand

2 SWS: (2 SWS + 1.5 x 2 SWS) x 15 + 15 h exam preparation = 90 h = 3 ECTS

Empfehlungen

Knowledge of Digital Design and Computer Architecture is helpful.

M**4.187 Modul: Text Indexing [M-INFO-107202]**

Verantwortung: Prof. Dr. Peter Sanders
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
5 LP	Zehntelnoten	Jedes Wintersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114226	Text-Indexing	4 LP	Sanders
T-INFO-114227	Text-Indexing Project/Experiment	1 LP	Sanders

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students acquire a systematic understanding of algorithmic issues and solution approaches in the area of text indexing, building on existing knowledge in the subject area of algorithms. They will also be able to apply learned techniques to related problems and interpret and comprehend current research topics in the area of text indexing.

Upon successful completion of the course, students will be able to:

- explain terms, structures, basic problem definitions, and algorithms from the lecture;
- select which algorithms and data structures are suitable for solving a problem and, if necessary, adapt them to the requirements of a specific problem;
- use algorithms and data structures, analyze them mathematically, and prove the algorithmic properties.

Inhalt

In this lecture we deal with algorithms and data structures for texts, especially text indices. Text indices are data structures that provide additional information about a text in order to accelerate queries regarding this text. These can be simple pattern matching queries ("Does a pattern occur in the text?") or more complex data mining queries ("Which pattern of a certain length occurs most often in the text?").

Furthermore, we deal with text compression. Here, we want to represent a text as space-efficiently as possible. However, we have to make sure that the original text can be reconstructed completely. Here, we speak of lossless compression. In the lecture, we will learn about techniques that are used in compression programs such as gzip.

Anmerkungen**Arbeitsaufwand**

The lectures including the project/experiment with 5 ECTS corresponds to 150 working hours, which are divided approximately as follows:

- ca. 30 hours attending lectures
- ca. 60 hours preparing and following-up lectures
- ca. 30 hours working on the project/experiment
- ca. 30 hours preparing for the examination

Empfehlungen

The lecture builds on parts of the contents of the lectures Algorithms I and Algorithms II. Corresponding knowledge is therefore helpful.

M**4.188 Modul: Theory of Satisfiability Algorithms [M-INFO-107639]**

Verantwortung: Prof. Dr.-Ing. Marvin Künnemann
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Algorithm Engineering](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Unregelmäßig	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114982	Theory of Satisfiability Algorithms	4 LP	Künnemann

Erfolgskontrolle(n)

See Partial Achievements (Teilleistung).

Voraussetzungen

See Partial Achievements (Teilleistung).

Qualifikationsziele

Students know the foundations of the rigorous theoretical analysis of algorithms for the Satisfiability problem and its many variants. They have an overview over the state of the art of worst-case time upper bounds. They understand further theoretical aspects such as combinatorial properties, hardness conjectures, etc. and are able to derive implications for the (sub)exponential-time solvability of related problems.

Inhalt

The Satisfiability (SAT) problem for Boolean formulas is fundamental for computer science, due to a variety of applications (cf. the practical success of SAT solvers) as well as its central role for the theory of computation. This course focuses on theoretical aspects of the problem, which includes topics among:

- exponential-time algorithms with provable worst-case guarantees (e.g., for k-SAT, CNF-SAT and further variants)
- tractable special cases of the problem
- classification of constraint satisfaction problems
- combinatorial insights (e.g., the algorithmic Lovasz Local Lemma)
- the sparsification lemma and its limits
- generalizations: MaxSAT, #SAT, Quantified SAT
- hardness hypotheses and their implications: ETH, SETH, NSETH, MASETH, Formula-SETH
- isolation lemma

Arbeitsaufwand

4 CP correspond to approx. 120 h of work, estimated as follows:

- ca. 30 h attendance in lectures
- ca. 60 h follow-up on lectures, solving exercises, independent review
- ca. 30 h preparation for exam

Empfehlungen

See Partial Achievements (Teilleistung).

M**4.189 Modul: Timed Systems [M-INFO-106293]****Verantwortung:** Jun.-Prof. Dr. Maike Schwammberger**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
6 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	3

Pflichtbestandteile			
T-INFO-112754	Timed Systems	6 LP	Schwammberger

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students can independently model and analyze software systems with time components. To this end, they can select the appropriate modeling method for the application area from a range of different modeling methods. With the help of formal methods and practical tools (UPPAAL), students analyze their modeling with regard to correctness. Students can transfer the methods they have learned to current problems.

Inhalt

Many of the (embedded) software systems we are confronted with in everyday life have time-critical functionalities. For example, an airbag should be activated within a certain, very short period of time in the event of an accident. Similarly, we expect fast response times from the various apps on our smartphones in order to use them conveniently and effectively.

"Time" is therefore a decisive factor when modeling software systems. This lecture describes various mechanisms for formalizing so-called real-time systems. In addition to modeling, the lecture also focuses on the analysis of systems. The following topics are covered in particular:

- Timed automata (an extension of finite automata by time)
- Model checking of timed automata with the help of UPPAAL
- Duration calculus (a logic that talks about time intervals)
- Extensions and applications of timed systems

The weekly lecture consists of both theoretical and applied parts. For application and transfer of the contents, voluntary exercises are offered, which are discussed in the bi-weekly exercise.

Arbeitsaufwand

4 SWS lecture

6 ECTS equals 180 working hours, of which

approx. 40 hours attending the lecture (theoretical and applied part)

approx. 70 hours preparation and follow-up

approx. 40 hours working on in-depth exercises

approx. 30 hours exam preparation

Empfehlungen

Basic knowledge in areas of theoretical computer science and modeling of (embedded) software systems is helpful (e.g. temporal logics, finite automata, predicate logic), but is not required.

Literatur

The book "E.-R. Olderog, H. Dierks: Real-Time Systems" is used as reading material for some of the lecture contents (<https://doi.org/10.1017/CBO9780511619953>).

M**4.190 Modul: Type Systems and Program Analysis [M-INFO-107664]****Verantwortung:** Prof. Dr. Sebastian Erdweg**Einrichtung:** KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Theoretical Foundations](#)
[Area of Specialization: Software Engineering and Compiler Construction](#)
[Elective Studies in Informatics](#)

Leistungspunkte
 6 LP

Notenskala
 Zehntelnoten

Turnus
 Unregelmäßig

Dauer
 1 Semester

Sprache
 Englisch

Level
 4

Version
 1

Pflichtbestandteile			
T-INFO-115016	Type Systems and Program Analysis	6 LP	Erdweg

Erfolgskontrolle(n)

See Partial Achievements (Teilleistung).

Voraussetzungen

See Partial Achievements (Teilleistung).

Qualifikationsziele

- Students can understand, design, and implement static type systems.
- Students can validate the soundness of static type systems.
- Students know the foundations of abstract interpretation.
- Students can understand, design, and implement data-flow analyses.
- Students can assess analyses based on their soundness, recall, and precision.

Inhalt

Static type systems and program analyses predict the behavior of programs through source-code analysis, but without executing the program. They are fundamental for isolating ill-defined program behavior (e.g., from type errors), but can also find bugs or also can provide guarantees (e.g., memory safety). Topics covered in this course include the design, definition, formalization, and application of type systems and static program analyses, including type inference, dependent typing, control flow, data flow, abstract interpretation, soundness, and precision.

Arbeitsaufwand

1. Attendance time 4 h x 15 weeks = 60 h
2. Self-study 7 h x 15 weeks = 105 h
3. Preparation for the exam 15 h

Empfehlungen

Knowledge of programming semantics is helpful but not required.
 Knowledge of logic inference systems is helpful but not required.

M**4.191 Modul: Ubiquitous Computing [M-INFO-107161]**

Verantwortung: Prof. Dr.-Ing. Michael Beigl
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [Area of Specialization: Telematics](#)
[Elective Studies in Informatics](#)

Leistungspunkte
5 LP

Notenskala
Zehntelnoten

Turnus
Unregelmäßig

Dauer
1 Semester

Sprache
Englisch

Level
4

Version
1

Pflichtbestandteile			
T-INFO-114188	Ubiquitous Computing	5 LP	Beigl

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

The aim of the lecture is to impart knowledge of the fundamentals and advanced methods and techniques of ubiquitous computing. After completing the lecture, students will be able to reproduce and discuss what they have learnt about existing ubiquitous computing systems.
 evaluate the general knowledge of ubiquitous systems and transfer statements and laws to special cases.
 evaluate and assess different methods for design processes and user studies and select suitable methods for the development of new solutions.
 invent, plan, design and evaluate new ubiquitous systems for use in everyday or industrial process environments and assess the costs and technical implications.

Inhalt

The lecture provides an overview of the history and teaches the concepts, theories and methods of ubiquitous information technology (ubiquitous computing). Based on the appliance concept, students then design their own appliances in the exercise, plan the construction and then develop them. The necessary technical and methodological basics such as hardware for ubiquitous systems, software for ubiquitous systems, principles of context recognition for ubiquitous systems, networking of ubiquitous systems and design of ubiquitous systems and in particular information appliances are discussed. Methods of design and testing for human-machine interaction and human-machine interfaces developed in ubiquitous computing are explained in detail. There is also an introduction to the economic aspects of a ubiquitous system.

In the practical part of the lecture, the understanding of ubiquitous systems is deepened through practical application of the knowledge base of the lecture. The students design and develop their own appliance and test it. The aim is to have gone through the steps towards a prototypical and possibly marketable appliance.

Arbeitsaufwand

The total workload for this course unit is approximately 150 hours (5.0 credits).

Activity Workload:

- Attendance time: Attendance of the lecture
15 x 90 min; 22 h 30 min
- Attendance time: Attendance of the exercise
15 x 45 min; 11 h 15 min
- Preparation / follow-up of the lecture and exercise
15 x 90 min; 22 h 30 min
- Developing a self-developed concept for an information appliance
33 h 45 min
- Go through set of slides 2x
2 x 12 h; 24 h 00 min
- Prepare exam
36 h 00 min

TOTAL
150 h 00 min

M**4.192 Modul: Wearable Robotic Technologies [M-INFO-107113]**

Verantwortung: Prof. Dr.-Ing. Tamim Asfour
Prof. Dr.-Ing. Michael Beigl

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [Area of Specialization: Robotics and Automation](#)
[Area of Specialization: Human-centred Machine Intelligence](#)
[Elective Studies in Informatics](#)

Leistungspunkte	Notenskala	Turnus	Dauer	Sprache	Level	Version
4 LP	Zehntelnoten	Jedes Sommersemester	1 Semester	Englisch	4	1

Pflichtbestandteile			
T-INFO-114145	Wearable Robotic Technologies	4 LP	Asfour, Beigl

Erfolgskontrolle(n)

See partial achievements (Teilleistung)

Voraussetzungen

See partial achievements (Teilleistung)

Qualifikationsziele

Students can explain the fundamentals, motivation, and applications of wearable robotic technologies including exoskeletons, orthoses, and prostheses. They have comprehensive knowledge of human anatomy, biomechanics, and the neuromuscular system relevant to wearable robot design.

Students understand the core components and challenges associated with the design of wearable robotic technologies, including mechanisms, actuation technologies, sensing approaches, and wearable computing concepts. They can analyze design requirements and challenges including kinematic compatibility and human-robot interfaces for wearable robots.

Students can explain control strategies for prosthetic and exoskeleton systems, understand both physical and cognitive aspects of human-robot interaction, and comprehend concepts such as body schema and symbiotic human-machine interaction in wearable systems.

Inhalt

The lecture provides a comprehensive introduction to wearable robotic technologies. It covers human anatomy and biomechanics as foundations for wearable robot design. Joint biomechanics, locomotion and gait analysis, the neuromuscular system, and models of the human body are discussed to understand how wearable robots can be designed and controlled.

Design principles and mechanisms of wearable robots are addressed. Actuation technologies are covered in depth: requirements analysis based on human motion and user needs, compliant actuation approaches (serial elastic, parallel elastic, variable stiffness actuation), and strategies for increasing efficiency such as energy harvesting, multi-articulation, and underactuation.

The lecture covers wearable computing principles such as body signals and phenomena, design principles for body-worn systems, attachment considerations based on body maps for heat tolerance, weight distribution, and flexibility, and fabric design with textile-embedded electronics.

Exoskeleton systems are presented, covering application areas, design challenges such as redundancy, kinematic compatibility, and misalignments, various types of exoskeletons, and the general design process. Prosthetic systems cover types of hand and lower limb prostheses, hand synergies for control, and mechanical interfaces through socket design.

Human-robot interaction addresses both physical and cognitive interfaces. Control strategies are presented for prosthetic systems (semi-autonomous grasping) and exoskeleton systems (hierarchical control, assistance evaluation). The concept of body schema from both neuroscience and robotics perspectives is discussed, emphasizing symbiotic human-machine interaction in tightly coupled wearable systems. State-of-the-art examples across various application domains illustrate current research and developments.

Arbeitsaufwand

Lecture with 2 SWS, 4 ECTS.

4 ECTS corresponds to ca. 120 hours, thereof

ca. $15 \cdot 2h = 30h$ attendance in the lecture

ca. $15 \cdot 3h = 45h$ preparation and follow-up of the lecture

ca. 45h preparation for the exam and exam itself

5 Teilleistungen

T

5.1 Teilleistung: Access Control Systems: Models and Technology [T-INFO-112775]

Verantwortung: Prof. Dr. Hannes Hartenstein

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-106303 - Access Control Systems: Models and Technology](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	5 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	2400147	Access Control Systems: Models and Technology	3 SWS	Vorlesung / Übung (VÜ) / 	Hartenstein, Hemken

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as a written examination (§ 4 Abs. 2 Nr. 1 SPO) lasting 60 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (§ 6 Abs. 3 SPO) whether the examination takes place

- in the form of an oral examination lasting 30 minutes pursuant to § 4 Abs. 2 Nr. 2 SPO or
- in the form of a written examination lasting 60 minutes in accordance with § 4 Abs. 2 Nr. 1 SPO.

Voraussetzungen

Keine.

Empfehlungen

Basics according to the lectures "Information Security" and "IT Security Management for Networked Systems" are recommended.

T

5.2 Teilleistung: Advanced Bayesian Data Analysis [T-INFO-113673]**Verantwortung:** Prof. Dr. Nadja Klein**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106812 - Advanced Bayesian Data Analysis](#)**Teilleistungsart**
Prüfungsleistung schriftlich**Leistungspunkte**
5 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400120	Advanced Bayesian Data Analysis	3 SWS	Vorlesung / Übung (VÜ) / ●	Klein

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 90 minutes.

A bonus can be acquired through successful participation in the exercise as a success control of a different kind (§4(2), 3 SPO 2008) or study performance (§4(3) SPO 2015). The exact criteria for awarding a bonus will be announced at the beginning of the lecture. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The bonus is only valid for the main and post exams of the semester in which it was earned. After that, the grade bonus expires.

Empfehlungen

- Knowledge in R or Python
- Mathematics-heavy lecture. The basics will be reviewed, but mathematical proficiency is helpful

T

5.3 Teilleistung: Advanced Data Structures [T-INFO-114223]

Verantwortung: Prof. Dr. Peter Sanders

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107200 - Advanced Data Structures](#)

Teilleistungsart
Prüfungsleistung mündlich

Leistungspunkte
4 LP

Notenskala
Drittelnoten

Turnus
Jedes Sommersemester

Version
1

Lehrveranstaltungen					
SS 2026	2400164	Advanced Data Structures	3 SWS	Vorlesung (V) / ●	Sanders, Hermann, Walzer, Groot Koerkamp

Legende: 📺 Online, 🔄 Präsenz/Online gemischt, ● Präsenz, ✕ Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Voraussetzungen

None.

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Die Teilleistung [T-INFO-114224 - Advanced Data Structures Project/Experiment](#) muss begonnen worden sein.

Empfehlungen

The lecture builds on parts of the contents of the lectures Algorithms I and Algorithms II. Corresponding knowledge is therefore helpful.

T**5.4 Teilleistung: Advanced Data Structures Project/Experiment [T-INFO-114224]**

Verantwortung: Prof. Dr. Peter Sanders
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107200 - Advanced Data Structures](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	1 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	2400164	Advanced Data Structures	3 SWS	Vorlesung (V) / ●	Sanders, Hermann, Walzer, Groot Koerkamp

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The examination takes place in the form of an examination of another type (§ 4 Abs. 2 No. 3 SPO) in form of a project/experiment.

An overall grade is awarded.

Voraussetzungen

None.

Empfehlungen

The lecture builds on parts of the contents of the lectures Algorithms I and Algorithms II. Corresponding knowledge is therefore helpful.

T**5.5 Teilleistung: Advanced Machine Learning and Data Science [T-WIWI-111305]****Verantwortung:** Prof. Dr. Maxim Ulrich**Einrichtung:** KIT-Fakultät für Wirtschaftswissenschaften**Bestandteil von:** [M-WIWI-105659 - Advanced Machine Learning and Data Science](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
9 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
5

Lehrveranstaltungen					
SS 2026	2500016	Advanced Machine Learning and Data Science	4 SWS	Projekt (PRO) / 	Ulrich

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

Die Erfolgskontrolle erfolgt in Form einer Prüfungsleistung anderer Art. Die Note wird auf der Grundlage der Zwischenpräsentationen während des Projekts, der Qualität der Implementierung, der schriftlichen Abschlussarbeit und einer Endpräsentation bewertet.

Anmerkungen

The course is aimed at students majoring in data science and/or machine learning. It offers students the opportunity to gain practical knowledge of new developments in the fields of data science and machine learning.

Kick-off Meeting for Winter Term 2025/26:

We will introduce project topics designed for students of computer science, engineering, and mathematics. The meeting will provide an overview of the course structure, project topics, and expectations for the semester.

- **Date:** Tuesday, October 28, 2025, Time: 5:00 PM – 7:00 PM
- **Format:** Online (MS Teams link: Kick-Off (WS-2025)-Tuesday, October 28, 2025 | Meeting-Join | Microsoft Teams)
- **Ms Team Meeting link:** <https://teams.microsoft.com/meet/3867172715541?p=GS2swv7ueF5gACIUEs>
- **Form link:** https://docs.google.com/forms/d/e/1FAIpQLSek36NGe9I7JyIlIc4iBnd-XEwbhexFS7Wn_P6Wga1-EKU8Qw/viewform?usp=dialog

After the meeting (within 48 hours): Please submit your Priority A and Priority B topic preferences via the form within 48 hours after the kick-off.

If you need more information, please visit <https://risk.fbv.kit.edu/2307.php>.

Arbeitsaufwand

270 Std.

T

5.6 Teilleistung: Algorithm Engineering [T-INFO-101332]

Verantwortung: Prof. Dr. Peter Sanders
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-100795 - Algorithm Engineering](#)

Teilleistungsart
 Prüfungsleistung mündlich

Leistungspunkte
 4 LP

Notenskala
 Drittelnoten

Turnus
 Jedes Sommersemester

Version
 4

Lehrveranstaltungen					
WS 25/26	2400021	Algorithm Engineering		Vorlesung (V) / 	Sanders, Seemaier
SS 2026	2400022	Algorithm Engineering	3 SWS	Vorlesung (V) / 	Sanders, Hermann

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination lasting approximately 20 minutes (§ 4 Abs. 2 Nr. 2 SPO).

Voraussetzungen

None.

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Die Teilleistung [T-INFO-111856 - Algorithm Engineering Pass](#) muss begonnen worden sein.

T**5.7 Teilleistung: Algorithm Engineering Pass [T-INFO-111856]**

Verantwortung: Prof. Dr. Peter Sanders
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-100795 - Algorithm Engineering](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	1 LP	Drittelnoten	Jedes Sommersemester	1

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 2 Abs. 2 Nr. 3).

The exercise can be evidenced by various performance records. This is determined individually during the lecture. Usually, the student prepares a seminar presentation and/or works on a practical tasks with written elaboration and evaluation (the main performance consists of the programming, documented by the source code that is to be handed in and supplemented by a short written report).

Students may redraw from the examination during the first four weeks after they have been assigned a task.

An overall grade is awarded.

Voraussetzungen

None.

T

5.8 Teilleistung: Algorithmic Graph Theory [T-INFO-113918]

Verantwortung: Dr. rer. nat. Torsten Ueckerdt

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-106960 - Algorithmic Graph Theory](#)

Teilleistungsart
Prüfungsleistung mündlich

Leistungspunkte
5 LP

Notenskala
Drittelnoten

Turnus
Unregelmäßig

Version
1

Lehrveranstaltungen					
SS 2026	2400028	Algorithmic Graph Theory - findet im SS 26 nicht statt	3 SWS	Vorlesung / Übung (VÜ) / 	Ueckerdt

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Voraussetzungen

None.

Empfehlungen

Knowledge of the basics of graph theory and algorithm technology is helpful

T

5.9 Teilleistung: Algorithms for Visualization of Graphs [T-INFO-113919]

Verantwortung: Dr. rer. nat. Torsten Ueckerdt

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-106961 - Algorithms for Visualization of Graphs](#)

Teilleistungsart
Prüfungsleistung mündlich

Leistungspunkte
5 LP

Notenskala
Drittelnoten

Turnus
Unregelmäßig

Version
1

Lehrveranstaltungen					
SS 2026	2400155	Graph Drawing	3 SWS	Vorlesung / Übung (VÜ) / 	Ueckerdt, Merker

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Voraussetzungen

None.

Empfehlungen

Knowledge of the basics of graph theory and algorithm technology is helpful.

T

5.10 Teilleistung: Algorithms II [T-INFO-114225]

Verantwortung: Prof. Dr. Peter Sanders
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107201 - Algorithms II](#)

Teilleistungsart
 Prüfungsleistung schriftlich

Leistungspunkte
 6 LP

Notenskala
 Drittelnoten

Turnus
 Jedes Wintersemester

Version
 1

Lehrveranstaltungen					
WS 25/26	2424079	Algorithms II	4 SWS	Vorlesung (V) / 	Sanders, Künnemann, Redzic, Stieß

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 120 minutes.

Voraussetzungen

None.

T**5.11 Teilleistung: Artificial Intelligence & IT-Security [T-INFO-113668]****Verantwortung:** Prof. Dr. Christian Wressnegger**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106810 - Artificial Intelligence & IT-Security](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	6 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	24000	Artificial Intelligence & IT-Security	4 SWS	Vorlesung / Übung (VÜ) / 	Wressnegger

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 120 minutes.

Depending on the number of participants, it will be announced six weeks before the examination performance (§ 6 Abs. 3 SPO) whether the performance review will be

- in the form of an oral examination according to § 4 Abs. 2 no. 2 SPO **or**
- in the form of a written examination according to § 4 Abs. 2 Nr. 1 SPO

takes place.

Voraussetzungen

None.

Empfehlungen

The basics of IT security and artificial intelligence are a prerequisite.

T

5.12 Teilleistung: Automated Planning and Scheduling [T-INFO-109085]**Verantwortung:** Prof. Dr. Peter Sanders**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-104447 - Automated Planning and Scheduling](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	5 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
WS 25/26	2400026	Nicht im WS 2025/26 - Automated Planning and Scheduling		Vorlesung / Übung (VÜ)	Schreiber, Sanders
SS 2026	2400108	Nicht im SoSe 2026! Automated Planning and Scheduling		Vorlesung / Übung (VÜ)	Schreiber, Sanders

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 30 minutes.

Voraussetzungen

None.

T**5.13 Teilleistung: Autonomous Learning for Intelligent Robot Perception [T-INFO-113327]****Verantwortung:** Prof. Dr. Rudolph Triebel**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106608 - Autonomous Learning for Intelligent Robot Perception](#)**Teilleistungsart**
Prüfungsleistung schriftlich**Leistungspunkte**
4 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400213	Autonomous Learning for Intelligent Robot Perception	2 SWS	Vorlesung / Übung (VÜ) / ●	Triebel

Legende: ■ Online, 🌀 Präsenz/Online gemischt, ● Präsenz, ✕ Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 120 minutes.

Voraussetzungen

None.

Empfehlungen

A basic understanding of probability theory and linear algebra is required

T**5.14 Teilleistung: Beating the Worst-Case in Practice: Unexpectedly Efficient Algorithms [T-INFO-114898]****Verantwortung:** TT-Prof. Dr. Thomas Bläsius**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107599 - Beating the Worst-Case in Practice: Unexpectedly Efficient Algorithms](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	6 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
WS 25/26	2400121	Beating the Worst Case in Practice: Unerwartet effiziente Algorithmen	4 SWS	Praktikum (P) / ●	Wilhelm, von der Heydt, Bläsius
WS 25/26	2400253	Beating the Worst-Case in Practice: Unexpectedly Efficient Algorithms	4 SWS	Praktikum (P) / ●	Bläsius, Wilhelm

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). An implementation and a written elaboration must be prepared.

Voraussetzungen

None.

T

5.15 Teilleistung: Biologically Inspired Robots [T-MACH-113856]

Verantwortung: Prof. Dr.-Ing. Arne Rönnau**Einrichtung:** KIT-Fakultät für Maschinenbau**Bestandteil von:** [M-MACH-106903 - Biologically Inspired Robots](#)**Teilleistungsart**
Prüfungsleistung
mündlich**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes
Sommersemester**Dauer**
1
Sem.**Version**
1**Lehrveranstaltungen**

SS 2026	2122330	Biologically Inspired Robots	2 SWS	Vorlesung (V) / 	Rönnau
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Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt**Erfolgskontrolle(n)**

Die Erfolgskontrolle erfolgt in Form einer mündlichen Prüfung (ca. 15-20min.)

Voraussetzungen

keine

Empfehlungen

Es ist empfehlenswert zuvor die LV „Robotik I“ zu hören.

Anmerkungen

Die Lehrveranstaltung wird in englischer Sprache angeboten.

Arbeitsaufwand

90 Std.

T**5.16 Teilleistung: Business Innovation in Optics and Photonics [T-ETIT-104572]****Verantwortung:** Prof. Dr. Werner Nahm**Einrichtung:** KIT-Fakultät für Elektrotechnik und Informationstechnik**Bestandteil von:** [M-ETIT-101834 - Business Innovation in Optics and Photonics](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	4 LP	Drittelnoten	Jedes Wintersemester	2

Lehrveranstaltungen					
WS 25/26	2305742	Business Innovation in Optics and Photonics	2 SWS	Vorlesung (V) / ●	Riedel, Nahm
WS 25/26	2305743	Exercise for 2305742 Business Innovation in Optics and Photonics	1 SWS	Übung (Ü) / ●	Riedel, Nahm

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

Type of Examination: examination of another type

Duration of Examination: 4 group presentations à 20 minutes (approx.)

Modality of Exam: The exam consists of four group presentations. 2nd day: Technology Presentation. 3rd day: Development plan presentation. 4th day: Business Canvas presentation. Final presentation: Business pitch. The overall impression is rated.

Voraussetzungen

none

Empfehlungen

Good knowledge in optics & photonics. Personal motivation and interest for getting deeper into business development aspects, methods and tools. Commitment to active, regular and continuous participation in the group work.

T**5.17 Teilleistung: CAD Engineering Project for Intelligent Systems [T-MACH-113857]****Verantwortung:** Prof. Dr.-Ing. Arne Rönnau**Einrichtung:** KIT-Fakultät für Maschinenbau**Bestandteil von:** [M-MACH-106905 - CAD Engineering Project for Intelligent Systems](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Dauer**
1 Sem.**Version**
2

Lehrveranstaltungen					
WS 25/26	2122331	CAD Engineering Project for Intelligent Systems	4 SWS	Projekt (PRO) / ●	Rönnau
SS 2026	2122331	CAD Engineering Project for Intelligent Systems	4 SWS	Projekt (PRO) / ●	Rönnau

Legende: ■ Online, 🔄 Präsenz/Online gemischt, ● Präsenz, ✕ Abgesagt

Erfolgskontrolle(n)

Prüfungsleistung anderer Art. Konstruktionsprojekt sowie schriftliche Ausarbeitung im Team und ein Abschlussvortrag. Der Gesamteindruck wird bewertet. Nähere Angaben erfolgen zu Beginn der Veranstaltung.

Voraussetzungen

keine

Anmerkungen

Die Lehrveranstaltung wird in englischer Sprache angeboten.

Arbeitsaufwand

180 Std.

T**5.18 Teilleistung: Channel Coding: Algebraic Methods for Communications and Storage [T-ETIT-111244]****Verantwortung:** Prof. Dr.-Ing. Laurent Schmalen**Einrichtung:** KIT-Fakultät für Elektrotechnik und Informationstechnik**Bestandteil von:** [M-ETIT-105616 - Channel Coding: Algebraic Methods for Communications and Storage](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Dauer	Version
Prüfungsleistung mündlich	3 LP	Drittelnoten	Jedes Sommersemester	1 Sem.	1

Lehrveranstaltungen					
SS 2026	2310546	Channel Coding: Algebraic Methods for Communications and Storage	2 SWS	Vorlesung (V) / 	Fengler, Schmalen

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The exam is held as an oral exam of approx. 20 min.

The module grade of the module corresponds to the grade of the oral exam.

Voraussetzungen

keine

Empfehlungen

Previous attendance of the lectures "Communication Engineering I" and "Probability Theory" is recommended.

T

5.19 Teilleistung: Channel Coding: Graph-Based Codes [T-ETIT-111245]**Verantwortung:** Prof. Dr.-Ing. Laurent Schmalen**Einrichtung:** KIT-Fakultät für Elektrotechnik und Informationstechnik**Bestandteil von:** [M-ETIT-105617 - Channel Coding: Graph-Based Codes](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Dauer	Version
Prüfungsleistung mündlich	6 LP	Drittelnoten	Jedes Wintersemester	1 Sem.	1

Lehrveranstaltungen					
WS 25/26	2310520	Channel Coding: Graph-Based Codes	3 SWS	Vorlesung (V) / 	Schmalen
WS 25/26	2310521	Exercise for 2310520 Channel Coding: Graph-Based Codes	1 SWS	Übung (Ü) / 	Schmalen

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 30 minutes in which preparatory tasks are solved.

Voraussetzungen

none

Empfehlungen

Previous attendance of the lectures "Communication Engineering I" and "Theory of Probability" is recommended. Knowledge from the lectures "Applied Information Theory" and "Verfahren der Kanalcodierung" is helpful.

T

5.20 Teilleistung: Compiler Design [T-INFO-113925]

Verantwortung: Prof. Dr. André Platzer
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-106966 - Compiler Design](#)

Teilleistungsart
 Prüfungsleistung schriftlich

Leistungspunkte
 9 LP

Notenskala
 Drittelnoten

Turnus
 Unregelmäßig

Version
 2

Lehrveranstaltungen					
SS 2026	2400055	Compiler Design (findet im SoSe 2026 nicht statt)	4 SWS	Vorlesung (V) / 	Platzer

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is usually carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 120 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take the form of an oral examination of approx.

- in the form of an oral examination of approx. 30 minutes in accordance with § 4 Para. 2 No. 2 SPO or
 - in the form of a written examination in accordance with § 4 Para. 2 No. 1 SPO
- takes place.

In order to receive a bonus, you must earn at least 50% of the points for solving the exercises. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4).

Voraussetzungen

None.

Empfehlungen

Students are expected to have significant experience in a high-level programming language. Students are also expected to follow the lecture notes.

T

5.21 Teilleistung: Computational Geometry [T-INFO-114251]

Verantwortung: TT-Prof. Dr. Thomas Bläsius

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107228 - Computational Geometry](#)

Teilleistungsart
Prüfungsleistung mündlich

Leistungspunkte
6 LP

Notenskala
Drittelnoten

Turnus
Unregelmäßig

Version
1

Lehrveranstaltungen					
SS 2026	2400119	Computational Geometry - findet im SS 26 nicht statt	4 SWS	Vorlesung / Übung (VÜ) / 	Bläsius, Yi, Wilhelm, von der Heydt

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

PLUS: The assessment is carried out in form of course work (German Studienleistung, § 4 Abs. 3 SPO). A total of two repetitions are possible.

Voraussetzungen

None.

Empfehlungen

Basic knowledge of algorithms and data structures (e.g., from the courses Algorithms 1 + 2) is expected.

T

5.22 Teilleistung: Computational Geometry - Pass [T-INFO-114252]

Verantwortung: TT-Prof. Dr. Thomas Bläsius**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107228 - Computational Geometry](#)**Teilleistungsart**
Studienleistung**Leistungspunkte**
0 LP**Notenskala**
best./nicht best.**Turnus**
Unregelmäßig**Version**
1

Lehrveranstaltungen					
SS 2026	2400119	Computational Geometry - findet im SS 26 nicht statt	4 SWS	Vorlesung / Übung (VÜ) / 	Bläsius, Yi, Wilhelm, von der Heydt

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out in form of course work (German Studienleistung, § 4 Abs. 3 SPO). A total of two repetitions are possible.

Voraussetzungen

None.

Empfehlungen

Basic knowledge of algorithms and data structures (e.g., from the courses Algorithms 1 + 2) is expected.

T

5.23 Teilleistung: Computational Imaging [T-INFO-112573]

Verantwortung: Dr.-Ing. Johannes Meyer
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-106190 - Computational Imaging](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	5 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2400173	Computational Imaging	3 SWS	Vorlesung / Übung (VÜ) / ●	Meyer, Beyerer

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment takes the form of a written examination, usually lasting 60 minutes in accordance with Section 4 (2) No. 1 SPO.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take place

- in the form of an oral examination in accordance with Section 4 (2) No. 2 SPO or
- in the form of a written examination in accordance with Section 4 (2) No. 1 SPO.

Voraussetzungen

None.

T**5.24 Teilleistung: Computational Risk and Asset Management [T-WIWI-102878]**

Verantwortung: Prof. Dr. Maxim Ulrich
Einrichtung: KIT-Fakultät für Wirtschaftswissenschaften
Bestandteil von: [M-WIWI-105032 - Data Science for Finance](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	4,5 LP	Drittelnoten	Unregelmäßig	5

Lehrveranstaltungen					
WS 25/26	2500015	Computational Risk and Asset Management	4 SWS	Vorlesung (V)	Ulrich

Erfolgskontrolle(n)

Die Modulprüfung erfolgt in Form einer Prüfungsleistung anderer Art. Die Prüfungsleistung anderer Art besteht aus einem Python-basierten "Takehome Exam". Am Ende der dritten Januarkalenderwoche bekommt der Student ein "Takehome Exam" ausgehändigt, welches er binnen 4 Stunden eigenständig und mittels Python bearbeitet und zurückschickt. Genaue Anweisungen werden zu Beginn der Lehrveranstaltung bekannt gegeben.

Die Prüfungsleistung anderer Art kann maximal einmal wiederholt werden. Eine fristgerechte Wiederholungsmöglichkeit findet am Ende der dritten Märzkalenderwoche des gleichen Jahres statt. Genauere Anweisungen werden zu Beginn der Lehrveranstaltung bekannt gegeben.

Voraussetzungen

Keine.

Empfehlungen

Grundkenntnisse der Kapitalmarkttheorie.

Anmerkungen

Lehr- und Lernform: Vorlesung und Übung

Arbeitsaufwand

135 Std.

T

5.25 Teilleistung: Constructive Logic [T-INFO-112704]

Verantwortung: Prof. Dr. André Platzer
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-106256 - Constructive Logic](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	5 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	2400097	Constructive Logic (findet im SoSe 26 nicht statt!)	4 SWS	Vorlesung (V) / 	Platzer

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is usually carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 120 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take the form of an oral examination of approx.

- in the form of an oral examination of approx. 30 minutes in accordance with § 4 Para. 2 No. 2 SPO or
- in the form of a written examination in accordance with § 4 Para. 2 No. 1 SPO takes place.

Voraussetzungen

None.

Empfehlungen

You will be expected to follow the lecture notes.

T**5.26 Teilleistung: Cryptographic Protocols II: Universal Composability [T-INFO-114602]****Verantwortung:** Prof. Dr. Jörn Müller-Quade**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107435 - Cryptographic Protocols II: Universal Composability](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	3 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2400022	Cryptographic Protocols II: Universal Composability	2 SWS	Vorlesung (V) / ● ^s	Müller-Quade, Mechler

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) approx. 30 minutes.

Voraussetzungen

None.

Empfehlungen

Students should be familiar with the contents of the modules "Theoretische Grundlagen der Kryptographie" (Theoretical Foundations of Cryptography) and „Kryptographische Protokolle“ (Cryptographic Protocols I).

T**5.27 Teilleistung: Data Science and Artificial Intelligence for Energy Systems [T-INFO-113402]****Verantwortung:** TT-Prof. Dr. Benjamin Schäfer**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106655 - Data Science and Artificial Intelligence for Energy Systems](#)**Teilleistungsart**
Prüfungsleistung mündlich**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400079	Data Science and Artificial Intelligence for Energy Systems	2 SWS	Vorlesung (V) / 	Schäfer
WS 25/26	2400082	Data Science and Artificial Intelligence for Energy Systems	2 SWS	Übung (Ü) / 	Schäfer, Nikoltchovska, Sadric
SS 2026	2400098	Data science and Artificial Intelligence for Energy Systems	2 SWS	Vorlesung (V) / 	Schäfer
SS 2026	2400173	Data science and Artificial Intelligence for Energy Systems (Exercise)	2 SWS	Übung (Ü) / 	Schäfer

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting about 30 minutes.

Voraussetzungen

None.

Empfehlungen

Knowledge of AI basics is very helpful.

Previous participation in "Energieinformatik 1" and/or "Energieinformatik 2" is beneficiary but not mandatory.

Knowledge of Python is highly recommended.

T**5.28 Teilleistung: Decentralized Systems: Fundamentals, Modeling, and Applications [T-INFO-110820]****Verantwortung:** Prof. Dr. Hannes Hartenstein**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-105334 - Decentralized Systems: Fundamentals, Modeling, and Applications](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	6 LP	Drittelnoten	Jedes Sommersemester	5

Lehrveranstaltungen					
SS 2026	2400089	Decentralized Systems: Fundamentals, Modeling, and Applications	4 SWS	Vorlesung / Übung (VÜ) / ●	Hartenstein, Jacob

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (§ 6 Abs. 3 SPO) whether the examination takes place

in the form of an oral examination lasting 30 minutes pursuant to § 4 Abs. 2 Nr. 2 SPO or

in the form of a written examination lasting 60 minutes in accordance with § 4 Abs. 2 Nr. 1 SPO.

Voraussetzungen

None.

Empfehlungen

Basics according to the lectures "Information Security" and "Introduction to Computer Networks" are recommended.

T

5.29 Teilleistung: Deep Learning and Neural Networks [T-INFO-114219]

Verantwortung: Prof. Dr. Jan Niehues**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107197 - Deep Learning and Neural Networks](#)**Teilleistungsart**
Prüfungsleistung schriftlich**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	2400024	Deep Learning and Neural Networks	4 SWS	Vorlesung (V) / 	Niehues

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Voraussetzungen

T-INFO-101383 - Neural networks must not be started.

Empfehlungen

Prior successful completion of the core module "Cognitive Systems" is recommended.

T**5.30 Teilleistung: Design and Architectures of Embedded Systems (ESII) [T-INFO-114254]****Verantwortung:** Prof. Dr.-Ing. Jörg Henkel**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107230 - Design and Architectures of Embedded Systems \(ESII\)](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	3 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2424106	Design and architectures of embedded systems (ES2)	2 SWS	Vorlesung (V)	Khdr, Henkel

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Voraussetzungen

None.

Empfehlungen

Knowledge of computer structures is helpful.

T

5.31 Teilleistung: Digital Marketing [T-WIWI-112693]

Verantwortung: Prof. Dr. Ann-Kristin Kupfer
Einrichtung: KIT-Fakultät für Wirtschaftswissenschaften
Bestandteil von: [M-WIWI-106258 - Digital Marketing](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	4,5 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	2571185	Digital Marketing	2 SWS	Vorlesung (V) / ●	Kupfer
SS 2026	2571186	Digital Marketing Exercise	1 SWS	Übung (Ü) / ●	Schmitt, Cinar, Kupfer

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

Success is assessed in the form of an examination of another type. The following aspects are included in the assessment:

- Elaboration and presentation of a group task
- Written exam

Further details on the organization of the performance and the points system for the assessment will be announced in the lecture.

Voraussetzungen

None

Empfehlungen

Students are highly encouraged to actively participate in class.

Arbeitsaufwand

135 Std.

T

5.32 Teilleistung: Digital Marketing and Sales in B2B [T-WIWI-106981]

Verantwortung: Prof. Dr. Martin Klarmann
Anja Konhäuser

Einrichtung: KIT-Fakultät für Wirtschaftswissenschaften

Bestandteil von: [M-WIWI-106258 - Digital Marketing](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	1,5 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	2571156	Digital Marketing and Sales in B2B	1 SWS	Sonstige (sonst.) / 	Konhäuser

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

Die Erfolgskontrolle erfolgt in Form einer Prüfungsleistung anderer Art nach § 4 Abs. 2 Nr. 3 SPO (Teampräsentation einer Case Study mit anschließender Diskussion im Umfang von insg. 30 Minuten).

Voraussetzungen

Keine.

Anmerkungen

Die Veranstaltung findet im Sommersemester 2023 leider nicht statt und wird voraussichtlich ab dem Sommersemester 2024 wieder regulär angeboten.

Für die Teilnahme an dieser Veranstaltung ist eine Bewerbung erforderlich. Die Bewerbungsphase findet in der Regel zu Beginn der Vorlesungszeit des jeweiligen Semesters statt. Nähere Informationen zum Bewerbungsprozess erhalten Sie in der Regel kurz vor Beginn der Vorlesungszeit im Wintersemester auf der Webseite der Forschungsgruppe Marketing und Vertrieb (marketing.iism.kit.edu). Diese Veranstaltung hat eine Teilnahmebeschränkung. Die Forschungsgruppe Marketing und Vertrieb ermöglicht typischerweise allen Studierenden den Besuch einer Veranstaltung mit 1,5 Leistungspunkten im entsprechenden Modul. Eine Garantie für den Besuch einer bestimmten Veranstaltung kann auf keinen Fall gegeben werden. Nähere Informationen erhalten Sie direkt bei der Forschergruppe Marketing und Vertrieb (marketing.iism.kit.edu). Bitte beachten Sie, dass nur eine der 1,5-LP-Veranstaltungen für das Modul angerechnet werden kann.

Arbeitsaufwand

45 Std.

T

5.33 Teilleistung: Distributed Computing [T-INFO-114235]

Verantwortung: Prof. Dr. Achim Streit
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107215 - Distributed Computing](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	4 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2400050	Distributed Computing	2 SWS	Vorlesung (V) / 	Streit, Krauß, Schnepf

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes. Depending on the number of participants it will be announced six weeks before the assessment (§3 Abs. 3 SPO) if the assessment is done

- as an oral examination according to § 4 Abs. 2 No. 2 SPO or
- as a written examination according to § 4 Abs. 2 No. 1 SPO.

Voraussetzungen

None.

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Die Teilleistung [T-INFO-101298 - Verteiltes Rechnen](#) darf nicht begonnen worden sein.

Empfehlungen

Knowledge in the area of computer networks helpful.

T

5.34 Teilleistung: Economic Decision Making [T-WIWI-114174]

Verantwortung: Prof. Dr. Benjamin Scheibehenne
Einrichtung: KIT-Fakultät für Wirtschaftswissenschaften
Bestandteil von: M-WIWI-106258 - Digital Marketing

Teilleistungsart Prüfungsleistung anderer Art	Leistungspunkte 4,5 LP	Notenskala Drittelnoten	Turnus Jedes Wintersemester	Dauer 1 Sem.	Version 1
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Lehrveranstaltungen					
WS 25/26	2500059	Economic Decision Making	3 SWS	Vorlesung / Übung (VÜ) / ●	Scheibehenne

Legende: ■ Online, 🔄 Präsenz/Online gemischt, ● Präsenz, ✕ Abgesagt

Erfolgskontrolle(n)

Alternative exam assessment. The course format consists of a lecture and a corresponding exercise. The grading includes the following aspects:

- A written exam (60 minutes)
- A presentation during the exercise
- Regular attendance
- In-class assignments

The scoring system for the grading will be announced at the beginning of the course.

Voraussetzungen

Registration via the WiWi Portal is required for participation in the Übung. The Übung is a prerequisite for the exam.

Anmerkungen

The judgments and decisions that we make can have long ranging and important consequences for our (financial) well-being and individual health. Hence, the goal of this lecture is to gain a better understanding of how people make judgments and decisions and the factors that influences their behavior. We will look into simple heuristics and mental shortcuts that decision makers use to navigate their environment, in particular so in an economic context. Following this, the lecture will provide an overview into social and emotional influences on decision making. In the second half of the semester we will look into some more specific topics including self-control, nudging, and food choice. The last part of the lecture will focus on risk communication and risk perception. We will address these questions from an interdisciplinary perspective at the intersection of Psychology, Behavioral Economics, Marketing, Cognitive Science, and Biology. Across all topics covered in class, we will engage with basic theoretical work as well as with groundbreaking empirical research and current scientific debates.

The workload of the class is 4.5 ECTS. This consists of 3 ETCS for the lecture and 1.5 ETCS for the Übung. Details about the Übung will be communicated at the first day of the class.

Arbeitsaufwand

135 Std.

T**5.35 Teilleistung: Edge-AI in Software and Sensor Applications [T-INFO-114258]****Verantwortung:** Dr. Victor Pankratius**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107234 - Edge-AI in Software and Sensor Applications](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	3 LP	Drittelnoten	Jedes Semester	1

Lehrveranstaltungen					
WS 25/26	2400124	EdgeAI in Software and Sensor Applications	2 SWS	Vorlesung (V) / 	Pankratius
SS 2026	2400006	EdgeAI in Software and Sensor Applications	2 SWS	Vorlesung (V) / 	Pankratius

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

Die Erfolgskontrolle erfolgt in Form einer schriftlichen Prüfung im Umfang von i.d.R. 60 Minuten nach § 4 Abs. 2 Nr. 1 SPO.

Voraussetzungen

Grundstudium Informatik

Empfehlungen

Hilfreich sind Kenntnisse z.B. aus Kognitive Systeme, Softwaretechnik, Algorithmen, Rechnernetze & -strukturen, Low-Power-Design

T

5.36 Teilleistung: Embedded Machine Learning Lab [T-INFO-111549]

Verantwortung: Prof. Dr.-Ing. Jörg Henkel

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-105775 - Embedded Machine Learning Lab](#)

Teilleistungsart
Prüfungsleistung anderer Art

Leistungspunkte
4 LP

Notenskala
Drittelnoten

Turnus
Jedes Semester

Version
1

Lehrveranstaltungen					
WS 25/26	2400295	Embedded Machine Learning Lab	4 SWS	Praktikum (P) / ●	Henkel, Ahmed, Dietrich
SS 2026	2400295	Embedded Machine Learning Lab	4 SWS	Praktikum (P) / ●	Henkel, Ahmed, Dietrich, Khdr

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO), in the form of a practical assignment, presentations and, if applicable, a written paper. The written paper, presentations and practical work are weighted according to the course.

Voraussetzungen

None.

Empfehlungen

This lab requires a basic (theoretic) knowledge about neural networks and training. Further knowledge of Linux environments and Python is strongly advised since they will be intensively used in the lab and are the de-facto industry standard for machine learning research.

T**5.37 Teilleistung: Empirical Software Engineering [T-INFO-115054]**

Verantwortung: Raziye Dehghani
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107685 - Empirical Software Engineering](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	4 LP	Drittelnoten	Jedes Semester	1

Erfolgskontrolle(n)

The assessment is carried out as an oral examination, lasting approx. 30 minutes in accordance with Section 4 (2) No. 2 SPO.

Voraussetzungen

None.

Empfehlungen

Familiarity with statistical analysis methods is highly recommended.

T

5.38 Teilleistung: Engineering Self-Adaptive Systems [T-INFO-113349]**Verantwortung:** Prof. Dr. Raffaella Mirandola**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106626 - Engineering Self-Adaptive Systems](#)**Teilleistungsart**
Prüfungsleistung mündlich**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400186	Engineering Self-Adaptive Systems		Vorlesung (V)	Mirandola

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Voraussetzungen

None.

T

5.39 Teilleistung: Entrepreneurship [T-WIWI-102864]

Verantwortung: Prof. Dr. Orestis Terzidis
Einrichtung: KIT-Fakultät für Wirtschaftswissenschaften
Bestandteil von: [M-ETIT-105073 - Student Innovation Lab](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	3 LP	Drittelnoten	Jedes Semester	1

Lehrveranstaltungen					
WS 25/26	2545001	Entrepreneurship	2 SWS	Vorlesung (V) / 	Malik, Terzidis, Dang
SS 2026	2545001	Entrepreneurship	2 SWS	Vorlesung (V) / 	Terzidis, Dang

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

Die Erfolgskontrolle erfolgt in Form einer schriftlichen Prüfung (60 min.) (nach §4(2), 1 SPO).

Die Note ist die Note der schriftlichen Prüfung.

Den Studierenden wird durch gesonderte Aufgabenstellungen die Möglichkeit geboten einen Notenbonus zu erwerben. Liegt die Note der schriftlichen Prüfung zwischen 4,0 und 1,3, so verbessert der Bonus die Note um maximal eine Notenstufe (0,3 oder 0,4). Die genauen Kriterien für die Vergabe eines Bonus werden zu Vorlesungsbeginn bekannt gegeben.

Voraussetzungen

Keine

Empfehlungen

Keine

T

5.40 Teilleistung: EU Data Protection Law [T-INFO-113887]**Verantwortung:** Gustavo Gil Gasiola**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107030 - EU Data Protection Law](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	3 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2424019	EU Data Protection Law	2 SWS	Vorlesung (V) / 	Gil Gasiola

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Voraussetzungen

None

Anmerkungen**Competency Goals:**

Students are able to comprehend the EU data protection regulation, including the General Data Protection Regulation and related EU data regulations.

They know the foundations of data protection rules, including fundamental concepts (e.g., “personal data”, “processing”, “data subject”). They are also familiar with the principles of personal data processing (lawfulness, limited purpose, transparency, accountability) as well as the rights of the data subject.

They can identify the main obligations of the controller and the processor.

Students understand the conditions for the transfer of personal data to third countries.

They can identify the other regulations that govern data in the European Union.

Students are able to read and understand legal text related to data regulation.

They can understand and solve simple data protection cases.

Content:

The General Data Protection Regulation (GDPR) of the European Union is a milestone in protecting individuals from the unlawful use of their data. In a data-driven society, economy, and government, this protection has become essential to guarantee fundamental rights. In addition to its direct impact on the legal systems of all Member States, the GDPR has a major influence on third countries that have adopted similar regulations (e.g. Switzerland, Argentina, Brazil, South Africa, and many others). In this way, the EU Data Protection Regulation has established itself as the “gold standard” of data protection, providing guidance to address the challenges posed by new technologies and new ways of creating, using and sharing personal data. Understanding the structure of data protection in the EU is therefore essential to grasp its impact on individual rights, public administration, business models, and even technological development.

This lecture aims to provide a structured overview of the EU Data Protection Regulation, and to offer tools to understand the regulatory structure of the EU Data Regulation. The lecture will cover the following topics:

- Introduction to EU law
- Development of the EU data protection regulation
- Legal structure of data protection in the EU
- Role of national and sectoral laws
- Data protection as fundamental right
- Principles of data protection
- Lawfulness of personal data processing
- Anonymization and pseudonymization of personal data
- Special categories of personal data
- Rights of the data subject
- Transfer of personal data to third countries
- Responsibility of the controller and the processor
- Security of personal data and personal data breach
- Open Data Directive
- Data Governance Act
- Data Act

Workload

- Attendance time to the lectures = 15 x 90 min = 22 h 30 min
- Self-study during the semester = 47 h 30 min
- Preparation for the exam = 20 h
- Total = 90 h

T

5.41 Teilleistung: Explainable Artificial Intelligence [T-INFO-112774]

Verantwortung: TT-Prof. Dr. Rudolf Lioutikov**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106302 - Explainable Artificial Intelligence](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	3 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	2400128	Explainable Artificial Intelligence	2 SWS	Vorlesung (V) / 	Lioutikov

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 120 minutes.

A bonus can be acquired through successful participation in the exercise as a success control of a different kind (§4(2), 3 SPO 2008) or study performance (§4(3) SPO 2015). The exact criteria for awarding a bonus will be announced at the beginning of the lecture. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The bonus is only valid for the main and post exams of the semester in which it was earned. After that, the grade bonus expires.

Voraussetzungen

None.

Empfehlungen

- Experience in Machine Learning is recommended, e.g. through prior coursework.
 - The Computer Science Department offers several great lectures e.g., “Maschinelles Lernen - Grundlagen und Algorithmen” and “Deep Learning”
- A good mathematical background will be beneficial
- Python / PyTorch experience could be beneficial when we discuss practical examples/implementations.

T**5.42 Teilleistung: Fine-Grained Complexity Theory & Algorithms [T-INFO-113391]****Verantwortung:** Prof. Dr.-Ing. Marvin Künnemann**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106644 - Fine-Grained Complexity Theory & Algorithms](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	6 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
WS 25/26	2400104	Fine-grained Algorithm Design and Engineering	4 SWS	Praktikum (P) / ●	Künnemann
SS 2026	2400152	Fine-Grained Complexity Theory & Algorithms	4 SWS	Vorlesung / Übung (VÜ) / ●	Künnemann

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20-30 minutes.

Voraussetzungen

None.

Empfehlungen

Basic knowledge of theoretical computer science and algorithm design is recommended.

T

5.43 Teilleistung: Geometric Deep Learning [T-INFO-112662]

Verantwortung: Jun.-Prof. Dr. Jan Stühmer**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106237 - Geometric Deep Learning](#)**Teilleistungsart**
Prüfungsleistung mündlich**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
2

Lehrveranstaltungen					
WS 25/26	2400179	Geometric Deep Learning	4 SWS	Vorlesung / Übung (VÜ) / 	Stühmer, Berndt

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Voraussetzungen

None.

Empfehlungen

Knowledge about the foundations of machine learning, group theory and linear algebra useful but not required.

T**5.44 Teilleistung: Graph Partitioning and Graph Clustering in Theory and Practice [T-INFO-114232]**

Verantwortung: Prof. Dr. Peter Sanders
Dr. rer. nat. Torsten Ueckerdt

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107211 - Graph Partitioning and Graph Clustering in Theory and Practice](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	4 LP	Drittelnoten	Jedes Sommersemester	1

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

The module grade is made up of the graded and weighted performance assessments (usually 80% of the oral examination and 20% of the other performance).

Voraussetzungen

None.

Empfehlungen

Knowledge of graph theory and algorithm technology is helpful.

T**5.45 Teilleistung: Graph Partitioning and Graph Clustering in Theory and Practice - Practical [T-INFO-114233]**

Verantwortung: Prof. Dr. Peter Sanders
Dr. rer. nat. Torsten Ueckerdt

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107211 - Graph Partitioning and Graph Clustering in Theory and Practice](#)

Teilleistungsart
Prüfungsleistung anderer Art

Leistungspunkte
1 LP

Notenskala
Drittelnoten

Turnus
Jedes Sommersemester

Version
1

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). (seminar paper/presentation/programming task or similar).

The module grade is made up of the graded and weighted performance assessments (usually 80% of the oral examination and 20% of the other performance). An overall grade is awarded.

Voraussetzungen

None.

Empfehlungen

Knowledge of graph theory and algorithm technology is helpful.

T

5.46 Teilleistung: Graphentheorie [T-MATH-102273]

Verantwortung: Prof. Dr. Maria Aksenovich
Einrichtung: KIT-Fakultät für Mathematik
Bestandteil von: [M-MATH-101336 - Graphentheorie](#)

Teilleistungsart
Prüfungsleistung schriftlich

Leistungspunkte
9 LP

Notenskala
Drittelnoten

Version
3

Lehrveranstaltungen					
WS 25/26	0104500	Graph Theory	4 SWS	Vorlesung (V)	Aksenovich, Liu
WS 25/26	0104510	Tutorial for 0104500 (Graph Theory)	2 SWS	Übung (Ü)	Aksenovich, Liu

Voraussetzungen
Keine

T**5.47 Teilleistung: Hands-on Bioinformatics Practical [T-INFO-103009]**

Verantwortung: Prof. Dr. Alexandros Stamatakis

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-101573 - Hands-on Bioinformatics Practical](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Unregelmäßig	3

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Practical tasks in the field of bioinformatics must be completed. The results must be presented in writing or orally.

Voraussetzungen

The exam in Introduction to Bioinformatics for Computer Scientists must have been passed in one of the preceding semesters.

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Das Modul [M-INFO-100749 - Introduction to Bioinformatics for Computer Scientists](#) muss erfolgreich abgeschlossen worden sein.

T

5.48 Teilleistung: Hardware Modeling and Simulation [T-ETIT-100672]

Verantwortung: Dr.-Ing. Jens Becker
Prof. Dr.-Ing. Jürgen Becker

Einrichtung: KIT-Fakultät für Elektrotechnik und Informationstechnik

Bestandteil von: [M-ETIT-100449 - Hardware Modeling and Simulation](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	4 LP	Drittelnoten	Jedes Wintersemester	2

Lehrveranstaltungen					
WS 25/26	2311608	Hardware Modeling and Simulation	2 SWS	Vorlesung (V) / 	Becker, Becker
WS 25/26	2311610	Tutorial for 2311608 Hardware Modeling and Simulation	1 SWS	Übung (Ü) / 	Sotiropoulos

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

Achievement is examined in the form of a written examination lasting 120 minutes.

Voraussetzungen

none

T

5.49 Teilleistung: Hardware Synthesis and Optimization [T-ETIT-113922]**Verantwortung:** Prof. Dr.-Ing. Jürgen Becker**Einrichtung:** KIT-Fakultät für Elektrotechnik und Informationstechnik**Bestandteil von:** [M-ETIT-106963 - Hardware Synthesis and Optimization](#)**Teilleistungsart**
Prüfungsleistung mündlich**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	2311619	Hardware Synthesis and Optimization	3 SWS	Vorlesung (V) / ●	Becker
SS 2026	2311621	Tutorial for 2311619 Hardware Synthesis and Optimization	1 SWS	Übung (Ü) / ●	Schmidt

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The examination takes place within the framework of an oral overall examination (approx. 30 minutes).
The module grade is the grade of the oral exam.

Voraussetzungen

none

T

5.50 Teilleistung: Hardware-efficient Artificial Intelligence [T-INFO-114628]

Verantwortung: Prof. Dr.-Ing. Jörg Henkel
Dr. Martin Rapp

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107448 - Hardware-efficient Artificial Intelligence](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Dauer	Version
Prüfungsleistung mündlich	3 LP	Drittelnoten	Jedes Wintersemester	1 Sem.	2

Lehrveranstaltungen					
WS 25/26	2400232	Hardware-effizientes Maschinelles Lernen	2 SWS	Vorlesung (V)	Rapp

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) approx. 30 minutes.

Voraussetzungen

None.

Empfehlungen

A basic understanding of deep learning concepts and neural networks is expected; familiarity with hardware architectures and embedded systems is helpful but not required.

T

5.51 Teilleistung: HRI and Social Robotics [T-INFO-113396]**Verantwortung:** TT-Prof. Dr. Barbara Bruno**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106650 - HRI and Social Robotics](#)**Teilleistungsart**
Prüfungsleistung schriftlich**Leistungspunkte**
4 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	2400159	HRI and Social Robotics	4 SWS	Vorlesung / Übung (VÜ) / ●	Bruno, Maure

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 120 minutes.

Voraussetzungen

None.

Empfehlungen

Knowledge of the content of modules Robotics I - Introduction to Robotics is helpful.

T**5.52 Teilleistung: HRI and Social Robotics - Pass [T-INFO-113397]****Verantwortung:** TT-Prof. Dr. Barbara Bruno**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106650 - HRI and Social Robotics](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
2 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	2400159	HRI and Social Robotics	4 SWS	Vorlesung / Übung (VÜ) / ●	Bruno, Maure

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Students must regularly submit exercise sheets. The number of exercise sheets and the scale for passing will be announced at the beginning of the course. The assessment can only be repeated once.

Empfehlungen

Knowledge of the content of modules Robotics I - Introduction to Robotics is helpful.

T

5.53 Teilleistung: Human-Computer-Interaction [T-INFO-114192]**Verantwortung:** Prof. Dr.-Ing. Michael Beigl**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107166 - Human Computer Interaction](#)**Teilleistungsart**
Prüfungsleistung schriftlich**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	24659	Human-Computer-Interaction	2 SWS	Vorlesung (V) / 	Beigl

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as a written exam (of approx. 60 minutes) as a digital exam according to §2 (3) of the Statute for the implementation of online examinations. The exam takes place ON SITE at KIT!

Voraussetzungen

Participation in the exercise is compulsory and the contents of the exercise are relevant for the examination.

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Die Teilleistung [T-INFO-114193 - Human-Computer-Interaction Pass](#) muss erfolgreich abgeschlossen worden sein.

T

5.54 Teilleistung: Human-Computer-Interaction Pass [T-INFO-114193]**Verantwortung:** Prof. Dr.-Ing. Michael Beigl**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107166 - Human Computer Interaction](#)**Voraussetzung für:** [T-INFO-114192 - Human-Computer-Interaction](#)**Teilleistungsart**
Studienleistung**Leistungspunkte**
0 LP**Notenskala**
best./nicht best.**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	2400095	Human-Computer-Interaction	1 SWS	Übung (Ü) / 	Beigl

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Exercise sheets must be handed in regularly to pass the course. The specific details will be announced in the lecture.

Voraussetzungen

None.

Anmerkungen

Participation in the exercise is compulsory and the contents of the exercise are relevant for the examination.

T**5.55 Teilleistung: Humanoid Robots - Locomotion and Whole-Body Control [T-INFO-113395]****Verantwortung:** Prof. Dr. Katja Mombaur**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106649 - Humanoid Robots - Locomotion and Whole-Body Control](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
SS 2026	2400135	Humanoid Robots – Locomotion and Whole-Body Control	4 SWS	Vorlesung (V) / 	Mombaur, Große Sundrup

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

The grade of the course is given based on the performance in an individual programming project on the topic of humanoid robots, which consists of the definition and solution of the project itself as well as a subsequent oral presentation in a block event and the submission of a written report. Project work starts in the exercise slots during the second half of the term and ends during the lecture free time.

As a prerequisite for the enrollment in the project, the students must regularly and successfully participate in the exercises and present their results for the exercise sheets during the first part of the term, according to the modalities announced at the beginning of the course.

Both components can be completed in the same group of two students. Withdrawal is possible until 2 weeks after enrollment in the project.

Active participation in the class is expected from all students and is a necessary requirement for the course.

Voraussetzungen

- Completion of module Robotics 1 or corresponding knowledge required.
- Programming skills

Empfehlungen

Attendance of the lectures Robotics I - Introduction to Robotics and Mechano-Informatics in Robotics is required.

Anmerkungen

Limitation to 30 participants

T**5.56 Teilleistung: Humanoid Robots - Locomotion and Whole-Body Control -Pass [T-INFO-114282]****Verantwortung:** Prof. Dr. Katja Mombaur**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106649 - Humanoid Robots - Locomotion and Whole-Body Control](#)**Teilleistungsart**
Studienleistung mündlich**Leistungspunkte**
0 LP**Notenskala**
best./nicht best.**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
SS 2026	2400135	Humanoid Robots – Locomotion and Whole-Body Control	4 SWS	Vorlesung (V) / 	Mombaur, Große Sundrup

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out in form of course work (German Studienleistung, § 4 Abs. 3 SPO).

The grade of the course is given based on the performance in in an individual programming project on the topic of humanoid robots, which consists of the definition and solution of the project itself as well as a subsequent oral presentation in a block event and the submission of a written report. Project work starts in the exercise slots during the second half of the term and ends during the lecture free time.

As a prerequisite for the enrollment in the project, the students must regularly and successfully participate in the exercises and present their results for the exercise sheets during the first part of the term, according to the modalities announced at the beginning of the course.

Both components can be completed in the same group of two students. Withdrawal is possible until 2 weeks after enrollment in the project.

Active participation in the class is expected from all students and is a necessary requirement for the course.

Voraussetzungen

- Completion of module Robotics 1 or corresponding knowledge required.
- Programing skills

Empfehlungen

Attendance of the lectures Robotics I - Introduction to Robotics and Mechano-Informatics in Robotics is required.

Anmerkungen

Limitation to 30 participants

T

5.57 Teilleistung: Humanoid Robots - Seminar [T-INFO-114170]**Verantwortung:** Prof. Dr.-Ing. Tamim Asfour**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107152 - Humanoid Robots - Seminar](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400048	Humanoid Robots - Seminar	2 SWS	Seminar (S) / ● ^s	Asfour

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). It includes a presentation at the end of the term and a term paper.

Voraussetzungen

None.

Empfehlungen

Attending the lectures Robotics I – Introduction to Robotics, Robotics II: Humanoid Robotics, Robotics III – Sensors and Perception in Robotics, Mechano-Informatics and Robotics and Wearable Robotic Technologies is recommended.

T**5.58 Teilleistung: Information, Wissenschaft und Verantwortung – aktuelle ethische Herausforderungen der IT [T-INFO-111839]****Verantwortung:** Angelika Kaplan**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107254 - Interdisciplinary Qualifications](#)**Teilleistungsart**
Studienleistung**Leistungspunkte**
1 LP**Notenskala**
best./nicht best.**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	2400094	Ethik der IT	2 SWS	Vorlesung (V) / 	Reussner, Bagattini

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

Die Erfolgskontrolle erfolgt in Form einer Studienleistung nach § 4 Abs. 3 SPO. Es besteht Anwesenheitspflicht zwecks Erfolgskontrolle. Es wird eine Teilnahmebestätigung ausgestellt.

Voraussetzungen

Keine.

Empfehlungen

Keine.

Anmerkungen

Selbstfahrende Autos, Pflegeroboter, Apps, Software für Einstellungsverfahren oder für die Anwendung in komplexen medizinischen Diagnoseverfahren wie dem MRT. Längst ist klar, dass viele neue Technologien im Bereich der IT gleichermaßen Vorteile aber auch Gefahren mit sich bringen. Heute sind wir fast alle persönlich von phishing emails und spam betroffen, und die adversen Effekte von social media – wie Vereinsamung und damit korrelierende psychische Erkrankungen – sind allgemein bekannt. Eine wesentlich größere Tragweite wird deutlich, wenn man an die Möglichkeiten moderner Überwachungssysteme, Gesichtserkennungstechnologien und Big Data Algorithmen in sozio-globalen Kontexten betrachtet, wo sie, wie beim Cambridge Analytica Skandal, politische Wahlen und sogar ganze politische Systeme unterminieren können. Diese Ambivalenz neuer Technologien führt auch zur Frage nach der Verantwortung für die mit ihnen verbundenen Folgen. Es greift hier sicherlich zu kurz, den einzelnen Informatiker allein in der Pflicht zu sehen, schließlich handelt es sich um ein gesamtgesellschaftliches Problemfeld, an dem beispielsweise politische Akteure und Firmen beteiligt sind, was sich angesichts der oftmals globalen Dimension kooperativer Projekte im Bereich der IT noch komplizierter darstellt. Allerdings sind Informatiker oftmals diejenigen, die als erste mit ethischen Problemen konfrontiert sind. Um hier eine eigene Position beziehen zu können, ist wichtig zu lernen, wie man sich souverän in den für ethische Fragestellungen typischen Grauzonen (in denen es oft kein klares Wahr oder Falsch gibt) argumentativ bewegen kann.

Das Ziel dieser Ringvorlesung ist es, hierzu einen Beitrag zu leisten, indem wir gemeinsam mit einschlägigen Expert*innen über grundsätzliche und anwendungsbezogene ethische Fragen im Bereich der IT diskutieren. Hierbei ist uns ein offener Diskurs wichtig, bei dem zu den oftmals kontroversen Themen alle Argumente gehört und bewertet werden können.

T**5.59 Teilleistung: Information, Wissenschaft und Verantwortung – aktuelle ethische Herausforderungen der IT [T-INFO-112148]****Verantwortung:** Angelika Kaplan**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107254 - Interdisciplinary Qualifications](#)**Teilleistungsart**
Studienleistung**Leistungspunkte**
2 LP**Notenskala**
best./nicht best.**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	2400094	Ethik der IT	2 SWS	Vorlesung (V) /	Reussner, Bagattini

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

Die Erfolgskontrolle erfolgt in Form einer Studienleistung nach § 4 Abs. 3 SPO. Es besteht Anwesenheitspflicht zwecks Erfolgskontrolle. Es wird eine Teilnahmebestätigung ausgestellt.

Voraussetzungen

Keine.

Empfehlungen

Keine.

Anmerkungen

Selbstfahrende Autos, Pflegeroboter, Apps, Software für Einstellungsverfahren oder für die Anwendung in komplexen medizinischen Diagnoseverfahren wie dem MRT. Längst ist klar, dass viele neue Technologien im Bereich der IT gleichermaßen Vorteile aber auch Gefahren mit sich bringen. Heute sind wir fast alle persönlich von phishing emails und spam betroffen, und die adversen Effekte von social media – wie Vereinsamung und damit korrelierende psychische Erkrankungen – sind allgemein bekannt. Eine wesentlich größere Tragweite wird deutlich, wenn man an die Möglichkeiten moderner Überwachungssysteme, Gesichtserkennungstechnologien und Big Data Algorithmen in sozio-globalen Kontexten betrachtet, wo sie, wie beim Cambridge Analytica Skandal, politische Wahlen und sogar ganze politische Systeme unterminieren können. Diese Ambivalenz neuer Technologien führt auch zur Frage nach der Verantwortung für die mit ihnen verbundenen Folgen. Es greift hier sicherlich zu kurz, den einzelnen Informatiker allein in der Pflicht zu sehen, schließlich handelt es sich um ein gesamtgesellschaftliches Problemfeld, an dem beispielsweise politische Akteure und Firmen beteiligt sind, was sich angesichts der oftmals globalen Dimension kooperativer Projekte im Bereich der IT noch komplizierter darstellt. Allerdings sind Informatiker oftmals diejenigen, die als erste mit ethischen Problemen konfrontiert sind. Um hier eine eigene Position beziehen zu können, ist wichtig zu lernen, wie man sich souverän in den für ethische Fragestellungen typischen Grauzonen (in denen es oft kein klares Wahr oder Falsch gibt) argumentativ bewegen kann.

Das Ziel dieser Ringvorlesung ist es, hierzu einen Beitrag zu leisten, indem wir gemeinsam mit einschlägigen Expert*innen über grundsätzliche und anwendungsbezogene ethische Fragen im Bereich der IT diskutieren. Hierbei ist uns ein offener Diskurs wichtig, bei dem zu den oftmals kontroversen Themen alle Argumente gehört und bewertet werden können.

T

5.60 Teilleistung: Innovation Lab [T-ETIT-110291]

Verantwortung: Prof. Dr.-Ing. Sören Hohmann
 Prof. Dr. Werner Nahm
 Prof. Dr.-Ing. Eric Sax
 Prof. Dr. Wilhelm Stork
 Prof. Dr.-Ing. Thomas Zwick

Einrichtung: KIT-Fakultät für Elektrotechnik und Informationstechnik

Bestandteil von: [M-ETIT-105073 - Student Innovation Lab](#)

Teilleistungsart
 Prüfungsleistung anderer Art

Leistungspunkte
 9 LP

Notenskala
 Drittelnoten

Turnus
 Jedes Semester

Dauer
 2 Sem.

Version
 1

Lehrveranstaltungen					
WS 25/26	2303192	Innovation Lab	2 SWS	Projekt (PRO) / ●	Hohmann, Zwick, Sax, Stork, Nahm, Schmalen, Rost
SS 2026	2303192	Innovation Lab	2 SWS	Projekt (PRO) / ●	Hohmann, Zwick, Sax, Stork, Terzidis

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

Success control takes place in the form of other types of examination and includes:

- Submission of a Technical Report with requirements list and system architecture
- Submission of the reflection of the Gate Plans
- Presentation of the High-fidelity

Voraussetzungen

none

T**5.61 Teilleistung: International Entrepreneurship and Brand Development [T-WIWI-114893]**

Verantwortung: Prof. Dr. Ann-Kristin Kupfer
Prof. Dr. Orestis Terzidis

Einrichtung: KIT-Fakultät für Wirtschaftswissenschaften

Bestandteil von: [M-WIWI-106258 - Digital Marketing](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	6 LP	Drittelnoten	siehe Anmerkungen	1

Lehrveranstaltungen					
WS 25/26	2572189	International Entrepreneurship and Brand Development	4 SWS	Block (B) / ●	Terzidis, Kupfer

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

Die Erfolgskontrolle erfolgt in Form einer Prüfungsleistung anderer Art (Präsentation). Die Note setzt sich aus der Leistung bei der Präsentation, der anschließenden Diskussion und der schriftlichen Ausarbeitung zusammen.

Anmerkungen

Aktuelle Informationen erhalten Sie bei der Forschungsgruppe Marketing und Vertrieb.

Arbeitsaufwand

180 Std.

T

5.62 Teilleistung: Internet of Everything [T-INFO-101337]

Verantwortung: Prof. Dr. Martina Zitterbart
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-100800 - Internet of Everything](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	4 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2424104	Internet of Everything	2 SWS	Vorlesung (V) / ●	Zitterbart, Hildenbrand, Mahrt, Neumeister

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take the form of an oral examination of approx.

- in the form of an oral examination of approx. 30 minutes in accordance with § 4 Para. 2 No. 2 SPO or
- in the form of a written examination in accordance with § 4 Para. 2 No. 1 SPO

takes place.

Voraussetzungen

None.

Empfehlungen

The contents of the lecture Introduction to Computer Networks are assumed to be known. Attendance of the lecture Telematics is strongly recommended, as the contents are an important basis for understanding and classifying the material.

T**5.63 Teilleistung: Introduction to Bioinformatics for Computer Scientists [T-INFO-101286]****Verantwortung:** Prof. Dr. Alexandros Stamatakis**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-100749 - Introduction to Bioinformatics for Computer Scientists](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	3 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2400055	Introduction to Bioinformatics for Computer Scientists	2 SWS	Vorlesung (V) / 	Stamatakis

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Voraussetzungen

None

Empfehlungen

Basic knowledge in the areas of theoretical computer science (algorithms, data structures) and technical computer science (sequential optimisation in C or C++, computer architectures, parallel programming, vector processors) will be beneficial.

T

5.64 Teilleistung: IT Security [T-INFO-113960]

Verantwortung: Prof. Dr. Jörn Müller-Quade
Prof. Dr. Christian Wressnegger

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-106998 - IT Security](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	6 LP	Drittelnoten	Jedes Wintersemester	3

Lehrveranstaltungen					
WS 25/26	2400010	IT Security	4 SWS	Vorlesung / Übung (VÜ) / ●	Müller-Quade, Wressnegger, Martin

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 90 minutes.

Voraussetzungen

None.

Empfehlungen

Students should be familiar with the content of the compulsory lecture "Informationssicherheit".

T

5.65 Teilleistung: Kombinatorik [T-MATH-105916]

Verantwortung: Prof. Dr. Maria Aksenovich
Einrichtung: KIT-Fakultät für Mathematik
Bestandteil von: [M-MATH-102950 - Kombinatorik](#)

Teilleistungsart
 Prüfungsleistung schriftlich

Leistungspunkte
 9 LP

Notenskala
 Drittelnoten

Version
 4

Lehrveranstaltungen					
SS 2026	0150300	Combinatorics	4 SWS	Vorlesung (V)	Aksenovich, Sagdeev
SS 2026	0150310	Tutorial for Combinatorics (0150300)	2 SWS	Übung (Ü)	Aksenovich, Sagdeev

Voraussetzungen
 Keine

T

5.66 Teilleistung: Lab Project: Speech Translation [T-INFO-114205]

Verantwortung: Prof. Dr. Jan Niehues**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107176 - Lab Project: Speech Translation](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400150	Lab Project: Speech Translation	4 SWS	Praktikum (P) / ●	Niehues, Dinh

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Voraussetzungen

None.

Empfehlungen

Students should have understood the theoretical principles as introduced in the lectures Deep Learning or Machine Translation.

T**5.67 Teilleistung: Lesegruppe Softwaretechnik [T-INFO-102051]**

Verantwortung: Prof. Dr. Ralf Reussner
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107254 - Interdisciplinary Qualifications](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Studienleistung mündlich	1 LP	best./nicht best.	Jedes Semester	1

Erfolgskontrolle(n)

Die Erfolgskontrolle erfolgt in Form einer Studienleistung nach § 4 Abs. 3 SPO.

Die Erfolgskontrolle erfolgt durch die Teilnahme an Diskussionen und Vorstellung eines Beitrages aus einer Fachzeitschrift bzw. aus einem Konferenzband.

Voraussetzungen

Keine.

T**5.68 Teilleistung: Localization of Mobile Agents [T-INFO-115014]****Verantwortung:** Prof. Dr.-Ing. Uwe Hanebeck**Einrichtung:** KIT-Fakultät für Geistes- und Sozialwissenschaften
KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107663 - Localization of Mobile Agents](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Dauer	Version
Prüfungsleistung mündlich	6 LP	Drittelnoten	Jedes Sommersemester	1 Sem.	3

Erfolgskontrolle(n)

The assessment is carried out as an oral examination, lasting 15-30 minutes in accordance with Section 4 (2) No. 2 SPO. Participants can contact the chair at any time to arrange an appointment for the oral examination. (§ 4 Abs. 2 Nr. 2 SPO)

Voraussetzungen

T-INFO-115015 Digital exercise must be successfully completed before exam.

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Die Teilleistung [T-INFO-115015 - Localization of Mobile Agents - Pass](#) muss begonnen worden sein.

Empfehlungen

Knowledge of a higher-level programming language with mature libraries for scientific and numerical computing (e.g., Julia, Matlab, Python) is an advantage.

T**5.69 Teilleistung: Localization of Mobile Agents - Pass [T-INFO-115015]**

Verantwortung: Prof. Dr.-Ing. Uwe Hanebeck
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107663 - Localization of Mobile Agents](#)

Teilleistungsart
Studienleistung

Leistungspunkte
0 LP

Notenskala
best./nicht best.

Turnus
Jedes Sommersemester

Dauer
1 Sem.

Version
1

Erfolgskontrolle(n)

The assessment is carried out in form of course work (German Studienleistung, § 4 Abs. 3 SPO). Students must complete a digital exercise in Ilias which is ungraded. The digital exercise can be repeated as often as desired.

Voraussetzungen

None.

Empfehlungen

Knowledge of a higher-level programming language with mature libraries for scientific and numerical computing (e.g., Julia, Matlab, Python) is an advantage.

T**5.70 Teilleistung: Logical Foundations of Cyber-Physical Systems [T-INFO-112360]****Verantwortung:** Prof. Dr. André Platzer**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106102 - Logical Foundations of Cyber-Physical Systems](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	6 LP	Drittelnoten	Jedes Wintersemester	3

Lehrveranstaltungen					
WS 25/26	2400161	Logical Foundations of Cyber-Physical Systems	4 SWS	Vorlesung (V) / 	Platzer

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is usually carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 120 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take the form of an oral examination of approx.

- in the form of an oral examination of approx. 30 minutes in accordance with § 4 Para. 2 No. 2 SPO or
- in the form of a written examination in accordance with § 4 Para. 2 No. 1 SPO takes place.

In order to receive a bonus, you must earn at least 50% of the points for solving the exercises. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4).

Voraussetzungen

None.

Empfehlungen

The course assumes prior exposure to basic computer programming and mathematical reasoning. This course covers the basic required mathematical and logical background of cyber-physical systems. You will be expected to follow the textbook as needed: André Platzer. Logical Foundations of Cyber-Physical Systems. Springer 2018. DOI:10.1007/978-3-319-63588-0

AnmerkungenCourse web page: <https://lfcps.org/course/lfcps.html>

T

5.71 Teilleistung: Low Power Design [T-INFO-101344]

Verantwortung: Prof. Dr.-Ing. Jörg Henkel
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-100807 - Low Power Design](#)

Teilleistungsart
 Prüfungsleistung mündlich

Leistungspunkte
 3 LP

Notenskala
 Drittelnoten

Turnus
 Jedes Sommersemester

Version
 1

Lehrveranstaltungen					
SS 2026	2424672	Low Power Design	2 SWS	Vorlesung (V) / 	Henkel, Nassar, Khdr

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination lasting 25-30 minutes, in accordance with Section 4 (2) No. 2 SPO.

Voraussetzungen

None.

Empfehlungen

- Basic knowledge from the modules "Design and Architectures of Embedded Systems (ESII)" and "Optimization and Synthesis of Embedded Systems (ESI)" are helpful but not essential for understanding of this lecture.
- The lecture is equally suitable for students from both computer science as well as electrical engineering department.
- The Lab of "Low Power Design and Embedded Systems" enables students to apply some of the theoretical knowledge gained from the lecture in practice.

T**5.72 Teilleistung: Machine Learning - Foundations and Algorithms [T-INFO-114197]**

Verantwortung: Prof. Dr. Gerhard Neumann
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107169 - Machine Learning - Foundations and Algorithms](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	6 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	2400018	Machine Learning – Foundations and Algorithms	4 SWS	Vorlesung / Übung (VÜ) / ●	Neumann

Legende: ■ Online, 🔄 Präsenz/Online gemischt, ● Präsenz, ✕ Abgesagt

Erfolgskontrolle(n)

The success control takes place in the form of a written exam, usually 90 minutes in length, according to § 4 Abs. 2 Nr. 1 SPO.

A bonus can be acquired through successful participation in the exercise as a success control of a different kind (§4(2), 3 SPO 2008) or study performance (§4(3) SPO 2015). The exact criteria for awarding a bonus will be announced at the beginning of the lecture. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The bonus is only valid for the main and post exams of the semester in which it was earned. After that, the grade bonus expires.

Voraussetzungen

None.

Empfehlungen

- Attendance of the lecture "Foundations of Artificial Intelligence" ("Grundlagen der Künstlichen Intelligence")
- Knowledge in python
- Mathematics-heavy lecture. The basics will be reviewed, but mathematical proficiency is helpful

T

5.73 Teilleistung: Machine Learning for Natural Sciences [T-INFO-113916]**Verantwortung:** Prof. Dr. Pascal Friederich**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106959 - Machine Learning for Natural Sciences](#)**Teilleistungsart**
Prüfungsleistung mündlich**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	2400008	Machine Learning for the Natural Sciences	2 SWS	Vorlesung (V) / 	Friederich

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

Lecture: The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 90 minutes.

Exercise: The assessment is carried out in form of course work (German Studienleistung, § 4 Abs. 3 SPO). Students must regularly submit exercise sheets. The number of exercise sheets and the scale for passing will be announced at the beginning of the course. The assessment can only be repeated once.

Voraussetzungen

None.

Empfehlungen

- Knowledge of the basics of machine learning is helpful but not required
- Interest in natural science topics is required
- Basic knowledge of python is recommended. It has to be acquired during the semester through self-study

T**5.74 Teilleistung: Machine Learning for Natural Sciences - Pass [T-INFO-113917]**

Verantwortung: Prof. Dr. Pascal Friederich
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-106959 - Machine Learning for Natural Sciences](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	2400034	Exercise for Machine Learning for the Natural Sciences	2 SWS	Vorlesung / Übung (VÜ) / 	Friederich, Torresi, Neubert, Eberhard, Schlöder, Metni, Teufel

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

Lecture: The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 90 minutes.

Exercise: The assessment is carried out in form of course work (German Studienleistung, § 4 Abs. 3 SPO). Students must regularly submit exercise sheets. The number of exercise sheets and the scale for passing will be announced at the beginning of the course. The assessment can only be repeated once.

Voraussetzungen

None.

Empfehlungen

- Knowledge of the basics of machine learning is helpful but not required
- Interest in natural science topics is required
- Basic knowledge of python is recommended. It has to be acquired during the semester through self-study

T**5.75 Teilleistung: Machine Learning in Climate and Environmental Sciences [T-INFO-113083]****Verantwortung:** TT-Prof. Dr. Peer Nowack**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106470 - Machine Learning in Climate and Environmental Sciences](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	6 LP	Drittelnoten	Jedes Sommersemester	2

Lehrveranstaltungen					
SS 2026	2400003	Machine Learning in Climate and Environmental Sciences	4 SWS	Vorlesung / Übung (VÜ) / ●	Nowack

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment of the lectures is likely carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60-120 minutes (exact duration to be confirmed).

Depending on the class size, this might be changed to an oral examination (lasting around 20 minutes, § 4 Abs. 2 No. 2 SPO). The exact type of assessment will be confirmed at least six weeks prior to the assessment.

Voraussetzungen

No strict prerequisites but several strong recommendations (see below).

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Die Teilleistung [T-INFO-113085 - Machine Learning in Climate and Environmental Sciences - Pass](#) muss begonnen worden sein.

Empfehlungen

- Previous programming experience, e.g. in scientific contexts or in computer science, is required.
- Knowledge of fundamentals about machine learning is an advantage.
- Knowledge of the Python programming language is an advantage.
- Good knowledge of mathematical concepts such as linear algebra is an advantage.
- An interest in scientific questions important for the climate- and environmental sciences.

T**5.76 Teilleistung: Machine Learning in Climate and Environmental Sciences - Pass [T-INFO-113085]****Verantwortung:** TT-Prof. Dr. Peer Nowack**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106470 - Machine Learning in Climate and Environmental Sciences](#)**Teilleistungsart**
Studienleistung**Leistungspunkte**
0 LP**Notenskala**
best./nicht best.**Turnus**
Jedes Sommersemester**Version**
2

Lehrveranstaltungen					
SS 2026	2400003	Machine Learning in Climate and Environmental Sciences	4 SWS	Vorlesung / Übung (VÜ) / ●	Nowack

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out in form of course work (German Studienleistung, § 4 Abs. 3 SPO). Students must regularly submit exercise sheets. The number of exercise sheets and the scale for passing will be announced at the beginning of the course. The assessment can only be repeated once.

Voraussetzungen

No strict prerequisites but several strong recommendations (see below).

Empfehlungen

- Previous programming experience, e.g. in scientific contexts or in computer science, is required.
- Knowledge of fundamentals about machine learning is an advantage.
- Knowledge of the Python programming language is an advantage.
- Good knowledge of mathematical concepts such as linear algebra is an advantage.
- An interest in scientific questions important for the climate- and environmental sciences.

T

5.77 Teilleistung: Market Research [T-WIWI-107720]

Verantwortung: Prof. Dr. Martin Klarmann
Einrichtung: KIT-Fakultät für Wirtschaftswissenschaften
Bestandteil von: [M-WIWI-106258 - Digital Marketing](#)

Teilleistungsart
 Prüfungsleistung schriftlich

Leistungspunkte
 4,5 LP

Notenskala
 Drittelnoten

Turnus
 Jedes Sommersemester

Version
 3

Lehrveranstaltungen					
SS 2026	2571150	Market Research	2 SWS	Vorlesung (V) / 	Klarmann
SS 2026	2571151	Market Research Tutorial	1 SWS	Übung (Ü) / 	Klarmann

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

Die Erfolgskontrolle erfolgt in Form einer Klausur (70 Minuten) mit zusätzlichen Hilfsmitteln im Sinne einer Open Book Klausur. Weitere Details zur Ausgestaltung der Erfolgskontrolle werden im Rahmen der Vorlesung bekannt gegeben.

Voraussetzungen

Keine

Empfehlungen

Keine

Anmerkungen

Diese Veranstaltung ist Voraussetzung für Studierende, die an Abschlussarbeiten bei der Forschungsgruppe "Marketing und Vertrieb" interessiert sind.

T

5.78 Teilleistung: Master's Thesis [T-INFO-113697]

Verantwortung: Prof. Dr. Gerhard Neumann
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-106828 - Master's Thesis](#)

Teilleistungsart
Abschlussarbeit

Leistungspunkte
30 LP

Notenskala
Drittelnoten

Turnus
Jedes Semester

Version
1

Erfolgskontrolle(n)

The Master's thesis is regulated in § 14 and § 19 of the SPO23 Master of Computer Science. The presentation should take place no later than four weeks after submission of the Master's thesis.

The Master's thesis is assessed in the form of a report. An overall assessment (including the presentation) must be written.

The presentation should take place no later than four weeks after submission of the Master's thesis. The presentation may also take place before submission.

Voraussetzungen

A prerequisite for admission to the Master's thesis is that students have generally already acquired 60 credit points, of which at least 15 credit points must come from one of the two specialization subjects. The application for admission to the Master's thesis must be submitted no later than three months after taking the last module examination.

Abschlussarbeit

Bei dieser Teilleistung handelt es sich um eine Abschlussarbeit. Es sind folgende Fristen zur Bearbeitung hinterlegt:

Bearbeitungszeit 6 Monate
Maximale Verlängerungsfrist 3 Monate
Korrekturfrist 8 Wochen

Die Abschlussarbeit ist genehmigungspflichtig durch den Prüfungsausschuss.

T

5.79 Teilleistung: Media Management [T-WIWI-112711]

Verantwortung: Prof. Dr. Ann-Kristin Kupfer
Einrichtung: KIT-Fakultät für Wirtschaftswissenschaften
Bestandteil von: [M-WIWI-106258 - Digital Marketing](#)

Teilleistungsart
 Prüfungsleistung anderer Art

Leistungspunkte
 4,5 LP

Notenskala
 Drittelnoten

Turnus
 Jedes Wintersemester

Version
 1

Lehrveranstaltungen					
WS 25/26	2572192	Media Management	2 SWS	Vorlesung (V) / ●	Kupfer
WS 25/26	2572193	Media Management Exercise	1 SWS	Übung (Ü) / ●	Kupfer, Mitarbeiter, Schmitt

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

Die Erfolgskontrolle erfolgt in Form einer Prüfungsleistung anderen Art. In die Bewertung fließen folgende Aspekte ein:

- Ausarbeitung und Präsentation einer Gruppenaufgabe
- Schriftliche Prüfung

Weitere Details zur Organisation der Leistungserbringung sowie zum Punktesystem der Bewertung werden in der Vorlesung bekannt gegeben.

Voraussetzungen

None

Empfehlungen

Studierende werden ausdrücklich dazu ermutigt, sich aktiv am Unterricht zu beteiligen.

Arbeitsaufwand

135 Std.

T

5.80 Teilleistung: Mobile Communication [T-INFO-114271]

Verantwortung: Prof. Dr. Oliver Waldhorst
Prof. Dr. Martina Zitterbart

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107245 - Mobile Communication](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	4 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2400015	Mobile Communication	2 SWS	Vorlesung (V)	Waldhorst, Mahrt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take the form of an oral examination of approx.

- in the form of an oral examination of approx. 30 minutes in accordance with § 4 Para. 2 No. 2 SPO or

- in the form of a written examination in accordance with § 4 Para. 2 No. 1 SPO

takes place.

Voraussetzungen

None.

Empfehlungen

The contents of the lecture Introduction to Computer Networks are assumed to be known. Attendance of the lecture Telematics is strongly recommended, as the contents are an important basis for understanding and classifying the material.

T

5.81 Teilleistung: Mobile Communications [T-ETIT-112127]**Verantwortung:** Prof. Dr.-Ing. Peter Rost**Einrichtung:** KIT-Fakultät für Elektrotechnik und Informationstechnik**Bestandteil von:** [M-ETIT-105971 - Mobile Communications](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	4 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2310523	Mobile Communications	2 SWS	Vorlesung (V) / 	Rost
WS 25/26	2310524	Exercises for 2310523 Mobile Communications	1 SWS	Übung (Ü) / 	Rost

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 15 minutes in which preparatory tasks are solved.

Voraussetzungen

none

T

5.82 Teilleistung: Model-Driven Software Development [T-INFO-113896]

Verantwortung: Dr.-Ing. Erik Burger
Prof. Dr. Ralf Reussner

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-106931 - Model-Driven Software Development](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	3 LP	Drittelnoten	Jedes Wintersemester	2

Lehrveranstaltungen					
WS 25/26	2424657	Model-Driven Software Development	2 SWS	Vorlesung (V) / 	Burger

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 25 minutes.

Voraussetzungen

None.

Empfehlungen

Basic knowledge from the lecture Software Engineering II is helpful.

T**5.83 Teilleistung: Moderne Methoden der Kombinatorik [T-MATH-113911]**

Verantwortung: Prof. Dr. Maria Aksenovich
Einrichtung: KIT-Fakultät für Mathematik
Bestandteil von: [M-MATH-106957 - Moderne Methoden der Kombinatorik](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	6 LP	Drittelnoten	Unregelmäßig	1

Erfolgskontrolle(n)
mündliche Prüfung (ca. 30 min)

Voraussetzungen
Keine

Arbeitsaufwand
180 Std.

T

5.84 Teilleistung: Motion in Human and Machine - Seminar [T-INFO-105140]**Verantwortung:** Prof. Dr.-Ing. Tamim Asfour**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-102555 - Motion in Human and Machine - Seminar](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Jedes Sommersemester	3

Lehrveranstaltungen					
SS 2026	2400063	Motion in Human and Machine	3 SWS	Seminar (S) / ●	Asfour

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). It includes a term paper and a final presentation.

Voraussetzungen

None.

Empfehlungen

Programming skills in C++, Python or Matlab are recommended.

Attendance of the lectures Robotics I – Introduction to Robotics, Robotics II – Humanoid Robotics, Robotics III – Sensors and Perception in Robotics, Mechano-Informatics and Robotics and Wearable Robot Technologies is recommended

Anmerkungen

This interdisciplinary seminar is conducted in cooperation with the University of Stuttgart and the University of Tübingen.

T

5.85 Teilleistung: Multimodal Artificial Intelligence [T-INFO-115041]

Verantwortung: Prof. Dr. Jan Niehues
Prof. Dr.-Ing. Rainer Stiefelhagen

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107676 - Multimodal Artificial Intelligence](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	6 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	2400141	Multimodal Artificial Intelligence	4 SWS	Vorlesung (V) / 	Niehues, Stiefelhagen

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) approx. 60 minutes.

Voraussetzungen

None.

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Die Teilleistung [T-INFO-114220 - Advanced Artificial Intelligence](#) darf nicht begonnen worden sein.

T

5.86 Teilleistung: Nano- and Quantum Electronics [T-ETIT-111232]

Verantwortung: Prof. Dr. Sebastian Kempf

Einrichtung: KIT-Fakultät für Elektrotechnik und Informationstechnik

Bestandteil von: [M-ETIT-105604 - Nano- and Quantum Electronics](#)

Teilleistungsart Prüfungsleistung schriftlich	Leistungspunkte 6 LP	Notenskala Drittelnoten	Turnus Jedes Sommersemester	Dauer 1 Sem.	Version 1
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Lehrveranstaltungen					
SS 2026	2312668	Nano- and Quantum Electronics	3 SWS	Vorlesung (V) / 	Kempf
SS 2026	2312670	Tutorial for 2312668 Nano- and Quantum Electronics	1 SWS	Übung (Ü) / 	Kempf, Mitarbeiter*innen

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment of success takes place in the form of a written examination lasting 120min. The grade corresponds to the result of the written examination.

Voraussetzungen

none

Empfehlungen

Successful completion of the modules "Superconductivity for Engineers" and „Einführung in die Quantentheorie für Elektrotechniker“ is recommended.

T

5.87 Teilleistung: Natural Language Processing [T-INFO-114207]**Verantwortung:** Prof. Dr. Jan Niehues**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107178 - Natural Language Processing](#)**Teilleistungsart**
Prüfungsleistung schriftlich**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400147	Natural Language Processing	4 SWS	Vorlesung (V) / 	Niehues, Liu, Züfle

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Voraussetzungen

None.

T**5.88 Teilleistung: Natural Language Processing and Software Engineering [T-INFO-114257]****Verantwortung:** Prof. Dr.-Ing. Anne Kozirolek**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107233 - Natural Language Processing and Software Engineering](#)**Teilleistungsart**
Prüfungsleistung mündlich**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	2424187	Natural Language Processing and Software Engineering	2 SWS	Vorlesung (V) / 	Hey, Kozirolek, Keim

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 25 minutes.

Voraussetzungen

None.

T**5.89 Teilleistung: Network Security: Architectures and Protocols [T-INFO-114238]****Verantwortung:** Prof. Dr. Martina Zitterbart**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107218 - Network Security: Architectures and Protocols](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	4 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	24601	Network Security: Architectures and Protocols	2 SWS	Vorlesung (V) / 	Baumgart, Bless, Zitterbart

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take the form of an oral examination of approx.

- in the form of an oral examination of approx. 30 minutes in accordance with § 4 Para. 2 No. 2 SPO or
- in the form of a written examination in accordance with § 4 Para. 2 No. 1 SPO

takes place.

Voraussetzungen

None.

Empfehlungen

The contents of the lecture Introduction to Computer Networks are assumed to be known. Attendance of the lecture Telematics is strongly recommended, as the contents are an important basis for understanding and classifying the material.

T

5.90 Teilleistung: Next Generation Internet [T-INFO-101321]

Verantwortung: Dr.-Ing. Roland Bless
Prof. Dr. Martina Zitterbart

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-100784 - Next Generation Internet](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	4 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	24674	Next Generation Internet	2 SWS	Vorlesung (V) / 	Bless

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take the form of an oral examination of approx.

- in the form of an oral examination of approx. 30 minutes in accordance with § 4 Para. 2 No. 2 SPO **or**
- in the form of a written examination in accordance with § 4 Para. 2 No. 1 SPO takes place.

Voraussetzungen

None.

Empfehlungen

The contents of the lecture Introduction to Computer Networks are assumed to be known. Attendance of the lecture Telematics is strongly recommended, as the contents are an important basis for understanding and classifying the material.

T**5.91 Teilleistung: Online-Konzepte für Karlsruher Innenstadthändler [T-WIWI-111848]**

Verantwortung: Prof. Dr. Ann-Kristin Kupfer
Einrichtung: KIT-Fakultät für Wirtschaftswissenschaften
Bestandteil von: [M-WIWI-106258 - Digital Marketing](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Jedes Sommersemester	2

Lehrveranstaltungen					
SS 2026	2571184	Online-Konzepte für Karlsruher Innenstadthändler	2 SWS	Sonstige (sonst.) / 	Kupfer

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt

Erfolgskontrolle(n)

Die Erfolgskontrolle erfolgt in Form einer Prüfungsleistung anderer Art:

- Präsentationen in Teams im Umfang von jeweils ca. 15 Minuten pro Team mit anschließender Diskussion
- Abgabe einer schriftlichen Ausarbeitung pro Team.

Anmerkungen

Bitte beachten Sie, dass für den Besuch dieser Veranstaltung eine Bewerbung erforderlich ist. Nähere Informationen zum Bewerbungsprozess erhalten Sie in der Regel kurz vor Beginn der Vorlesungszeit im Sommersemester auf der Webseite der Forschungsgruppe Marketing und Vertrieb (marketing.iism.kit.edu).

Arbeitsaufwand

90 Std.

T

5.92 Teilleistung: Optical Engineering [T-ETIT-100676]

Verantwortung: Prof. Dr. Wilhelm Stork**Einrichtung:** KIT-Fakultät für Elektrotechnik und Informationstechnik**Bestandteil von:** [M-ETIT-100456 - Optical Engineering](#)**Teilleistungsart**
Prüfungsleistung mündlich**Leistungspunkte**
4 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	2311629	Optical Engineering	2 SWS	Vorlesung (V) / x	Stork
WS 25/26	2311631	Tutorial for 2311629 Optical Engineering	1 SWS	Übung (Ü) / x	Fan

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt**Erfolgskontrolle(n)**

Die Erfolgskontrolle erfolgt im Rahmen einer mündlichen Prüfung (ca. 20 Minuten).

Voraussetzungen

keine

T**5.93 Teilleistung: Optimization and Synthesis of Embedded Systems (ESI) [T-INFO-114253]****Verantwortung:** Prof. Dr.-Ing. Jörg Henkel**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107229 - Optimization and Synthesis of Embedded Systems \(ESI\)](#)**Teilleistungsart**
Prüfungsleistung mündlich**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	2424143	Optimierung und Synthese Eingebetteter Systeme (ES1)	2 SWS	Vorlesung (V) / 	Henkel, Nassar

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting approx. 20 minutes.

Voraussetzungen

None.

Empfehlungen

Knowledge of computer structures is helpful.

The prerequisites, if any, are explained in more detail in the module description.

T**5.94 Teilleistung: Optimization Methods for Machine Learning and Engineering [T-INFO-110809]**

Verantwortung: Prof. Dr.-Ing. Jürgen Beyerer
Julius Pfrommer

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-105329 - Optimization Methods for Machine Learning and Engineering](#)

Teilleistungsart
Prüfungsleistung schriftlich

Leistungspunkte
5 LP

Notenskala
Drittelnoten

Turnus
Jedes Wintersemester

Version
2

Lehrveranstaltungen					
WS 25/26	2400280	Optimization Methods for Machine Learning and Engineering	2 SWS	Vorlesung (V) / 	Pfrommer, Beyerer
WS 25/26	2400281	Optimization Methods for Machine Learning and Engineering	1 SWS	Übung (Ü) / 	Pfrommer, Beyerer

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take the form of an oral examination of approx.

- in the form of an oral examination of approx. 30 minutes in accordance with § 4 Para. 2 No. 2 SPO **or**
- in the form of a written examination in accordance with § 4 Para. 2 No. 1 SPO

takes place.

Voraussetzungen

None.

T

5.95 Teilleistung: Parallel Algorithms [T-INFO-114221]

Verantwortung: Prof. Dr. Peter Sanders
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107199 - Parallel Algorithms](#)

Teilleistungsart
 Prüfungsleistung mündlich

Leistungspunkte
 4 LP

Notenskala
 Drittelnoten

Turnus
 Jedes Wintersemester

Version
 1

Lehrveranstaltungen					
WS 25/26	2400053	Parallel Algorithms	3 SWS	Vorlesung (V) / 	Sanders, Uhl, Laupichler

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Voraussetzungen

T-INFO-114222 - Parallel Algorithms Pass must be started.

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Die Teilleistung [T-INFO-101333 - Parallele Algorithmen](#) darf nicht begonnen worden sein.
2. Die Teilleistung [T-INFO-114222 - Parallel Algorithms Pass](#) muss begonnen worden sein.

Empfehlungen

Knowledge from lectures such as Algorithms I/II is recommended.

T**5.96 Teilleistung: Parallel Algorithms Pass [T-INFO-114222]**

Verantwortung: Prof. Dr. Peter Sanders
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107199 - Parallel Algorithms](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	1 LP	Drittelnoten	Jedes Wintersemester	1

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

The exercise can be proven via various performance records (usually exercise sheets). This will be determined individually during the lecture.

An overall grade will be awarded.

Voraussetzungen

None.

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Die Teilleistung [T-INFO-111857 - Parallele Algorithmen Übung](#) darf nicht begonnen worden sein.

Empfehlungen

Knowledge from lectures such as Algorithms I/II is recommended.

T

5.97 Teilleistung: Parameterized Algorithms [T-INFO-114194]

Verantwortung: TT-Prof. Dr. Thomas Bläsius
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107167 - Parameterized Algorithms](#)

Teilleistungsart
 Prüfungsleistung mündlich

Leistungspunkte
 6 LP

Notenskala
 Drittelnoten

Version
 1

Lehrveranstaltungen					
SS 2026	2400178	Parameterized Algorithms	4 SWS	Vorlesung / Übung (VÜ) / ●*	Bläsius

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Voraussetzungen

None.

Empfehlungen

Basic knowledge of algorithms and data structures (e.g. from the lectures Algorithms 1 + 2) is helpful.

T

5.98 Teilleistung: Parameterized Algorithms - Pass [T-INFO-114195]

Verantwortung: TT-Prof. Dr. Thomas Bläsius
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107167 - Parameterized Algorithms](#)

Teilleistungsart
Studienleistung

Leistungspunkte
0 LP

Notenskala
best./nicht best.

Version
1

Lehrveranstaltungen					
SS 2026	2400178	Parameterized Algorithms	4 SWS	Vorlesung / Übung (VÜ) / ●	Bläsius

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A total of two repetitions are possible.

Voraussetzungen

None.

Empfehlungen

Basic knowledge of algorithms and data structures (e.g. from the lectures Algorithms 1 + 2) is helpful.

T

5.99 Teilleistung: Participatory Technology Design [T-INFO-114199]**Verantwortung:** Prof. Dr. Kathrin Gerling**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107170 - Participatory Technology Design](#)**Teilleistungsart**
Prüfungsleistung schriftlich**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400189	Participatory Technology Design (V/Ü)	4 SWS	Vorlesung / Übung (VÜ)	Gerling

Erfolgskontrolle(n)

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 90 minutes.

Voraussetzungen

None.

Empfehlungen

Knowledge of the basics of human-machine interaction is helpful.

T**5.100 Teilleistung: Participatory Technology Design - Pass [T-INFO-114200]****Verantwortung:** Prof. Dr. Kathrin Gerling**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107170 - Participatory Technology Design](#)**Teilleistungsart**
Studienleistung**Leistungspunkte**
0 LP**Notenskala**
best./nicht best.**Turnus**
Jedes Wintersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400189	Participatory Technology Design (V/Ü)	4 SWS	Vorlesung / Übung (VÜ)	Gerling

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). A total of two repetitions are possible.

Voraussetzungen

None.

Empfehlungen

Knowledge of the basics of human-machine interaction is helpful.

T**5.101 Teilleistung: Post-quantum Cryptography from Lattices [T-INFO-114987]****Verantwortung:** Prof. Dr. Jörn Müller-Quade**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107642 - Post-quantum Cryptography from Lattices](#)**Teilleistungsart**
Prüfungsleistung mündlich**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Dauer**
1 Sem.**Version**
1

Lehrveranstaltungen					
SS 2026	2400076	Post-quantum Cryptography from Lattices	2 SWS	Vorlesung (V) / ● ^s	Klooß, Kühn, Brede

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) approx. 45 minutes.

Voraussetzungen

None.

Empfehlungen

Basic familiarity with cryptography as introduced in the module “Theoretische Grundlagen der Kryptographie” is recommended, in particular, familiarity with security notions for basic primitives (IND-CPA, EUF-CMA) and security reductions (including hybrid arguments and guessing).

T**5.102 Teilleistung: Practical Course on Network Security Research [T-INFO-114270]**

Verantwortung: Mario Hock
Prof. Dr. Martina Zitterbart

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107244 - Practical Course on Network Security Research](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
SS 2026	2400222	Practical Course on Network Security Research		Praktikum (P) / ●	Zitterbart

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Among other things, implementation, documentation, presentation in the colloquium and the research report to be prepared are included in the assessment of success.

Withdrawal is possible up to two weeks after the first (online) presentation event.

Voraussetzungen

None.

Empfehlungen

The module Network Security: Architectures and Protocols [M-INFO-100782] should have been started or completed.

T**5.103 Teilleistung: Practical Course on Telematics Research [T-INFO-114239]****Verantwortung:** Prof. Dr. Martina Zitterbart**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107220 - Practical Course on Telematics Research](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Unregelmäßig	1

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Among other things, implementation, documentation, presentation in the colloquium and the research report to be prepared are included in the assessment of success.

Voraussetzungen

None.

Empfehlungen

A pronounced scientific interest in the topics of network security is a prerequisite: no prefabricated exercises are worked on, instead the internship requires a high degree of personal initiative.

T**5.104 Teilleistung: Practical Course: Advanced Topics in High Performance Computing, Data Management and Analytics [T-INFO-111803]****Verantwortung:** Prof. Dr. Achim Streit**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-105870 - Practical Course: Advanced Topics in High Performance Computing, Data Management and Analytics](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400043	Advanced Topics in High Performance Computing, Data Management and Analytics	3 SWS	Praktikum (P)	Farhadi, Streit
SS 2026	2400068	Advanced Topics in High Performance Computing, Data Management and Analytics	3 SWS	Praktikum (P)	Streit, Schlitter

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). The examination can consist of experiments or projects, each with a concluding presentation. Students may redraw from the assigned topic during the first two weeks after the topic has been communicated.

Voraussetzungen

None.

Empfehlungen

Knowledge in the area of databases, data management, data analytics, parallel computing is helpful.

T**5.105 Teilleistung: Practical Course: AI for Climate and Weather Predictions [T-INFO-113659]****Verantwortung:** TT-Prof. Dr. Peer Nowack**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106800 - Practical Course: AI for Climate and Weather Predictions](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	6 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
WS 25/26	2400064	AI for climate and weather predictions	2 SWS	Praktikum (P) / ● ^s	Nowack, Amiramjadi
SS 2026	2400082	AI for climate and weather predictions	3 SWS	Praktikum (P) / ● ^s	Nowack, Amiramjadi

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Es müssen eine schriftliche Ausarbeitung erstellt und eine Präsentation gehalten werden. Ein Rücktritt ist innerhalb von zwei Wochen nach Vergabe des Themas möglich.

Voraussetzungen

- Previous programming experience, e.g., in scientific contexts or in computer science, is required.
- Students should have previous experience in the theory and implementation of machine learning models.

Empfehlungen

- Knowledge of the Python programming language.
- Good knowledge of mathematical concepts such as linear algebra is an advantage.
- An interest in scientific questions around climate science and weather forecasting.

T**5.106 Teilleistung: Practical Course: Application Security [T-INFO-113958]****Verantwortung:** Prof. Dr. Jörn Müller-Quade**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106996 - Practical Course: Application Security](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
4 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400114	Application Security Lab	4 SWS	Praktikum (P) / ●	Müller-Quade, Mechler, Dörre, Noppel, Wressnegger
SS 2026	2400117	Application Security Lab	4 SWS	Praktikum (P) / ●	Müller-Quade, Wressnegger, Sperle, Gharibyar

Legende: ■ Online, 🔄 Präsenz/Online gemischt, ● Präsenz, ✕ Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). Students have to solve different tasks. An overall grade is awarded.

Voraussetzungen

None.

Empfehlungen

The basics of IT security are assumed.

The content of the lectures "Computer Organization" and "Operating Systems" should be known.

T**5.107 Teilleistung: Practical Course: Artificial Intelligence & Security Lab (AISEC-Lab) [T-INFO-113760]****Verantwortung:** Prof. Dr. Christian Wressnegger**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106867 - Practical Course: Artificial Intelligence & Security Lab \(AISEC-Lab\)](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
4 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	24002	Practical Course: Artificial Intelligence & Security (AISEC-Lab)	4 SWS	Praktikum (P) / ●	Wressnegger, Zhao

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

At least one assignment from each unit must be successfully completed (comparable results to other students).

Voraussetzungen

Keine.

Empfehlungen

The basics of IT security and artificial intelligence are a prerequisite.

T

5.108 Teilleistung: Practical Course: Building PETs [T-INFO-114550]

Verantwortung: Daniel Schadt
Prof. Dr. Thorsten Strufe

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107412 - Practical Course: Building PETs](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Dauer	Version
Prüfungsleistung anderer Art	4 LP	Drittelnoten	Jedes Wintersemester	1 Sem.	1

Lehrveranstaltungen					
WS 25/26	2400056	Praktikum: Building PETs	4 SWS	Praktikum (P) / ●	Strufe

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

The students must hand in the source code of their programs to be examined. Additionally, each student must explain their solutions to the tasks. An overall grade is awarded.

Voraussetzungen

None.

Empfehlungen

Familiarity with a programming language and a HTTP library is recommended.

T

5.109 Teilleistung: Practical Course: Chip Design I [T-INFO-114298]**Verantwortung:** Prof. Dr. Mehdi Baradaran Tahoori**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107265 - Practical Course: Chip Design I](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400216	Practical Course: Chip Design I	4 SWS	Praktikum (P) / ●	Tahoori
SS 2026	2400104	Practical Course: Chip Design I	4 SWS	Praktikum (P) / ●	Tahoori

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

The overall impression is evaluated. The grading is based on the results of the practical work (80 %) and the final presentation(20%). An overall grade is awarded.

Students may redraw from the examination during the first two weeks after the topic has been communicated. The assessment can be repeated once.

Voraussetzungen

None.

Empfehlungen

The requirements are individual to each of the offered projects.
Knowledge of HDL is helpful.

T

5.110 Teilleistung: Practical Course: Chip Design II [T-INFO-114299]**Verantwortung:** Prof. Dr. Mehdi Baradaran Tahoori**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107266 - Practical Course: Chip Design II](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400219	Practical Course: Chip Design II	4 SWS	Praktikum (P) / ●	Tahoori
SS 2026	2400107	Practical Course: Chip Design II	4 SWS	Praktikum (P) / ●	Tahoori

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

The overall impression is evaluated. The grading is based on the results of the practical work (80 %) and the final presentation(20%). An overall grade is awarded.

Students may redraw from the examination during the first two weeks after the topic has been communicated. The assessment can be repeated once.

Voraussetzungen

None.

Empfehlungen

The requirements are individual to each of the offered projects.
Knowledge of HDL is helpful.

T**5.111 Teilleistung: Practical Course: Continuous Software Engineering [T-INFO-115043]****Verantwortung:** Prof. Dr.-Ing. Anne Koziolk**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107677 - Practical Course: Continuous Software Engineering](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	6 LP	Drittelnoten	Jedes Semester	1

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO), in the form of predominantly practical tasks and a final presentation.

Voraussetzungen

None.

Empfehlungen

Solid programming skills are required in order to successfully complete the lab course within the specified workload.

Anmerkungen

In earlier semesters, this course was named „Praktikum: Werkzeuge für agile Modellierung“.

T**5.112 Teilleistung: Practical Course: Digital Design & Test Automation Flow [T-INFO-105565]****Verantwortung:** Prof. Dr. Mehdi Baradaran Tahoori**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-102570 - Practical Course: Digital Design & Test Automation Flow](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	2424318	Digital Design & Test Automation Flow	4 SWS	Praktikum (P) / ● ^s	Tahoori

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out in form of an examination of another type (§ 4 Abs. 2 No. 3 SPO). Students must give a presentation.

The module grade is made up of 80% of the work completed in the practical course and 20% of the presentation.

An overall grade is awarded.

Voraussetzungen

None.

Empfehlungen

Knowledge of "Dependable Computing" and "Fault Tolerant Computing" and Computer Architecture is helpful.

T

5.113 Teilleistung: Practical Course: Efficient Parallel C++ [T-INFO-114228]

Verantwortung: Prof. Dr. Peter Sanders**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107203 - Practical Course: Efficient Parallel C++](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	6 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
WS 25/26	2400195	Lab: Efficient Parallel C++	4 SWS	Praktikum (P) / 	Sanders, Schimek, Williams, Seemaier

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). Students have to solve multiple programming tasks in C++. An overall grade is awarded.

Voraussetzungen

None.

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Die Teilleistung [T-INFO-106992 - Praktikum: Effizientes paralleles C++](#) darf nicht begonnen worden sein.

Empfehlungen

At least basic knowledge of the C++ language is necessary for participation in the course. Students should be able to implement given algorithms.

T**5.114 Teilleistung: Practical Course: Engineering Approaches to Software Development [T-INFO-115044]****Verantwortung:** Prof. Dr. Ralf Reussner**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107678 - Practical Course: Engineering Approaches to Software Development](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400093	Praktikum Ingenieurmäßige Software-Entwicklung	4 SWS	Praktikum (P)	Reussner
SS 2026	24908	Praktikum Ingenieurmäßige Software-Entwicklung	4 SWS	Praktikum (P) / 	Reussner

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO), in the form of predominantly practical tasks and a final presentation.

Voraussetzungen

None.

Empfehlungen

Solid programming skills are required in order to successfully complete the lab course within the specified workload.

Anmerkungen

In earlier semesters, this course was named „Praktikum Ingenieurmäßige Software-Entwicklung“.

T**5.115 Teilleistung: Practical Course: Fine-grained Algorithm Design and Engineering [T-INFO-113635]****Verantwortung:** Prof. Dr.-Ing. Marvin Künnemann**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106784 - Practical Course: Fine-grained Algorithm Design and Engineering](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	6 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
WS 25/26	2400104	Fine-grained Algorithm Design and Engineering	4 SWS	Praktikum (P) / ● ^s	Künnemann

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). The overall performance is evaluated, which includes the quality of the produced results, the project report and the presentation.

Voraussetzungen

None.

Empfehlungen

- Basic knowledge of algorithms and data structures is assumed.
- Knowledge of fine-grained complexity is helpful, but not required.

T**5.116 Teilleistung: Practical Course: FPGA Programming [T-INFO-105576]****Verantwortung:** Prof. Dr. Mehdi Baradaran Tahoori**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-102661 - Practical Course: FPGA Programming](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400106	FPGA Programming	4 SWS	Praktikum (P) / ●	Tahoori
SS 2026	2400106	FPGA Programming	4 SWS	Praktikum (P) / ●	Tahoori

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out in form of an examination of another type (§ 4 Abs. 2 No. 3 SPO). Students must give a presentation.

The module grade is made up of 80% of the work completed in the practical course and 20% of the presentation.

An overall grade is awarded.

Voraussetzungen

None.

Empfehlungen

Knowledge of "Dependable Computing" and "Fault Tolerant Computing" and Computer Architecture is helpful.

T**5.117 Teilleistung: Practical Course: General-Purpose Computation on Graphics Processing Units [T-INFO-109914]****Verantwortung:** Prof. Dr.-Ing. Carsten Dachsbacher**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-100724 - Practical Course: General-Purpose Computation on Graphics Processing Units](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	2424297	Practical course General-Purpose Computation on Graphics Processing Units	2 SWS	Praktikum (P) / ●	Klepikov, Alber, Lerzer, Dachsbacher
SS 2026	24911	Practical course General-Purpose Computation on Graphics Processing Units	2 SWS	Praktikum (P) / ●	Lerzer, Klepikov, Dachsbacher, Dereviannykh

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Performance is assessed continuously for the individual projects and in a final presentation.

Voraussetzungen

None.

Empfehlungen

It is recommended to have attended relevant lectures in the specialisation area of computer graphics.

T**5.118 Teilleistung: Practical Course: Human-Centred Robotics [T-INFO-113393]****Verantwortung:** Prof. Dr. Katja Mombaur**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106646 - Practical Course: Human-Centred Robotics](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400193	Praktikum: Human-Centred Robotics Projects	4 SWS	Praktikum (P) / ● ^s	Mombaur, Lee
SS 2026	2400149	Praktikum: Human-Centred Robotics	4 SWS	Praktikum (P) / ● ^s	Mombaur, Lee

Legende: ■ Online, 🔄 Präsenz/Online gemischt, ● Präsenz, x Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

This includes the preparation of a project report (ca. 10 pages and an oral presentation with slides and hardware demonstration (30 Min + 15 min questions). Students may withdraw from the examination during the first two weeks after the topic has been communicated.

Voraussetzungen

Programming skills are required..

Empfehlungen

Knowledge in Robotics (e.g. from the class Robotics 1 and follow-ups) are very helpful. h.

Anmerkungen

Limited number of projects and participants. Specific project topics will be different each term and will be announced in a presentation during the first semester week. .

T**5.119 Teilleistung: Practical Course: Interactive Theorem Proving [T-INFO-114691]**

Verantwortung: Prof. Dr. Bernhard Beckert
Terru Stübinger

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107494 - Practical Course: Interactive Theorem Proving](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Dauer	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Unregelmäßig	1 Sem.	1

Lehrveranstaltungen					
WS 25/26	2400229	"Practical Course: Interactive Theorem Proving"	2 SWS	Praktikum (P) / ●	Beckert, Stübinger, Färber
SS 2026	2400216	Interactive Theorem Proving	2 SWS	Praktikum (P) / ●	Erdweg, Xu, Bitar, Lego

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). Students are graded on the formalisation (e.g. Isabelle theory files) produced during the project phase, along with a presentation on their completed project (duration approx. 20min). Students may redraw from the examination until two weeks before the final presentations. The assessment can be repeated once.

Voraussetzungen

None.

Empfehlungen

- Knowledge of functional programming is assumed.
- Some knowledge of logic (e.g. via courses such as Formale Systeme and/or Constructive Logic) will be helpful, but is not required

T

5.120 Teilleistung: Practical Course: Internet of Things (IoT) [T-INFO-107493]**Verantwortung:** Prof. Dr.-Ing. Jörg Henkel**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-103706 - Practical Course: Internet of Things \(IoT\)](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
4 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	2424304	Internet of Things (IoT) Lab	4 SWS	Praktikum (P) /	Mentzos, Henkel, Tobar
SS 2026	2424304	Internet of Things (IoT) Lab	4 SWS	Praktikum (P) /	Henkel, Mentzos, Tobar

Legende: Online, Präsenz/Online gemischt, Präsenz, x Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO), in the form of a practical assignment, presentations and, if necessary, a written report. Written reports, presentations and practical work are weighted depending on the event.

Voraussetzungen

Basic skills in **C** or **C++** programming.

Empfehlungen

- Familiarity with other (than C) languages like Python could be helpful as well.
- Basic knowledge from the modules "Design and Architectures of Embedded Systems (ESII)" and "Optimization and Synthesis of Embedded Systems (ESI)" are helpful but not essential for understanding the lab.

T**5.121 Teilleistung: Practical Course: Low Power Design and Embedded Systems [T-INFO-108323]****Verantwortung:** Prof. Dr.-Ing. Jörg Henkel**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-104031 - Practical Course: Low Power Design and Embedded Systems](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
4 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
2

Lehrveranstaltungen					
SS 2026	2424811	Low Power Design and Embedded Systems	4 SWS	Praktikum (P) / 	Henkel, Khdr, Sikal, Mentzos

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

The overall impression is evaluated.

The grading will be based on multiple exercises and a final report.

Details of the grading scale will be announced during the course.

Voraussetzungen

None.

Empfehlungen

Students should be familiar with software development practices under Linux-based systems. Practical knowledge in C/C++ as well as Python is required.

T**5.122 Teilleistung: Practical Course: Mathematical and Computational Methods in Robotics and AI [T-INFO-114911]**

Verantwortung: Prof. Dr. Katja Mombaur
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107609 - Practical Course: Mathematical and Computational Methods in Robotics and AI](#)

Teilleistungsart Prüfungsleistung anderer Art	Leistungspunkte 6 LP	Notenskala Drittelnoten	Turnus Jedes Semester	Version 2
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Lehrveranstaltungen					
WS 25/26	2400181	Praktikum: Mathematical and Computational Methods in Robotics & AI	4 SWS	Praktikum (P) / 	Mombaur, Große Sundrup
SS 2026	2400202	Praktikum: Mathematical and Computational Methods in Robotics & AI	4 SWS	Praktikum (P) / 	Mombaur, Große Sundrup

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

This includes the preparation of a project report (ca. 10 pages and an oral presentation (30 Min + 15 min questions). Students may withdraw from the examination during the first two weeks after the topic has been communicated.

Voraussetzungen

Programming skills

Empfehlungen

Basic mathematical knowledge will be helpful, depending on topic. For Projects in the context of robotics, knowledge of robotics is recommended, but not strictly necessary.

Anmerkungen

Limited number of projects and participants. Specific project topics will be different each term and will be announced in a presentation during the first semester week or can be individually agreed upon with people supervising the practical. Students can propose topics on their own as well.

T**5.123 Teilleistung: Practical Course: Model-Driven Software Development [T-INFO-113897]**

Verantwortung: Dr.-Ing. Erik Burger
Prof. Dr. Ralf Reussner

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-106932 - Practical Course: Model-Driven Software Development](#)

Teilleistungsart
Prüfungsleistung anderer Art

Leistungspunkte
6 LP

Notenskala
Drittelnoten

Turnus
Jedes Sommersemester

Version
2

Lehrveranstaltungen					
SS 2026	2400091	Praktikum Modellgetriebene Software-Entwicklung	4 SWS	Praktikum (P) / 	Burger

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO), in the form of predominantly practical tasks.

Voraussetzungen

None.

Empfehlungen

Attending the lectures Software Engineering II and Model-Driven Software Development is helpful.

T**5.124 Teilleistung: Practical Course: Movement and Technology [T-INFO-113394]****Verantwortung:** Prof. Dr. Katja Mombaur**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106648 - Practical Course: Movement and Technology](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
2

Lehrveranstaltungen					
SS 2026	2400151	Praktikum: Movement and Technology	4 SWS	Praktikum (P) / ●	Mombaur, Lau

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

This includes the preparation of a project report (ca. 10 pages and an oral presentation of the project topics and results with slides. Students may withdraw from the examination during the first two weeks after the topic has been communicated.

Voraussetzungen

Programming skills are required.

Empfehlungen

Knowledge in Robotics (e.g. from the class Robotics 1 and follow-ups) are very helpful.

Programming skills.

Anmerkungen

Limited number of projects and participants. Specific project topics will be different each term and will be announced in a presentation during the first semester week.

T**5.125 Teilleistung: Practical Course: Natural Language Dialog Systems [T-INFO-114206]****Verantwortung:** Prof. Dr. Jan Niehues**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107177 - Practical Course: Natural Language Dialog Systems](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	6 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
SS 2026	2400187	Praktikum Natural Language Dialog Systems	2 SWS	Praktikum (P) / ● ^s	Niehues, Koneru

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Voraussetzungen

None.

T**5.126 Teilleistung: Practical Course: Real-world Vulnerability Discovery and Exploits [T-INFO-113350]****Verantwortung:** Prof. Dr. Christian Wressnegger**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106627 - Practical Course: Real-world Vulnerability Discovery and Exploits](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
4 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	241337	Practical Course: Real-world Vulnerability Discovery and Exploits	4 SWS	Praktikum (P) / ●	Wressnegger
SS 2026	241337	Practical Course: Real-world Vulnerability Discovery and Exploits	2 SWS	Praktikum (P) / ●	Wressnegger

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

Die Erfolgskontrolle erfolgt in Form einer Prüfungsleistung anderer Art nach §4 Abs. 2 Nr. 3 SPO

Voraussetzungen

Keine

Empfehlungen

Praktikum Anwendungssicherheit

T**5.127 Teilleistung: Practical Course: Robotics [T-INFO-114172]****Verantwortung:** Prof. Dr.-Ing. Tamim Asfour**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107155 - Robotics - Practical Course](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	6 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	24870	Robotics - Practical Course	4 SWS	Praktikum (P) / ●	Asfour

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). It is composed of several sub-tasks.

Voraussetzungen

Knowledge of the programming language C++ is required.

Empfehlungen

Attending the lectures Robotics I – Introduction to Robotics, Robotics II: Humanoid Robotics, Robotics III - Sensors and Perception in Robotics and Mechano-Informatics and Robotics is recommended.

T**5.128 Teilleistung: Practical Course: Route Planning [T-INFO-114899]**

Verantwortung: TT-Prof. Dr. Thomas Bläsius
Dr. rer. nat. Torsten Ueckerdt

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107600 - Practical Course: Route Planning](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	6 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
WS 25/26	2400252	Practical Course: Route Planning	4 SWS	Praktikum (P) / ●	Bläsius, Zündorf, Feilhauer, Laupichler
WS 25/26	2424305	Praktikum Algorithm Engineering-Routenplanung	4 SWS	Praktikum (P) / ●	Feilhauer, Bläsius, Zündorf

Legende: ■ Online, 🔄 Präsenz/Online gemischt, ● Präsenz, ✕ Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO)

The grade is made up of the completion of the programming task, a written evaluation of the results (approx. 10 pages) and the final presentation.

Voraussetzungen

None.

T**5.129 Teilleistung: Practical Course: Security, Usability and Society [T-INFO-110990]****Verantwortung:** Prof. Dr. Thorsten Strufe**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-105453 - Practical Course: Security, Usability and Society](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	4 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
WS 25/26	2512554	Praktikum Security, Usability and Society (Bachelor)	3 SWS	Praktikum (P) / 	Volkamer, Strufe, Berens, Fallahi, Ballreich, Hennig, Länge, Mossano, Hilt
WS 25/26	2512555	Praktikum Security, Usability and Society (Master)	3 SWS	Praktikum (P) / 	Volkamer, Strufe, Berens, Fallahi, Ballreich, Hennig, Länge, Mossano, Hilt
SS 2026	2512554	Praktikum Security, Usability and Society (Bachelor)	3 SWS	Praktikum (P) / 	Volkamer, Strufe, Hennig, Länge, Fallahi, Ballreich, Sikra, Hilt, Reimer, Algedri, Mack

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Voraussetzungen

None.

T**5.130 Teilleistung: Practical Course: Smart Energy System Lab [T-INFO-112030]****Verantwortung:** Dr.-Ing. Richard Jumar**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-105955 - Practical Course: Smart Energy System](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400230	Praktikum: Smart Energy System Lab	4 SWS	Praktikum (P) / ● ^s	Hagenmeyer, Jumar, Fernengel
SS 2026	2400170	Praktikum: Smart Energy System Lab	4 SWS	Praktikum (P) / ● ^s	Hagenmeyer, Waczowicz, Jumar, Aghaei Attar

Legende: Online, Präsenz/Online gemischt, Präsenz, x Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

An overall grade is awarded which consists of:

40% 10 min presentation (per team) + 5 min of discussion

60% Write up in the style of a research paper of around 5-6 pages (for solo participants) or 7-8 pages (for teams of two)

The students may redraw from the examination before the end of the first laboratory week. The assessment can be repeated once.

Voraussetzungen

None.

Empfehlungen

- Knowledge of the fundamentals of energy informatics is a prerequisite.
- Knowledge of the fundamentals of electrical engineering and energy technology is required.
- Knowledge of the basics of mechatronics, data analysis and signal processing is helpful.
- Knowledge of power systems or power electronics is helpful.

T**5.131 Teilleistung: Practical Course: Software Defined Networking [T-INFO-114240]****Verantwortung:** Prof. Dr. Martina Zitterbart**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107221 - Practical Course: Software Defined Networking](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	6 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	2400223	Practical Course Software Defined Networking	4 SWS	Praktikum (P) / 	König, Seehofer, Zitterbart

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Voraussetzungen

None.

Empfehlungen

Knowledge of a programming language (Java, C++, Python, ...) and the contents of the telematics lectures are assumed. Previous knowledge of SDN is not mandatory: the topic will be introduced in an introductory task at the beginning of the practical course. Note: Successful participation in the introductory assignment is a prerequisite for further participation in the practical course.

T**5.132 Teilleistung: Practical Course: Software Development and Application of Mobile, Bio-Inspired Robots [T-MACH-113854]**

Verantwortung: Prof. Dr.-Ing. Arne Rönnau
Einrichtung: KIT-Fakultät für Maschinenbau
Bestandteil von: [M-MACH-106904 - Practical Course: Software Development and Application of Mobile, Bio-Inspired Robots](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Dauer	Version
Prüfungsleistung anderer Art	6 LP	Drittelnoten	Jedes Wintersemester	1 Sem.	1

Lehrveranstaltungen					
WS 25/26	2121342	Practical Course: Software Development and Application of Mobile, Bio-Inspired Robots	4 SWS	Praktikum (P) / ●	Rönnau

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

Die Erfolgskontrolle erfolgt in Form einer Prüfungsleistung anderer Art nach § 4 Abs. 2 Nr. 3 SPO.

Es müssen Code erzeugt, eine schriftliche Ausarbeitung erstellt und eine Präsentation gehalten werden.

Voraussetzungen

Keine

Empfehlungen

Kenntnisse zu Grundlagen der Robotik aus Robotik 1 sind hilfreich. Grundkenntnisse im Umgang mit C++ oder Python und Linux werden vorausgesetzt.

Anmerkungen

Die Lehrveranstaltung wird in englischer Sprache angeboten.

Arbeitsaufwand

180 Std.

T**5.133 Teilleistung: Practical Course: Visual Computing [T-INFO-103000]****Verantwortung:** Prof. Dr.-Ing. Carsten Dachsbacher**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-101567 - Practical Course: Visual Computing](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
4

Lehrveranstaltungen					
WS 25/26	2424283	Practical course GPU-Computing	4 SWS	Praktikum (P) / ●	Klepikov, Alber, Lerzer, Dachsbacher
SS 2026	2424902	Practical course GPU-Computing	4 SWS	Praktikum (P) / ●	Lerzer, Klepikov, Dereviannykh, Dachsbacher

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO) in the form of practical work, presentations and, if applicable, a written paper

Written papers, presentations and practical work are weighted according to the course.

Voraussetzungen

None.

Empfehlungen

Programming skills in C/C++ are recommended.

T**5.134 Teilleistung: Practical Introduction to Hardware Security [T-INFO-114267]****Verantwortung:** Prof. Dr. Mehdi Baradaran Tahoori**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107241 - Practical Introduction to Hardware Security](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400033	Practical Introduction to Hardware Security	4 SWS	Vorlesung / Übung (VÜ) / ● ^s	Gnad, Tahoori
SS 2026	2400009	Practical Introduction in Hardware Security	4 SWS	Vorlesung / Übung (VÜ) / ● ^s	Tahoori, Gnad

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). 4 topics will be covered in this lecture. After each topic the student will receive an assignment. The quality of his tasks will be evaluated afterwards of its correctness.

Voraussetzungen

None.

Empfehlungen

Knowledge of Digital Design (lecture TI)

Practical Course "FPGA Programming"

T**5.135 Teilleistung: Practical SAT Solving [T-INFO-114262]**

Verantwortung: Dr. Tomas Balyo
 Dr. Ashlin Iser
 Prof. Dr. Peter Sanders
 Dr. Dominik Schreiber

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107238 - Practical SAT Solving](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	5 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
SS 2026	2400218	Practical SAT Solving	3 SWS	Vorlesung / Übung (VÜ)	Sanders, Iser, Schreiber

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) usually lasting 30 minutes.

Voraussetzungen

None.

Empfehlungen

Relevant literature will be announced in the lecture.

T**5.136 Teilleistung: Privacy Enhancing Technologies [T-INFO-110989]****Verantwortung:** Prof. Dr. Thorsten Strufe**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-105452 - Privacy Enhancing Technologies](#)**Teilleistungsart**
Prüfungsleistung mündlich**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	2400088	Privacy Enhancing Technologies	3 SWS	Vorlesung (V) / ●	Strufe

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Voraussetzungen

None.

T**5.137 Teilleistung: Probability and Computing [T-INFO-114196]**

Verantwortung: TT-Prof. Dr. Thomas Bläsius
Dr. Maximilian Katzmann
Prof. Dr. Peter Sanders

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107168 - Probability and Computing](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	5 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2400153	Probability and Computing	3 SWS	Vorlesung / Übung (VÜ) / ●	Sanders, Walzer, Hermann

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Voraussetzungen

None.

Empfehlungen

Basic knowledge of algorithms and data structures (e.g. from the lectures Algorithms 1 + 2) as well as basic knowledge of probability theory (e.g. from the lecture Introduction to Stochastics) are helpful.

T**5.138 Teilleistung: Public International Law [T-INFO-113381]****Verantwortung:** TT-Prof. Dr. Frederike Zufall**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107029 - Public International Law with an Economic Law Focus](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	3 LP	Drittelnoten	Jedes Sommersemester	2

Lehrveranstaltungen					
SS 2026	2400172	Public International Law with an Economic Law Focus	2 SWS	Vorlesung (V) / 	Kasper-Schöniger

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (§ 6 (3) SPO) whether the performance assessment is carried out

- as an oral examination (duration approx. 20 mins.) (§ 4 Abs. 2 Nr. 2 SPO) or
- as a written examination (lasting 60 mins.) (§ 4 Abs. 2 No. 1 SPO).

Voraussetzungen

None.

Empfehlungen

- General knowledge of (public) law (eg, through participating in public law or EU law modules) is helpful but not necessary.
- Interest in international affairs and politics is welcomed.

Anmerkungen

Competency Goals:

- Participating students will be able to navigate the plethora of multilateral treaties to detect relevant international law for specific cases.
- They can develop solutions for legal problems based on case law of international courts and tribunals.
- Students will be able to read and comprehend international treaties and case law.
- They will have a fundamental understand of the interplay between various subfields of public international law.
- Students can identify and explain current issues in public international law.

Content:

The lecture is designed to provide participating students with a general understanding of the foundations, subjects, and sources of public international law, its interplay with national legal regimes, and more detailed knowledge of particular subfields of public international law.

Since the lecture targets students of information systems, particular focus will be given to economic topics in international law, such as investment and trade law aspects. Due to the general importance of climate change for today's (economic) law, international climate change law and environmental law will form further focus areas.

In addition, a concise overview on human rights law, the law on State responsibility, and the peaceful settlement of disputes will be provided.

Throughout the lecture, important case law will be referenced and students are expected to read relevant cases in part to facilitate a discussion of such cases and their relevance for a subject field. Although the United Nations, including its principal judicial organ, the International Court of Justice, is one of the, if not the, key international organization in public international law, further international organizations (eg, Council of Europe, World Trade Organization) and their respective law(s) will also be touched.

Students are advised to have a statute book at hand that includes the most important international treaties and conventions (eg, Evans, Blackstone's International Law Documents, currently 15th ed 2021).

Conducting the lecture in English intends to facilitate students to link their ideas and arguments to current debates in international law.

T**5.139 Teilleistung: Python for Computational Risk and Asset Management [T-WIWI-110213]**

Verantwortung: Prof. Dr. Maxim Ulrich
Einrichtung: KIT-Fakultät für Wirtschaftswissenschaften
Bestandteil von: [M-WIWI-105032 - Data Science for Finance](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	4,5 LP	Drittelnoten	Jedes Wintersemester	3

Lehrveranstaltungen					
WS 25/26	2500016	Python for Computational Risk and Asset Management	2 SWS	Praktikum (P)	Ulrich

Erfolgskontrolle(n)

Die Prüfung erfolgt in Form einer Prüfungsleistung anderer Art. Die Prüfungsleistung anderer Art besteht aus einem Python-basierten "Takehome Exam". Am Ende der dritten Januarkalenderwoche bekommt der Student ein "Takehome Exam" ausgehändigt, welches er binnen 4 Stunden eigenständig und mittels Python bearbeitet und zurückschickt. Genaue Anweisungen werden zu Beginn der Lehrveranstaltung bekannt gegeben.

Die Prüfungsleistung anderer Art kann maximal einmal wiederholt werden. Eine fristgerechte Wiederholungsmöglichkeit findet am Ende der dritten Märzkalenderwoche des gleichen Jahres statt. Genauere Anweisungen werden zu Beginn der Lehrveranstaltung bekannt gegeben.

Voraussetzungen

Keine.

Empfehlungen

Gute Statistikkenntnisse und grundlegende Programmierkenntnisse werden empfohlen.

Arbeitsaufwand

135 Std.

T

5.140 Teilleistung: Reinforcement Learning [T-INFO-111255]

Verantwortung: TT-Prof. Dr. Rudolf Lioutikov
Prof. Dr. Gerhard Neumann

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-105623 - Reinforcement Learning](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	6 LP	Drittelnoten	Jedes Wintersemester	2

Lehrveranstaltungen					
WS 25/26	2400163	Reinforcement Learning		Vorlesung / Übung (VÜ) / ●	Neumann, Lioutikov

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The success control takes place in the form of a written exam, usually 90 minutes in length, according to § 4 Abs. 2 Nr. 1 SPO.

A bonus can be acquired through successful participation in the exercise as a success control of a different kind (§4(2), 3 SPO 2008) or study performance (§4(3) SPO 2015). The exact criteria for awarding a bonus will be announced at the beginning of the lecture. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The bonus is only valid for the main and post exams of the semester in which it was earned. After that, the grade bonus expires.

Voraussetzungen

None.

Empfehlungen

- It is recommended that students are familiar with the content of the "Foundations of Artificial Intelligence" lecture.
- Good Python knowledge is recommended.
- Good mathematical background knowledge is recommended.

T

5.141 Teilleistung: Reliable Computing I [T-INFO-101387]

Verantwortung: Prof. Dr. Mehdi Baradaran Tahoori

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-100850 - Reliable Computing I](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	3 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
WS 25/26	2424071	Reliable Computing I	2 SWS	Vorlesung (V) / 	Tahoori

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Voraussetzungen

None.

Empfehlungen

Knowledge of Digital Design and Computer Architecture is helpful.

T**5.142 Teilleistung: Research Focus Class: Blockchain & Cryptocurrencies [T-INFO-113400]****Verantwortung:** Prof. Dr. Hannes Hartenstein**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106654 - Research Focus Class: Blockchain & Cryptocurrencies](#)**Teilleistungsart**
Studienleistung**Leistungspunkte**
3 LP**Notenskala**
best./nicht best.**Turnus**
Unregelmäßig**Version**
1

Lehrveranstaltungen					
SS 2026	2400184	Research Focus Class: Blockchain & Cryptocurrencies Seminar	2 SWS	Seminar (S) / ●	Hartenstein, Spiesberger, Grundmann
SS 2026	2400185	Research Focus Class: Blockchain & Cryptocurrencies	1 SWS	Vorlesung (V) / ●	Hartenstein, Spiesberger, Grundmann

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out in form of course work (German Studienleistung, § 4 Abs. 3 SPO). A presentation must be given.

Voraussetzungen

None.

T**5.143 Teilleistung: Research Focus Class: Blockchain & Cryptocurrencies - Seminar [T-INFO-113401]****Verantwortung:** Prof. Dr. Hannes Hartenstein**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106654 - Research Focus Class: Blockchain & Cryptocurrencies](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
SS 2026	2400184	Research Focus Class: Blockchain & Cryptocurrencies Seminar	2 SWS	Seminar (S) / 	Hartenstein, Spiesberger, Grundmann
SS 2026	2400185	Research Focus Class: Blockchain & Cryptocurrencies	1 SWS	Vorlesung (V) / 	Hartenstein, Spiesberger, Grundmann

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A written paper must be prepared and a presentation given. Withdrawal is possible within two weeks of the topic being assigned.

Voraussetzungen

None.

T**5.144 Teilleistung: Research Practical Course on Behavior Foundation Models [T-INFO-115118]**

Verantwortung: TT-Prof. Dr. Rudolf Lioutikov
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107728 - Research Practical Course on Behavior Foundation Models](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	6 LP	Drittelnoten	Jedes Sommersemester	1

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). Presentation on the chosen topic at the end of the semester and written elaboration.

Voraussetzungen

None.

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Die Teilleistung [T-INFO-112772 - Research Practical Course: Interactive Learning](#) darf nicht begonnen worden sein.

Empfehlungen

We highly recommend to take this research project in combination with the Seminar on Behavior Foundation Models.

It is highly recommended to attend the “Explainable Artificial Intelligence” lecture in parallel or prior to this project.

- Experience in Machine Learning is recommended, e.g. through prior coursework.
 - The Computer Science Department offers several great lectures e.g., “Maschinelles Lernen - Grundlagen und Algorithmen” and “Deep Learning”
- A good mathematical background will be beneficial
- Python experience is recommended
- We might use the PyTorch deep learning library in the exercises. Some prior knowledge in this is helpful but not necessary.

T**5.145 Teilleistung: Research Practical Course: Artificial Intelligence & Security [T-INFO-113759]****Verantwortung:** Prof. Dr. Christian Wressnegger**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106866 - Research Practical Course: Artificial Intelligence & Security](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	24042	Research Lab: Artificial Intelligence & Security	4 SWS	Praktikum (P) / 	Wressnegger

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

Die Erfolgskontrolle erfolgt in Form einer Prüfungsleistung anderer Art nach § 4 Abs. 2 Nr. 3 SPO .

Voraussetzungen

Keine.

Empfehlungen

Grundlagen der IT-Sicherheit und der künstlichen Intelligenz werden vorausgesetzt.

T**5.146 Teilleistung: Research Practical Course: Explainable Artificial Intelligence [T-INFO-113114]****Verantwortung:** TT-Prof. Dr. Rudolf Lioutikov**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106495 - Research Practical Course: Explainable Artificial Intelligence](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	6 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2400142	Forschungspraktikum: Explainable Artificial Intelligence	4 SWS	Praktikum (P) / ●	Lioutikov, Li, Mattes

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). The overall impression is evaluated. The following partial aspects are included in the grading: Term paper (approx. 10 pages), Presentation (duration approx. 20 min.), Practical training (protocols)

Vortrag zum gewählten Thema am Ende des Semesters und schriftliche Ausarbeitung.

Voraussetzungen

None

Empfehlungen

We recommend taking this Praktikum after attending the “Explainable Artificial Intelligence” lecture in the summer semester.

We highly recommend to take this research project in combination with the “Explainable Artificial Intelligence” seminar.

- Experience in Machine Learning is recommended, e.g. through prior coursework.
 - The Computer Science Department offers several great lectures e.g., “Maschinelles Lernen - Grundlagen und Algorithmen” and “Deep Learning”
- A good mathematical background will be beneficial
- Python experience is recommended
- We might use the PyTorch deep learning library in the exercises. Some prior knowledge in this is helpful but not necessary.

Anmerkungen

It is only possible to resign within two weeks after assignment of the topic.

T**5.147 Teilleistung: Research Project Deep Learning for Robotics [T-INFO-114203]****Verantwortung:** Prof. Dr. Gerhard Neumann**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107174 - Research Project Deep Learning for Robotics](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	6 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2400091	Deep Learning for Robotics	4 SWS	Praktikum (P) / ●	Neumann

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

It is only possible to resign within two weeks after assignment of the topic.

Es müssen eine schriftliche Ausarbeitung erstellt und eine Präsentation gehalten werden.

- The discussed algorithms have to be implemented successfully.
- The experiments need to be conducted scientifically and need to be well documented.
- The final report is well written and well structured
- The final presentation is well prepared

Voraussetzungen

None.

Empfehlungen

- Experience in Machine Learning is recommended.
- Python experience is recommended
- We will use the PyTorch deep learning library. Some prior knowledge in this is helpful but not necessary.

T**5.148 Teilleistung: Research Project: Generative AI for Autonomous Agents [T-INFO-114189]****Verantwortung:** Prof. Dr. Gerhard Neumann**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107163 - Research Project: Generative AI for Autonomous Agents](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	2400049	Research Project: Generative AI for Autonomous Agents	4 SWS	Praktikum (P) / ●	Neumann

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

- The discussed algorithms have to be implemented successfully.
- The experiments need to be conducted scientifically and need to be well documented.
- The final report is well written and well structured
- The final presentation is well prepared

Voraussetzungen

None.

Empfehlungen

- Experience in Machine Learning is recommended.
- Python experience is recommended
- We will use the PyTorch deep learning library. Some prior knowledge in this is helpful but not necessary.

T

5.149 Teilleistung: Resilient Networking [T-INFO-111209]

Verantwortung: Prof. Dr. Thorsten Strufe
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-105591 - Resilient Networking](#)

Teilleistungsart
 Prüfungsleistung mündlich

Leistungspunkte
 6 LP

Notenskala
 Drittelnoten

Turnus
 Jedes Wintersemester

Version
 1

Lehrveranstaltungen					
WS 25/26	2400134	Resilient Networking	3 SWS	Vorlesung (V) / ☞	Strufe
WS 25/26	2400136	Resilient Networking	1 SWS	Übung (Ü) / ☞	Strufe

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Voraussetzungen

None.

Empfehlungen

Basics from cryptography and computer networks are helpful.

T**5.150 Teilleistung: Robotic Intelligence: Navigation and Mobile Manipulation [T-MACH-114952]****Verantwortung:** Prof. Dr.-Ing. Arne Rönnau**Einrichtung:** KIT-Fakultät für Maschinenbau**Bestandteil von:** [M-MACH-107621 - Robotic Intelligence: Navigation and Mobile Manipulation](#)

Teilleistungsart Prüfungsleistung schriftlich	Leistungspunkte 6 LP	Notenskala Drittelnoten	Turnus Jedes Sommersemester	Dauer 1 Sem.	Version 1
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Lehrveranstaltungen					
SS 2026	2121333	Robotic Intelligence: Navigation and Mobile Manipulation	2 SWS	Vorlesung (V) / 	Rönnau
SS 2026	2121334	Robotic Intelligence: Navigation and Mobile Manipulation (Exercise)	1 SWS	Übung (Ü) / 	Rönnau

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

Schriftliche Klausur: Dauer ca. 60 Minuten. Ist die schriftliche Prüfung bestanden, kann durch erfolgreiche Bearbeitung der Übungen ein Notenbonus erreicht werden.

Voraussetzungen

Die Teilleistung "T-MACH-114034 – Robotic Intelligence for Mobile Systems" darf nicht begonnen sein.

Anmerkungen

Die Lehrveranstaltung wird in englischer Sprache angeboten.

Arbeitsaufwand

180 Std.

T**5.151 Teilleistung: Robotic Intelligence: Perception and Scene Understanding [T-MACH-114813]****Verantwortung:** Prof. Dr.-Ing. Arne Rönnau**Einrichtung:** KIT-Fakultät für Maschinenbau**Bestandteil von:** [M-MACH-107558 - Robotic Intelligence: Perception and Scene Understanding](#)**Teilleistungsart**
Prüfungsleistung schriftlich**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Dauer**
1 Sem.**Version**
1

Lehrveranstaltungen					
WS 25/26	2121331	Robotic Intelligence: Perception and Scene Understanding	2 SWS	Vorlesung (V) / 	Rönnau
WS 25/26	2121332	Robotic Intelligence: Perception and Scene Understanding (Exercise)	1 SWS	Übung (Ü) / 	Rönnau

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

Schriftliche Klausur: Dauer 60 Minuten. Durch erfolgreiche Bearbeitung der Übungen kann ein Notenbonus erreicht werden.

Voraussetzungen

"T-MACH-114034 - Robotic Intelligence for Mobile Systems" darf nicht begonnen sein.

Anmerkungen

Die Lehrveranstaltung wird in englischer Sprache angeboten.

Arbeitsaufwand

180 Std.

T**5.152 Teilleistung: Robotics I - Introduction to Robotics [T-INFO-114190]****Verantwortung:** Prof. Dr.-Ing. Tamim Asfour**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107162 - Robotics I - Introduction to Robotics](#)**Teilleistungsart**
Prüfungsleistung schriftlich**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	2424152	Robotics I - Introduction to Robotics	4 SWS	Vorlesung (V) / 	Asfour, Mombaur

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 120 minutes.

Voraussetzungen

none.

T**5.153 Teilleistung: Robotics II - Humanoid Robotics [T-INFO-114152]****Verantwortung:** Prof. Dr.-Ing. Tamim Asfour**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107123 - Robotics II - Humanoid Robotics](#)**Teilleistungsart**
Prüfungsleistung schriftlich**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	2400074	Robotics II: Humanoid Robotics	2 SWS	Vorlesung (V) / 	Asfour

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) of, in general, 60 minutes.

Voraussetzungen

- M-INFO-100816 - Robotics II - Learning and planning robots Module must not have been started.
- T-INFO-101391 - Anthropomatics: Humanoid Robotics Partial work must not have been started.

Empfehlungen

Having visited the lecture on Robotics I – Introduction to Robotics is recommended

T**5.154 Teilleistung: Robotics III - Sensors and Perception in Robotics [T-INFO-114155]****Verantwortung:** Prof. Dr.-Ing. Tamim Asfour**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107130 - Robotics III - Sensors and Perception in Robotics](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	3 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	2400067	Robotics III: Sensors and Perception in Robotics	2 SWS	Vorlesung (V) / 	Asfour, Triebel

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Voraussetzungen

none.

Empfehlungen

Having visited the lectures on Robotics I – Introduction to Robotics is recommended.

T**5.155 Teilleistung: Robotics Institute Germany - Lecture Series [T-INFO-114895]**

Verantwortung: Prof. Dr.-Ing. Tamim Asfour
 TT-Prof. Dr. Barbara Bruno
 Prof. Dr. Jan Niehues

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107598 - Robotics Institute Germany - Lecture Series](#)

Teilleistungsart	Leistungspunkte	Notenskala	Dauer	Version
Prüfungsleistung mündlich	3 LP	Drittelnoten	1 Sem.	1

Lehrveranstaltungen					
WS 25/26	2400246	RIG Lecture Series	2 SWS	Vorlesung (V) / 	Bruno, Niehues, Asfour

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

Depending on the number of attending students, the assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting approx. 20 minutes or as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting approx. 120 minutes. The art of assessment will be announced at least six weeks before the end of the course. The default type of examination is oral.

Voraussetzungen

None.

Empfehlungen

Knowledge of the content of module Robotics I - Introduction to Robotics is helpful.

T**5.156 Teilleistung: Sampling Methods for Machine Learning [T-INFO-114133]****Verantwortung:** Prof. Dr.-Ing. Uwe Hanebeck**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107090 - Sampling Methods for Machine Learning](#)**Teilleistungsart**
Prüfungsleistung mündlich**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	2400194	Sampling Methods for Machine Learning	3 SWS	Vorlesung / Übung (VÜ) / 	Hanebeck, Frisch

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an oral examination, lasting 20 minutes in accordance with Section 4 (2) No. 2 SPO.

Voraussetzungen

Additional certificate for digital exercise (Übungsschein)

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Die Teilleistung [T-INFO-114134 - Sampling Methods for Machine Learning - Pass](#) muss begonnen worden sein.

Empfehlungen

Knowledge of a higher programming language with sophisticated libraries for scientific-numerical computing (e.g. Julia, Matlab, Python) is advantageous.

T**5.157 Teilleistung: Sampling Methods for Machine Learning - Pass [T-INFO-114134]****Verantwortung:** Prof. Dr.-Ing. Uwe Hanebeck**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107090 - Sampling Methods for Machine Learning](#)**Teilleistungsart**
Studienleistung**Leistungspunkte**
0 LP**Notenskala**
best./nicht best.**Turnus**
Jedes Sommersemester**Version**
1**Lehrveranstaltungen**

SS 2026	2400194	Sampling Methods for Machine Learning	3 SWS	Vorlesung / Übung (VÜ) / 	Hanebeck, Frisch
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Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

Digital exercise:

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Empfehlungen

Knowledge of a higher programming language with sophisticated libraries for scientific-numerical computing (e.g. Julia, Matlab, Python) is advantageous

T**5.158 Teilleistung: Scientific Methods to Design and Analyze Secure Decentralized Systems [T-INFO-111568]**

Verantwortung: Prof. Dr. Hannes Hartenstein
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-105780 - Scientific Methods to Design and Analyze Secure Decentralized Systems](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	5 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2400009	Scientific Methods to Design and Analyze Secure Decentralized Systems / wird im WS 2025/26 nicht angeboten	3 SWS	Vorlesung / Übung (VÜ) / ●	Hartenstein, Jacob

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take the form of an oral examination of approx.

- in the form of an oral examination of approx. 30 minutes in accordance with § 4 Para. 2 No. 2 SPO

or

- in the form of a written examination in accordance with § 4 Para. 2 No. 1 SPO
takes place.

Voraussetzungen

None.

Empfehlungen

Prior knowledge on the abstract concepts as well as concrete use cases of decentralized systems is strongly recommended. The "Decentralized Systems: Fundamentals, Modeling, and Applications" lecture covers all necessary aspects, but equivalent lectures and / or self-study can also be sufficient.

T**5.159 Teilleistung: Selbstverbuchung-HOC-SPZ-FORUM-PEBA-benotet [T-INFO-111475]****Verantwortung:** Lena Coerdts**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107254 - Interdisciplinary Qualifications](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	2 LP	Drittelnoten	Jedes Semester	1

Verbuchung von ÜQ-Leistungen

Diese Teilleistung eignet sich zur Selbstverbuchung von SQ/ÜQ-Leistungen durch Studierende. Es können Leistungen der folgenden Anbieter ohne Antrag verbucht werden:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Personalentwicklung und Berufliche Ausbildung

Anmerkungen

Überfachliche Qualifikationen (ÜQ), die am House-of-Competence (HoC), Zentrum für Angewandte Kulturwissenschaft (ZAK) oder am Sprachenzentrum (SpZ) erbracht wurden, können im Selfservice zugeordnet werden.

Wählen Sie dazu zunächst in Ihrem

Studienablaufplan eine Selbstverbuchungsteilleistung und ordnen Sie dann über den Reiter "ÜQ-Leistungen" eine ÜQ-Leistung zu.

T**5.160 Teilleistung: Selbstverbuchung-HOC-SPZ-FORUM-PEBA-benotet [T-INFO-115027]****Verantwortung:** Lena Coerdts**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107254 - Interdisciplinary Qualifications](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	2 LP	Drittelnoten	Jedes Semester	1

Verbuchung von ÜQ-Leistungen

Diese Teilleistung eignet sich zur Selbstverbuchung von SQ/ÜQ-Leistungen durch Studierende. Es können Leistungen der folgenden Anbieter ohne Antrag verbucht werden:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Personalentwicklung und Berufliche Ausbildung

Anmerkungen

Überfachliche Qualifikationen (ÜQ), die am House-of-Competence (HoC), Zentrum für Angewandte Kulturwissenschaft (ZAK) oder am Sprachenzentrum (SpZ) erbracht wurden, können im Selfservice zugeordnet werden.

Wählen Sie dazu zunächst in Ihrem

Studienablaufplan eine Selbstverbuchungsteilleistung und ordnen Sie dann über den Reiter "ÜQ-Leistungen" eine ÜQ-Leistung zu.

T**5.161 Teilleistung: Selbstverbuchung-HOC-SPZ-FORUM-PEBA-benotet [T-INFO-111476]****Verantwortung:** Lena Coerdts**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107254 - Interdisciplinary Qualifications](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Jedes Semester	1

Verbuchung von ÜQ-Leistungen

Diese Teilleistung eignet sich zur Selbstverbuchung von SQ/ÜQ-Leistungen durch Studierende. Es können Leistungen der folgenden Anbieter ohne Antrag verbucht werden:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Personalentwicklung und Berufliche Ausbildung

Anmerkungen

Überfachliche Qualifikationen (ÜQ), die am House-of-Competence (HoC), Zentrum für Angewandte Kulturwissenschaft (ZAK) oder am Sprachenzentrum (SpZ) erbracht wurden, können im Selfservice zugeordnet werden.

Wählen Sie dazu zunächst in Ihrem

Studienablaufplan eine Selbstverbuchungsteilleistung und ordnen Sie dann über den Reiter "ÜQ-Leistungen" eine ÜQ-Leistung zu.

T**5.162 Teilleistung: Selbstverbuchung-HOC-SPZ-FORUM-PEBA-benotet [T-INFO-111474]****Verantwortung:** Lena Coerdts**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107254 - Interdisciplinary Qualifications](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	1 LP	Drittelnoten	Jedes Semester	1

Verbuchung von ÜQ-Leistungen

Diese Teilleistung eignet sich zur Selbstverbuchung von SQ/ÜQ-Leistungen durch Studierende. Es können Leistungen der folgenden Anbieter ohne Antrag verbucht werden:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- FORUM (ehem. ZAK) Begleitstudium
- Personalentwicklung und Berufliche Ausbildung

Anmerkungen

Überfachliche Qualifikationen (ÜQ), die am House-of-Competence (HoC), Zentrum für Angewandte Kulturwissenschaft (ZAK) oder am Sprachenzentrum (SpZ) erbracht wurden, können im Selfservice zugeordnet werden.

Wählen Sie dazu zunächst in Ihrem

Studienablaufplan eine Selbstverbuchungsteilleistung und ordnen Sie dann über den Reiter "ÜQ-Leistungen" eine ÜQ-Leistung zu.

T**5.163 Teilleistung: Selbstverbuchung-HOC-SPZ-FORUM-PEBA-benotet [T-INFO-115028]****Verantwortung:** Lena Coerdts**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107254 - Interdisciplinary Qualifications](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	1 LP	Drittelnoten	Jedes Semester	1

Verbuchung von ÜQ-Leistungen

Diese Teilleistung eignet sich zur Selbstverbuchung von SQ/ÜQ-Leistungen durch Studierende. Es können Leistungen der folgenden Anbieter ohne Antrag verbucht werden:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Personalentwicklung und Berufliche Ausbildung

Anmerkungen

Überfachliche Qualifikationen (ÜQ), die am House-of-Competence (HoC), Zentrum für Angewandte Kulturwissenschaft (ZAK) oder am Sprachenzentrum (SpZ) erbracht wurden, können im Selfservice zugeordnet werden.

Wählen Sie dazu zunächst in Ihrem

Studienablaufplan eine Selbstverbuchungsteilleistung und ordnen Sie dann über den Reiter "ÜQ-Leistungen" eine ÜQ-Leistung zu.

T**5.164 Teilleistung: Selbstverbuchung-HOC-SPZ-FORUM-PEBA-unbenotet [T-INFO-111478]****Verantwortung:** Lena Coerdts**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107254 - Interdisciplinary Qualifications](#)**Teilleistungsart**
Studienleistung**Leistungspunkte**
2 LP**Notenskala**
best./nicht best.**Turnus**
Jedes Semester**Version**
1**Verbuchung von ÜQ-Leistungen**

Diese Teilleistung eignet sich zur Selbstverbuchung von SQ/ÜQ-Leistungen durch Studierende. Es können Leistungen der folgenden Anbieter ohne Antrag verbucht werden:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Personalentwicklung und Berufliche Ausbildung

Anmerkungen

Überfachliche Qualifikationen (ÜQ), die am House-of-Competence (HoC), Zentrum für Angewandte Kulturwissenschaft (ZAK) oder am Sprachenzentrum (SpZ) erbracht wurden, können im Selfservice zugeordnet werden.

Wählen Sie dazu zunächst in Ihrem

Studienablaufplan eine Selbstverbuchungsteilleistung und ordnen Sie dann über den Reiter "ÜQ-Leistungen" eine ÜQ-Leistung zu.

T**5.165 Teilleistung: Selbstverbuchung-HOC-SPZ-FORUM-PEBA-unbenotet [T-INFO-111479]****Verantwortung:** Lena Coerdts**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107254 - Interdisciplinary Qualifications](#)**Teilleistungsart**
Studienleistung**Leistungspunkte**
3 LP**Notenskala**
best./nicht best.**Turnus**
Jedes Semester**Version**
1**Verbuchung von ÜQ-Leistungen**

Diese Teilleistung eignet sich zur Selbstverbuchung von SQ/ÜQ-Leistungen durch Studierende. Es können Leistungen der folgenden Anbieter ohne Antrag verbucht werden:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Personalentwicklung und Berufliche Ausbildung

Anmerkungen

Überfachliche Qualifikationen (ÜQ), die am House-of-Competence (HoC), Zentrum für Angewandte Kulturwissenschaft (ZAK) oder am Sprachenzentrum (SpZ) erbracht wurden, können im Selfservice zugeordnet werden.

Wählen Sie dazu zunächst in Ihrem

Studienablaufplan eine Selbstverbuchungsteilleistung und ordnen Sie dann über den Reiter "ÜQ-Leistungen" eine ÜQ-Leistung zu.

T**5.166 Teilleistung: Selbstverbuchung-HOC-SPZ-FORUM-PEBA-unbenotet [T-INFO-111477]****Verantwortung:** Lena Coerdts**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107254 - Interdisciplinary Qualifications](#)**Teilleistungsart**
Studienleistung**Leistungspunkte**
1 LP**Notenskala**
best./nicht best.**Turnus**
Jedes Semester**Version**
1**Verbuchung von ÜQ-Leistungen**

Diese Teilleistung eignet sich zur Selbstverbuchung von SQ/ÜQ-Leistungen durch Studierende. Es können Leistungen der folgenden Anbieter ohne Antrag verbucht werden:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Personalentwicklung und Berufliche Ausbildung

Anmerkungen

Überfachliche Qualifikationen (ÜQ), die am House-of-Competence (HoC), Zentrum für Angewandte Kulturwissenschaft (ZAK) oder am Sprachenzentrum (SpZ) erbracht wurden, können im Selfservice zugeordnet werden.

Wählen Sie dazu zunächst in Ihrem

Studienablaufplan eine Selbstverbuchungsteilleistung und ordnen Sie dann über den Reiter "ÜQ-Leistungen" eine ÜQ-Leistung zu.

T**5.167 Teilleistung: Selbstverbuchung-HOC-SPZ-FORUM-PEBA-unbenotet [T-INFO-115077]****Verantwortung:** Lena Coerdts**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107254 - Interdisciplinary Qualifications](#)**Teilleistungsart**
Studienleistung**Leistungspunkte**
4 LP**Notenskala**
best./nicht best.**Turnus**
Jedes Semester**Version**
1**Verbuchung von ÜQ-Leistungen**

Diese Teilleistung eignet sich zur Selbstverbuchung von SQ/ÜQ-Leistungen durch Studierende. Es können Leistungen der folgenden Anbieter ohne Antrag verbucht werden:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Personalentwicklung und Berufliche Ausbildung

Anmerkungen

Überfachliche Qualifikationen (ÜQ), die am House-of-Competence (HoC), Zentrum für Angewandte Kulturwissenschaft (ZAK) oder am Sprachenzentrum (SpZ) erbracht wurden, können im Selfservice zugeordnet werden.

Wählen Sie dazu zunächst in Ihrem

Studienablaufplan eine Selbstverbuchungsteilleistung und ordnen Sie dann über den Reiter "ÜQ-Leistungen" eine ÜQ-Leistung zu.

T**5.168 Teilleistung: Selbstverbuchung-HOC-SPZ-FORUM-PEBA-unbenotet [T-INFO-115030]****Verantwortung:** Lena Coerdts**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107254 - Interdisciplinary Qualifications](#)**Teilleistungsart**
Studienleistung**Leistungspunkte**
1 LP**Notenskala**
best./nicht best.**Turnus**
Jedes Semester**Version**
1**Verbuchung von ÜQ-Leistungen**

Diese Teilleistung eignet sich zur Selbstverbuchung von SQ/ÜQ-Leistungen durch Studierende. Es können Leistungen der folgenden Anbieter ohne Antrag verbucht werden:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Personalentwicklung und Berufliche Ausbildung

Anmerkungen

Überfachliche Qualifikationen (ÜQ), die am House-of-Competence (HoC), Zentrum für Angewandte Kulturwissenschaft (ZAK) oder am Sprachenzentrum (SpZ) erbracht wurden, können im Selfservice zugeordnet werden.

Wählen Sie dazu zunächst in Ihrem

Studienablaufplan eine Selbstverbuchungsteilleistung und ordnen Sie dann über den Reiter "ÜQ-Leistungen" eine ÜQ-Leistung zu.

T**5.169 Teilleistung: Selbstverbuchung-HOC-SPZ-FORUM-PEBA-unbenotet [T-INFO-115029]****Verantwortung:** Lena Coerdts**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107254 - Interdisciplinary Qualifications](#)**Teilleistungsart**
Studienleistung**Leistungspunkte**
2 LP**Notenskala**
best./nicht best.**Turnus**
Jedes Semester**Version**
1**Verbuchung von ÜQ-Leistungen**

Diese Teilleistung eignet sich zur Selbstverbuchung von SQ/ÜQ-Leistungen durch Studierende. Es können Leistungen der folgenden Anbieter ohne Antrag verbucht werden:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Personalentwicklung und Berufliche Ausbildung

Anmerkungen

Überfachliche Qualifikationen (ÜQ), die am House-of-Competence (HoC), Zentrum für Angewandte Kulturwissenschaft (ZAK) oder am Sprachenzentrum (SpZ) erbracht wurden, können im Selfservice zugeordnet werden.

Wählen Sie dazu zunächst in Ihrem

Studienablaufplan eine Selbstverbuchungsteilleistung und ordnen Sie dann über den Reiter "ÜQ-Leistungen" eine ÜQ-Leistung zu.

T**5.170 Teilleistung: Seminar Dependable Computing [T-INFO-105577]****Verantwortung:** Prof. Dr. Mehdi Baradaran Tahoori**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-102662 - Seminar Dependable Computing](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400030	Dependable Computing	2 SWS	Seminar (S) / 	Tahoori
SS 2026	2400030	Dependable Computing	2 SWS	Seminar (S) / 	Tahoori

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A written paper must be prepared and a presentation given. Withdrawal is possible within two weeks of the topic being assigned.

The module grade is made up of 50% of the presentation and 50% of the written paper.

Voraussetzungen

None.

Empfehlungen

Knowledge of "Dependable Computing" and "Fault Tolerant Computing" and computer architecture is helpful.

T**5.171 Teilleistung: Seminar in Privacy [T-INFO-114268]**

Verantwortung: Prof. Dr. Thorsten Strufe
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107242 - Seminar in Privacy](#)

Teilleistungsart
 Prüfungsleistung anderer Art

Leistungspunkte
 4 LP

Notenskala
 Drittelnoten

Turnus
 Jedes Sommersemester

Version
 1

Lehrveranstaltungen					
SS 2026	2400087	Seminar Privacy und Technischer Datenschutz	2 SWS	Seminar (S) / 	Strufe, Guerra Balboa

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A written paper must be prepared and a presentation given; in addition, preliminary papers must be submitted and commented on in a peer review between fellow students. Withdrawal is possible within two weeks of the topic being assigned.

Voraussetzungen

None.

Empfehlungen

Fundamentals of IT security, computer networks and distributed systems are required

T**5.172 Teilleistung: Seminar on Behavior Foundation Models [T-INFO-115119]****Verantwortung:** TT-Prof. Dr. Rudolf Lioutikov**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107729 - Seminar on Behavior Foundation Models](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	3 LP	Drittelnoten	Jedes Sommersemester	1

Erfolgskontrolle(n)

Die Erfolgskontrolle erfolgt in Form einer Prüfungsleistung anderer Art nach § 4 Abs. 2 Nr. 3 SPO .
Vortrag zum gewählten Thema am Ende des Semesters und schriftliche Ausarbeitung.

Voraussetzungen

None.

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Die Teilleistung [T-INFO-112773 - Seminar: Interactive Learning](#) darf nicht begonnen worden sein.

Empfehlungen

We highly recommend to take this seminar in combination with the Research Practical Course on Behavior Foundation Models (Forschungspraktikum).

It is highly recommended to attend the "Explainable Artificial Intelligence" lecture in parallel or prior to this seminar.

- Experience in Machine Learning is recommended, e.g. through prior coursework.
 - The Computer Science Department offers several great lectures e.g., "Maschinelles Lernen - Grundlagen und Algorithmen" and "Deep Learning "
- A good mathematical background will be beneficial
- Python experience is recommended
- We might use the PyTorch deep learning library In the exercises. Some prior knowledge in this is helpful but not necessary.

T**5.173 Teilleistung: Seminar: Advanced Topics in Natural Language Processing [T-INFO-114433]****Verantwortung:** Prof. Dr. Jan Niehues**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107353 - Seminar: Advanced Topics in Natural Language Processing](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400074	Advanced Topics in Natural Language Processing	2 SWS	Seminar (S)	Niehues, Li
SS 2026	2400217	Advanced Topics in Natural Language Processing	2 SWS	Seminar (S)	Niehues

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A written paper must be prepared and a presentation given. Withdrawal is possible within two weeks of the topic being assigned.

Voraussetzungen

None.

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Die Teilleistung [T-INFO-105653 - Seminar Advanced Topics in Machine Translation](#) darf nicht begonnen worden sein.

Empfehlungen

Knowledge from the lectures Machine Translation and Cognitive Systems

T**5.174 Teilleistung: Seminar: Advanced Topics on SAT Solving [T-INFO-114231]**

Verantwortung: Dr. Ashlin Iser
Prof. Dr. Peter Sanders

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107209 - Seminar: Advanced Topics on SAT Solving](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2400020	Advanced Topics on SAT Solving	2 SWS	Seminar (S) / 	Sanders, Iser, Schreiber

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A presentation must be given. Withdrawal is possible within two weeks of the topic being assigned.

Voraussetzungen

None.

Empfehlungen

Knowledge of the basics from "SAT Solving in Practice" is helpful.

T**5.175 Teilleistung: Seminar: Algorithm Engineering [T-INFO-112312]****Verantwortung:** Prof. Dr. Peter Sanders**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106086 - Seminar: Algorithm Engineering](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	4 LP	Drittelnoten	Unregelmäßig	1

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A written paper must be prepared and a presentation given. Withdrawal is possible within two weeks of the topic being assigned.

Voraussetzungen

None.

Empfehlungen

Knowledge of algorithms is an advantage. Exemplary lectures are Algorithms I, Algorithms II, Algorithm Engineering and Parallel Algorithms.

T**5.176 Teilleistung: Seminar: Applications and Extensions of Timed Systems [T-INFO-113132]****Verantwortung:** Jun.-Prof. Dr. Maike Schwammberger**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106512 - Seminar: Applications and Extensions of Timed Systems](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	4 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
WS 25/26	2400196	Seminar: Applications and Extensions of Timed Systems	2 SWS	Seminar (S) / 	Schwammberger

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Paper and presentation. The main language of the seminar will be English, but it is possible to write the paper either in German or English. The same holds for the presentation.

Voraussetzungen

None.

Empfehlungen

Knowledge in areas of theoretical computer science and modeling of (embedded) software systems is helpful (e.g. CTL, finite automata, first order logic). It is also helpful, but not at all necessary, to have knowledge of the topics of the summer term lecture „Timed Systems“. Necessary topics from that lecture will also be introduced in the beginning of the winter term, if necessary.

T**5.177 Teilleistung: Seminar: Artificial Intelligence for Energy Systems [T-INFO-113110]****Verantwortung:** TT-Prof. Dr. Benjamin Schäfer**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106490 - Seminar: Artificial Intelligence for Energy Systems](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	4 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2400175	Seminar: Artificial Intelligence for Energy Systems		Seminar (S) / 	Schäfer

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO), consisting of a Term paper (max. 15 pages) and a Presentation (duration approx. 30 min.)

The grading scale will be announced in the course.

Students may redraw from the examination during the first two weeks after the topic has been communicated. The assessment can be repeated once.

Voraussetzungen

None.

Empfehlungen

Previous participation in "Energieinformatik 1" and/or "Energieinformatik 2" is beneficiary but not mandatory.

T**5.178 Teilleistung: Seminar: Bionic Algorithms and Robot Technologies [T-MACH-113842]****Verantwortung:** Prof. Dr.-Ing. Arne Rönnau**Einrichtung:** KIT-Fakultät für Maschinenbau**Bestandteil von:** [M-MACH-106902 - Seminar: Bionic Algorithms and Robot Technologies](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Dauer**
1 Sem.**Version**
1

Lehrveranstaltungen					
WS 25/26	2121343	Seminar: Bionic Algorithms and Robot Technologies	2 SWS	Seminar (S) / 	Rönnau
SS 2026	2121343	Seminar: Bionic Algorithms and Robot Technologies	2 SWS	Seminar (S) / 	Rönnau

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

Die Erfolgskontrolle erfolgt durch Ausarbeiten einer schriftlichen Seminararbeit sowie deren Präsentation als Erfolgskontrolle anderer Art.

Voraussetzungen

Keine

Empfehlungen

Der Besuch der Vorlesung "Biologisch Inspirierte Roboter" ist hilfreich.

Anmerkungen

Die Lehrveranstaltung wird in englischer Sprache angeboten.

Arbeitsaufwand

90 Std.

T**5.179 Teilleistung: Seminar: Continuous Software Engineering [T-INFO-110794]****Verantwortung:** Prof. Dr.-Ing. Anne Koziolk**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-105309 - Seminar: Continuous Software Engineering](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
4 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
2

Lehrveranstaltungen					
WS 25/26	2400108	Continuous Software Engineering	2 SWS	Seminar (S)	Koziolk
SS 2026	2400166	Seminar Continuous Software Engineering	2 SWS	Seminar (S)	Koziolk

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

- the preparation of a written paper (50%)
- the assessment of two seminar papers as part of a peer review (10%)
- the preparation of presentation slides and giving a presentation (20%)
- punctuality of submissions (20%)

Voraussetzungen

None.

T

5.180 Teilleistung: Seminar: Critical Topics in AI [T-INFO-113915]**Verantwortung:** Prof. Dr. Pascal Friederich**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106958 - Seminar: Critical Topics in AI](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
WS 25/26	2400231	Seminar: Critical topics in AI	2 SWS	Seminar (S) / 	Friederich, Zhou, Torresi, Neubert, Eberhard, Schlöder
SS 2026	2400215	Seminar: Critical topics in AI	2 SWS	Seminar (S) / 	Friederich, Torresi, Neubert, Schlöder, Metni

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). The following partial aspects are included in the grading: Term paper (approx. 10-15 pages), presentation (duration 30+15 min.). The grading scale will be announced in the course. Students may redraw from the examination during the first two weeks after the topic has been communicated. The assessment can be repeated once.

Voraussetzungen

Basic knowledge in AI and Machine Learning, e.g.

- BA Informatics: Introduction to artificial intelligence

Empfehlungen

Interest in social topics and research questions is required

T**5.181 Teilleistung: Seminar: Current Trends in Theoretical Computer Science [T-INFO-114091]****Verantwortung:** Prof. Dr.-Ing. Marvin Künnemann**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107027 - Seminar: Current Trends in Theoretical Computer Science](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	4 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
SS 2026	2400101	Current Trends in Theoretical Computer Science		Seminar (S) / ●	Künnemann

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO) and consists of the overall impression during the seminar, including the presentation as session leader and a scientific report at the end of the seminar.

Voraussetzungen

None.

Empfehlungen

Basic knowledge of theoretical computer science and algorithm design is recommended.

T**5.182 Teilleistung: Seminar: Deep Learning for Robotics [T-INFO-114204]****Verantwortung:** Prof. Dr. Gerhard Neumann**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107175 - Seminar: Deep Learning for Robotics](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400099	Deep Learning for Robotics	2 SWS	Seminar (S) / 	Neumann

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Presentation on the chosen topic at the end of the semester and written elaboration

Withdrawal is possible within two weeks of the topic being assigned.

Voraussetzungen

None.

Empfehlungen

Attendance of the lecture "Machine Learning - Fundamentals and Algorithms" is recommended.

T

5.183 Teilleistung: Seminar: Embedded Systems I [T-INFO-114255]**Verantwortung:** Prof. Dr.-Ing. Jörg Henkel**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** M-INFO-107231 - Seminar: Embedded Systems I**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400092	Internet of Things		Seminar (S) / ☞	Henkel
WS 25/26	2400137	Embedded Machine Learning Seminar		Seminar (S) / ☞	Sikal, Ahmed, Dietrich, Demirdag, Henkel, Khdr, Mentzos
WS 25/26	2400148	Embedded Security and Architectures		Seminar (S) / ☞	Nassar, Khdr, Sikal, Henkel, Alsharkawy, Tobar
SS 2026	2400129	Internet of Things		Seminar (S) / ☞	Henkel
SS 2026	2400137	Embedded Machine Learning		Seminar (S) / ☞	Henkel, Sikal, Khdr, Ahmed, Demirdag, Mentzos
SS 2026	2400148	Embedded Security and Architectures		Seminar (S) / ☞	Henkel, Nassar, Khdr, Sikal, Tobar, Alsharkawy, Dietrich

Legende: 📺 Online, ☞ Präsenz/Online gemischt, ● Präsenz, ✕ Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). A written paper must be prepared and a presentation given. Withdrawal is possible within two weeks of the topic being assigned

Voraussetzungen

None.

Empfehlungen

Knowledge of IoT and embedded systems

T**5.184 Teilleistung: Seminar: Embedded Systems II [T-INFO-114256]****Verantwortung:** Prof. Dr.-Ing. Jörg Henkel**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107232 - Seminar: Embedded Systems II](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Version**
1

Lehrveranstaltungen					
SS 2026	2400129	Internet of Things		Seminar (S) / 	Henkel
SS 2026	2400137	Embedded Machine Learning		Seminar (S) / 	Henkel, Sikal, Khdr, Ahmed, Demirdag, Mentzos
SS 2026	2400148	Embedded Security and Architectures		Seminar (S) / 	Henkel, Nassar, Khdr, Sikal, Tobar, Alsharkawy, Dietrich

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). A written paper must be prepared and a presentation given. Withdrawal is possible within two weeks of the topic being assigned.

Voraussetzungen

None.

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Die Teilleistung [T-INFO-114256 - Seminar: Embedded Systems II](#) darf nicht begonnen worden sein.

Empfehlungen

Knowledge of IoT and embedded systems

Anmerkungen

This is identical to the module 'Seminars: Embedded Systems I' and enables participation in a second seminar at the CES Chair.

T**5.185 Teilleistung: Seminar: Exoskeletons & Motion Capture [T-INFO-113892]****Verantwortung:** Prof. Dr. Katja Mombaur**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106927 - Seminar: Exoskeletons & Motion Capture](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2400187	Seminar: Exoskeletons & Motion Capture	2 SWS	Seminar (S) / 	Mombaur, Lau

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

This includes the preparation of a term paper in form of a scientific paper (6 pages double column) an oral presentation with slides (30 Min + 15 Min discussion) . Students may redraw from the examination during the first two weeks after the topic has been communicated.

Participation in the block seminar is mandatory.

Students have to actively participate in all discussions.

Voraussetzungen

None.

Empfehlungen

Knowledge in Robotics (e.g. from the class Robotics 1 and follow-ups) is helpful

T**5.186 Teilleistung: Seminar: Explainable Artificial Intelligence [T-INFO-113115]****Verantwortung:** TT-Prof. Dr. Rudolf Lioutikov**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106497 - Seminar: Explainable Artificial Intelligence](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2400143	Seminar: Explainable Artificial Intelligence	2 SWS	Seminar (S) / 	Lioutikov, Mattes, Li

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Presentation on the chosen topic at the end of the semester and written paper.

Voraussetzungen

None.

Empfehlungen

We recommend taking this research project after attending the “Explainable Artificial Intelligence” lecture in the summer semester.

We highly recommend to take this seminar in combination with the “Explainable Artificial Intelligence” research project (Forschungspraktikum).

- Experience in Machine Learning is recommended, e.g. through prior coursework.
 - The Computer Science Department offers several great lectures e.g., “Maschinelles Lernen - Grundlagen und Algorithmen” and “Deep Learning”
- A good mathematical background will be beneficial
- Python experience is recommended
- We will use the PyTorch deep learning library. Some prior knowledge in this is helpful but not necessary.

Anmerkungen

It is only possible to resign within two weeks after assignment of the topic.

T**5.187 Teilleistung: Seminar: Exploring Robotics - Insights from Science Fiction, Research and Society [T-INFO-113398]****Verantwortung:** TT-Prof. Dr. Barbara Bruno**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106651 - Seminar: Exploring Robotics - Insights from Science Fiction, Research and Society](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	2400161	Exploring Robotics: Insights from Science Fiction, Research and Society	2 SWS	Seminar (S) / ●	Bruno, Maure, Yang

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

The overall impression is evaluated. The following partial aspects are included in the grading: Term paper (approx. 6 pages in double-column format), Presentation (duration approx. 10+10 min.).

Voraussetzungen

None.

Empfehlungen

Knowledge of the content of modules Robotics I - Introduction to Robotics, Robotics II: Humanoid Robotics, Robotics III - Sensors and Perception in Robotics is helpful.

T**5.188 Teilleistung: Seminar: Failing Forward: Robotics, Critical Reading and the Making of Good Research [T-INFO-114604]**

Verantwortung: TT-Prof. Dr. Barbara Bruno
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107436 - Seminar: Failing Forward: Robotics, Critical Reading and the Making of Good Research](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Dauer	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Jedes Wintersemester	1 Sem.	1

Lehrveranstaltungen					
WS 25/26	2400194	Failing Forward: Robotics, Critical Reading and the Making of Good Research - Seminar	2 SWS	Seminar (S) / ●	Bruno, Maure, Neef

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). The overall impression is evaluated. The following partial aspects are included in the grading: Term paper (approx. 6 pages in double-column format), Presentation (duration approx. 10+10 min.).

Voraussetzungen

None.

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Die Teilleistung [T-INFO-113116 - Seminar: Human-Robot Interaction](#) darf nicht begonnen worden sein.

T**5.189 Teilleistung: Seminar: Fine-Grained Complexity Theory & Algorithms [T-INFO-113392]****Verantwortung:** Prof. Dr.-Ing. Marvin Künnemann**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106645 - Seminar: Fine-Grained Complexity Theory & Algorithms](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	4 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
SS 2026	2400153	Fine-Grained Complexity Theory & Algorithms	2 SWS	Seminar (S) / ● [§]	Künnemann

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO) and consists of a presentation and a scientific report.

Voraussetzungen

None.

Empfehlungen

Basic knowledge of theoretical computer science and algorithm design is recommended.

Concurrent or previous attendance of the lecture "Fine-Grained Complexity Theory & Algorithms" is helpful, but not required. This seminar can be attended independently.

T**5.190 Teilleistung: Seminar: Hot Topics in Artificial Intelligence & Security 1 [T-INFO-113761]****Verantwortung:** Prof. Dr. Christian Wressnegger**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106868 - Seminar: Hot Topics in Artificial Intelligence & Security 1](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
4 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	24004	Seminar: Hot Topics in Security of Machine Learning	2 SWS	Seminar (S) / 	Wressnegger, Zhao
WS 25/26	24005	Seminar: Hot Topics in Cyber-Physical Systems Security	2 SWS	Seminar (S) / 	Wressnegger

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A written paper must be prepared and a presentation given. One repetition is possible.

Voraussetzungen

None.

Empfehlungen

The basics of IT security and artificial intelligence are a prerequisite.

T**5.191 Teilleistung: Seminar: Hot Topics in Artificial Intelligence & Security 2 [T-INFO-113762]****Verantwortung:** Prof. Dr. Christian Wressnegger**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106869 - Seminar: Hot Topics in Artificial Intelligence & Security 2](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
4 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	24004	Seminar: Hot Topics in Security of Machine Learning	2 SWS	Seminar (S) / 	Wressnegger, Zhao
WS 25/26	24005	Seminar: Hot Topics in Cyber-Physical Systems Security	2 SWS	Seminar (S) / 	Wressnegger

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A written elaboration must be prepared and a presentation must be given. Withdrawal is possible within two weeks after assignment of the topic. One repetition is possible.

Voraussetzungen

None.

Anmerkungen

The basics of IT security and artificial intelligence are a prerequisite.

T**5.192 Teilleistung: Seminar: Hot Topics in Bioinformatics [T-INFO-101287]****Verantwortung:** Prof. Dr. Alexandros Stamatakis**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-100750 - Seminar: Hot Topics in Bioinformatics](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
2

Lehrveranstaltungen					
SS 2026	2400011	Hot Topics in Bioinformatics	2 SWS	Seminar (S) / 	Stamatakis

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO). (Weighting of presentation and written report: 50% each)

Voraussetzungen

The exam in Introduction to Bioinformatics for Computer Scientists must have been passed in one of the preceding semesters.

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Das Modul [M-INFO-100749 - Introduction to Bioinformatics for Computer Scientists](#) muss erfolgreich abgeschlossen worden sein.

Empfehlungen

Basic knowledge in the areas of theoretical computer science (algorithms, data structures) and technical computer science (sequential optimisation in C or C++, computer architectures, parallel programming, vector processors) will be beneficial.

T**5.193 Teilleistung: Seminar: Hot Topics in Decentralized Systems [T-INFO-109922]**

Verantwortung: Prof. Dr. Hannes Hartenstein
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-104891 - Seminar: Hot Topics in Decentralized Systems](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Unregelmäßig	1

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A written paper must be prepared and a presentation given. Withdrawal is possible within two weeks of the topic being assigned.

Voraussetzungen

None.

Empfehlungen

Knowledge of the basics of IT security management for networked systems and the basic security module is helpful.

T**5.194 Teilleistung: Seminar: Hot Topics in Explainable Artificial Intelligence (XAI) [T-INFO-112917]****Verantwortung:** Prof. Dr. Christian Wressnegger**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106392 - Seminar: Hot Topics in Explainable Artificial Intelligence \(XAI\)](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
4 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
2

Lehrveranstaltungen					
SS 2026	24005	Seminar: Hot Topics in Explainable Artificial Intelligence (XAI)	2 SWS	Seminar (S) / 	Wressnegger, Noppel

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A written paper (seminar paper) must be prepared and a presentation must be given.

Voraussetzungen

None.

T**5.195 Teilleistung: Seminar: Information Systems [T-INFO-115055]****Verantwortung:** Prof. Dr.-Ing. Klemens Böhm**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107686 - Seminar: Information Systems](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Semester**Dauer**
1 Sem.**Version**
1

Lehrveranstaltungen					
SS 2026	2400131	Advances in Data Science	2 SWS	Seminar (S) / ●	Böhm, Matteucci, Friederich

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

Success is assessed by preparing a written seminar paper and presenting it as a graded assessment of a different kind in accordance with Section 4 (2) No. 3 SPO. The seminar grade corresponds to the written performance, but can be lowered or raised by up to two grade levels by the presentation performance. If the seminar paper is discontinued after the topic has been issued, the seminar is assessed with a grade of 5.0.

The coursework is passed if the written paper and the presentations have each been passed individually; regular attendance at the sessions and active participation in the content-related discussions are also required to pass.

Voraussetzungen

None.

Empfehlungen

Lectures of the Chair of Information Management Systems that match the topic of the seminar are strongly recommended.

T**5.196 Teilleistung: Seminar: Interpretability and Causality in Machine Learning [T-INFO-114237]****Verantwortung:** Jun.-Prof. Dr. Jan Stühmer**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107217 - Seminar: Interpretability and Causality in Machine Learning](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	2400181	Interpretability and Causality in Machine Learning	2 SWS	Seminar (S) / 	Stühmer

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A written elaboration must be prepared and a presentation must be given. Students may redraw from the examination during the first two weeks after the topic has been communicated. The assessment can be repeated once.

Voraussetzungen

None.

Empfehlungen

Attendance of the lecture "Machine Learning - Fundamentals and Algorithms" is recommended.

T

5.197 Teilleistung: Seminar: Law and Legal Studies [T-INFO-114094]

Verantwortung: Gustavo Gil Gasiola
Maitrayee Pathak

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107028 - Seminar: Law and Legal Studies](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Jedes Semester	1

Lehrveranstaltungen					
WS 25/26	2400177	Human vs. Machine: (Un-)Biased Legal Decision-Making?	2 SWS	Seminar (S) / 	Holzhausen
WS 25/26	2400184	EU Digital Regulatory Framework	2 SWS	Seminar (S) / 	Zufall
WS 25/26	2400203	EU Fundamental Rights and EU Economic Law in Smart Cities	2 SWS	Seminar (S) / 	Kasper-Schöniger
SS 2026	2400171	Regulating AI: from ethics to law	2 SWS	Seminar (S) / 	Gil Gasiola
SS 2026	2400177	Designing Data Governance of Digital Systems	2 SWS	Seminar (S) / 	Pathak
SS 2026	2400190	EU Digital Regulatory Framework	2 SWS	Seminar (S) / 	Zufall

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Voraussetzungen

None.

T**5.198 Teilleistung: Seminar: Machine Learning in Climate and Environmental Sciences [T-INFO-113519]****Verantwortung:** TT-Prof. Dr. Peer Nowack**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106719 - Seminar: Machine Learning in Climate and Environmental Sciences](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
WS 25/26	2400207	Seminar Machine Learning in Climate and Environmental Sciences (findet im WS25/26 nicht statt)	2 SWS	Seminar (S) / 	Nowack, Amiramjadi

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

In the form of a written seminar paper and the presentation of the same.

Voraussetzungen

- Familiarity with machine learning concepts and techniques.
- Basic knowledge of climate and environmental science is advantageous but not mandatory.

Empfehlungen

- An interest in climate and environmental sciences topics is a prerequisite.

T**5.199 Teilleistung: Seminar: Operating Systems [T-INFO-114230]****Verantwortung:** Prof. Dr.-Ing. Frank Bellosa**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107205 - Seminar: Operating Systems](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
WS 25/26	2400017	Hot Topics in Modern Operating Systems	2 SWS	Seminar (S) / ●	Bellosa, Khalil
SS 2026	24346	Seminar Hot Topics in Modern Operating Systems	2 SWS	Seminar (S) / ●	Bellosa, Khalil

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO), by preparing a written seminar paper and the presentation of the same.

The overall grade is made up of the graded and weighted performance assessments (usually 50 % seminar paper, 50 % presentation). An overall grade is awarded.

Voraussetzungen

None.

T**5.200 Teilleistung: Seminar: Post-Quantum Cryptography [T-INFO-111200]****Verantwortung:** Prof. Dr. Jörn Müller-Quade**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-105585 - Seminar: Post-Quantum Cryptography](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
WS 25/26	2400126	Seminar Post-Quantum Cryptography	2 SWS	Seminar (S) /	Müller-Quade, Hellmich
SS 2026	2400002	Post-Quantum Cryptography	2 SWS	Seminar (S) /	Kloof, Müller-Quade

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A written paper must be prepared and a presentation given. Withdrawal is possible within two weeks of the topic being assigned

Voraussetzungen

None.

Empfehlungen

Basic knowledge of IT-Security and cryptography are recommended.

T**5.201 Teilleistung: Seminar: Practical Graph Algorithms [T-INFO-114297]****Verantwortung:** Prof. Dr. Henning Meyerhenke**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107264 - Seminar: Practical Graph Algorithms](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	4 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
SS 2026	2400196	Practical Graph Algorithms	2 SWS	Seminar (S)	Meyerhenke

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A written paper must be prepared and a presentation given. Withdrawal is possible within two weeks of the topic being assigned.

Voraussetzungen

None.

Empfehlungen

Knowledge of algorithms, in particular graph algorithms, is a clear advantage. Exemplary lectures are Algorithms I and Algorithms II.

T**5.202 Teilleistung: Seminar: Privacy and Security [T-INFO-114236]****Verantwortung:** Prof. Dr. Thorsten Strufe**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107216 - Seminar: Privacy and Security](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	4 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2400118	Seminar Privatsphäre und Sicherheit	2 SWS	Seminar (S)	Strufe, Guerra Balboa

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A written paper must be prepared and a presentation given; in addition, preliminary papers must be submitted and commented on in a peer review between fellow students. Withdrawal is possible within two weeks of the topic being assigned.

Voraussetzungen

None.

Empfehlungen

Fundamentals of IT security, computer networks and distributed systems are required

T

5.203 Teilleistung: Seminar: Proofs from THE BOOK [T-INFO-106604]**Verantwortung:** Prof. Dr. Peter Sanders**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-103306 - Seminar: Proofs from THE BOOK](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
SS 2026	2400033	Nicht im SoSe 2026! Seminar: Proofs from THE BOOK	2 SWS	Seminar (S) / ●	Sanders, Walzer, Lehmann

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

The student must present multiple proofs over the course of the semester and moderate the ensuing discussion about those proofs. No written documents are required. Students may redraw from their participation until the end of the second seminar date.

Voraussetzungen

None.

Empfehlungen

The German version "Das Buch der Beweise" is available online at the KIT library within the KIT network. The English version "Proofs from THE BOOK" is available as a physical copy at the KIT library. We recommend having a look inside either version before registering for this seminar.

T**5.204 Teilleistung: Seminar: Quantum Information Theory [T-INFO-110904]****Verantwortung:** Prof. Dr. Jörn Müller-Quade**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-105408 - Seminar: Quantum Information Theory](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
SS 2026	2400085	Quantum Information Theory	2 SWS	Seminar (S) / 	Müller-Quade, Fruböse, Meier, Jiang

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A presentation must be given and a written elaboration of exercises must be prepared.

Withdrawal is possible within two weeks of the topic being assigned.

Voraussetzungen

None.

Empfehlungen

Basic knowledge of IT-Security and linear algebra are recommended.

T**5.205 Teilleistung: Seminar: Recent Highlights in Algorithms [T-INFO-114201]**

Verantwortung: TT-Prof. Dr. Thomas Bläsius
Prof. Dr. Peter Sanders
Dr. rer. nat. Torsten Ueckerdt

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107172 - Seminar: Recent Highlights in Algorithms](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	4 LP	Drittelnoten	Unregelmäßig	1

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO), by preparing a written seminar paper and the presentation of the same

Voraussetzungen

None.

Empfehlungen

Knowledge of the basics of graph theory and algorithm technology is helpful.

T**5.206 Teilleistung: Seminar: Secure Multiparty Computation [T-INFO-111501]****Verantwortung:** Prof. Dr. Jörn Müller-Quade**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-105761 - Seminar: Secure Multiparty Computation](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
WS 25/26	2400088	Seminar Secure Multiparty Computation	2 SWS	Seminar (S) / ●	Raiber, Jiang, Müller-Quade

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

A written paper must be prepared and/or a presentation must be given. Withdrawal is possible within two weeks of the topic being assigned.

Voraussetzungen

None.

Empfehlungen

Knowledge of the content of the lecture Cryptographic Protocols is assumed.

T**5.207 Teilleistung: Seminar: Software Architecture, Security and Privacy [T-INFO-114260]****Verantwortung:** Prof. Dr. Ralf Reussner**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107236 - Seminar: Software Architecture, Security and Privacy](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	4 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
WS 25/26	2400060	Daten in software-intensiven technischen Systemen – Modellierung – Analyse – Schutz	2 SWS	Seminar (S) / ●	Reussner, Raabe, Werner, Müller-Quade

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

- the preparation of a written paper (50%)
- the assessment of two seminar papers as part of a peer review (10%)
- the preparation of presentation slides and giving a presentation (20%)
- punctuality of submissions (20%)

Voraussetzungen

None.

T**5.208 Teilleistung: Seminar: Speech-to-Speech Translation [T-INFO-114208]****Verantwortung:** Prof. Dr. Jan Niehues**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107179 - Seminar: Speech-to-Speech Translation](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	3 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	2400032	Seminar Speech-to-Speech Translation	2 SWS	Seminar (S) / 	Niehues

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Voraussetzungen

None.

T**5.209 Teilleistung: Seminar: Type Systems and Static Program Analysis [T-INFO-114955]**

Verantwortung: Prof. Dr. Sebastian Erdweg
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107625 - Seminar: Type Systems and Static Program Analysis](#)

Teilleistungsart
 Prüfungsleistung anderer Art

Leistungspunkte
 4 LP

Notenskala
 Drittelnoten

Version
 1

Lehrveranstaltungen					
WS 25/26	2400267	Type Systems and Static Program Analysis	2 SWS	Seminar (S) / 	Erdweg, Lego, Marx

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Students must write a seminar paper, deliver a presentation, and participate in the discussion. Withdrawal is possible within two weeks after the topic has been assigned.

Voraussetzungen

None.

Empfehlungen

Students should be familiar with the foundational knowledge about type systems (e.g., typing rules, type checking, polymorphism) or static program analysis (e.g., data flow, abstract interpretation).

T

5.210 Teilleistung: Service Design Thinking [T-WIWI-102849]

Verantwortung: Prof. Dr. Gerhard Satzger
Prof. Dr. Orestis Terzidis

Einrichtung: KIT-Fakultät für Wirtschaftswissenschaften

Bestandteil von: [M-WIWI-101503 - Service Design Thinking](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung anderer Art	9 LP	Drittelnoten	Unregelmäßig	5

Lehrveranstaltungen					
WS 25/26	2595600	Service Design Thinking	2 SWS	Vorlesung (V) / 	Osaro, Terzidis, Feldmann
SS 2026	2595600	Service Design Thinking	2 SWS	Vorlesung (V) / 	Osaro, Terzidis, Feldmann

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

Die Erfolgskontrolle erfolgt in Form einer Prüfungsleistung anderer Art (Fallstudie, Workshops, Abschlusspräsentation). Die Gewichtung dieser Bestandteile für die Notenbildung wird zu Beginn der Veranstaltung bekannt gegeben.

Voraussetzungen

Keine

Empfehlungen

Diese Veranstaltung findet in englischer Sprache statt – Teilnehmer sollten sicher in Schrift und Sprache sein.

Unsere bisherigen Teilnehmer fanden es empfehlenswert, das Modul zu Beginn des Master-Programms zu belegen.

Anmerkungen

Aufgrund der Projektarbeit ist die Zahl der Teilnehmer beschränkt. Das Modul (und auch die Teilleistung) geht über zwei Semester. Es startet jedes Jahr Ende September und läuft bis Ende Juni des darauffolgenden Jahres. Ein Einstieg ist nur zu Programmstart im September (Bewerbung im Mai/Juni) möglich. Weitergehende Informationen zum Bewerbungsprozess und dem Programm selbst finden Sie in der Teilleistungsbeschreibung sowie über die Website des Programms (<https://sdtkarlsruhe.de/>). Ferner führen die Dozenten jedes Jahr im Mai eine Informationsveranstaltung zum Programm durch.

T**5.211 Teilleistung: SIL Entrepreneurship Projekt [T-WIWI-110166]**

Verantwortung: Prof. Dr. Orestis Terzidis
Einrichtung: KIT-Fakultät für Wirtschaftswissenschaften
Bestandteil von: [M-ETIT-105073 - Student Innovation Lab](#)

Teilleistungsart
 Prüfungsleistung anderer Art

Leistungspunkte
 3 LP

Notenskala
 Drittelnoten

Turnus
 Jedes Wintersemester

Version
 1

Lehrveranstaltungen					
WS 25/26	2545082	SIL Entrepreneurship Projekt	4 SWS	Seminar (S)	Terzidis
SS 2026	2545082	SIL Entrepreneurship Projekt		Seminar (S) / 	Mitarbeiter

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

Die Erfolgskontrolle erfolgt in Form einer Prüfungsleistung anderer Art. Die Note ergibt sich aus der Bewertung von zwei Seminararbeiten. Nähere Angaben erfolgen zu Beginn der Veranstaltung. Das Punkteschema für die Bewertung der beiden Seminararbeiten legt der/die Dozent/in der Lehrveranstaltung fest. Es wird zu Beginn der Lehrveranstaltung bekannt gegeben.

Voraussetzungen

Keine

Empfehlungen

Keine

Arbeitsaufwand

90 Std.

T**5.212 Teilleistung: Simulation and Optimization in Robotics and Biomechanics [T-INFO-113123]****Verantwortung:** Prof. Dr. Katja Mombaur**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106504 - Simulation and Optimization in Robotics and Biomechanics](#)**Teilleistungsart**
Prüfungsleistung mündlich**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
6

Lehrveranstaltungen					
WS 25/26	2400160	Simulation and Optimization in Robotics and Biomechanics	4 SWS	Vorlesung / Übung (VÜ) / ●	Mombaur, Ackermann

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 No. 1 SPO) lasting up to 30 minutes. It can be repeated once.

As a prerequisite for the participation in the oral exam, students must regularly and successfully participate in the exercises. Students must regularly submit exercise sheets. The number of exercise sheets and the scale for passing will be announced at the beginning of the course.

Voraussetzungen

T-INFO-114838 – Simulation and Optimization in Robotics and Biomechanics – Pass grade is required. Completion of module Robotics 1 or corresponding knowledge required
 Programing skills in C/C++

Modellierte Voraussetzungen

Es müssen die folgenden Bedingungen erfüllt werden:

1. Es muss eine von 2 Bedingungen erfüllt werden:
 1. Die Teilleistung T-INFO-101465 - Robotik I - Einführung in die Robotik muss erfolgreich abgeschlossen worden sein.
 2. Die Teilleistung T-INFO-108014 - Robotik I - Einführung in die Robotik muss erfolgreich abgeschlossen worden sein.
2. Es müssen die folgenden Bedingungen erfüllt werden:
 1. Die Teilleistung [T-INFO-114838 - Simulation and Optimization in Robotics and Biomechanics - Pass](#) muss erfolgreich abgeschlossen worden sein.

Anmerkungen

Limitation to 30 participants

T**5.213 Teilleistung: Simulation and Optimization in Robotics and Biomechanics - Pass [T-INFO-114838]**

Verantwortung: Prof. Dr. Katja Mombaur
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-106504 - Simulation and Optimization in Robotics and Biomechanics](#)
Voraussetzung für: [T-INFO-113123 - Simulation and Optimization in Robotics and Biomechanics](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Dauer	Version
Studienleistung mündlich	0 LP	best./nicht best.	Jedes Wintersemester	1 Sem.	1

Erfolgskontrolle(n)

Success is assessed in the form of coursework in accordance with Section 4 (3) SPO (ungraded exercise certificate).
 Exercise sheets must be handed in regularly to pass the course. The specific details will be announced in the lecture.

Anmerkungen

Participation in the exercise is compulsory and the contents of the exercise are relevant for the examination.

T**5.214 Teilleistung: Software Architecture and Quality [T-INFO-114261]****Verantwortung:** Prof. Dr. Ralf Reussner**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107237 - Software Architecture and Quality](#)**Teilleistungsart**
Prüfungsleistung schriftlich**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
3

Lehrveranstaltungen					
WS 25/26	2424667	Software Architecture and Quality	2 SWS	Vorlesung (V) / 	Reussner

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

VoraussetzungenThis lecture and the lectures *Component-Based Software Development* and *Software Architecture* are mutually exclusive.

T

5.215 Teilleistung: Software Engineering II [T-INFO-114259]

Verantwortung: Prof. Dr.-Ing. Anne Koziolk
 Prof. Dr. Raffaella Mirandola
 Prof. Dr. Ralf Reussner

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107235 - Software Engineering II](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	6 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	24076	Software Engineering II	4 SWS	Vorlesung (V) / ● ^s	Reussner, Mirandola, Dehghani

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 90 minutes.

Voraussetzungen

None.

Empfehlungen

The course *Software Engineering I* should already have been attended.

T**5.216 Teilleistung: Software Product Line Engineering [T-INFO-114234]****Verantwortung:** Prof. Dr.-Ing. Ina Schaefer**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107212 - Software Product Line Engineering](#)**Teilleistungsart**
Prüfungsleistung mündlich**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	2400050	Software Product Line Engineering	2 SWS	Vorlesung / Übung (VÜ) / 	Feichtinger

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an oral examination, usually lasting 25 minutes in accordance with Section 4 (2) No. 2 SPO.

Depending on the number of attending students, it will be announced six weeks before the examination (§ 6 Para. 3 SPO) whether the performance assessment will take place

- in the form of an oral examination in accordance with Section 4 (2) No. 2 SPO (as described above) or
- in the form of a written examination lasting 90 minutes in accordance with Section 4 (2) No. 1 SPO.

Voraussetzungen

None.

Empfehlungen

Basic knowledge from the lectures Software Engineering II [T-INFO-101370] and Formal Systems [T-INFO-101336] is helpful.

T**5.217 Teilleistung: Software Security Engineering [T-INFO-112862]**

Verantwortung: Dr. Christopher Gerking
Prof. Dr. Ralf Reussner

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-106344 - Software Security Engineering](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	3 LP	Drittelnoten	Jedes Sommersemester	2

Lehrveranstaltungen					
SS 2026	2400059	Software Security Engineering	2 SWS	Vorlesung (V) / 	Gerking

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting approx. 90 minutes.

Voraussetzungen

None.

Empfehlungen

Knowledge of Software Engineering I and Software Engineering II is recommended.

T**5.218 Teilleistung: Software-Evolution [T-INFO-101256]**

Verantwortung: Prof. Dr. Ralf Reussner
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-100719 - Software-Evolution](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	3 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2424164	Software-Evolution	2 SWS	Vorlesung (V) / 	Lange

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 25 minutes.

Voraussetzungen

None.

Empfehlungen

Knowledge of software technology and software architectures is helpful.

T**5.219 Teilleistung: Softwaretest and Quality Management (SQM) [T-INFO-114263]****Verantwortung:** Prof. Dr.-Ing. Ina Schaefer**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107239 - Softwaretest and Quality Management \(SQM\)](#)**Teilleistungsart**
Prüfungsleistung schriftlich**Leistungspunkte**
5 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400112	Softwaretest and Quality Management (SQM)	2 SWS	Vorlesung (V) / ●	Heydari Tabar, Schaefer
WS 25/26	2400113	Softwaretest and Quality Management (SQM) - Exercise	2 SWS	Übung (Ü) / ●	Schaefer, Heydari Tabar

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 90 minutes.

Voraussetzungen

None.

Anmerkungen

At the end of the course there is also the opportunity to be certified as an "ISTQB - Certified Tester - Foundation Level". A date and the modalities for the exam will be agreed on in the lecture.

T**5.220 Teilleistung: Systems and Software Engineering [T-ETIT-100675]****Verantwortung:** Prof. Dr.-Ing. Eric Sax**Einrichtung:** KIT-Fakultät für Elektrotechnik und Informationstechnik**Bestandteil von:** [M-ETIT-100537 - Systems and Software Engineering](#)**Teilleistungsart**
Prüfungsleistung schriftlich**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
3

Lehrveranstaltungen					
WS 25/26	2311605	Systems and Software Engineering	2 SWS	Vorlesung (V) / ●	Sax
WS 25/26	2311607	Übungen zu 2311605 Systems and Software Engineering	2 SWS	Übung (Ü) / ☞	Nägele, Kechler

Legende: ■ Online, ☞ Präsenz/Online gemischt, ● Präsenz, x Abgesagt

Erfolgskontrolle(n)

Written exam of 90 minutes.

Students are given the opportunity to earn a grade bonus through separate task assignments. If the grade of the written exam is between 4.0 and 1.3, the bonus improves the grade by a maximum of one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the lecture. Bonus points do not expire and remain valid for exams taken at a later date.

The grade is determined by the written exam and the bonus points.

Voraussetzungen

none

T**5.221 Teilleistung: Systems Engineering for Automotive Electronics [T-ETIT-100677]****Verantwortung:** Prof. Dr.-Ing. Jürgen Bortolazzi**Einrichtung:** KIT-Fakultät für Elektrotechnik und Informationstechnik**Bestandteil von:** [M-ETIT-100462 - Systems Engineering for Automotive Electronics](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	4 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	2311642	Systems Engineering for Automotive Electronics	2 SWS	Vorlesung (V) / 	Bortolazzi
SS 2026	2311644	Tutorial for 2311642 Systems Engineering for Automotive Electronics	1 SWS	Übung (Ü) / 	Beck

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

Die Art und Weise (schriftliche oder mündliche Prüfung) der Erfolgskontrolle wird zu Beginn der Veranstaltung bekanntgegeben.

Die Prüfung findet ohne Hilfsmittel statt.

Voraussetzungen

keine

Empfehlungen

Empfohlen wird der Besuch der Vorlesung SE (23611)

Anmerkungen

Die Art und Weise (schriftliche oder mündliche Prüfung) der Erfolgskontrolle wird zu Beginn der Veranstaltung bekanntgegeben.

Die Vorlesung wird im Haupttermin schriftlich geprüft, für den Nachholtermin kann die Prüfung auch mündlich erfolgen.

Die Prüfung findet ohne Hilfsmittel statt.

Der Besuch von Labor / Übung zur Vorlesung ist Voraussetzung für die Zulassung zur Prüfung.

T

5.222 Teilleistung: Telematics [T-INFO-114269]

Verantwortung: Prof. Dr. Martina Zitterbart
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: M-INFO-107243 - Telematics

Teilleistungsart
 Prüfungsleistung schriftlich

Leistungspunkte
 6 LP

Notenskala
 Drittelnoten

Turnus
 Jedes Wintersemester

Version
 1

Lehrveranstaltungen					
WS 25/26	2400007	Telematics	4 SWS	Vorlesung (V) / ●	Zitterbart, Helmig, Mahrt, Seehofer

Legende: ■ Online, 🔄 Präsenz/Online gemischt, ● Präsenz, x Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 90 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take place

- in the form of an oral examination of approx. 30 minutes in accordance with § 4 Para. 2 No. 2 SPO
- or in the form of a written examination in accordance with § 4 Para. 2 No. 1 SPO

Voraussetzungen

None.

Empfehlungen

- Contents of the lecture **Introduction to computer networks** or comparable lectures are a prerequisite.
- Attendance of the module-accompanying **basic practical course Protocol Engineering** is recommended.

T

5.223 Teilleistung: Testing Digital Systems I [T-INFO-101388]

Verantwortung: Prof. Dr. Mehdi Baradaran Tahoori
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-100851 - Testing Digital Systems I](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	3 LP	Drittelnoten	Unregelmäßig	1

Lehrveranstaltungen					
WS 25/26	2400151	Testing Digital Systems I	2 SWS	Vorlesung (V) / 	Tahoori

Legende:  Online,  Präsenz/Online gemischt,  Präsenz, x Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Voraussetzungen

None.

Empfehlungen

Knowledge of Digital Design and Computer Architecture is helpful.

T**5.224 Teilleistung: Testing Digital Systems II [T-INFO-105936]****Verantwortung:** Prof. Dr. Mehdi Baradaran Tahoori**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-102962 - Testing Digital Systems II](#)**Teilleistungsart**
Prüfungsleistung mündlich**Leistungspunkte**
3 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
1

Lehrveranstaltungen					
SS 2026	2400014	Testing Digital Systems II (findet im SS 2026 nicht statt)	2 SWS	Vorlesung (V) / 	Tahoori

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Voraussetzungen

None.

Empfehlungen

Knowledge of Digital Design and Computer Architecture is helpful.

T

5.225 Teilleistung: Text-Indexing [T-INFO-114226]

Verantwortung: Prof. Dr. Peter Sanders
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107202 - Text Indexing](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	4 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2400005	Nicht im WS 2025/26! Text Indexing	3 SWS	Vorlesung (V) / 	Kurpicz

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment consists of an oral exam (generally 15 minutes) according to § 4 Abs. 2 Nr. 2 SPO.

The examination takes place in the form of an oral examination and a project/experiment as an examination of success of a different kind.

Weighting: 80% oral examination, 20% project/experiment. An overall grade is awarded.

Voraussetzungen

None.

Empfehlungen

The lecture builds on parts of the contents of the lectures Algorithms I and Algorithms II. Corresponding knowledge is therefore helpful.

Anmerkungen

T**5.226 Teilleistung: Text-Indexing Project/Experiment [T-INFO-114227]****Verantwortung:** Prof. Dr. Peter Sanders**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107202 - Text Indexing](#)**Teilleistungsart**
Prüfungsleistung anderer Art**Leistungspunkte**
1 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Wintersemester**Version**
1

Lehrveranstaltungen					
WS 25/26	2400005	Nicht im WS 2025/26! Text Indexing	3 SWS	Vorlesung (V) / 	Kurpicz

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt**Erfolgskontrolle(n)**

The assessment is carried out in form of course work (German Studienleistung, § 4 Abs. 3 SPO). Students must regularly submit exercise sheets. The number of exercise sheets and the scale for passing will be announced at the beginning of the course. The assessment can only be repeated once.

The examination takes place in the form of an oral examination and a project/experiment as an examination of success of another type.

Weighting: 80% oral examination, 20% project/experiment. An overall grade is awarded.

Voraussetzungen

None.

Empfehlungen

The lecture builds on parts of the contents of the lectures Algorithms I and Algorithms II. Corresponding knowledge is therefore helpful.

T**5.227 Teilleistung: Theory of Satisfiability Algorithms [T-INFO-114982]**

Verantwortung: Prof. Dr.-Ing. Marvin Künnemann
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107639 - Theory of Satisfiability Algorithms](#)

Teilleistungsart
 Prüfungsleistung mündlich

Leistungspunkte
 4 LP

Notenskala
 Drittelnoten

Dauer
 1 Sem.

Version
 1

Lehrveranstaltungen					
SS 2026	2400077	Theory of Satisfiability Algorithms	2 SWS	Vorlesung / Übung (VÜ) / ●*	Künnemann

Legende: Online, Präsenz/Online gemischt, Präsenz, Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting approx. 20-30 minutes.

Voraussetzungen

None.

Empfehlungen

Basic knowledge of algorithms and data structures (e.g., Algorithms 1) and theoretical computer science (e.g., TGI) is assumed.

Basic knowledge in the analysis of randomized algorithms may be helpful (e.g., Probability and Computing), but is not a strict requirement.

T

5.228 Teilleistung: Timed Systems [T-INFO-112754]**Verantwortung:** Jun.-Prof. Dr. Maike Schwammberger**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-106293 - Timed Systems](#)**Teilleistungsart**
Prüfungsleistung mündlich**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Jedes Sommersemester**Version**
4

Lehrveranstaltungen					
SS 2026	2400146	Timed Systems	4 SWS	Vorlesung (V)	Schwammberger, Hamarneh

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take the form of an oral examination of approx.

- in the form of an oral examination of approx. 30 minutes in accordance with § 4 Para. 2 No. 2 SPO or
- in the form of a written examination in accordance with § 4 Para. 2 No. 1 SPO

takes place.

Voraussetzungen

None.

Empfehlungen

Basic knowledge in areas of theoretical computer science and modeling of (embedded) software systems is helpful (e.g. temporal logics, finite automata, predicate logic), but is not required.

Anmerkungen

The book "E.-R. Olderog, H. Dierks: Real-Time Systems" is used as reading material for some of the lecture contents (<https://doi.org/10.1017/CBO9780511619953>).

T**5.229 Teilleistung: Type Systems and Program Analysis [T-INFO-115016]****Verantwortung:** Prof. Dr. Sebastian Erdweg**Einrichtung:** KIT-Fakultät für Informatik**Bestandteil von:** [M-INFO-107664 - Type Systems and Program Analysis](#)**Teilleistungsart**
Prüfungsleistung schriftlich**Leistungspunkte**
6 LP**Notenskala**
Drittelnoten**Turnus**
Unregelmäßig**Dauer**
1 Sem.**Version**
1

Lehrveranstaltungen					
SS 2026	2400188	Type Systems and Static Program Analysis	2 SWS	Vorlesung (V) / 🌀	Erdweg, Lego, Greule
SS 2026	2400192	Übungen zur Vorlesung zu Type Systems and Static Program Analysis	2 SWS	Übung (Ü) / 🌀	Erdweg, Lego, Marx

Legende: 📺 Online, 🌀 Präsenz/Online gemischt, 🟢 Präsenz, ✕ Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 120 minutes.

Voraussetzungen

None.

Empfehlungen

Knowledge of programming semantics is helpful but not required.

Knowledge of logic inference systems is helpful but not required.

T

5.230 Teilleistung: Ubiquitous Computing [T-INFO-114188]

Verantwortung: Prof. Dr.-Ing. Michael Beigl
Einrichtung: KIT-Fakultät für Informatik
Bestandteil von: [M-INFO-107161 - Ubiquitous Computing](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung mündlich	5 LP	Drittelnoten	Jedes Wintersemester	1

Lehrveranstaltungen					
WS 25/26	2424146	Ubiquitous Computing		Vorlesung / Übung (VÜ)	Beigl, Röddiger

Erfolgskontrolle(n)

The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Voraussetzungen

None.

T

5.231 Teilleistung: Wearable Robotic Technologies [T-INFO-114145]

Verantwortung: Prof. Dr.-Ing. Tamim Asfour
Prof. Dr.-Ing. Michael Beigl

Einrichtung: KIT-Fakultät für Informatik

Bestandteil von: [M-INFO-107113 - Wearable Robotic Technologies](#)

Teilleistungsart	Leistungspunkte	Notenskala	Turnus	Version
Prüfungsleistung schriftlich	4 LP	Drittelnoten	Jedes Sommersemester	1

Lehrveranstaltungen					
SS 2026	2400062	Wearable Robotic Technologies	2 SWS	Vorlesung (V) / 	Asfour, Beigl
SS 2026	5016643	BUT - Anziehbare Robotertechnologien		Vorlesung (V) / 	Asfour

Legende:  Online,  Präsenz/Online gemischt,  Präsenz,  Abgesagt

Erfolgskontrolle(n)

The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.